Informatica

Informatica® Data Privacy Management
10.5.10

Administrator Guide

Informatica Data Privacy Management Administrator Guide
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Preface

See the *Data Privacy ManagementAdministrator Guide* to learn how to manage and configure Data Privacy Management. Understand how to configure the Data Privacy Management Service, user roles and privileges, data store groups, connectivity to data sources, subject registry, and encryption. View system-defined data domains and classification policies.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

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Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the Support option.

Part I: Service and Security

This part contains the following chapters:

- Data Privacy Management Service, 18
- Security, 36

CHAPTER 1

Data Privacy Management Service

This chapter includes the following topics:

- Data Privacy Management Service Overview, 18
- Creating the Data Privacy Management Service, 18
- Data Privacy Management Service Properties, 22
- Data Privacy Management Service Process Properties, 27
- SAML Authentication, 30
- Data Privacy Management Service Logs, 31
- Re-encrypting Repository Content, 32
- Disabling, Enabling, and Recycling the Data Privacy Management Service, 33

Data Privacy Management Service Overview

The Data Privacy Management Service is an application service that runs the Data Privacy Management application in an Informatica domain. Create a Data Privacy Management Service in the domain to access the application.

If you install Data Privacy Management in an existing domain with Enterprise Data Catalog, the Informatica domain and Enterprise Data Catalog must have SSL authentication enabled.

Creating the Data Privacy Management Service

Use the service creation wizard in the Administrator tool to create the service.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Click the domain name in the Domain Navigator pane.
4. Click the **Actions** menu in the Domain Navigator pane and select **New** > **Data Privacy Management Service**.

   The **New Data Privacy Management Service** dialog box appears.

5. On the **Step 1 of 4** page, enter the following properties:

Property	Description
Name	Name of the service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [You cannot change the name of the service after you create it.
Description	Description of the service. The description cannot exceed 765 characters.
Location	Domain and folder where the service is created. Click **Browse** to choose a different folder. You can move the service after you create it.
License	License object that allows use of the service. You cannot edit this property.
Node	Node on which the service runs.

6. Click **Next**.

   The **Step 2 of 4** page appears.

7. Enter the following properties for the Data Privacy Management repository database:

Property	Description
Database Type	The type of the repository database.
URL	The JDBC connection string used to connect to the repository database. Use the following syntax for each supported database: - IBM Db2. "jdbc:informatica:db2://<host name>:<port number>;DatabaseName=<database name>;BatchPerformanceWorkaround=true;DynamicSections=3000" - Microsoft SQL Server that uses the default instance. "jdbc:informatica:sqlserver://<host name>:<port number>;DatabaseName=<database name>;SnapshotSerializable=true" - Microsoft SQL Server that uses a named instance. "jdbc:informatica:sqlserver://<host name>\<named instance name>;DatabaseName=<database name>;SnapshotSerializable=true" - Microsoft SQL Server that uses the default instance with Windows NT credentials. "jdbc:informatica:sqlserver://<host name>:<port number>;DatabaseName=<database name>;SnapshotSerializable=true;authenticationMethod=NTLM" - Microsoft SQL Server that uses a named instance with Windows NT credentials. "jdbc:informatica:sqlserver://<host name>\<named instance name>;DatabaseName=<database name>;SnapshotSerializable=true;authenticationMethod=NTLM" - Azure SQL Server. "jdbc:informatica:sqlserver://<host name>:<port number>;DatabaseName=<database name>;SnapshotSerializable=true;SnapshotSerializable=true;EncryptionMethod=SSL;HostNameInCertificate=*.<hostnameincertificate>;ValidateServerCertificate=true" - Azure SQL Database with Active Directory authentication. "jdbc:informatica:sqlserver://<host_name>:<port_number>;database=<database_name>;encrypt=true;AuthenticationMethod=ActiveDirectoryPassword;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=<seconds>" - Oracle. "jdbc:informatica:oracle://<host name>:<port number>;SID=<database name>;MaxPooledStatements=20;CatalogOptions=0;BatchPerformanceWorkaround=true" To connect to Oracle using Oracle Connection Manager, use the following connection string: " jdbc:Informatica:oracle:TNSNamesFile=<fully qualified path to the tnsnames.ora file>;TNSServerName=<TNS server name>; " - PostgreSQL. "jdbc:informatica:postgresql://<host name>:<port number>;DatabaseName= "
Secure JDBC Parameters	The secure database parameters. If the repository database is secured with the SSL protocol, you must enter the secure database parameters. Enter the parameters as name=value pairs separated by semicolon characters (;). For example: param1=value1;param2=value2
User Name	The database user name for the repository.

Property	Description
Password	Repository database password for the database user.
Schema	The schema name for a particular database.
Tablespace	The tablespace name for a particular database. For a multi-partition IBM Db2 database, the tablespace must span a single node and a single partition.

8. Click **Test Connection** to verify that you can connect to the database.
9. Select the following option to create content for the repository:
   - **No content exists under specified connection string. Create new content.**

   Note: The Content Management Service must be running to create content for the Data Privacy Management Service. If you create content for the after you create the service, you must first disable the service. If you do not disable the service, you cannot create content.
10. Select **Enable the Data Privacy Management Service** to automatically enable the service after you create the service. If disabled, you must manually enable the service after you create the service.
11. Click **Next**.

    The **Step 3 of 4** page appears.
12. Enter the following properties for the associated application services:

Property	Description
Catalog Service Name	Name of the Catalog Service that you want to associate with the Data Privacy Management Service. The Catalog Service is an application service that runs Enterprise Data Catalog in the Informatica domain.
User Name	User name that the Data Privacy Management Service can use to access the Catalog Service.
Password	Password for the Catalog Service user.
Vibe Data Stream Service Name	Name of the Vibe Data Stream Service that you want to associate with the Data Privacy Management Service.
User Name	User name to access the Vibe Data Stream Service.
Password	Password for the Vibe Data Stream Service user.

13. Click **Next**.

    The **Step 4 of 4** page appears.

14. Enter the following HTTP configuration and SSL configuration properties:

Property	Description
HTTP Port	Port number on which the Data Privacy Management application runs. Default is 6200.
Enable Secure Communication	Enables secure communication for the Data Privacy Management Service in the domain.
HTTPS Port	Port number to use for a secure connection to the service. Use a different port number than the HTTP port number.
Keystore File	Directory that contains the keystore file that has the digital certificates.
Keystore Password	Password for the keystore file.

The domain creates the Data Privacy Management Service, creates content for the Data Privacy Management repository in the specified database, and enables the service.

15. Click **Finish**.

After you create the service through the wizard, you can edit the properties or configure other properties.

Data Privacy Management Service Properties

To view and update the Data Privacy Management Service properties, select the service in the Domain Navigator and click the **Properties** view. You can change the properties while the service is running, but you must restart the service for the properties to take effect.

You can configure the following types of Data Privacy Management Service properties:

- General
- Data Privacy Management Repository
- Associated Services
- User Activity Configuration
- Advanced Service
- Email Server Configuration

General Properties for the Data Privacy Management Service

The general properties of a Data Privacy Management Service include name, license, and node assignment.

The following table describes the general properties that you can configure for the service:

Property	Description
Name	Name of the service. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, begin with the @ symbol, or contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [After you create the service, you cannot change the service name.
Description	Description of the service that does not exceed 765 characters.
License	License object that allows you to use the service. You cannot edit this property.
Node	Node on which the service runs.

Repository Properties for the Data Privacy Management Service

The following table describes the Data Privacy Management repository properties you can configure for the Data Privacy Management Service:

Property	Description
Database Type	The type of repository database.
URL	The JDBC connection string to connect to the repository database. Use the following syntax for each supported database: - IBM Db2. `jdbc:informatica:db2://<host name>:<port number>;DatabaseName=<database name>;BatchPerformanceWorkaround=true;DynamicSections=3000` - Microsoft SQL Server that uses the default instance. `jdbc:informatica:sqlserver://<host name>:<port numbe name>;SnapshotSerializable=true` - Microsoft SQL Server that uses a named instance. `jdbc:informatica:sqlserver://<host name>\<named instan name>;SnapshotSerializable=true` - Azure SQL Server. `jdbc:informatica:sqlserver://<host name>:<port number>;DatabaseName=<database n SnapshotSerializable=true;EncryptionMethod=SSL;HostNameInCertificate=*.<hostnameincertificate` - Oracle. `jdbc:informatica:oracle://<host name>:<port number>;SID=<database name>;MaxPooledStatements=20;CatalogOptions=0;BatchPerformanceWorkaround=true` - PostgreSQL. `jdbc:informatica:postgresql://<host name>:<port number>;DatabaseName=`
Secure JDBC Parameters	If the repository database is secured with the SSL protocol, you must enter the secure database parameters. Enter the param semicolon characters (;). For example: `param1=value1;param2=value2`
User Name	The database user name for the repository.
Password	Repository database password for the database user.
Schema	The schema name for a particular database.
Tablespace	The tablespace name for a particular database. For a multi-partition IBM Db2 database, the tablespace must span a single n

Associated Services for the Data Privacy Management Service

The following table describes the properties you can configure for services associated with the Data Privacy Management Service:

Property	Description
Catalog Service Name	Name of the Catalog Service that you want to associate with the Data Privacy Management Service. The Catalog Service is an application service that runs Enterprise Data Catalog in the Informatica domain. Select a service from the list.
Persistent Masking Service Name	Name of the Persistent Masking Service that you want to associate with the Data Privacy Management Service. Select a service from the list.
User Name	User name for the Data Privacy Management Service to access the Catalog Service and Persistent Masking Service.
Password	Password for the Catalog Service and Persistent Masking Service user.

User Activity Configuration Properties for the Data Privacy Management Service

The following table describes the user activity properties that you can configure for the Data Privacy Management Service:

Property	Description
Enable User Activity	When enabled, ensures user activity data is streamed to Data Privacy Management. Default is False. **Note:** Although the TCP protocol ensures reliable transmission at the network layer, when messages are sent to the TCP listener at a very high rate, the application layer buffer on the listener might fill up with messages faster than the listener can process. In this event, some incoming messages might be dropped at the listener and not streamed to Data Privacy Management.
Event Details Retention Period (in Days)	Determines the number of days to retain user activity details and anomalies in the user activity store. The Data Privacy Management Service runs a daily retention job that purges expired data from the user activity store.
Event File Shared Location	The mount location on the domain machine where you want to store streamed user activity event messages.

Advanced Properties for the Data Privacy Management Service

The following table describes the advanced properties you can configure for the Data Privacy Management Service:

Property	Description
Minimum Conformance Percentage	Specifies the minimum percentage of values in a field that must match the data domain data match condition for Data Privacy Management to identify the field as sensitive. Default is 80.
User Activity Application Port Range	Specifies the port range for user activity applications. The range must include at least 10 ports. Enter the minimum and maximum port numbers in the range separated by a hyphen. Default is 40000 - 50000.
Cryptography Service User PIN	Enables the Soft Hardware Security Module (SoftHSM) key management tool through a command line interface (CLI) utility. Specifies a numeric, nine-digit PIN to access the key management tool. The CLI utility generates encryption keys that you can specify in Data Privacy Management encryption rule definitions for data domains and in encryption task protection properties. To change the PIN, select **Modify Cryptography Service User PIN**, enter a new PIN, and save the changes.

Email Server Configuration Properties for the Data Privacy Management Service

The following table describes the email server configuration properties for the Data Privacy Management Service:

Property	Description
Server Host Name	The SMTP outbound mail server host name. For example, enter the Microsoft Exchange Server for Microsoft Outlook.
Server Port	Port number used by the outbound SMTP mail server. Valid values are from 1 to 65535.
User Name	User name for authentication, if required by the outbound SMTP mail server.
Password	Password for authentication, if required by the outbound SMTP mail server.
Authentication Enabled	Indicates that the SMTP server is enabled for authentication. If selected, the outbound mail server requires a user name and password.
Use Security	Indicates that the SMTP server uses SSL or TLS protocol.
Security Protocol	The SSL or TLS port number for the SMTP server port property.
Sender Email Address	The email address that the Data Privacy Management Service displays in the **From** field when the service sends notification emails.

Custom Properties for the Data Privacy Management Service

You can configure custom properties to optimize performance in Data Privacy Management.

To add a custom property, click the **Edit** icon. In the **Edit Custom Properties** dialog, click **New**. In the **New Custom Property** dialog, enter a name and value.

The following table describes the advanced properties you can configure for the Data Privacy Management Service:

Purpose	Syntax
Change the amount of time to test a remote agent connection before the request times out. Enter the time to test a remote agent connection in milliseconds. Default remote agent timeout to test a connection is 10 seconds (10,000 milliseconds).	`AGENT_TESTCONN_TIMEOUT=<integer value>`
Configure a custom YARN queue for ElasticSearch, Percolator, Augmenter, and UBA Manager.	`DPM_YARN_QUEUE_NAME=<name of the Data Privacy Management YARN queue>`
Configure a custom ElasticSearch data path.	`DPM_ES_DATA_PATH=<Elasticsearch data path>.`
Set the maximum number of containers.	`UA_MAX_THREADS=<integer value>.`
Configure the number of levels of nested compressed files to which you drill down during a scan. Data Privacy Management can drill down compressed files with specific extensions.	`SatsAgentProfilingCompressedFilelevelsLimit=<integer value>`
Customize the header logo to use in PDF DSAR reports.	`DSAR_HEADER_LOGO_PATH=<Complete path and file name of the custom image to use in the template header>`
Customize the footer logo to use in PDF DSAR reports.	`DSAR_POWERED_BY_LOGO_PATH=<Complete path and file name of the custom image to use in the template footer>.`
Customize the template to use for PDF DSAR reports.	`DSAR_TEMPLATE_PATH=<Complete path and file name of the template file that you create to customize the report format>`

Data Privacy Management Service Process Properties

View the Data Privacy Management Service process nodes on the **Processes** tab.

You can edit the following types of Data Privacy Management Service process properties:

- Security Configuration
- Environment Variables
- Logger Options
- Advanced Process Configuration

If you update the logger options, the modifications take effect immediately. If you update the other process properties, restart the Data Privacy Management Service for the modifications to take effect.

Security Configuration for the Data Privacy Management Service Process

The following table describes the security properties you can configure for the Data Privacy Management Service process:

Property	Description
HTTP Port	Unique HTTP port number for the Data Privacy Management Service. Default is 6200.
Enable Secure Communication	When enabled, ensures secure and encrypted communication with the Data Privacy Management Service. When you enable secure communication, you must provide a HTTPS port and the keystore file and password. Default is true. **Note:** After you enable secure communication, you can no longer switch to HTTP mode.
HTTPS Port	Unique HTTPS port number to use for a secure connection to the Data Privacy Management Service. Use a different port number than the HTTP port number. Required if you select **Enable Secure Communication**.
Keystore File	Path and file name of the keystore file that contains the private or public key pairs and associated certificates. Required if you select **Enable Secure Communication**. You can create a keystore file with a keytool. A keytool is a utility that generates and stores private or public key pairs and associated certificates in a keystore file. You can use the self-signed certificate or a certificate that a certificate authority signed.
Keystore Password	Plain-text password for the keystore file. **Note:** To change the password for the keystore file, select the **Modify Keystore Password** check box, enter a new password, and save the changes.

Environment Variables for the Data Privacy Management Service Process

The following table describes the environment variable property that you can configure for the Data Privacy Management Service process:

Property	Description
Environment Variable	Environment variables defined for the Data Privacy Management Service process.

Logger Options for the Data Privacy Management Service Process

The following table describes the logger options that you can configure for the Data Privacy Management Service process:

Option	Description
Logging Level	The severity level for Data Privacy Management logs. Valid values are off, error, info, debug, trace, all. Default is info.
Maximum Log File Size	The maximum size for a log file in KB, MB, or GB. Configure a maximum size to enable log file rollover by file size. When the log file reaches the maximum size, the Data Privacy Management Service creates a new log file. The Data Privacy Management Service can create up to five backup log files. Default is 200M. Enter a number followed by one of the following units of measure: - K. Kilobytes - M. Megabytes - G. Gigabytes Do not use a space after the number. If you do not enter a unit of measurement, the Data Privacy Management Service uses megabytes.
Advanced Logger Options	Custom log4j.logger properties. For example, you might set the following property: `log4j.logger.org.apache.jasper.servlet.jspservlet=INFO`

Advanced Properties for the Data Privacy Management Service Process

The following table describes the advanced properties that you can configure for the Data Privacy Management Service process:

Property	Description
Number of Parallel Jobs	The maximum number of parallel jobs. Default is 100.
Additional JVM Options	You can add the following custom Java Virtual Machine (JVM) properties: **Note:** Set the heap size in the Maximum Heap Size property, not in the Additional JVM Options property. - -DdisableBlazeMode. Determines the engine that a Hadoop Scan job uses to run mappings. By default, a Hadoop Scan job uses the Blaze engine to run mappings. To enable a Hadoop scan job to use the Hive engine to run mappings, add the property in the following format: `-DdisableBlazeMode=true` - -DHOSTNAME, -DHadoopGatewayHost, and -DUserName. To enable Subject Registry when Data Privacy Management is installed on a Hadoop external cluster, add the following properties: - `-DHOSTNAME=<domain name>` - `-DHadoopGatewayHost=<Hadoop cluster gateway host name>` - `-DUserName=<domain user name>` - -DIncludeUnknownConnections. Enables an Informatica PowerCenter Scan job to import data stores that contain an unsupported data store type. When the Scan job imports an unsupported data store in Data Privacy Management, the data store is listed as a JDBC data store type. Enter the data store properties and configure JDBC connectivity to the source. After you scan the imported data stores, the Scan job can identify the proliferation information. Add the property in the following format: `-DIncludeUnknownConnections=Y` - -DLDAP_USE_SAMACCOUNTNAME. The user account name in Windows 2000 and earlier versions. If you use the `samAccountName` user account in LDAP and want the user to encrypt and decrypt data, add the property in the following format: `-DLDAP_USE_SAMACCOUNTNAME=true` When set to `true`, the property enables Data Privacy Management to import the `samAccountName` account name from LDAP as `<username>`. When the property is not set or set to `false`, Data Privacy Management imports the account name as `<username>@<DOMAIN-NAME>`. The `@<DOMAIN-NAME>` extension prevents the user from decrypting data. - -DmaxProfilingPoolConnections. Specifies the maximum number of profile mappings that the Data Privacy Management Service can execute concurrently for one Scan job. The number of profile mappings that a scan job creates depends on the type of profile in the scan. A Data Profile Scan job creates one mapping for each table in the data store. A Metadata Profile Scan job creates one mapping regardless of the number of tables in the data store. Each mapping uses one unit of the execution pool of the Data Integration Service. If you set the property to the execution pool size, the mappings from a single Data Profile Scan job might use the total execution pool. To allow multiple Scan jobs to run concurrently, minimize the number of execution pool units one Scan job can use. Informatica recommends that you set the value for the property to half or one-third of the value specified in the Maximum Profile Execution Pool Size property of the Data Integration Service. Add the property in the following format: `-DmaxProfilingPoolConnections=<number of profile mappings>` - -DSATS_THREAD_COUNT. Specifies the number of the threads that the Data Privacy Management Service uses for the Evaluate Classification Policies job and the Collect Row Count and Evaluate Classification Policies Scan job steps. Default is 10. Add the property in the following format: `-DSATS_THREAD_COUNT=<number of threads>`

Property	Description
	- -DAugmenter.PolltimeInMins. Specifies the interval at which to retrieve the status of augmenter spark jobs. Specify the time in minutes. - -DPercolator.PolltimeInMins. Specifies the interval at which to retrieve the status of percolator spark jobs. Specify the time in minutes. - -DUBAManager.PolltimeInMins. Specifies the interval at which to retrieve the status of UBA Manager spark jobs. Specify the time in minutes.
Maximum Heap Size	Maximum JVM heap size. Default is 1024.
Maximum Statements in Cache	Maximum number of cached SQL statements stored in the Data Privacy Management repository.
Shared Directory	The directory that Data Privacy Management shares with the other Informatica services. The default is: `<Informatica installation>\server\infa_shared \DataPrivacyManagementService`

SAML Authentication

You can configure Security Assertion Markup Language (SAML) authentication for Data Privacy Management.

Security Assertion Markup Language is an XML-based data format for exchanging authentication information between a service provider and an identity provider. In an Informatica domain, each web application is a service provider.

If you enable a domain to use SAML authentication, Data Privacy Management and all web applications that run in the domain use the identity provider you configure in the domain by default. You can configure web applications that run in a domain to use different identity providers.

For more information about SAML authentication in an Informatica domain, see the *Informatica Security Guide*.

SAML Configuration

Data Privacy Management can use a SAML identity provider after you configure SAML options in the domain.

To edit the Data Privacy Management Service SAML configuration, select the Data Privacy Management Service in the Informatica Administrator navigator panel, select the **Processes** tab, and then click **Edit** in the SAML Configuration section.

You can set the following properties in the SAML Configuration section:

Property	Description
Web Application ID	Optional. The ID of web application. For example, enter DPM to specify Data Privacy Management.
Identity Provider URL	Optional. The URL for the identity provider server. You must specify the complete URL string.
Service Provider ID	Optional. The relying party trust name or the service provider identifier for the domain as defined in the identity provider.
Assertion Signing Certificate Alias	Optional. The alias name specified when importing the identity provider assertion signing certificate into the truststore file used for SAML authentication. If you change the alias name, import the corresponding certificate into the truststore file on each gateway node, and then restart the node.
Clock Skew Tolerance	Optional. The allowed time difference between the identity provider host system clock and the system clock on the master gateway node. Optional. The lifetime of SAML tokens issued by the identity provider by is set according to the identity provider host system clock. The lifetime of a SAML token issued by the identity provider is valid if the start time or end time set in the token is within the specified number seconds of the system clock on the master gateway node. Values must be from 0 through 600 seconds. Default is 120 seconds.

Data Privacy Management Service Logs

The Data Privacy Management Service logs contain diagnostic information for system and user operations. Data Privacy Management uses a custom logger to log events in log files. Use the logs to view information about the tasks that the Service performs and to view messages that determine the cause of errors.

You configure the directory where the logger creates the log files. When the Data Privacy Management Service runs, the logger adds events to the log for the process that is running. You can access the log files from the directory, or you can download the logs from the Informatica Administrator.

In the properties of the Data Privacy Management Service process, you can configure the logging level to specify the severity of the events that you want to log. For example, you can set the logging level to include error, informational, debug, and trace events. You can also configure the maximum log file size.

Log Management

The logger continually logs events to a log file. As a result, the log file can contain a large number of events.

You can configure a maximum size for the log files. The logger writes log events to the log file until the maximum log file size is reached. You configure the maximum log file size in the properties of the Data Privacy Management Service process.

When the log file reaches the maximum size, the logger creates another log file and renames the original log file. The logger uses the following syntax to rename the original log file:
`DataPrivacyManagementService.log.<number>`

For example, the logger renames `DataPrivacyManagementService.log` to `DataPrivacyManagementService.log.1` after the log file reaches the maximum size.

The logger can create a maximum of five backup log files. The logger overwrites the oldest backup log file when the maximum number of log files is reached.

Downloading the Data Privacy Management Service Logs

From the Informatica Administrator, you can download the Data Privacy Management Service logs to any machine. You can download the current log file or all log files. The Data Privacy Management Service compresses the log files into a .zip file. You can extract the log files for analysis or send to Informatica Global Customer Support.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Select the Data Privacy Management Service.
4. From the Actions menu, select **Download Logs for Service**.

   The **Download the Data Privacy Management Service Logs** dialog box appears.
5. Select one of the following options:
   - **Current log file**. Downloads the most recent log file in compressed format.
   - **All log files**. Downloads the most recent log file and the backup log files in compressed format.
6. Click **OK**.

   The Informatica Administrator compresses the logs in a .zip file and downloads the .zip file to the default browser download directory.
   - The Informatica Administrator uses the following naming convention for the log files: `<DataPrivacyManagementService>currentLog.zip`
   - The Informatica Administrator uses the following naming convention to download all the log files: `<DataPrivacyManagement_Service>allLogs.zip`

Re-encrypting Repository Content

Re-encrypt secure content in the Data Privacy Management repository if the domain encryption key changes. The Data Privacy Management Service uses the domain encryption key file, named *siteKey*, to encrypt sensitive data in the repository. If the domain encryption key changes, the secure data in the repository is no longer valid. Re-encrypt the secure content in the repository with the new encryption key.

When you create the Data Privacy Management Service, the hash value of the domain encryption key is computed and stored in the repository. The Data Privacy Management Service uses the hash value to verify changes to the domain encryption key. The Data Privacy Management Service verifies encryption key changes when you enable the Data Privacy Management Service, when you log in to Informatica Administrator, or when you re-load the Informatica Administrator.

To re-encrypt the secure repository content, the Data Privacy Management Service must be disabled. By default, if an administrator runs the `infacmd migrateEncryptionKey` command to change the encryption key, the Data Privacy Management Service is disabled. The administrator must restart the domain after the command because of the detected encryption key change. If you do not re-encrypt the repository content, you cannot enable the Data Privacy Management Service.

To re-encrypt the secure repository content, select the service in the Domain Navigator. From the **Actions** menu, select **Re-encrypt Contents**. The menu option is enabled if the Data Privacy Management Service detects an encryption key change and the Data Privacy Management Service is disabled.

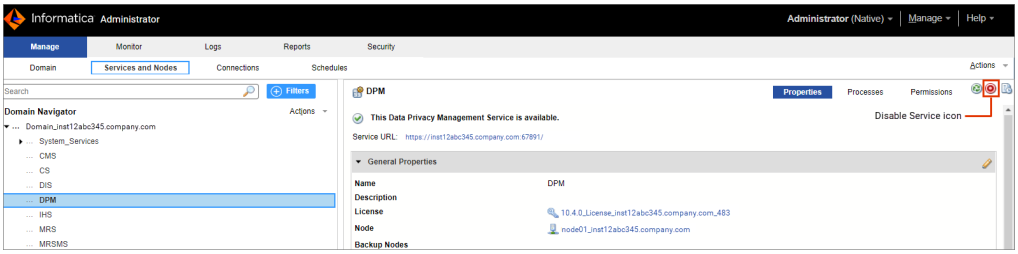
Disabling, Enabling, and Recycling the Data Privacy Management Service

You can disable, enable, and recycle the Data Privacy Management Service from the Administrator tool. Disable the service to perform maintenance or temporarily restrict users from using Data Privacy Management. Enable the service to restart it and make it available to users. To apply the updates you make to the service and service process properties, recycle the Data Privacy Management Service. When you recycle the service, the Data Privacy Management Service is disabled and enabled.

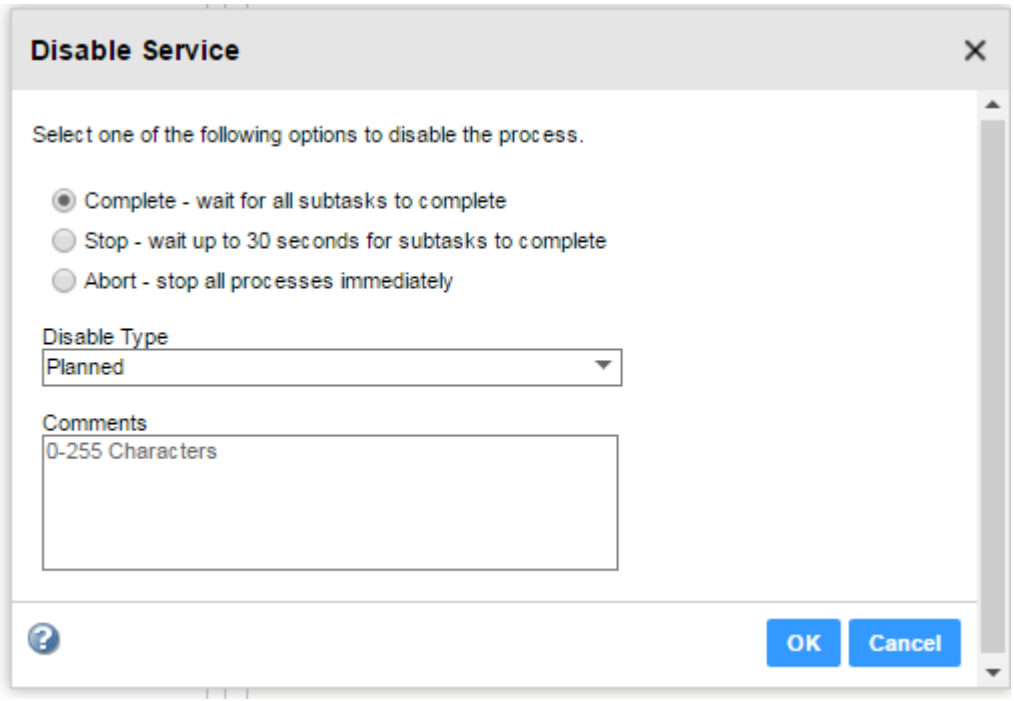
Disabling the Data Privacy Management Service

To perform maintenance or to temporarily restrict users from accessing Data Privacy Management, disable the Data Privacy Management Service .

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Select the Data Privacy Management Service in the Domain Navigator pane.
4. Click the **Disable Service** icon.



The **Disable Service** window appears.



5. Select one of the following options:
   - **Complete**. Stops all Data Privacy Management components such as the augmenter, percolator, and receiver. Also stops the user activity store. User activity updates are interrupted. Any jobs in progress either pause, fail, or stop. Resume the affected jobs after you enable the Data Privacy Management Service.
   - **Stop**. Stops all Data Privacy Management components such as the augmenter, percolator, and receiver. The user activity store is not affected. User activity updates are interrupted. Any jobs in progress either pause, fail, or stop. Resume the affected jobs after you enable the Data Privacy Management Service.
   - **Abort**. Stops the Data Privacy Management Service and components except for the user activity components. The user activity store is not affected. User activity updates are not interrupted. Any jobs in progress either pause, fail, or stop. Resume the affected jobs after you enable the Data Privacy Management Service.
6. Optionally, select a disable type.
7. Optionally, enter a comment such as the reason for disabling the service.
8. Click **OK**.

   The Data Privacy Management Service shuts down. When a service is disabled, the **Enable** option is available.

Enabling the Data Privacy Management Service

Enable the Data Privacy Management Service so users can log in to Data Privacy Management.

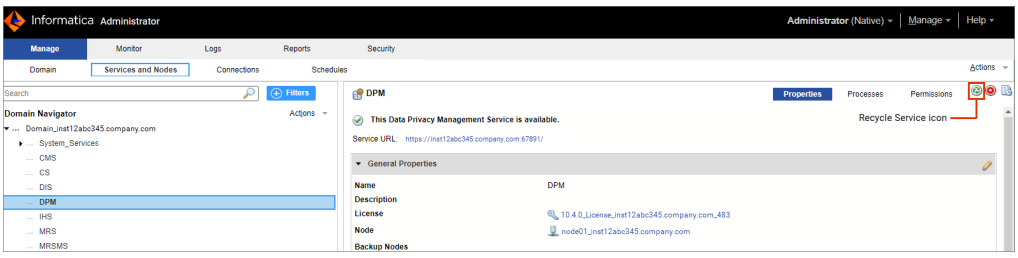
1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Select the Data Privacy Management Service in the Domain Navigator pane.
4. Click **Actions** > **Enable**.

   The Data Privacy Management Service restarts.

Recycling the Data Privacy Management Service

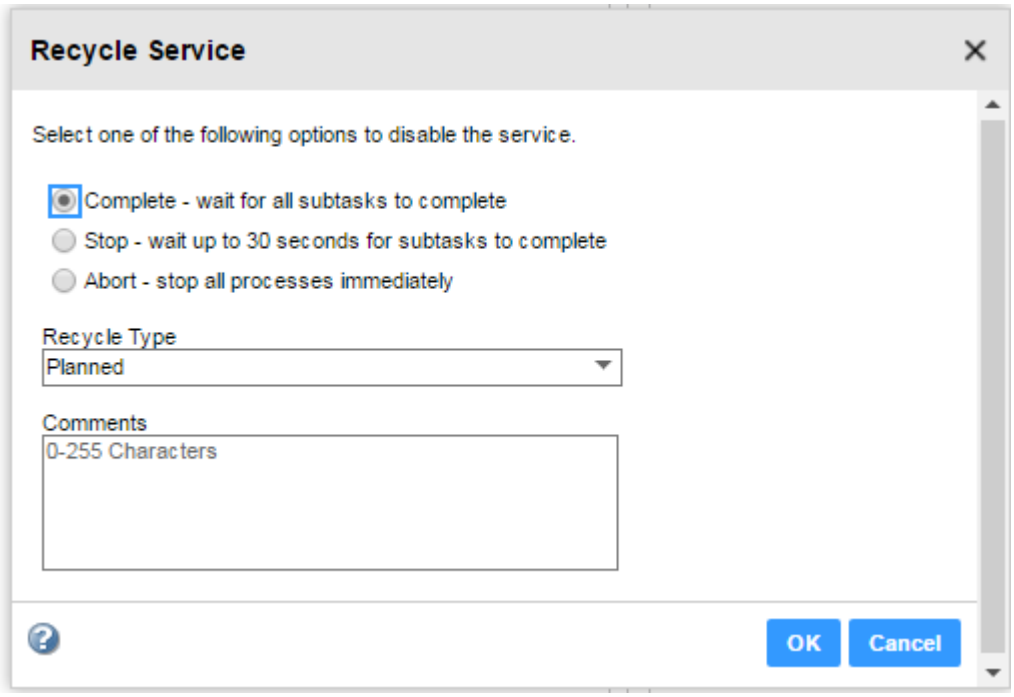
To apply the updates you make to the service and the service process properties, recycle the Data Privacy Management Service. You must also recycle the service when you update the password. When you recycle the service, the Data Privacy Management Service is disabled and then enabled.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Select the Data Privacy Management Service in the Domain Navigator pane.
4. Click the **Recycle Service** icon.



The **Recycle Service** window appears.

5. Select one of the following options:
   - **Complete**. Allows the jobs to run to completion and then shuts down the service, user activity applications, and Spark components.
   - **Stop**. Stops the jobs after 30 seconds and then shuts down the service and Spark components.
   - **Abort**. Stops all jobs immediately and then shuts down the service.



6. Optionally, select a recycle type.
7. Optionally, enter a comment such as the reason for recycling the service.
8. Click **OK**.

   The Data Privacy Management Service shuts down and restarts.

CHAPTER 2

Security

This chapter includes the following topics:

- Security Overview, 36
- Operational Security, 37
- Data Store Access, 37
- Data Privacy Management Service Privileges , 39
- Data Privacy Management Service Custom Roles, 45
- Domain Privileges, 52
- Domain System-Defined Role, 52

Security Overview

The Data Privacy Management Service is an application service in the Informatica domain. You can secure the Informatica domain for threat protection from within and external to the network where the domain runs.

The Informatica domain includes infrastructure and operational security. Infrastructure security includes user and service authentication, secure communication within the domain, and secure data storage. Operational security controls access to the data and services in the Informatica domain.

For more information about domain security, see the *Informatica Security Guide*.

Operational Security

Operational security controls access to the data and services in the Informatica domain and application clients, such as Data Privacy Management.

The following table describes the key components of operational security:

Component	Description
Users	A user account allows users to access the application services and objects in the Informatica domain and access to the application clients. You set up user accounts in the Informatica domain. You can assign groups, privileges, and roles to users. Groups, privileges, and roles determine the tasks that users can perform and the data that users can access.
Groups	A group is a set of related users that have the same authorization. You can set up groups of users and assign different roles and privileges to each group. All users in the group inherit the roles and privileges assigned to the group. The roles and privileges assigned to the group determine the tasks that users in the group can perform. You can nest groups within a group to create a hierarchy of groups. Users in the nested groups inherit the roles and privileges from the parent groups. Groups determine the data stores that users have access to. When you create a data store, you assign one or more groups to the data store. All users in the group and in the nested groups have access to the data store.
Privileges	A privilege determines an action that users can perform. For example, the export privilege allows a user to export data from Data Privacy Management. The Data Privacy Management Service includes system-defined privileges. You can assign a set of privileges to a user, group, or a role. Alternatively, you can assign custom roles to users or groups. A custom role includes system-defined privileges based on the tasks that users in the role perform. For example, the security manager role has privileges to create and review reports, view user activity and proliferation, author policies, and run scans.
Roles	A role is a set of privileges that you can assign to users or groups. You can use roles to easily manage assignments of privileges to users. You can create roles with related privileges to assign to users and groups that require the same level of access. The Data Privacy Management Service includes custom roles with assigned privileges based on the tasks that users in the role perform, such as Security Analyst, Security Manager, and Technical Administrator.

You create users, groups, and roles in the Informatica Administrator. For more information, see the *Informatica Security Guide*.

Data Store Access

The user group or user role assignment determines the data stores that users can access.

A group or a role determines access to a data store. The user privileges control the data store actions that the user can perform. For example, the privileges determine whether the user can display, edit, or delete a data store.

Group Access to Data Stores

Informatica domain groups determine the data stores that users can access in Data Privacy Management. In the Informatica Administrator, you can create groups and assign users to groups.

In Data Privacy Management, you can assign Informatica domain groups to a data store. When you assign a group to a data store, all users in the group and associated nested groups can access the data store.

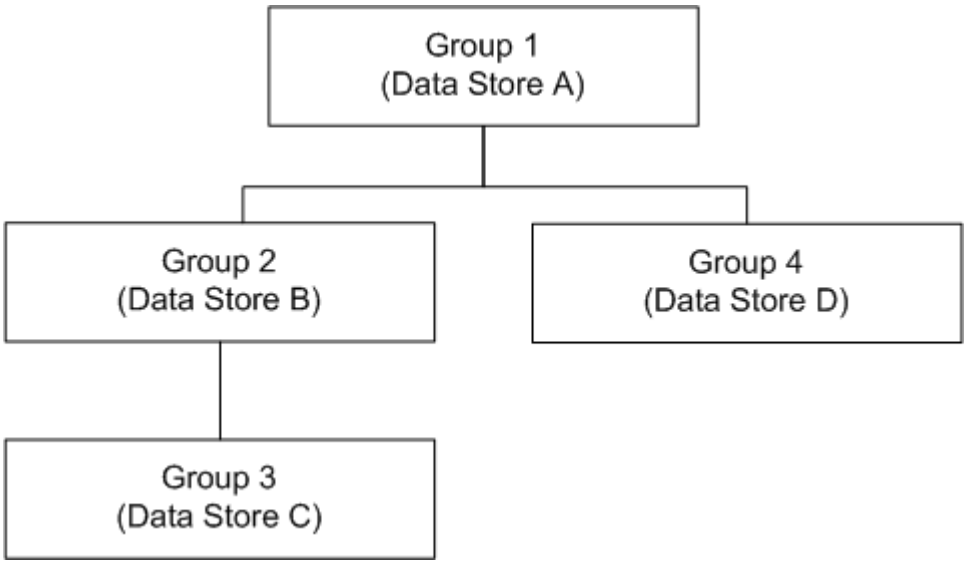
The following table describes the default groups in the Informatica domain:

Group	Description
Administrator group	The Administrator group has administrator privileges on the domain and all privileges on the Data Privacy Management Service. The administrator account created during installation is assigned to the Administrator group. Users in the Administrator group can access all data stores on the **Data Stores** workspace in Data Privacy Management, regardless of the group assigned to the data store. For all other workspaces, users in the Administrator group can access a data store only if the users belong to the group that is assigned to the data store.
Everyone group	The Everyone group includes all users in the domain. Users in the Everyone group can access a data store if the Everyone group is assigned to the data store.

Nested Group Access to Data Stores

You can nest groups to maintain a hierarchy of groups. The nested groups inherit the data store access from the parent groups.

The following image shows an example of nested groups and data store assignments:



Group 1 is the parent group to nested groups 2, 3, and 4. Group 1 is assigned to data store A. Users in groups 1, 2, 3, and 4 have access to data store A.

Group 2 is nested within group 1. Group 2 is the parent group to nested group 3. Group 2 is assigned to data store B. Users in group 2 have access to data stores A and B.

Group 3 is nested within group 2. Group 3 is assigned to data store C. Users in group 3 have access to data stores A, B, and C.

Group 4 is nested within group 1. Group 4 is assigned to data store D. Users in group 4 have access to data stores A and D.

Rules and Guidelines for Managing Groups

Consider the following rules and guidelines for managing groups in the Informatica domain:

- The domain configuration repository stores the Informatica domain users and groups. The Data Privacy Management repository stores the current assignment of a data store to a group. The repository does not store the assignment history if you change the group assignment for a data store.
- If you change user group assignments for a user who is logged in to Data Privacy Management, the changes do not go into effect until the next time the user interface is refreshed.
  For example, you add user A to group B. Group B is assigned to data store C. User A does not have access to data store C until the next time the user interface is refreshed.
- When you delete a group, the group assignment for a data store is not deleted in the Data Privacy Management repository. Before you delete a group, update the group assignment for the data stores that are in the group. If you do not remove the group assignment and you create another group with the same name, the users in the new group have access to the data store.
- When you delete a group, you delete associated nested groups.
- If you delete the groups to which a data store is assigned, only the users assigned to the Informatica domain Administrator role or the Administrator group can access the data store.

Role Access to Data Stores

Users with the default Informatica domain Administrator role can access all data stores on the **Data Stores** workspace in Data Privacy Management.

For all other workspaces, users with the default Informatica domain Administrator role can access a data store only if the users belong to the group assigned to the data store.

For all other system-defined and user-defined roles, including Data Privacy Management Administrator roles, the user group assignment controls data store access.

Data Privacy Management Service Privileges

The Data Privacy Management privileges determine the actions that users can perform in Data Privacy Management.

The following table describes each privilege group for the Data Privacy Management Service:

Privilege Group	Description
Classification	Includes privileges to manage custom risk factors, data domains, classification policies, and scans.
Dashboard	Includes privileges to view subject registry and to manage dashboard views, users, user activity, anomaly suppression, locations, proliferation, heat maps, departments, jobs, and advanced analytics.

Privilege Group	Description
Data Protection	Includes privileges to view subject registry and to manage protection extensions, tasks, encryption rules, remote agents, and risk simulation.
Discovery	Includes privileges to manage data stores.
Integration and APIs	Includes privileges to manage import, export, and API tasks.
Security Policy	Includes privileges to manage actions, security policies, and security policy violations.

For information about privileges required to perform specific tasks in Data Privacy Management, see the How-To article.

Classification Privilege Group

The privileges in the Classification privilege group determine the tasks that users can perform on data, data domains, classification policies, and scans.

The following table lists the actions that users can perform for the privileges in the Classification group:

Privilege	User Action
Add or Modify Data Domains	- Add, edit, and import data domains. - Associate protection rules to data domains. - Remove protection rules from data domains.
Add or Modify Policies	Add and edit classification policies.
Add or Modify Scans	Add and manage scans.
Customize Risk Scoring	Specify the weight of each risk factor in the risk score calculation.
Customize Severity Levels	Specify system-wide data sensitivity levels.
Delete Data Domains	Delete data domains.
Delete Policies	Delete classification policies.
Delete Scans	Delete scans.
Execute Scans	Schedule scan jobs.
Manage Custom Risk Factors	Manage custom factors to determine sensitive data risk.
Validate Classification Exceptions*	Review reports and validate the classification of objects.
View Data Domains	- View data domains. - View protection rules associated with data domains. - View data store information on the Privacy Dashboard.
View Policies	View classification policies.

Privilege	User Action
View Scans	View scan status.
* The user cannot perform the corresponding action in Data Privacy Management.	

Dashboard Privilege Group

The privileges in the Dashboard privilege group determine the dashboard tasks that users can perform.

The following table lists the actions that users can perform for the privileges in the Dashboard group:

Privilege	User Action
Access the Dashboard	View the Data Privacy Management dashboards.
Add or Modify Anomaly Suppression**	Create or edit rules to suppress an anomaly factor from triggering an anomaly.
Add or Modify Reports	Add and edit reports.
Customize Advanced Analytics Configuration	View the Anomaly Factor Weights section on the **Settings** workspace. Edit the weight of each anomaly factor that calculates the anomaly score.
Delete Anomaly Suppression	Delete an anomaly suppression rule.
Manage Jobs	Manage jobs on the **Jobs** workspace. Download job logs to view informational, warning, and error messages that a job encountered. You can pause, resume, terminate, and stop jobs.
Modify Dashboard	Rearrange and expand or compress indicators on the **Overview** workspace.
Modify Protection Field Settings	Modify protection field settings in a data store.
Modify Sensitive Field Settings	Modify sensitive field settings in a data store.
Monitor Jobs	View, open, pause, resume, skip and resume, terminate, and close jobs.
View Advanced Analytics	View anomalies.
View Anomaly Suppression	View anomaly suppression rules.
View Data Stores in Dashboard	View data stores on the **Overview** workspace.
View Departments	View departments.
View Heat Maps	View heat maps.
View Locations	View and import locations. View data in the Locations indicators on the dashboards.
View Proliferation	View data proliferation statistics.
View Subject Registry	Search for a data subject and view search results on the **Subject Registry** workspace. View data on the Privacy Dashboard.

Privilege	User Action
View User Accessibility	View user privileges.
View User Activity	View the **Top Users** indicator and the related pages on the **Overview** workspace.
View User Activity Requests*	View the user activity query in the user interface. Without this privilege, the query appears masked.
* The user cannot perform the corresponding action in Data Privacy Management. ** The user must also have the View Anomaly Suppression privilege to perform the action.	

Data Protection Privilege Group

The privileges in the Data Protection privilege group determine what users can view and the actions users can perform on extensions, tasks, remote agents, and rules.

The following table lists the actions that users can perform for the privileges in the Data Protection group:

Privilege	User Action
Add or Modify Encryption Rules	Create and edit encryption rules on the **Encryption Rules** workspace. - Create and edit encryption rules to specify the encryption technique. - Use an encryption rule to encrypt sensitive fields.
Add or Modify Proxies	Create and edit the connection to a remote agent on the **Remote Agents** workspace. - Create and edit connection properties for a subject registry remote agent and specify the unstructured data stores. - Create and edit connection properties for a protection remote agent and specify the Cloudera Hive data stores.
Add or Modify Task	- Create and edit custom, DSAR, email, protection, service management, and system log tasks for data domains, data stores, proliferations, classification policies, anomalies, and security policies. - Assign a task to another user.
Close Task	Close a custom, DSAR, email, protection, service management, or system log task.
Delete Encryption Rules	Delete an encryption rule.
Delete Proxies	Delete the connection to a subject registry or protection remote agent.
Delete Task	Delete a custom, DSAR, email, protection, service management, or system log task.
Execute Task	Schedule a protection job for a task, or run a custom, email, service management, or system log task.
Manage Extensions	Add, edit, and delete custom, email, protection, service management, and system log extensions.
Manage Protection Task Configurations	- Mark a configuration as complete. - Change the protection extension for a data store. - Assign protection rules to sensitive columns. - Configure protection rules after assignment.

Privilege	User Action
Risk Simulation and Manage Plans	Manage plans and indicators for simulating sensitive data risk.
View Encryption Rules	View the data type and encryption technique in an encryption rule.
View Extensions	View custom, email, protection, service management, and system log extensions.
View Proxies	View the connection details and the associated data stores for a subject registry or protection remote agent.
View Subject Index	Search for a data subject and view search results on the **Subject Registry** workspace.
View Task	- View tasks. - Add notes to tasks. - Upload attachments to tasks. - Reassign or reject a task. - View the Data Store Detail report. - View subject request information on the Privacy Dashboard.

Discovery Privilege Group

The privileges in the Discovery privilege group determine the tasks that users can perform on data stores.

The following table lists the actions that users can perform for the privileges in the Discovery group:

Privilege	User Action
Add or Modify Data Stores	Add and edit data stores.
Delete Data Stores	Delete data stores. **Note:** When you delete a data store, you delete the data associated with the data store. Reset classification results. When you reset classification results, you delete the column-to-data domain match results of a scan, but retain all other information.
Manage Data Store Groups	Create data store groups in a hierarchy.
Merge Data Stores	Merge data stores.
View Data Stores	View a list of data stores.

Integration and APIs Privilege Group

The privileges in the Integration and APIs privilege group determine the import and export tasks that users can perform.

The following table lists the actions that users can perform for the privileges in the Integration and APIs group:

Privilege	User Action
Access Embedded URL*	Embed the Data Privacy Management URL to link dashboard widgets.
Export Analysis	Export data from Data Privacy Management.
Import Audit Logs*	Import audit logs.
Import Classification Rules	Import data domains and classification policies. For example, you can export a list of data domains and classification policies from Data Quality and import the list into Data Privacy Management.
Import Data Owners	Import a list of data owners.
Import Data Stores	Import data stores. You can import multiple data stores at a time.
Import Enterprise User and Accessibility Information: From Active Directory or LDAP	Import user names and privileges from Active Directory or LDAP.
Import Enterprise User and Accessibility Information: From Database Catalog*	Import user names and privileges from a database catalog.
Import Enterprise User and Accessibility Information: From File	Import user names and privileges from a flat file.
Import Enterprise User and Accessibility Information: From Packaged Applications	Import user names and privileges from a packaged application.
Import Proliferation**	Import a list of data that proliferates between data stores.
Import Protection Status	Import a list of sensitive data with information about how the data is protected.
Import Sensitive Data Locations and Data Store Groups**	Import a list of the locations of sensitive data.
* The user cannot perform the corresponding action in Data Privacy Management. ** Data Privacy Management does not check user privileges when the action is performed.	

Security Policy Privilege Group

The privileges in the Security Policy privilege group determine the tasks that users can perform on the **Actions**, **Security Policy**, and **Security Policy Violations** workspaces.

The following table lists the actions that users can perform for the privileges in the Security Policy group:

Privilege	User Action
Add or Modify Security Policy	Create or edit security policies.
Add or Modify Security Responses	Create or edit actions.
Delete Alerts	Delete security policy violations.
Delete Security Policy	Delete security policies.
Delete Security Responses	Delete actions.
View Alerts	View security policy violations.
View Security Policies	View security policies.
View Security Responses	View actions.

Data Privacy Management Service Custom Roles

Data Privacy Management Service custom roles include the Data Owner, External API, Operator, Policy Author, Security Analyst, Security Manager, and Technical Administrator.

The following table describes the Data Privacy Management Service custom roles:

Role	Description
Data Owner	Knows where sensitive data is located and has privileges to view reports and validate objects as sensitive or not sensitive.
External API	Determines how to view Data Privacy Management data in an external API. Has privileges to view dashboards, user activity, and heat maps, to export data, and to embed the Data Privacy Management URL to link dashboard widgets.
Operator	Monitors routine operations that identify and protect sensitive data. Has privileges to view dashboards, encryption rules, jobs, remote agents, scans, and tasks, and can run tasks and scans.
Policy Author	Understands security and regulatory requirements and best practices, ensures that security policies and legal compliance requirements are met, and defines security, privacy, and governance policies in the organization. Has privileges to define categories of sensitive information, define security policies, and validate objects as sensitive or not sensitive.
Security Analyst	Understands regulatory and security policies, discovers sensitive information or high-risk situations, and monitors security events and reports. Has privileges to create and view reports, analyze profile results, and apply policies.

Role	Description
Security Manager	Ensures that regulatory, security, and compliance policies are met with security, auditing, compliance, and monitoring tools. Has privileges to create and review reports, view user activity and proliferation, author policies, and run scans.
Technical Administrator	Administers the daily operations of IT and security tools. Has privileges to manage services and nodes, oversee security and user management, migrate data, and manage licenses.

Data Owner

The Data Owner user knows where sensitive data is located and has privileges to view reports and validate objects as sensitive or not sensitive.

When you assign the Data Owner role to a user, to include all the default privileges, select the role from each of the following folders:

- Model Repository Service
- Data Privacy Management Service

The following table lists the default privileges assigned to the Data Owner custom role:

Privilege Group	Privilege Name
Classification	- Validate Classification Exceptions* - View Data Domains - View Policies - View Scans
Dashboard	- Access the Dashboard - Monitor Jobs - View Data Stores in Dashboard - View Heat Maps - View Proliferation - View User Activity - View User Activity Requests
Data Protection	- Add or Modify Encryption Rules - Add or Modify Task - Close Task - Execute Task - Manage Protection Task Configurations - View Encryption Rules - View Extensions - View Proxies - View Task
Discovery	- Add or Modify Data Stores - View Data Stores
Model Repository Service Administration	Access Developer
* The user cannot perform the corresponding action in Data Privacy Management.	

External API

The External API user determines how to view Data Privacy Management data in an external API. The External API user has access to view dashboards, user activity, and heat maps, and to export data and embed the Data Privacy Management URL to link dashboard widgets.

To assign the External API role to a user, select the role from the Data Privacy Management Service folder.

The following table lists the default privileges assigned to the External API custom role:

Privilege Group	Privilege Name
Integration and APIs	- Access Embedded URL* - Export Analysis * The user cannot perform the corresponding action in Data Privacy Management.
Dashboard	- Access the Dashboard - View Advanced Analytics - View Departments - View Heat Maps - View Locations - View Proliferation - View User Accessibility - View User Activity

Operator

The Operator user monitors routine operations that identify and protect sensitive data. The Operator has privileges to view dashboards, encryption rules, jobs, remote agents, scans, and tasks. The Operator can also run tasks and scans.

To assign the Operator role to a user, select the role from the Data Privacy Management Service folder.

The following table lists the default privileges assigned to the Operator custom role:

Privilege Group	Privilege Name
Classification	- View Scans - Execute Scans
Dashboard	- View Data Stores in Dashboard - Monitor Jobs
Data Protection	- Execute Task - View Encryption Rules - View Proxies - View Task
Discovery	- View Data Stores

Policy Author

The Policy Author user understands security and regulatory requirements and best practices, ensures that security policies and legal compliance requirements are met, and defines security, privacy, and governance policies in the organization. The Policy Author has privileges to define categories of sensitive information, define security policies, and validate objects as sensitive or not sensitive.

To assign the Policy Author role to a user, select the role from the Data Privacy Management Service folder.

The following table lists the default privileges assigned to the Policy Author custom role:

Privilege Group	Privilege Name
Classification	- Add or Modify Data Domains - Add or Modify Policies - Add or Modify Scans - Customize Risk Scoring - Customize Severity Levels - Delete Data Domains - Delete Policies - Delete Scans - Execute Scans - Validate Classification Exceptions* - View Data Domains - View Policies - View Scans * The user cannot perform the corresponding action in Data Privacy Management.
Dashboard	- Monitor Jobs
Data Protection	- View Encryption Rules

Security Analyst

The Security Analyst user understands regulatory and security policies, discovers sensitive information or high-risk situations, and monitors security events and reports. The Security Analyst has privileges to create and view reports, analyze profile results, and apply policies.

To assign the Security Analyst role to a user, select the role from the Data Privacy Management Service folder.

The following table lists the default privileges assigned to the Security Analyst custom role:

Privilege Group	Privilege Name
Classification	- View Data Domains - View Policies - View Scans
Dashboard	- Access the Dashboard - Add or Modify Reports - Add or Modify Anomaly Suppression - Delete Anomaly Suppression - Modify Dashboard - Monitor Jobs - View Advanced Analytics - View Anomaly Suppression - View Data Stores in Dashboard - View Departments - View Heat Maps - View Locations - View Proliferation - View User Accessibility - View User Activity - View User Activity Requests* * The user cannot perform the corresponding action in Data Privacy Management.
Data Protection	- Add or Modify Encryption Rules - Add or Modify Task - Close Task - Delete Encryption Rules - Execute Task - Manage Protection Task Configurations - Risk Simulation and Manage Plans - View Extensions - View Proxies - View Task
Discovery	View Data Stores
Security Policy	- Delete Alerts - View Alerts - View Security Policies - View Security Responses

Security Manager

The Security Manager user ensures that regulatory, security, and compliance policies are met with security, auditing, compliance, and monitoring tools. The Security Manager has privileges to create and review reports, view user activity and proliferation, author policies, and run scans.

To assign the Security Manager role to a user, select the role from the Data Privacy Management Service folder.

The following table lists the default privileges assigned to the Security Manager custom role:

Privilege Group	Privilege Name
Classification	- Execute Scans - View Data Domains - View Policies - View Scans
Dashboard	- Access the Dashboard - Add or Modify Anomaly Suppression - Add or Modify Reports - Customize Advanced Analytics Configuration - Delete Anomaly Suppression - Modify Dashboard - Modify Protection Field Settings - Modify Sensitive Field Settings - Monitor Jobs - View Advanced Analytics - View Anomaly Suppression - View Data Stores in Dashboard - View Departments - View Heat Maps - View Locations - View Proliferation - View Subject Registry - View User Accessibility - View User Activity - View User Activity Requests* * The user cannot perform the corresponding action in Data Privacy Management.
Data Protection	- Add or Modify Task - Close Task - Execute Task - Manage Protection Task Configurations - Risk Simulation and Manage Plans - View Encryption Rules - View Extensions - View Proxies - View Subject Index - View Task
Discovery	View Data Stores
Security Policy	- Add or Modify Security Policy - Add or Modify Security Responses - Delete Alerts - Delete Security Policy - Delete Security Responses - View Alerts - View Security Policies - View Security Responses

Technical Administrator

The Technical Administrator user administers the daily operations of IT and security tools. The Technical Administrator has privileges to manage services and nodes, oversee security and user management, migrate data, and manage licenses.

To include the default privileges when you assign the Data Privacy Management Technical Administrator role to a user, select the role from each of the following folders:

- Model Repository Service
- Data Privacy Management Service

The following table lists the default privileges assigned to the Technical Administrator custom role:

Privilege Group	Privilege Name
Classification	- Customize Risk Scoring - Customize Severity Levels - Manage Custom Risk Factors
Dashboard	- Customize Advanced Analytics Configuration - Manage Jobs
Data Protection	- Add or Modify Encryption Rules - Add or Modify Proxies - Delete Encryption Rules - Delete Proxies - Delete Task - Manage Extensions - View Encryption Rules - View Extensions - View Proxies - View Task
Discovery	- Add or Modify Data Stores - Delete Data Stores - Manage Data Store Groups - Merge Data Stores - View Data Stores
Integration and APIs	- Access Embedded URL* - Export Analysis - Import Audit Logs* - Import Classification Rules - Import Data Owners - Import Data Stores - Import Enterprise User and Accessibility Information: From Active Directory or LDAP - Import Enterprise User and Accessibility Information: From Database Catalog* - Import Enterprise User and Accessibility Information: From File - Import Enterprise User and Accessibility Information: From Packaged Applications - Import Proliferation** - Import Protection Status - Import Sensitive Data Locations and Data Store Groups** * The user cannot perform the corresponding action in Data Privacy Management. ** Data Privacy Management does not check user privileges when the action is performed.
Model Repository Service Administration	Access Developer

Domain Privileges

The Domain privileges determine the actions that users can perform in the Informatica domain and Data Privacy Management.

The following table describes each privilege group for the Informatica domain:

Privilege Group	Description
Security Administration	Includes privileges to manage users, groups, roles, and privileges.
Domain Administration	Includes privileges to manage the domain, folders, nodes, grids, licenses, application services, connections, and cluster configurations.
Monitoring	Includes privileges to configure monitoring statistics and reports, view monitoring for integration objects, and access monitoring.
Tools	Includes privileges to log in to the Administrator tool.
Cloud Administration	Includes privileges to add Informatica Cloud organizations in the Administrator tool and view them.

For detailed descriptions of the Domain privilege groups, see the *Informatica Security Guide*.

Domain System-Defined Role

Data Privacy Management includes the Domain Administrator system-defined role.

The following table describes the Domain Administrator role:

Role	Description
Domain Administrator	The Domain Administrator is responsible for Data Privacy Management user tasks in the Administrator tool. The Domain Administrator has privileges to grant privileges and roles, and manage users, groups, and roles.

Domain Administrator

To assign the Domain Administrator role to a user, expand the **Domain** folder and select **Domain Administrator** from the **System Defined Roles** list.

The following table lists the default privileges assigned to the Domain Administrator system-defined role:

Privilege Group	Privilege Name
Security Administration	- Grant Privileges and Roles - Manage Users, Groups, and Roles
Domain Administration	- Manage Connections - Manage Domain Folders - Manage Nodes and Grids - Manage Service Execution - Manage Services
Monitoring	- Access Monitoring: Access from Administrator Tool - Access Monitoring: Access from Analyst Tool - Access Monitoring: Access from Developer Tool - Manage Monitoring: Monitoring Configuration - Manage Monitoring: Report and Statistic Settings - Perform Actions on Jobs - View: View Jobs of All the Users in the Groups That the User Belongs To - View: View Jobs of Other Users - View: View Reports - View: View Statistics
Tools	Access Informatica Administrator
Cloud Administration	- Manage Organization - View Organization

Part II: Configuring Connectivity to Application Sources

This part contains the following chapters:

- Configuring Connectivity to Active Directory Databases, 55
- Configuring Connectivity to SAP Sources, 57
- Configuring Connectivity to SAP S/4HANA Sources, 62

CHAPTER 3

Configuring Connectivity to Active Directory Databases

This chapter includes the following topics:

- Configuring Active Directory Database Connectivity Overview, 55
- Step 1. Download the CData Driver JAR Files and Update the Entry, 55
- Step 2. Create an Active Directory Data Store in Data Privacy Management, 56

Configuring Active Directory Database Connectivity Overview

Before you scan an Active Directory database, configure connectivity to the database. If you do not correctly configure the connectivity, the scan job can fail.

To configure connectivity, complete the following steps:

1. Download the required driver JAR files to the required locations on the Data Privacy Management machine and add an entry for the driver JAR files.
2. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Download the CData Driver JAR Files and Update the Entry

Before you can create an Active Directory data store in Data Privacy Management, you must install the required drivers and install the SAP HANA client.

1. Download and install the CData JDBC driver and license file from the CData web site.

   Download the following files:

   - cdata.jdbc.activedirectory.jar

- cdata.jdbc.activedirectory.lic

2. Copy the JAR file to the following locations:
   - <Data Privacy Management installation directory>/externaljdbcjars
   - <Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip
     Add the file to the genericJDBC.zip file. Create the ZIP file if it does not exist.
3. Add an entry for the genericJDBC.zip file in the <Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml file.

   The following code is an example of an entry in the file:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="genericJDBC.zip">
               <PropertyName>JDBCScanner_DriverLocation</PropertyName>
               <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue>
   ```
4. Verify that you can connect to the database.
5. In a new session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

Step 2. Create an Active Directory Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Active Directory databases.

When you create the data store, specify the category **Application** and the data store type **Active Directory**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 4

Configuring Connectivity to SAP Sources

This chapter includes the following topics:

- Configuring Connectivity to SAP Sources for Domain Discovery, 57
- Configuring Connectivity to SAP for Subject Registry Scans, 59

Configuring Connectivity to SAP Sources for Domain Discovery

Configuring SAP Connectivity Overview

Before you run a scan on an SAP data store, configure connectivity to the source. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity to SAP, complete the following steps:

1. Install SAP Transports
2. Install SAP JCo
3. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the source.

After you configure the connectivity, you can scan the data store.

Step 1. Install SAP Transports

The SAP transports are included in the Data Privacy Management installation package. To install the transports, you must extract the transports to your SAP system.

Prerequisites

Before you import the SAP transports, perform the following tasks:

1. Access the SAP transports in the Data Privacy Management installation folder: `/Server/LDM/saptrans/mySAP/UC`
2. Examine the transports for the SAP version table that corresponds to your SAP version.

3. Delete the prefix from the name of each data file and cofile.
   For example, remove `ZINFABC_RUN_` from the cofile name `ZINFABC_RUN_K900183.R46` to rename it to `K900183.R46`.
4. Change the permission of the cofiles from read-only to write.

Steps to Install SAP Transports

Import the following SAP transports to your SAP system. Use the SAP transaction code `STMS`. Import the transports in the order listed:

1. `ZINFABC_RUN` run-time transport.
2. `TBL_READ_RUN` run-time transport.
3. `TRANS_VER_RUN` run-time transport.
4. `TBL_READ_RUN_CMP` transport.
5. `TBL_DESIGN_PROGINFO` design time transport.
6. `TBL_DESIGN` design-time transport.
7. `TBL_READ_RUN_V2` transport. This transport is available in the following Data Privacy Management installation folder `/Server/LDM/saptrans/mySAP/common`.

Example

The following table shows the SAP transport request number to import if you are using Secure@Source 4.0:

Transport Request Number	Cofile Name	Datafile Name
U47K900245	ZINFABC_RUN_K900245.U47	ZINFABC_RUN_R900245.U47
DV2K900370	TBL_READ_RUN_K9000370.DV2	TBL_READ_RUN_R9000370.DV2
DV2K900367	TRANS_VER_RUN_K900367.DV2	TRANS_VER_RUN_R900367.DV2
EC5K900448	TBL_READ_RUN_CMP_K900448.EC5	TBL_READ_RUN_CMP_R900448.EC5
UC5K900051	TBL_DESIGN_PROGINFO_K900051.UC5	TBL_DESIGN_PROGINFO_R900051.UC5
DV2K900289	TBL_DESIGN_K900289.DV2	TBL_DESIGN_K900289.DV2
DU5K900047	TBL_READ_RUN_V2_K900047.DU5	TBL_READ_RUN_V2_R900047.DU5

Step 2. Install the SAP Java Connector (SAP JCo)

Perform the following steps to install SAP JCo 3:

1. Log in to the Administrator Tool.
2. Disable the Catalog Service.
3. Download the following files from the SAP Service Marketplace:
   - sapjco3.jar
   - sapjco3.so
4. Extract the SAP JCo files to the `SAPJCo` folder.

5. Copy the `sapjco3.jar` files to the following locations:
   - `<Install directory>/services/CatalogService>/access/WEB-INF>/lib`
   - `<Install_directory>/services/shared/jars/thirdparty`
   - `<Install_directory>/externaljdbcjars/`
6. Copy the `libsapjco3.so` file to the following locations:
   - `<Install_directory>/server/bin`
   - `<Install_directory>/services/shared/bin`
7. Include the `libsapjco3.jar` file in the `SAPJCO.zip` file.
8. Copy the `SAPJCO.zip` file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`.
9. Restart the Informatica domain.

Step 3. Create an SAP Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to SAP. When you create the data store, specify the category **Application** and the data store type **SAP**.

You can include the data store in a scan. When you run a scan on the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the connection properties in the data store to connect to SAP.

For more information, see the *Informatica Data Privacy Management User Guide*.

Configuring Connectivity to SAP for Subject Registry Scans

Configuring SAP Connectivity for Subject Registry Overview

Before you run a Subject Registry scan on an SAP data store, configure connectivity to the source. If you do not correctly configure the connectivity, the Subject Registry scan job can fail.

To configure connectivity to SAP for Subject Registry, complete the following steps:

1. Download the required driver JAR files to the required locations on the Data Privacy Management machine and add an entry for the driver JAR files.
2. Install SAP JCo.
3. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the source.

After you configure connectivity and create the data store, you can run Subject Registry scans on the data store.

Step 1. Download the CDATA Driver JAR Files for SAP and Update the File Entry

Before you can create an SAP data store, you must install the required drivers.

1. Download and install the CData driver for SAP from the CData web site. Download the following files:
   - cdata.jdbc.saperp.jar
   - cdata.jdbc.saperp.lic
2. Copy the license and JAR files to the following locations:
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
     Add the file to the genericJDBC.zip file. Create the ZIP file if it does not exist.
3. Add an entry for the genericJDBC.zip file in the <Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml file.

   The following code is an example of an entry in the file:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="genericJDBC.zip">
               <PropertyName>JDBCScanner_DriverLocation</PropertyName>
               <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue>
   ```
4. Verify that you can connect to the database.
5. In a new session, restart the Informatica domain.

Step 2. Install SAPJCo

To configure connectivity, you require the SAP Java Connector library files.

1. Log in to the Administrator tool.
2. Disable the Catalog Service.
3. Download the SAP JCo 3.x file from the SAP Service Marketplace.
4. Extract the SAP JCo files to the SAPJCo folder.
5. Copy the sapjco3.jar files to the following locations:
   - `<Install directory>/services/CatalogService>/access/WEB-INF>/lib`
   - `<Install_directory>/services/shared/jars/thirdparty`
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
     Add the file to the genericJDBC.zip file.
6. Copy the libsapjco3.so file to the following locations:
   - `<Install_directory>/server/bin`
   - `<Install_directory>/services/shared/bin`
7. Include the libsapjco3.so file in the SAPJCO.zip file.
8. Copy the SAPJCO.zip file to the following location: `<Install_directory>/services/CatalogService/ScannerBinaries`.
9. Restart the Informatica domain.

Step 3. Create an SAP Data Store for Subject Registry in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to SAP. When you create the data store, specify the category **Application** and the data store type **SAP**.

You can include the data store in a scan. When you run a scan on the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the connection properties in the data store to connect to SAP.

For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 5

Configuring Connectivity to SAP S/4HANA Sources

This chapter includes the following topics:

- Configuring Connectivity to SAP S/4HANA Sources for Domain Discovery, 62
- Configuring Connectivity to SAP S/4HANA for Subject Registry Scans, 64

Configuring Connectivity to SAP S/4HANA Sources for Domain Discovery

Configuring SAP S/4HANA Connectivity Overview

Before you scan an SAP S/4HANA data store, configure connectivity to the source. Improper configuration can cause the scan job to fail.

To configure connectivity to SAP S/4HANA, complete the following steps:

1. Install Scanner transports.
2. Configure a user-authorization profile.
3. Install SAP Java Connector (SAP JCo).
4. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the SAP S/4HANA source.

After you configure the connectivity, you can scan the data store.

Step 1. Install Scanner Transports

To extract metadata from SAP S/4HANA, you must import the Scanner transports that Enterprise Data Catalog uses to extract metadata from SAP S/4HANA. To import scanner transports, download the `SAP_Scanner_Binaries.zip` file from the Informatica installer location.

Note: To extract profiling data from SAP S/4HANA, you must import the Scanner transports located in the following folder where you downloaded PowerCenter: `<downloadlocation>\saptrans\mySAP`.

Step 2. Configure User-Authorization Profiles

To access metadata from SAP S/4HANA, the SAP S/4HANA administrator needs to create the following user authorization profiles:

Authorization Object	Description	Field Values
S_RFC	Authorization check for RFC access	ACTVT: 16 (Execute) RFC_NAME: DDIF_FIELDINFO_GET RFCPING RFC_GET_FUNCTION_INTERFACE RFC_METADATA_GET SEU_COMPONENT ZINFA_ERP RFC_TYPE: FUGR, FUNC

Step 3. Install SAP Java Connector (SAP JCo)

Perform the following steps to install SAP Java Connector (SAP JCo) 3:

1. Log in to the Administrator tool.
2. Disable the Catalog Service.
3. Download the following files from the SAP Service Marketplace:
   - sapjco3.jar
   - sapjco3.so
4. Extract the SAP JCo files to the `SAPJCo` folder.
5. Copy the `sapjco3.jar` files to the following installation locations:
   - `<Data Privacy Management install directory>/services/CatalogService>/access/WEB-INF>/lib`
   - `<Data Privacy Management install directory>/services/shared/jars/thirdparty`
6. Copy the `libsapjco3.so` file to the following locations:
   - `<Data Privacy Management install directory>/server/bin`
   - `<Data Privacy Management install directory>/services/shared/bin`
7. Include the `libsapjco3.jar` file in the `SAPJCO.zip` file.
8. Copy the `SAPJCO.zip` file to the following location: `<Data Privacy Management install directory>/services/CatalogService/ScannerBinaries`.
9. Enable the Catalog Service.

   Note: If you use the SAP S/4HANA resource to extract profiling data, you must restart the Informatica domain.

Step 4. Create an SAP S/4HANA Data Store in Data Privacy Management

Perform the following steps to create an SAP S/4HANA Data Store in Data privacy Management:

1. In the Data Stores workspace of Data Privacy Management, create a new data store.

2. Specify the **Category** as Application and the **Data Store Type** as SAP S/4HANA.
3. Configure the connection properties to connect to SAP S/4HANA.

   After creating the data store, you can include it in a scan. Running the scan triggers a database scan job that uses the connection properties to connect to SAP S/4HANA.
   For more details about configuring connection properties for SAP S/4HANA, see the *Data Privacy Management User Guide*.

Configuring Connectivity to SAP S/4HANA for Subject Registry Scans

A Subject Registry Scan Job in Data Privacy Management is a process that scans data stores to identify data subjects. To run a Subject Registry scan successfully on an SAP S/4HANA data store, you must set up the required driver files, install SAPJCo, and create a data store with the correct connection settings.

Configuring SAP S/4HANA Connectivity for Subject Registry Overview

Before you run a Subject Registry scan on an SAP S/4HANA data store, configure connectivity to the source. Improper configuration can cause the scan job to fail.

To configure connectivity to SAP S/4HANA for Subject Registry, complete the following steps:

1. Download and register the required CDATA JAR files on the Data Privacy Management machine.
2. Install SAP Java Connector (SAP JCo).
3. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the SAP S/4HANA source.

After you configure connectivity and create the data store, you can run Subject Registry scans on the data store.

Note: SAP S/4HANA includes link subject scan.

Step 1. Download the CDATA Driver JAR Files for SAP S/4HANA and Update the Entry

Before you can create an SAP S/4HANA data store, you must install the required drivers.

1. Download and install the CData driver for SAP from the CData website including the following files:
   - `cdata.jdbc.saperp.jar`
   - `cdata.jdbc.saperp.lic`
2. Copy the license and JAR files to the following locations:
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - Extract or add to `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
     Create the ZIP file if it does not exist.

3. Add an entry for the `genericJDBC.zip` file in the `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml` file.

   Example entry:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="genericJDBC.zip">
               <PropertyName>JDBCScanner_DriverLocation</PropertyName>
               <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue>
   ```

4. Verify the database connection.
5. In a new session, restart the Informatica domain.

Step 2. Install SAPJCo for SAP S/4HANA Subject Registry

To configure connectivity, you require the SAP Java Connector library files.

1. Log in to the Administrator tool.
2. Disable the Catalog Service.
3. Download SAP JCo 3.x files from the SAP Service Marketplace.
4. Extract the files to the SAPJCo folder.
5. Copy `sapjco3.jar` to the following locations:
   - `<Install directory>/services/CatalogService>/access/WEB-INF>/lib`
   - `<Install directory>/services/shared/jars/thirdparty`
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - Add to `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
6. Copy the libsapjco3.so file to the following locations:
   - `<Install directory>/server/bin`
   - `<Install directory>/services/shared/bin`
7. Include the `libsapjco3.so` file in the `SAPJCO.zip` file.
8. Copy the `SAPJCO.zip` file to the following location: `<Install directory>/services/CatalogService/ScannerBinaries`.
9. Enable the Catalog service.

   Note: If you use the SAP S/4HANA resource to extract profiling data, you must restart the Informatica domain.

Step 4. Create an SAP S/4HANA Data Store in Data Privacy Management for Subject Registry Scan

Perform the following steps to create an SAP S/4HANA Data Store in Data privacy Management:

1. In the Data Stores workspace of Data Privacy Management, create a new data store.
2. Specify the **Category** as Application and the **Data Store Type** as SAP S/4HANA.
3. Configure the connection properties to connect to SAP S/4HANA.

   After creating the data store, you can include it in a scan. Running the scan triggers a database scan job that uses the connection properties to connect to SAP S/4HANA.
   For more details about configuring connection properties for SAP S/4HANA, see the *Data Privacy Management User Guide*.

Part III: Configuring Connectivity to Big Data Sources

This part contains the following chapters:

- Configuring Connectivity to Hadoop Distributed File System Sources, 67
- Configuring Connectivity to Hive Sources, 73

CHAPTER 6

Configuring Connectivity to Hadoop Distributed File System Sources

This chapter includes the following topics:

- Configuring HDFS Connectivity Overview, 67
- Step 1. Update the Domain Machines to Access Kerberos-Enabled HDFS Sources, 68
- Step 2. Create an HDFS Data Store in Data Privacy Management, 69
- Storage Type: Azure WASB, 69

Configuring HDFS Connectivity Overview

Before you scan a Hadoop Distributed File System (HDFS) data store, you must configure connectivity to the source.

If you do not correctly configure connectivity to the HDFS source, the Scan job might fail.

1. For HDFS sources on a Kerberos-enabled Hadoop cluster, update the client machines.
2. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the source.

After you configure the connectivity, you can scan the data store.

Data stores within the Big Data category require a third-party integration package to connect to the Hadoop or Databricks environment and to process complex files. The Informatica installer does not include the integration packages. You can download the integration packages before you run the installer and install the distributions when you install Informatica services. If you didn't install the distributions when you installed Informatica services, you can install them through the Integration Package Manager. For information about how to run Integration Package Manager, see "Appendix: Integration Package Management" in *Informatica Installation for Data Privacy Management*.

Step 1. Update the Domain Machines to Access Kerberos-Enabled HDFS Sources

To scan HDFS sources on a Kerberos-enabled Hadoop cluster, copy the keytab and Kerberos configuration files from the Hadoop cluster. Use the Kerberos kinit utility to generate the credentials cache file for the user account that runs the scan job.

Note: You do not need to update the domain machines for HDFS sources on a highly available Hadoop cluster.

1. Copy the keytab file from the Hadoop cluster for the Hive source to any directory in the machine that hosts the Informatica domain and Informatica Hadoop cluster.
   Example: `/opt/<keytabfilename>`

   Important: Ensure that the keytab file can be accessed by the Informatica domain and the Hadoop cluster.

   When you create a HDFS data store, you provide the full path to the file in the **Keytab File** connection property.
2. Copy the following Kerberos configuration file from the Hadoop cluster:

   ```
   /etc/krb5.conf
   ```

   - On the machine that hosts the associated Data Integration Service of the Data Privacy Management Service, copy the file to the following locations:

     ```
     /$INFA_HOME/services/shared/security
     /$INFA_HOME/java/lib/security
     ```

   - On the machine that hosts the Informatica domain and the machine that hosts the Hadoop cluster, copy the file to the same location on both machines. Alternatively, copy the file to a shared location that is accessible to both the Informatica domain and the Hadoop cluster.
3. Generate a valid Kerberos ticket for the user that the scan job uses to connect to HDFS. You specify the user that connects to HDFS when you configure the connection properties for the data store. To generate a valid ticket, run the kinit utility on the machine that hosts the Informatica domain, on each of the nodes of the cluster, and on the HDFS source. Give the path of the keytab and the user principal that you are using.
   For example, run the following command:

   ```
   kinit -kt <keytab file> <user principal name>
   ```

   Where:

   - The keytab file is the path to the keytab file on the machine that hosts the Informatica domain. Provide the directory to which you copied the file in step 1.
   - The user principal name is *<user name>@<realm name>*. Enter the realm name in uppercase letters.
4. From the Administrator tool, update the following Data Integration Service properties:

   - **Hadoop Kerberos Service Principal Name**. The Service Principal Name (SPN) of the Data Integration Service to connect to a Hadoop cluster that uses Kerberos authentication.
     Enter the name in the following format: `<SPN>/<DIS domain name>@<realm name>`

     For example: `joe/domain12345@HADOOP-MIT-REALM`
   - **Hadoop Kerberos Keytab**. The full path and file name of the Kerberos keytab file on the machine on which the Data Integration Service runs.
     For example: `<Informatica Installation Directory>/isp/config/keys/infa_hadoop.keytab`
5. Restart the Data Integration Service.

Step 2. Create an HDFS Data Store in Data Privacy Management

When you create an HDFS data store on the **Data Stores** workspace in Data Privacy Management, configure properties to connect to HDFS. Specify the category **Big Data** and the data store type **Hadoop Distributed File System**.

You can include an HDFS data store in a Big Data scan. The Big Data Scan job uses the connection properties in the data store to connect to HDFS.

For more information, see the *Informatica Data Privacy Management User Guide*.

Storage Type: Azure WASB

Step 1. Add HDInsight Host Entries to the Informatica Domain and the Hadoop Cluster Machines

1. Log in to the HDInsight Ambari server that is associated with the Hadoop cluster.
2. Click the **Hosts** tab.
3. Make a note of the host names and the IP address for each host.
4. Log in to the machine where the Informatica domain is installed and navigate to the `/etc/hosts` file.
5. Add the HDInsight Ambari cluster hosts to the file in the following format: `<IP Address> <Host Name> <Host Alias>`

   For example: `10.75.169.88 hn0-secure.qrg2yxlb0aouniiuvfp3betl3d.ix.internal.cloudapp.net hn0-secure`
6. Log in to the machine where the Informatica Hadoop cluster is installed.
7. Navigate to the `/etc/hosts` file and add the HDInsight Ambari cluster hosts to the file in the following format: `<IP Address> <Host Name> <Host Alias>`

   For example: `10.75.169.88 hn0-secure.qrg2yxlb0aouniiuvfp3betl3d.ix.internal.cloudapp.net hn0-secure`

Step 2. Create and Configure a Cluster Configuration Object

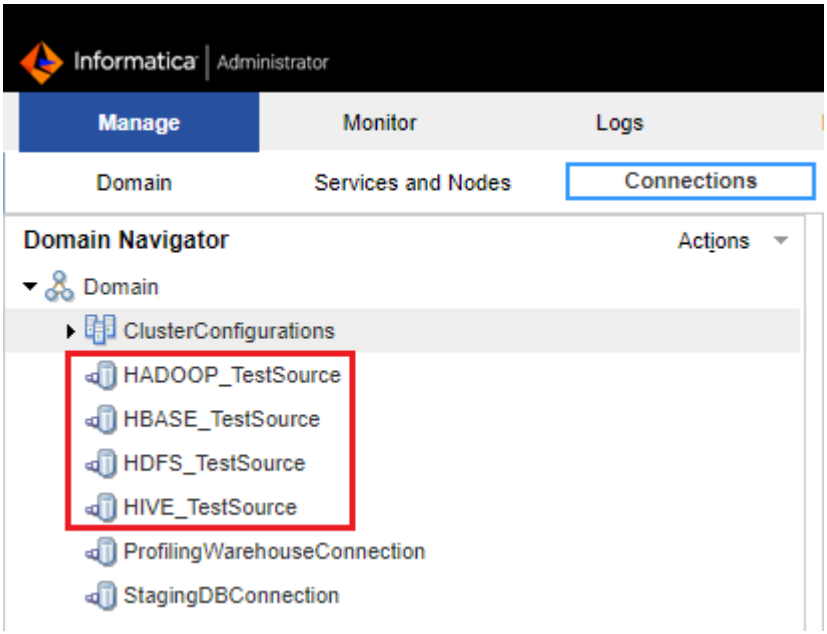
1. From the Administrator tool, select the **Connections** tab.
2. Expand **Domain** on the **Domain Navigator** pane.
3. Select **ClusterConfigurations** and click **Actions** > **New** > **Cluster Configuration**.
4. Enter a name for the cluster configuration.
5. Optionally, enter a description.
6. From the **Distribution Type** list, select **Azure HDInsight**.
7. In the **Method to Import the Cluster Configuration** field, select **Import from archive file**.

8. Log in to HDInsight Ambari and perform the following steps:
    a. Select **HDFS** on the left pane.
    b. From the **Service Actions** menu, select **Download Client Configs**.
    c. Repeat steps a and b to download the client configurations from **YARN**, **MapReduce2**, **Tez** and **Hive** services on the left pane.
    d. Verify that the following .tar files downloaded:
        - `HDFS_CLIENT-configs.tar.gz`
        - `YARN_CLIENT-configs.tar.gz`
        - `MAPREDUCE2_CLIENT-configs.tar.gz`
        - `TEZ_CLIENT-configs.tar.gz`
        - `HIVE_CLIENT-configs.tar.gz`
    e. Extract each file.

        Verify that the following files extracted:
        - `hdfs-site.xml` and `core-site.xml` from the `HDFS_CLIENT-configs.tar.gz` file.
        - `yarn-site.xml` from the `YARN_CLIENT-configs.tar.gz` file.
        - `hive-site.xml` from the `HIVE_CLIENT-configs.tar.gz` file.
        - `mapred-site.xml` from the `MAPREDUCE2_CLIENT-configs.tar.gz` file.
        - `tez-site.xml` from the `tez_CLIENT-configs.tar.gz` file.
    f. Select the extracted files and compress the files in a new .zip file.
9. Go back to the Administrator tool.
10. From the **Create Cluster Configuration** wizard, click **Browse** in the **Upload configuration archive file** field and upload the .zip file.
11. Check the **Create connections** option.
12. Click **Next**.
13. Review the information and click **Finish**.

    The Hadoop, HDFS, Hive, and HBase connections are created with the following name format:
    - `HADOOP_<Cluster Configuration Name>`
    - `HBASE_<Cluster Configuration Name>`
    - `HDFS_<Cluster Configuration Name>`
    - `HIVE_<Cluster Configuration Name>`

For example, if the cluster configuration name is `TestSource`, the connection names appear as shown in the following image:



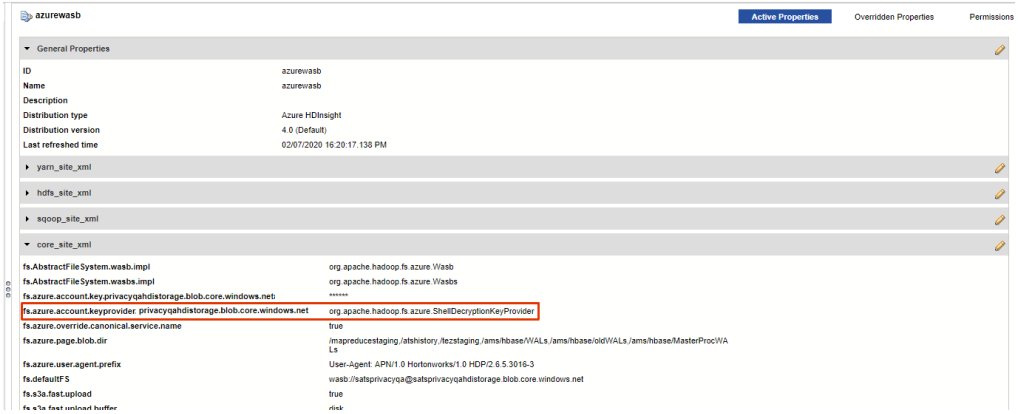
Note: If you want to delete the cluster configuration object, you must first delete the four connections.

Step 3. Verify and Update the core_site_xml Property Values

Update the core_site_xml properties in the cluster configuration object.

1. On the Domain Navigator, select the cluster configuration object created.
2. On the **Active Properties** tab, expand the **core_site_xml** section.
3. Verify that the `fs.azure.account.keyprovider.<your_account_name>.blob.core.windows.net` property contains the following value: `org.apache.hadoop.fs.azure.ShellDecryptionKeyProvide`

   The property might contain a different value. For example:
   `org.apache.hadoop.fs.azure.SimpleKeyProvider`
   The following image shows the property with the required value:



   a. If the property contains the value `org.apache.hadoop.fs.azure.SimpleKeyProvider`, use the **Import from Archive File option** and create the cluster configuration object.

4. Add the following property and value:`fs.azure.shellkeyprovider.script = /usr/lib/hdinsight-common/scripts/decrypt.sh`

Step 4. Copy the HDInsight Common Folder to the Informatica Domain and Hadoop Cluster Machines

1. On the Azure WASB cluster machine, browse to the `/usr/lib/hdinsight-common` folder.
2. Copy all the files and folders within the hdinsight-common folder.
3. Log in to the machine where the Informatica domain is installed and copy the folder with the same folder structure.

   Assign 777 permission recursively on the folder.
4. Log in to the machine where the Informatica Hadoop cluster is installed and copy the folder with the same folder structure.

   Assign 777 permission recursively on the folder.

CHAPTER 7

Configuring Connectivity to Hive Sources

This chapter includes the following topics:

- Configuring Hive Connectivity Overview, 73
- Hadoop Distribution Type: Amazon EMR, 74
- Hadoop Distribution Type: Azure HDInsight, 83
- Hadoop Distribution Type: Cloudera, 92
- Hadoop Distribution Type: Hortonworks, 103
- Hadoop Distribution Type: MapR, 113
- Hadoop Profile Execution Engine, 124
- Hive Profile Execution Engine, 124

Configuring Hive Connectivity Overview

Configure connectivity to Hive data stores before performing other operations in Data Privacy Management, such as running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

To configure connectivity, follow the steps in the topic for the specific Hadoop distribution type. For example, if the distribution type for the Hive data store is Cloudera, follow the steps in the *Hadoop Distribution Type: Cloudera* topic.

Data stores within the Big Data category require a third-party integration package to connect to the Hadoop or Databricks environment and to process complex files. The Informatica installer does not include the integration packages. You can download the integration packages before you run the installer and install the distributions when you install Informatica services. If you didn't install the distributions when you installed Informatica services, you can install them through the Integration Package Manager. For information about how to run Integration Package Manager, see "Appendix: Integration Package Management" in *Informatica Installation for Data Privacy Management*.

Important:

- For Kerberos-enabled Hive sources, if you change the distribution type of the data store after a scan, you must update the properties and recycle the Data Integration Service before you run the next scan.

- For Hive sources that you plan to associate with protection remote agents to encrypt and decrypt sensitive data, set the `Set hive.fetch.task.conversion` Hive parameter to `more` after you configure connectivity to the Hive source.

Hadoop Distribution Type: Amazon EMR

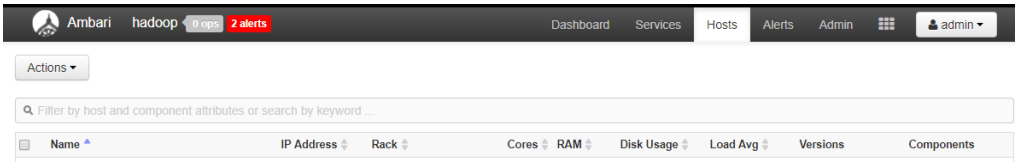
Before you perform other operations in Data Privacy Management, configure connectivity to data sources.

To connect to a Hive data source with Amazon EMR as the Hadoop distribution type, perform the following steps:

1. Add the Ambari cluster hosts to the Informatica domain and Hadoop cluster.
2. Create a cluster configuration object.
3. Install the database client software.
4. Configure the environment variables.
5. Create the Hive data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Add Ambari Cluster Hosts to the Informatica Domain and the Hadoop Cluster

1. Log in to the Ambari server that is associated with the Amazon EMR cluster.
2. Click the **Hosts** tab.



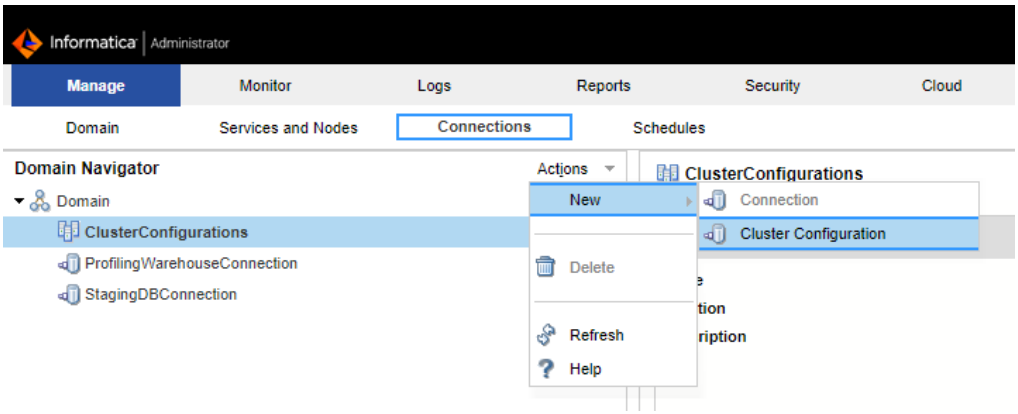
3. Make a note of the host names and the IP address for each host.
4. Log in to the machine where the Informatica Domain is installed and navigate to the `/etc/hosts` file.
5. Add the Ambari cluster hosts to the file in the following format: `<IP Address> <Host Name> <Host Alias>`

   For example : `10.75.169.88 hn0-secure.qrg2yxlb0aouniiuvfp3betl3d.ix.internal.cloudapp.net hn0-secure`
6. Log in to the machine where the Informatica Hadoop cluster is installed.
7. Navigate to the `/etc/hosts` file and add the Ambari cluster hosts to the file in the following format: `<IP Address> <Host Name> <Host Alias>`

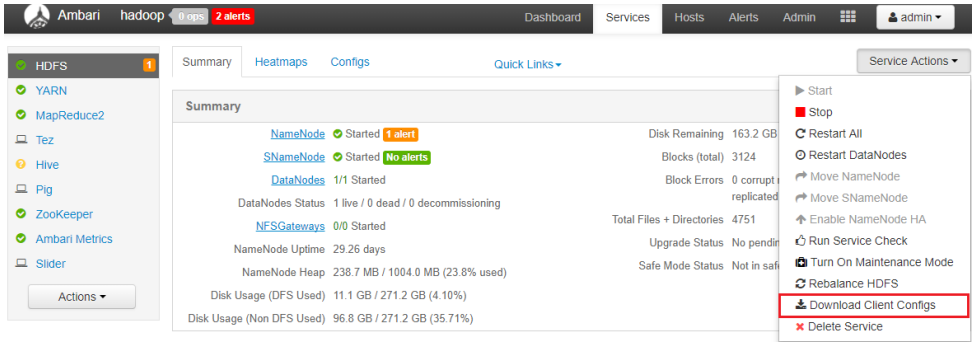
   For example : `10.75.169.88 hn0-secure.qrg2yxlb0aouniiuvfp3betl3d.ix.internal.cloudapp.net hn0-secure`

Step 2. Create a Cluster Configuration Object

1. From the Administrator tool, select the **Connections** tab.
2. Expand **Domain** on the **Domain Navigator** pane.
3. Select **ClusterConfigurations** and click **Actions** > **New** > **Cluster Configuration**.



4. Enter a name for the cluster configuration.
5. Optionally, enter a description.
6. From the **Distribution Type** list, select **Amazon EMR**.
7. In the **Method to Import the Cluster Configuration** field, select **Import from archive file**.
8. Log in to Ambari and perform the following steps:
   a. Select **HDFS** on the left pane.
   b. From the **Service Actions** menu, select **Download Client Configs**.



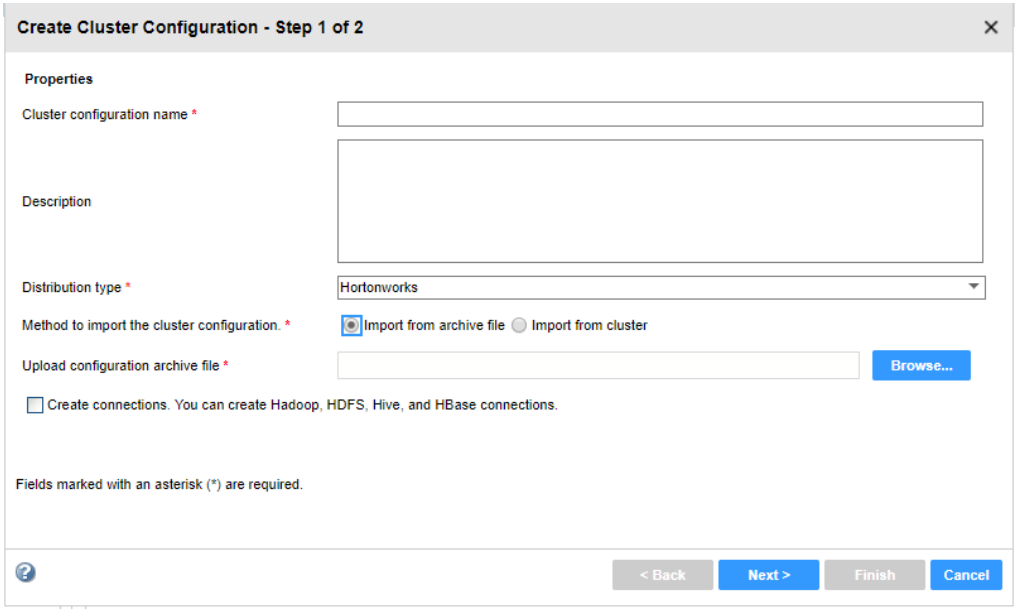
   c. Repeat steps a and b to download the client configurations from **YARN**, **MapReduce2**, **Tez** and **Hive** services on the left pane.
   d. Verify that the following .tar files downloaded:
      - `HDFS_CLIENT-configs.tar.gz`
      - `YARN_CLIENT-configs.tar.gz`
      - `MAPREDUCE2_CLIENT-configs.tar.gz`
      - `TEZ_CLIENT-configs.tar.gz`
      - `HIVE_CLIENT-configs.tar.gz`
   e. Extract each file.

      Verify that the following files extracted:
      - `hdfs-site.xml` and `core-site.xml` from the `HDFS_CLIENT-configs.tar.gz` file.
      - `yarn-site.xml` from the `YARN_CLIENT-configs.tar.gz` file.

- `hive-site.xml` from the `HIVE_CLIENT-configs.tar.gz` file.
- `mapred-site.xml` from the `MAPREDUCE2_CLIENT-configs.tar.gz` file.
- `tez-site.xml` from the `tez_CLIENT-configs.tar.gz` file.

f. Select the extracted files and compress the files in a new .zip file.

9. Go back to the Administrator tool.
10. From the **Create Cluster Configuration** wizard, click **Browse** in the **Upload configuration archive file** field and upload the .zip file.

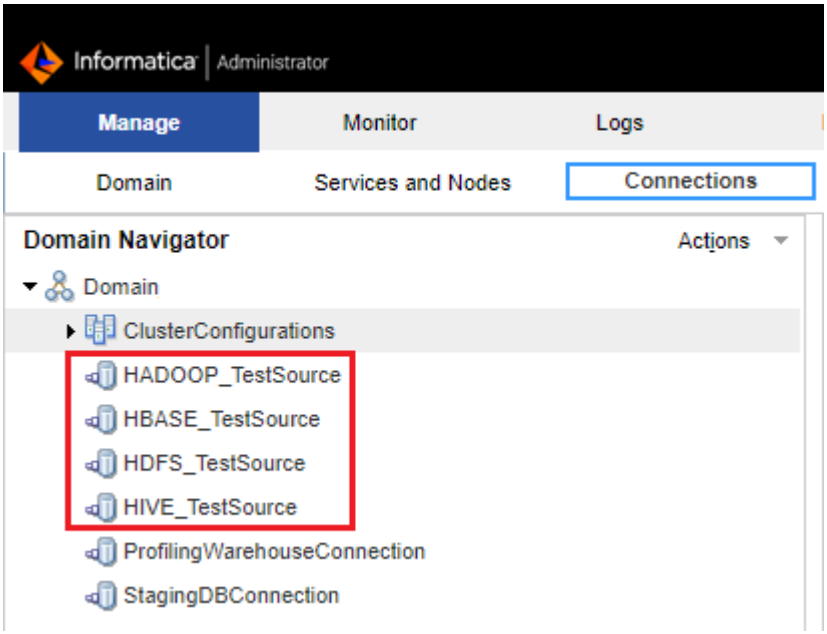


11. Check the **Create connections** option.
12. Click **Next**.
13. Review the information and click **Finish**.

    The Hadoop, HDFS, Hive, and HBase connections are created with the following name format:

    - `HADOOP_<Cluster Configuration Name>`
    - `HBASE_<Cluster Configuration Name>`
    - `HDFS_<Cluster Configuration Name>`
    - `HIVE_<Cluster Configuration Name>`

For example, if the cluster configuration name is `TestSource`, the connection names appear as shown in the following image:



Note: If you want to delete the cluster configuration object, you must first delete the four connections.

Step 3. Install the Database Client Software

If the Profiling Warehouse database in the associated Data Integration Service is on Oracle, install the database client software on the nodes within the Hive source cluster.

Note: If the Profiling Warehouse database in the associated Data Integration Service is on IBM Db2 or Microsoft SQL Server, use a JDBC connection and skip this step.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. Install the Oracle database client software in the same Oracle client path on all of the nodes within the Hive source cluster.
2. Add an entry in the `tnsnames.ora` file for the Profiling Warehouse database connection.

   For example:

```
ORAMAT=
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST =localhost)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = orcl)
    )
  )
```

Step 4. Configure the Environment Variables

Configure environment variables only if the Profiling Warehouse connection is Oracle and the profile execution engine for the data store is Hadoop or Hive.

Note: Do not configure environment variables in the following situations:

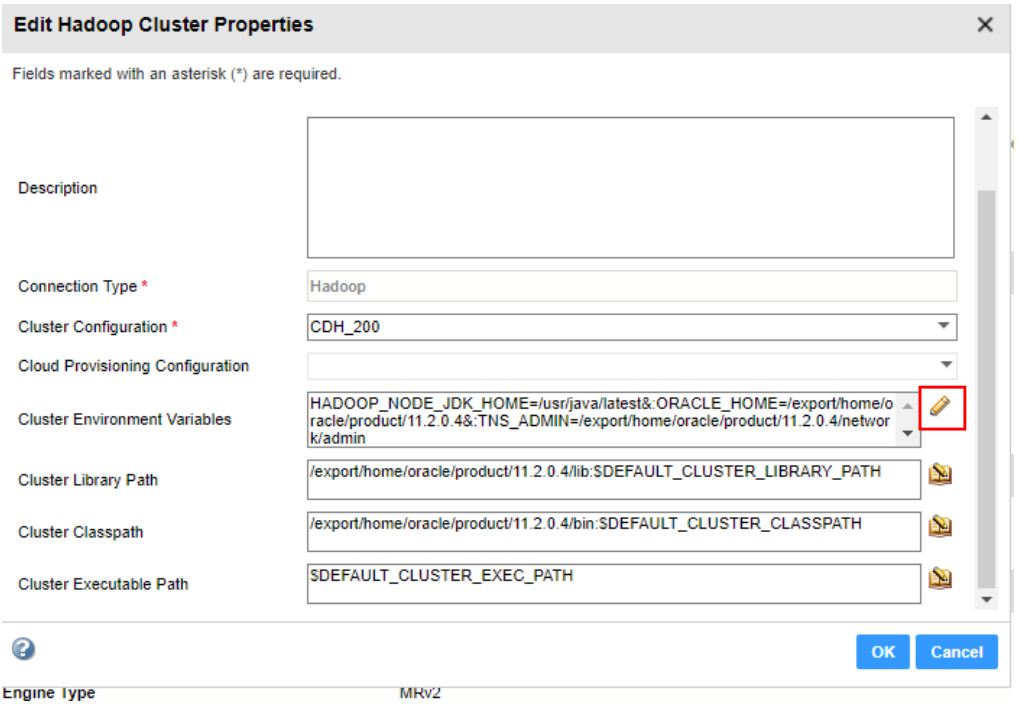
- The Profiling Warehouse connection is JDBC.
- The profile execution engine for the data store is native.

1. From the Administrator tool, click the **Manage** > **Connections** tab.
2. From the **Domain Navigator** pane on the left, select the Hadoop cluster connection.

   The Hadoop connection properties appear on the right.
3. Locate the **Hadoop Cluster Properties** section and click the **Edit** icon.



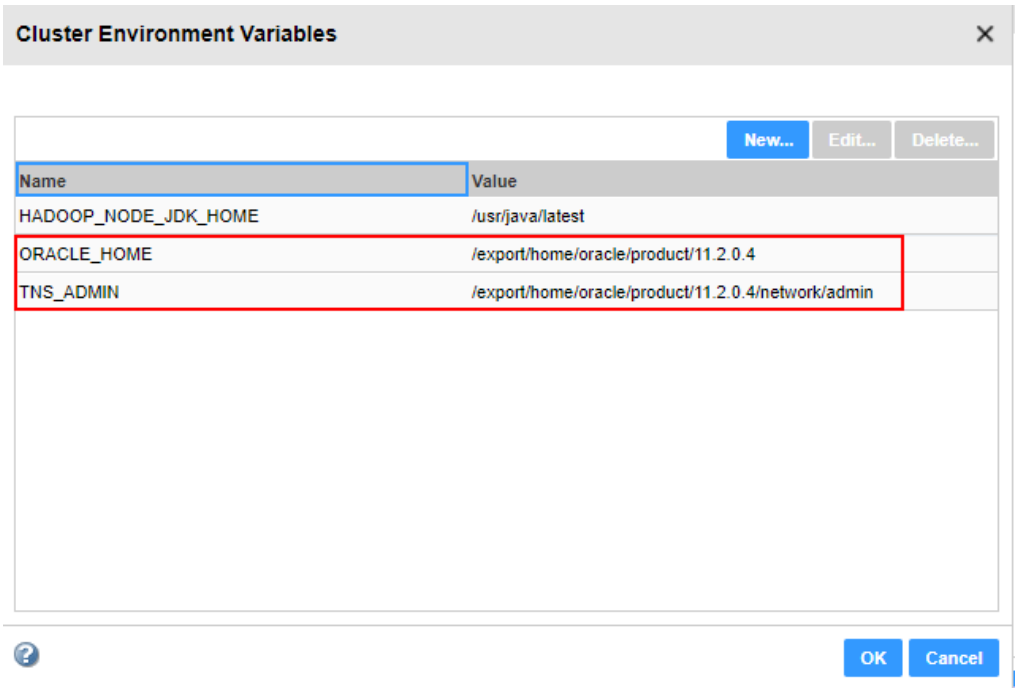
   The **Edit Hadoop Cluster Properties** window appears.
4. Click the **Edit** icon for the **Cluster Environment Variables** property.



   The **Cluster Environment Variables** window appears.
5. Click **New**.

6. Add the following environment variables:

Name	Value
ORACLE_HOME	`/export/home/oracle/product/11.2.0.4`
TNS_ADMIN	`/export/home/oracle/product/11.2.0.4/network/admin`



7. Click **OK**.

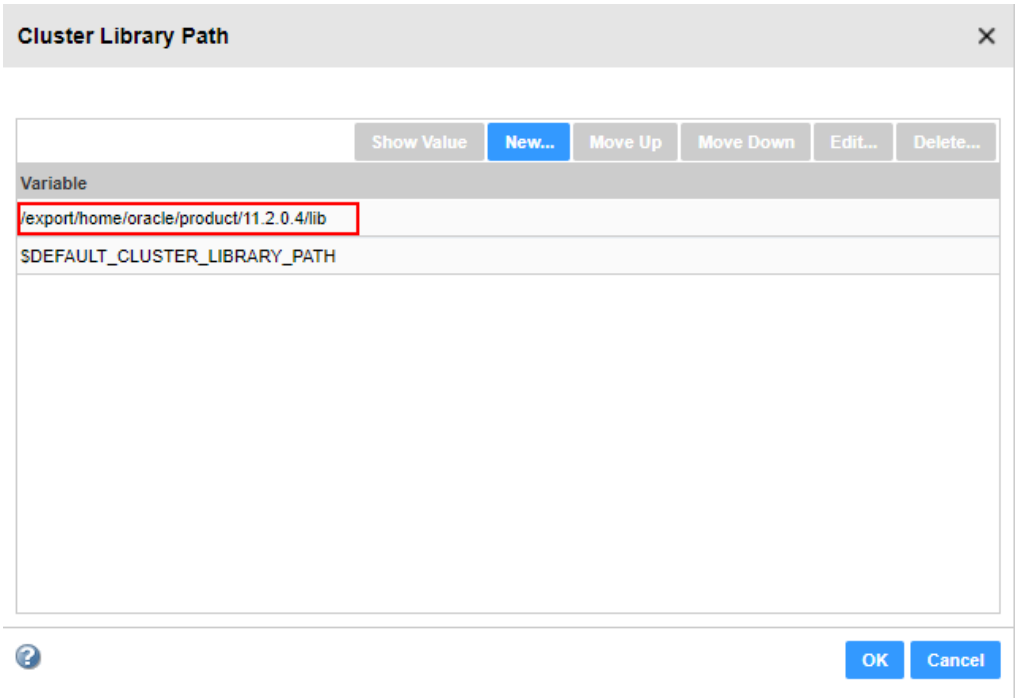
   The **Cluster Environment Variables** window closes.

8. Click the **Edit** icon for the **Cluster Library Path** property.

   The **Cluster Library Path** window appears.

9. Click **New**.

10. Enter the Oracle library path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/lib`



11. Click **OK**.

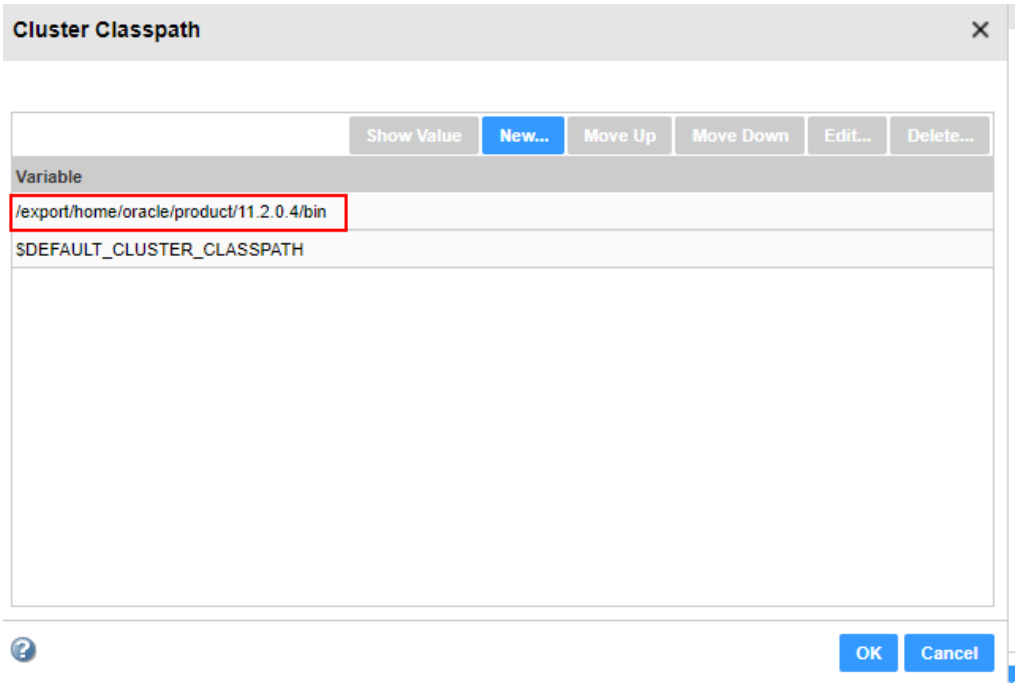
    The **Cluster Library Path** window closes.

12. Click the **Edit** icon for the **Cluster Classpath** property.

    The **Cluster Classpath** window appears.

13. Click **New**.

14. Enter the Oracle class path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/bin`



15. Click **OK**.

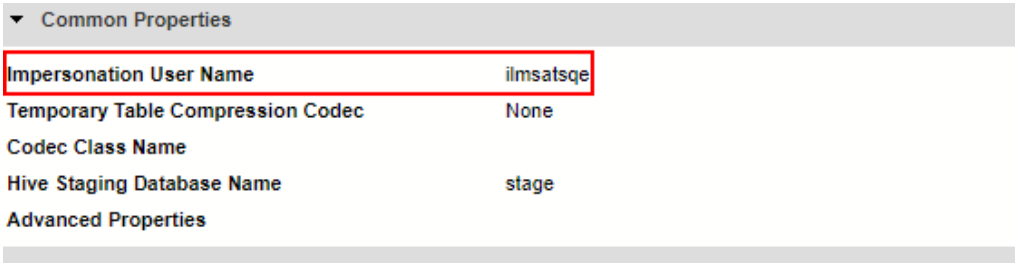
    The **Cluster Library Path** window closes.

16. Click **OK**.

    The **Edit Hadoop Cluster Properties** window closes.

17. Locate the **Common Properties** section and click the **Edit** icon.

18. In the **Impersonation User Name** field, enter the name of the user who can access the Hive database.



19. Click **OK**.

Step 5. Create a Hive Data Store in Data Privacy Management

To connect to the Amazon EMR Hive source, create a data store in Data Privacy Management and configure properties.

When you scan the Amazon EMR Hive data store, the Data Privacy Management Service creates a Big Data Scan job that connects to the data store. For more information, see the *Informatica Data Privacy Management User Guide*.

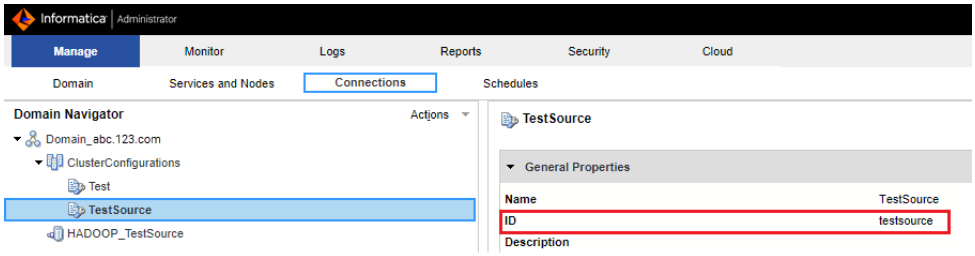
1. Log in toData Privacy Management and click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. Click **New**.

   The **New Data Store** page appears.
3. Enter a name for the data store.
4. Select **Big Data** as the **Category**.
5. Select **Hive** as the **Data Store Type**.
6. Select **Amazon EMR** as the **Hadoop Distribution**.
7. Enter or select values in the required fields.
8. To determine the **Cluster Configuration ID**, perform the following steps:
   a. In the Administrator tool, select the **Manage** > **Connections** tab.
   b. In the **Domain Navigator** pane, select the cluster configuration object you created.

      The **General Properties** section appears on the right.
   c. Make a note of the ID.



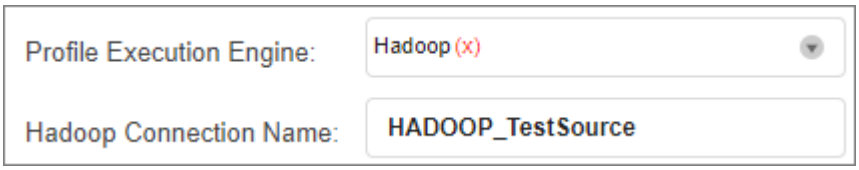
   Note: The **ID** value is automatically generated as the lowercase version of the cluster configuration name. For example, if the cluster configuration name is `TestSource`, then the ID is `testsource`.

   In Data Privacy Management, enter the ID in the **Cluster Configuration Id** field.
9. Specify the **Profile Execution Engine**. Select one of the following options:
   - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. See the "Hadoop Profile Execution Engine" on page 124 section for configuration steps.
   - **Hive**. Data Privacy Management uses the Hive engine to run profiles.

     Note: Informatica does not recommend this option because it slows performance.

     See the "Hive Profile Execution Engine" on page 124 section for prerequisite steps.
   - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. This is the default option and does not require any additional configuration.
10. If you selected **Hadoop** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



11. If you selected **Hive** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



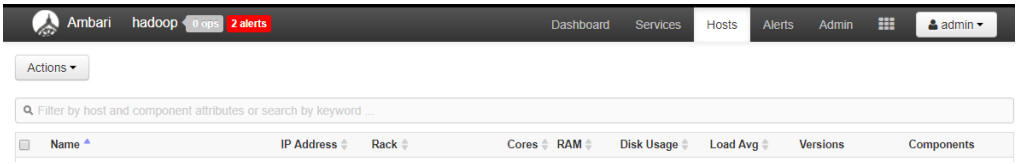
Hadoop Distribution Type: Azure HDInsight

Perform the following steps to connect to a Hive source with Azure HDInsight as the distribution type.

1. Add the Ambari cluster hosts to the Informatica domain and Hadoop cluster.
2. Create a cluster configuration object.
3. Install the database client software.
4. Configure the environment variables.
5. Create the Hive data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Add Ambari Cluster Hosts to the Informatica Domain and the Hadoop Cluster

1. Log in to the Ambari server that is associated with the Azure HDInsight cluster.
2. Click the **Hosts** tab.



3. Make a note of all the host names and their IP addresses.
4. Log in to the machine where the Informatica Domain is installed and navigate to the `/etc/hosts` file.
5. Add the Ambari cluster hosts to the file in the following format: `<IP Address> <Host Name> <Host Alias>`

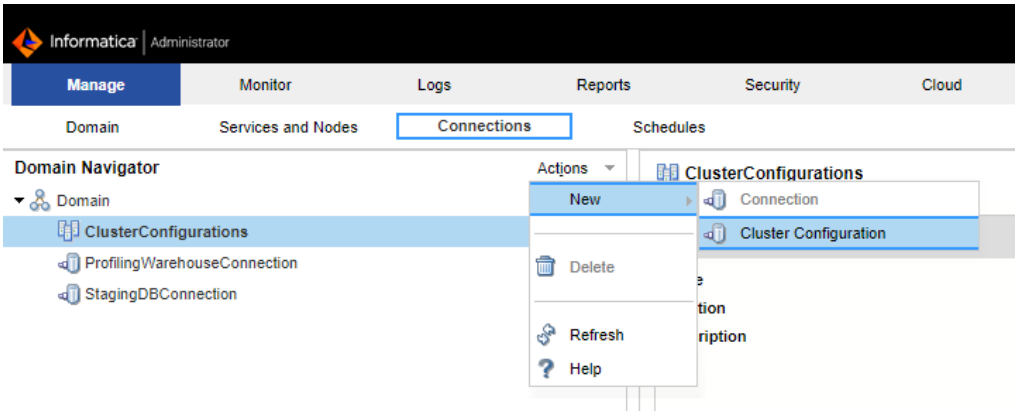
   For example : `10.75.169.88 hn0-secure.qrg2yxlb0aouniiuvfp3betl3d.ix.internal.cloudapp.net hn0-secure`
6. Log in to the machine where the Informatica Hadoop cluster is installed.
7. Navigate to the `/etc/hosts` file and add the Ambari cluster hosts to the file in the following format: `<IP Address> <Host Name> <Host Alias>`

   For example : `10.75.169.88 hn0-secure.qrg2yxlb0aouniiuvfp3betl3d.ix.internal.cloudapp.net hn0-secure`

Step 2. Create a Cluster Configuration Object

1. From the Administrator tool, select the **Connections** tab.

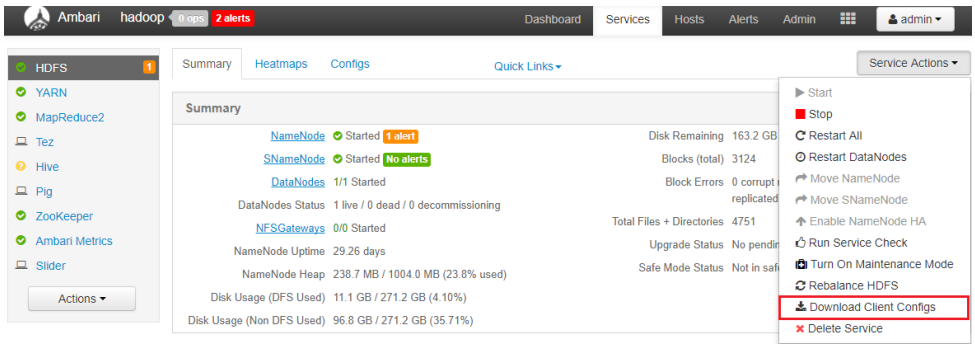
2. Expand **Domain** on the **Domain Navigator** pane.
3. Select **ClusterConfigurations** and click **Actions** > **New** > **Cluster Configuration**.



4. Enter a name for the cluster configuration.
5. Optionally, enter a description.
6. From the **Distribution Type** list, select **Hortonworks**.
7. Select a method to import cluster configuration details. You can choose from one of the following options:
   - **Import from archive file**
   - **Import from cluster**

   If you selected **Import from archive file**, go to Step 8.

   If you selected **Import from cluster**, skip to Step 9.
8. To import from an archive file, log in to Ambari and perform the following steps:
   a. Click **HDFS** on the left pane.
   b. From the **Service Actions** menu, select **Download Client Configs**.



   c. Repeat steps a and b to download the Client Configs from **YARN**, **MapReduce2**, **Tez** and **Hive** services on the left pane.
   d. Verify that the following .tar files downloaded:
      - `HDFS_CLIENT-configs.tar.gz`
      - `YARN_CLIENT-configs.tar.gz`
      - `MAPREDUCE2_CLIENT-configs.tar.gz`

- `TEZ_CLIENT-configs.tar.gz`
- `HIVE_CLIENT-configs.tar.gz`

e. Extract each file.

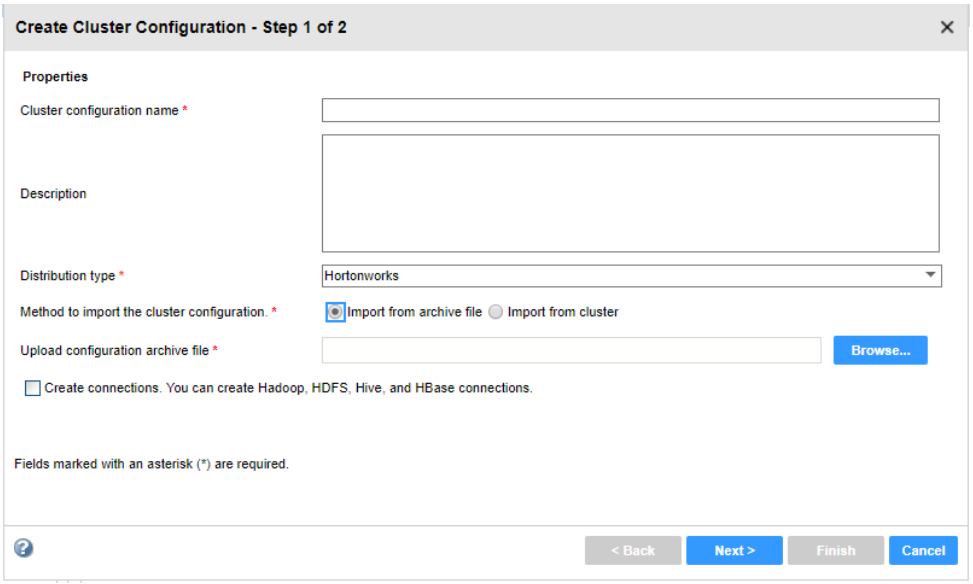
Verify that the following files extracted:

- `hdfs-site.xml` and `core-site.xml` from the `HDFS_CLIENT-configs.tar.gz` file.
- `yarn-site.xml` from the `YARN_CLIENT-configs.tar.gz` file.
- `hive-site.xml` from the `HIVE_CLIENT-configs.tar.gz` file.
- `mapred-site.xml` from the `MAPREDUCE2_CLIENT-configs.tar.gz` file.
- `tez-site.xml` from the `tez_CLIENT-configs.tar.gz` file.

f. Select the extracted files and compress the files in a new .zip file.

g. Go back to the Administrator tool.

h. From the **Create Cluster Configuration** wizard, click **Browse** in the **Upload configuration archive file** field and upload the .zip file.



9. To import from the cluster, enter values for the following properties:

Name	Description
Ambari Host	The hostname of the machine on which Ambari is running.
Ambari Port	The port on which Ambari is running. Default is 8080.
Ambari user ID	The user name for logging into Ambari.
Ambari password	The password for the Ambari user.
Cluster name	Optional field to provide the cluster name in Ambari. Do not provide the value if Ambari is not configured for multiple clusters. If you still want to provide a value, get the exact cluster name from your administrator.

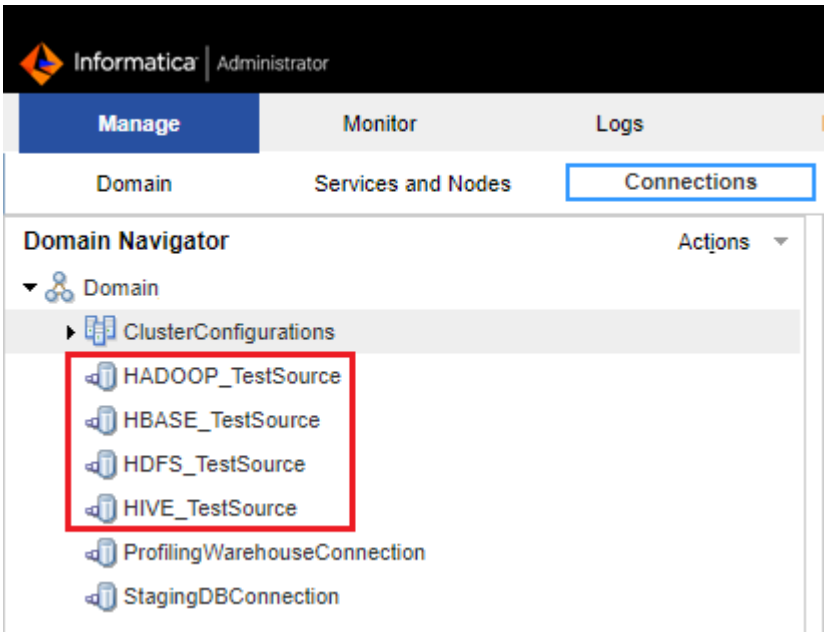
10. Check the **Create connections** option.

11. Click **Next**.
12. Review the information and click **Finish**.

    The Hadoop, HDFS, Hive, and HBase connections are created with the following name format:

    - `HADOOP_<Cluster Configuration Name>`
    - `HBASE_<Cluster Configuration Name>`
    - `HDFS_<Cluster Configuration Name>`
    - `HIVE_<Cluster Configuration Name>`

    For example, if the cluster configuration name is `TestSource`, the connection names would appear as in the following image:



    Note: If you want to delete the cluster configuration object, you must first delete the four connections.

Step 3. Install the Database Client Software

If the Profiling Warehouse database in the associated Data Integration Service is on Oracle, install the database client software on the nodes within the Hive source cluster.

Note: If the Profiling Warehouse database in the associated Data Integration Service is on IBM Db2 or Microsoft SQL Server, use a JDBC connection and skip this step.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. Install the Oracle database client software in the same Oracle client path on all of the nodes within the Hive source cluster.
2. Add an entry in the `tnsnames.ora` file for the Profiling Warehouse database connection.

   For example:

```
ORAMAT=
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST =localhost)(PORT = 1521))
    (CONNECT_DATA =
```

```
    (SERVER = DEDICATED)
    (SERVICE_NAME = orcl)
  )
)
```

Step 4. Configure the Environment Variables

Configure environment variables only if the Profiling Warehouse connection is Oracle and the profile execution engine for the data store is Hadoop or Hive.
Note: Do not configure environment variables in the following situations:

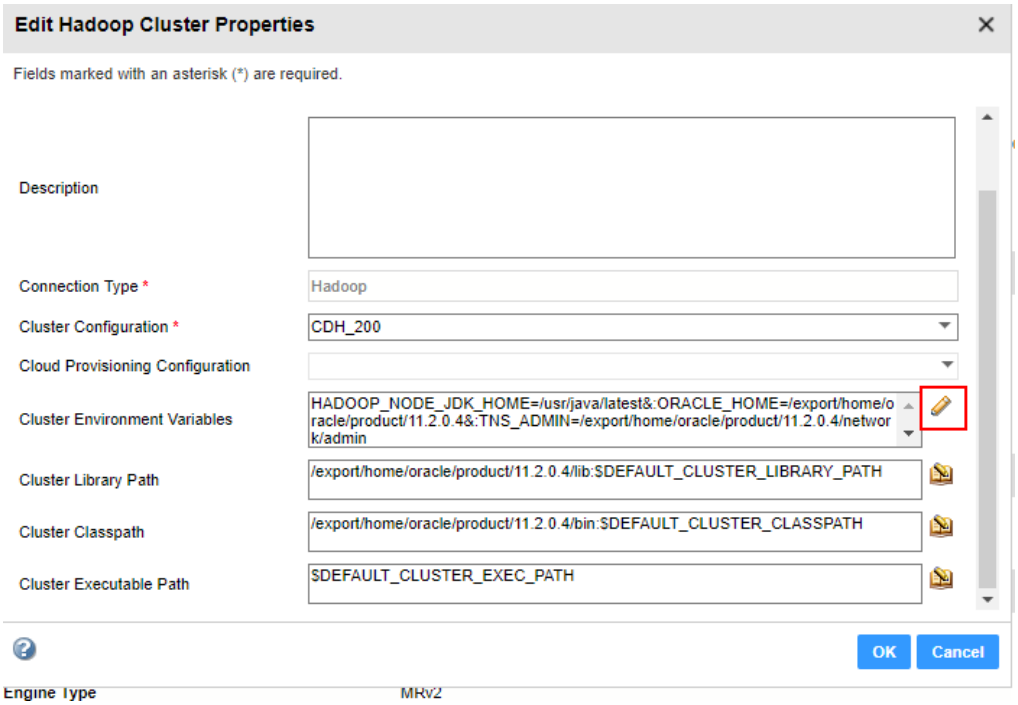
- The Profiling Warehouse connection is JDBC.
- The profile execution engine for the data store is native.

1. From the Administrator tool, click the **Manage** > **Connections** tab.
2. From the **Domain Navigator** pane on the left, select the Hadoop cluster connection.

   The Hadoop connection properties appear on the right.
3. Locate the **Hadoop Cluster Properties** section and click the **Edit** icon.



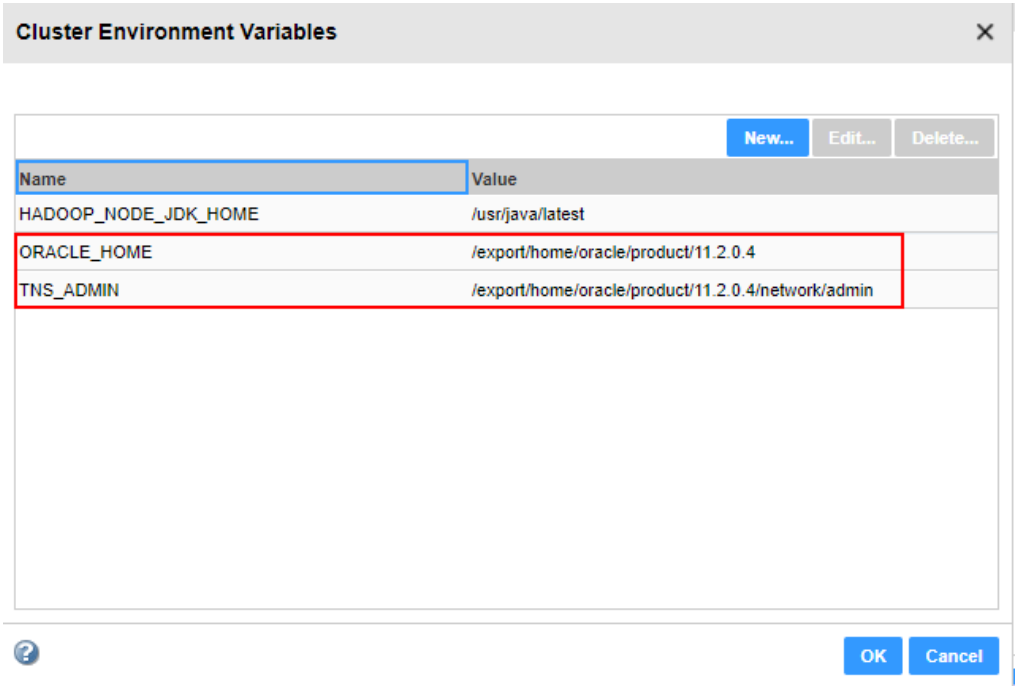
   The **Edit Hadoop Cluster Properties** window appears.
4. Click the **Edit** icon for the **Cluster Environment Variables** property.



   The **Cluster Environment Variables** window appears.

5. Click **New**.
6. Add the following environment variables:

Name	Value
ORACLE_HOME	/export/home/oracle/product/11.2.0.4
TNS_ADMIN	/export/home/oracle/product/11.2.0.4/network/admin

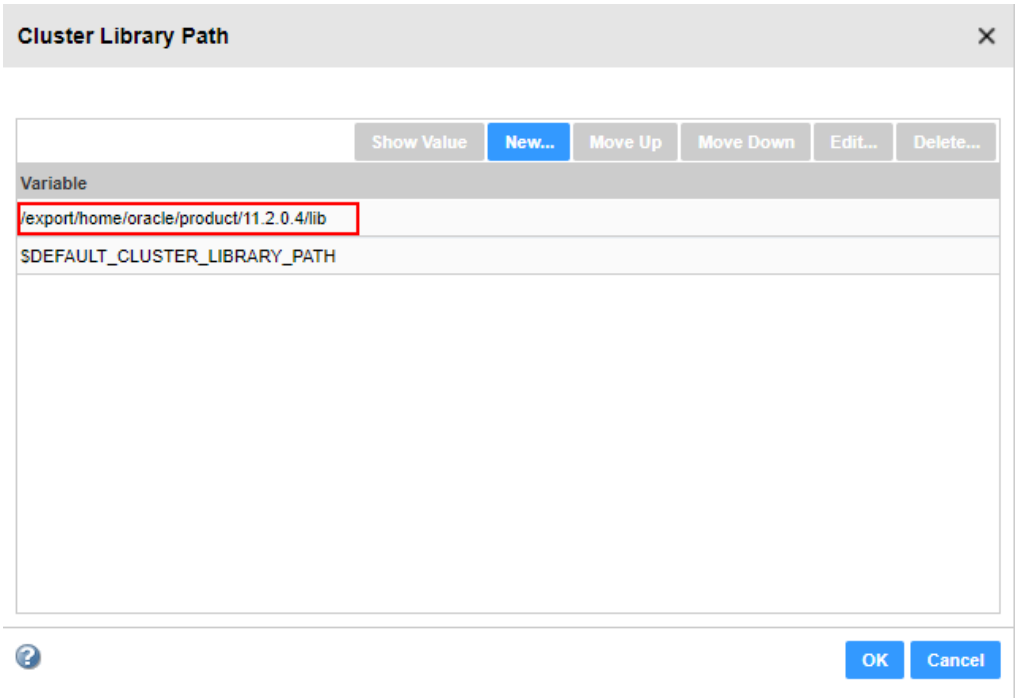


7. Click **OK**.

   The **Cluster Environment Variables** window closes.
8. Click the **Edit** icon for the **Cluster Library Path** property.

   The **Cluster Library Path** window appears.
9. Click **New**.

10. Enter the Oracle library path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/lib`



11. Click **OK**.

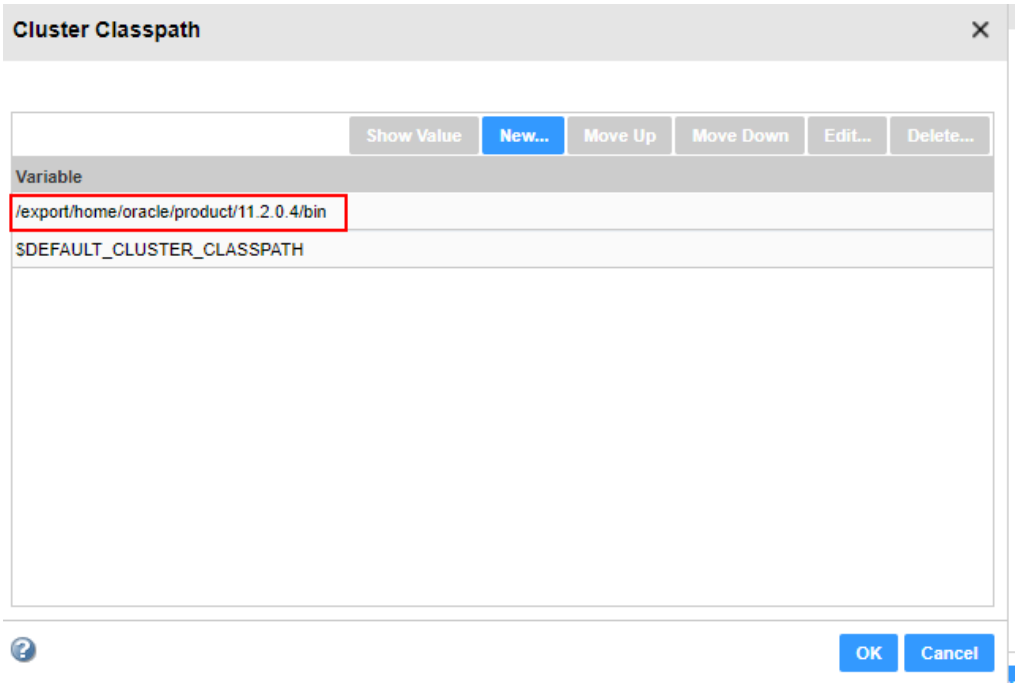
    The **Cluster Library Path** window closes.

12. Click the **Edit** icon for the **Cluster Classpath** property.

    The **Cluster Classpath** window appears.

13. Click **New**.

14. Enter the Oracle class path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/bin`



15. Click **OK**.

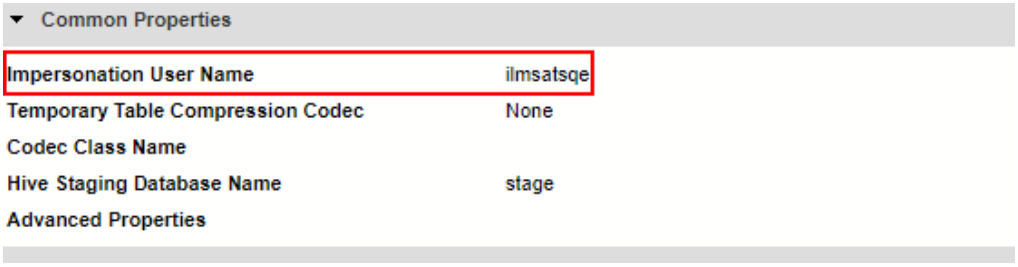
    The **Cluster Library Path** window closes.

16. Click **OK**.

    The **Edit Hadoop Cluster Properties** window closes.

17. Locate the **Common Properties** section and click the **Edit** icon.

18. In the **Impersonation User Name** field, enter the name of the user who can access the Hive database.



19. Click **OK**.

Step 5. Create a Hive Data Store in Data Privacy Management

Create a data store in Data Privacy Management and configure properties to connect the Hive source.

You can include the data store in a scan. When you scan the data store, the Data Privacy Management Service creates a Big Data Scan job. The Scan job uses the connection properties in the data store to connect to the Hive data store. For more information, see the *Informatica Data Privacy Management User Guide*.

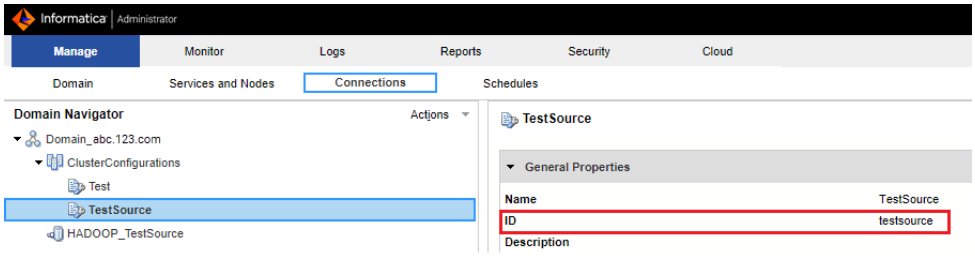
1. Go to the Data Privacy Management user interface and click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. Click **New**.

   The **New Data Store** page appears.
3. Enter a name for the data store.
4. Select **Big Data** as the **Category**.
5. Select **Hive** as the **Data Store Type**.
6. Select **Azure HDInsight** as the **Hadoop Distribution**.
7. Enter or select values for all required fields.
8. To determine the **Cluster Configuration ID**, perform the following steps:
   a. From the Administrator tool, select the **Manage** > **Connections** tab.
   b. From the **Domain Navigator** pane, select the cluster configuration object you created.

      The **General Properties** section appears on the right.
   c. Make a note of the ID.



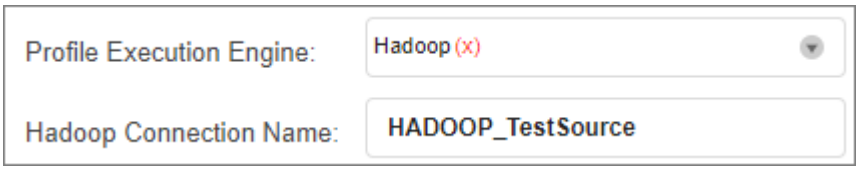
   Note: The **ID** value is automatically generated as the lowercase version of the cluster configuration name. For example, if the cluster configuration name is `TestSource`, then the ID is `testsource`.

   In Data Privacy Management, enter the ID in the **Cluster Configuration Id** field.
9. Specify the **Profile Execution Engine**. Select one of the following options:
   - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. See the "Hadoop Profile Execution Engine" on page 124 section for configuration steps.
   - **Hive**. Data Privacy Management uses the Hive engine to run profiles.

     Note: Informatica does not recommend this option because it slows down performance.

     See the "Hive Profile Execution Engine" on page 124 section for prerequisite steps.
   - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. This is the default option and does not require any additional configuration.
10. If you selected **Hadoop** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



11. If you selected **Hive** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



Hadoop Distribution Type: Cloudera

Before you perform other operations in Data Privacy Management, configure connectivity to data sources.

To connect to a Hive data source with Cloudera as the Hadoop distribution type, perform the following steps:

1. If the Cloudera Hive data store is Kerberos-enabled, add the keytab and Kerberos configuration files to the Informatica domain and Hadoop cluster.
2. Create a cluster configuration object.
3. Install the database client software.
4. Configure the environment variables.
5. Create the Cloudera Hive data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Add the Keytab File to the Informatica Domain and the Hadoop Cluster

If the Hive data store is Kerberos-enabled, add the keytab and Kerberos configuration files to the Informatica domain and Hadoop cluster.

1. Get the keytab file for a user that has access to Hive and the Hive source.
2. Copy the keytab file to a common directory on the Data Privacy Management node and all nodes of the external cluster that Data Privacy Management uses. The following example shows a common directory and keytab file name: `/opt/keytabs/<hiveuser>.keytab`

   Important:

   - The directory and file name must be the same on all Data Privacy Management nodes and Data Privacy Management Hadoop cluster nodes.
   - Verify that the file is readable. For example, you can enter the following command to check the readability of the keytab file: `chmod a+r hive.keytab`
3. Copy the `krb5.conf` file to the Data Privacy Management node in the following directories:

   ```
   /etc/krb5.conf
       <INFA_HOME>/services/shared/security/krb5.conf
       <INFA_HOME>/java/lib/security/krb5.conf
   ```
4. Add the `krb5.conf` file to the following directory on all Hadoop nodes of the external cluster:

   ```
   /etc/krb5.conf
   ```
5. Enter the following command to verify that the KDC server is known and accessible from all Data Privacy Management nodes and all Data Privacy Management Hadoop cluster nodes:

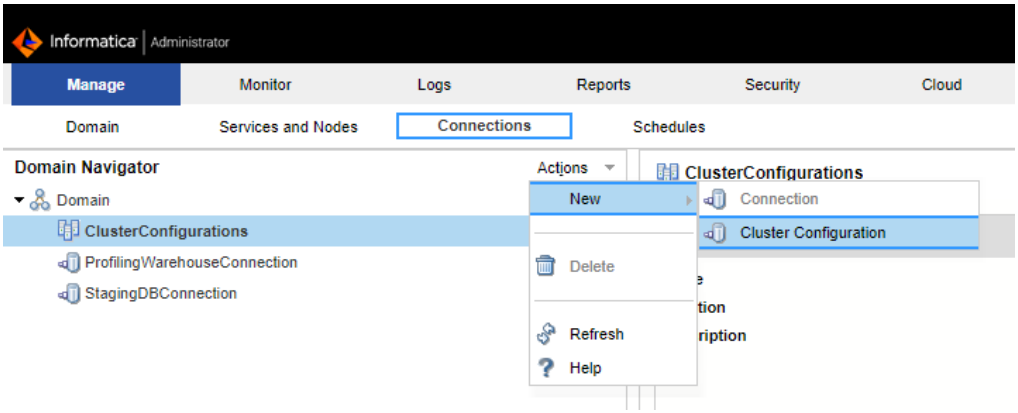
   ```
   update/etc/hosts
   ```

6. If you create a new JDBC connection to profile a database in Hadoop mode in the Administrator tool, specify the user ID and password for the database connection.

Step 2. Create a Cluster Configuration Object

Create a cluster configuration to create corresponding Hadoop, HBase, HDFS, and Hive service connections.

1. In the Administrator tool, select the **Connections** tab.
2. Expand **Domain** on the **Domain Navigator** pane.
3. Select **ClusterConfigurations** and click **Actions** > **New** > **Cluster Configuration**.



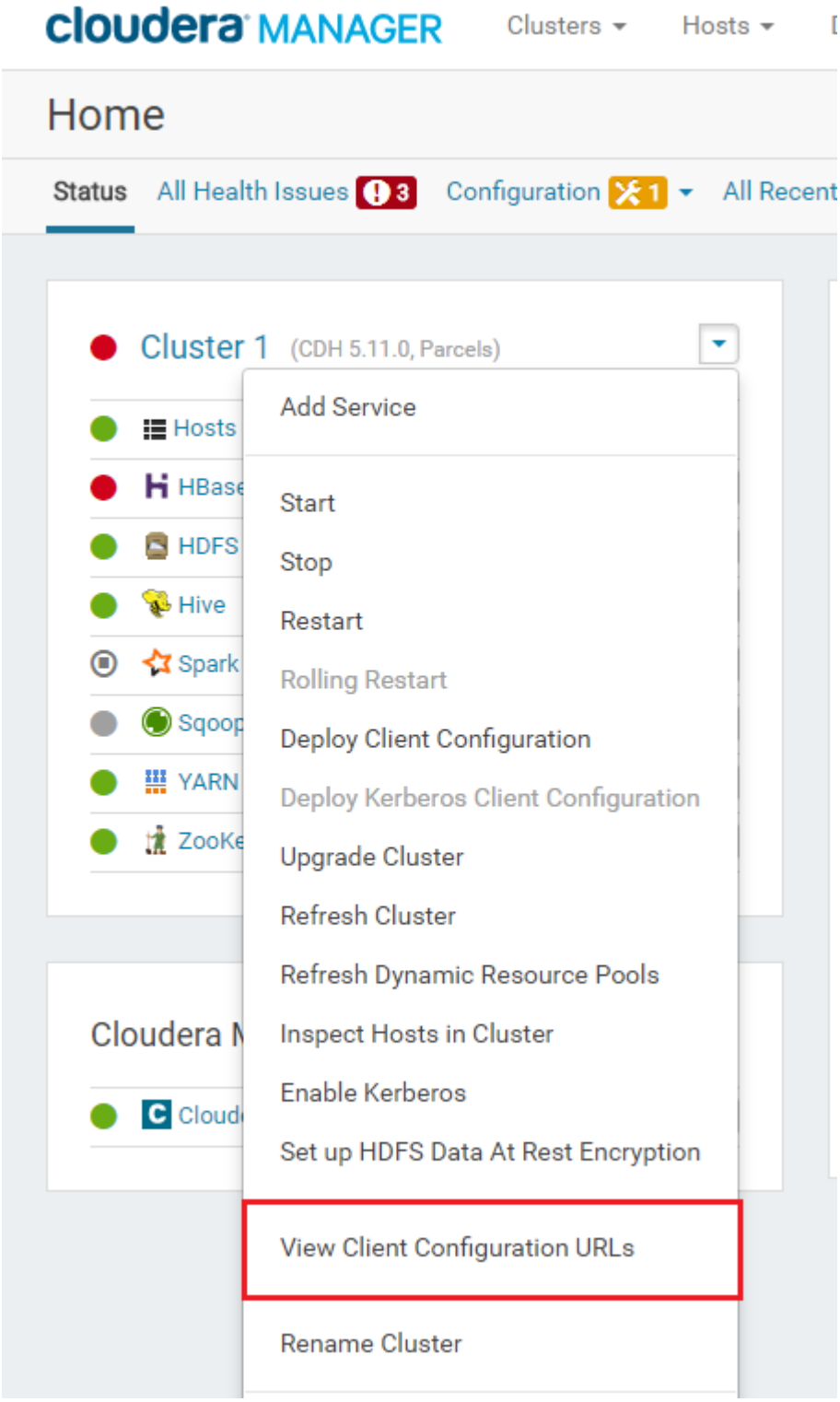
The **Create Cluster Configuration** window appears.

4. Enter a name for the cluster configuration.
5. Optionally, enter a description.
6. From the **Distribution Type** list, select **Cloudera**.
7. Select one of the following methods to import cluster configuration details:
   - **Import from archive file**
   - **Import from cluster**

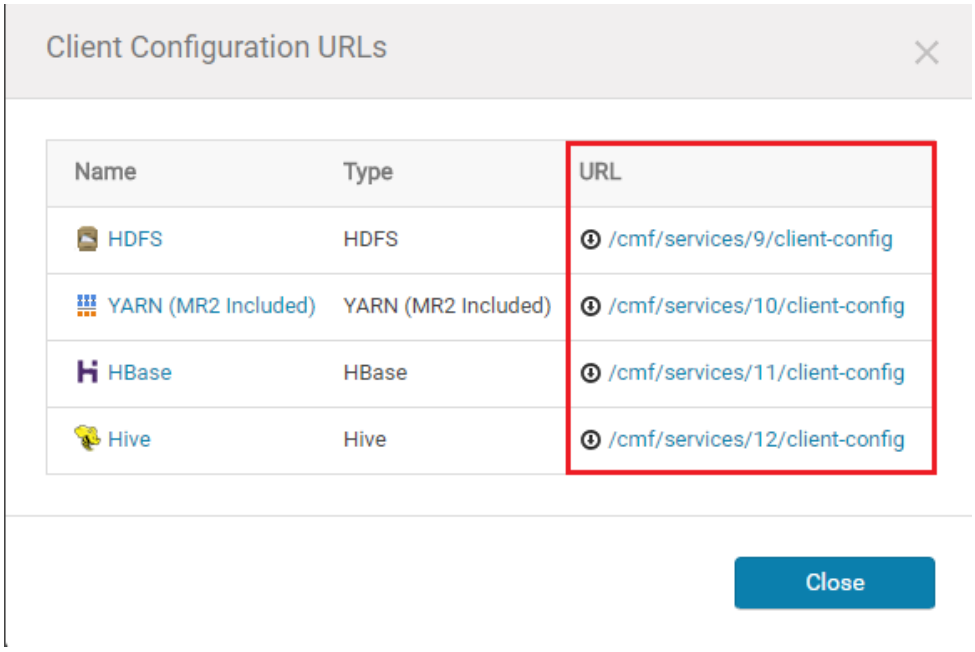
   If you selected **Import from archive file**, go to Step 8.

   If you selected **Import from cluster**, skip to Step 9.
8. To import from an archive file, perform the following steps:
   a. Log in to Cloudera Manager.

b. Click the arrow next to the Cluster name and select **View Client Configuration URLs**.



A list of client configuration URLs appears.



c. Click each URL to download the configuration file.

   The following .zip files download:

   - `hdfs-clientconfig.zip`
   - `yarn-clientconfig.zip`
   - `hbase-clientconfig.zip`
   - `hive-clientconfig.zip`

d. Extract each .zip file.

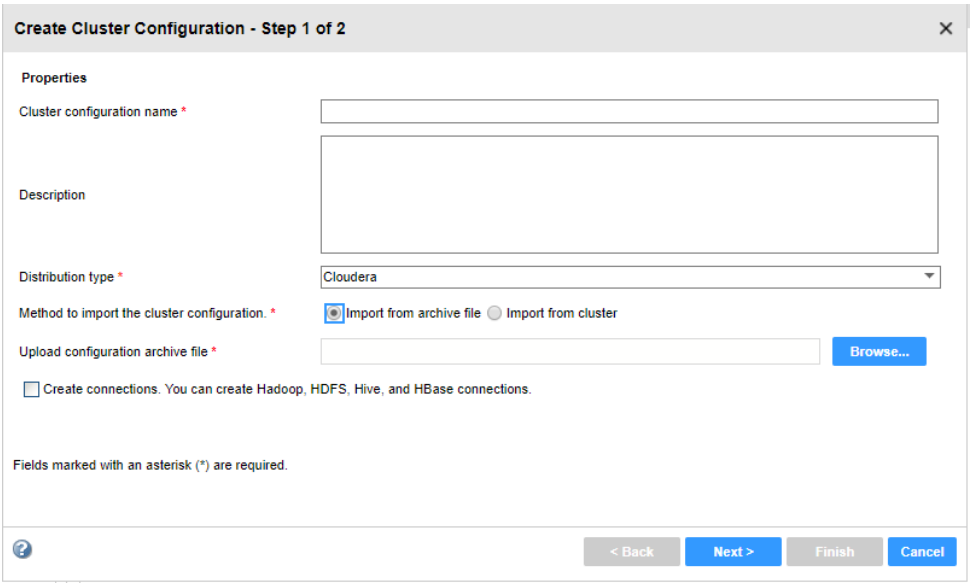
   Verify that the following files extracted:

   - `hdfs-site.xml` and `core-site.xml` from the `hdfs-clientconfig.zip` file.
   - `yarn-site.xml` from the `yarn-clientconfig.zip` file.
   - `hbase-site.xml` from the `hbase-clientconfig.zip` file.
   - `hive-site.xml` and `mapred-site.xml` from the `hive-clientconfig.zip` file.

e. Select the extracted files and compress the files in a new .zip file.

f. Go back to the Administrator tool.

g. From the **Create Cluster Configuration** wizard, click **Browse** in the **Upload configuration archive file** field and upload the new .zip file you created.



9. To import from the cluster, enter the following values:

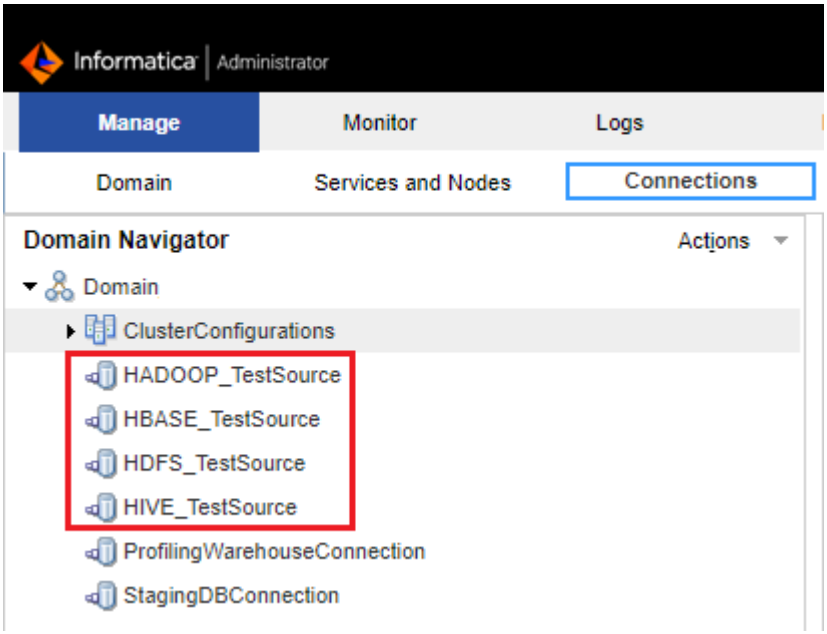
Name	Description
Cloudera Manager host	The hostname of the machine on which Cloudera Manager is running.
Cloudera Manager port	The port on which Cloudera Manager is running. Default is 7180.
Cloudera Manager user ID	The user name for logging into Cloudera Manager.
Cloudera Manager password	The password for the Cloudera Manager user.
Cluster name	Optional field to provide the cluster name in Cloudera. Do not provide the value if Cloudera is not configured for multiple clusters. If you still want to provide a value, get the exact cluster name from your administrator.

10. Check the **Create connections** option.
11. Click **Next**.
12. Review the information and click **Finish**.

    The Hadoop, HBase, HDFS, and Hive connections are created with the following name format:

    - `HADOOP_<Cluster_Configuration_Name>`
    - `HBASE_<Cluster_Configuration_Name>`
    - `HDFS_<Cluster_Configuration_Name>`
    - `HIVE_<Cluster_Configuration_Name>`

For example, if the cluster configuration name is `TestSource`, the connection names appear as shown in the following image:



Note: To delete the cluster configuration object, you must first delete the four services.

Step 3. Install the Database Client Software

If the Profiling Warehouse database in the associated Data Integration Service is on Oracle, install the database client software on the nodes within the Hive source cluster.

Note: If the Profiling Warehouse database in the associated Data Integration Service is on IBM Db2 or Microsoft SQL Server, use a JDBC connection and skip this step.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. Install the Oracle database client software in the same Oracle client path on all of the nodes within the Hive source cluster.
2. Add an entry in the `tnsnames.ora` file for the Profiling Warehouse database connection.

   For example:

```
ORAMAT=
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST =localhost)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = orcl)
    )
  )
```

Step 4. Configure the Environment Variables

If the Profiling Warehouse connection is Oracle and the profile execution engine for the data store is Hadoop or Hive, configure environment variables.

1. In the Administrator tool, click the **Manage** > **Connections** tab.

2. In the **Domain Navigator** pane on the left, select the Hadoop cluster configuration.

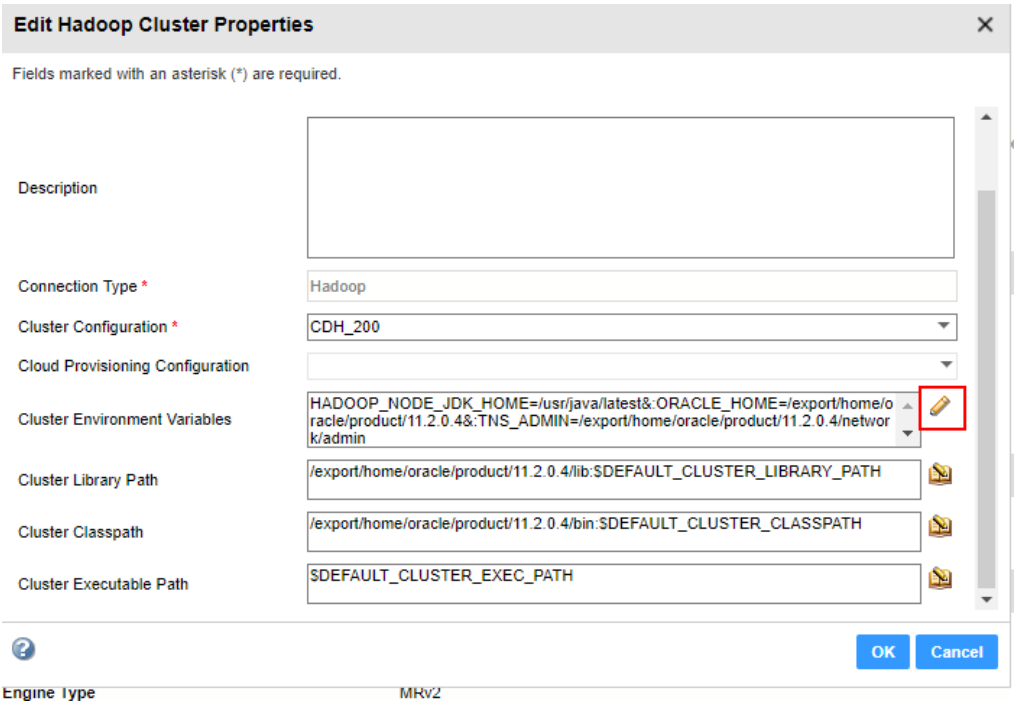
   The Hadoop cluster configuration properties appear on the right.

3. Locate the **Hadoop Cluster Properties** section and click the **Edit** icon.



   The **Edit Hadoop Cluster Properties** window appears.

4. Click the **Edit** icon for the **Cluster Environment Variables** property.

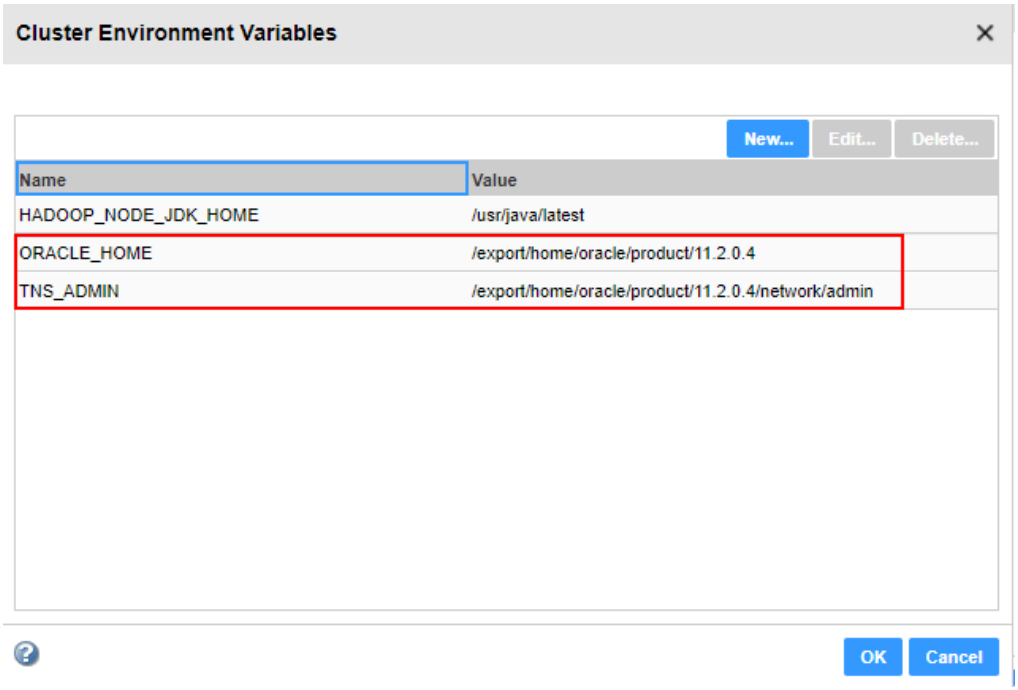


   The **Cluster Environment Variables** window appears.

5. Click **New**.

6. Add the following environment variables:

Name	Value
ORACLE_HOME	`/export/home/oracle/product/11.2.0.4`
SOFTHSM2_CONF **Note:** Required for Cloudera Hive connections with sensitive fields that you protect with encryption keys. For more information, see the "Encryption and Decryption Configuration" chapter in the *Informatica Data Privacy Management Installation and Configuration Guide*.	`/<Cloudera Hive machine path>/tokens/softhsm2.conf`
TNS_ADMIN	`/export/home/oracle/product/11.2.0.4/network/admin`



7. Click **OK**.

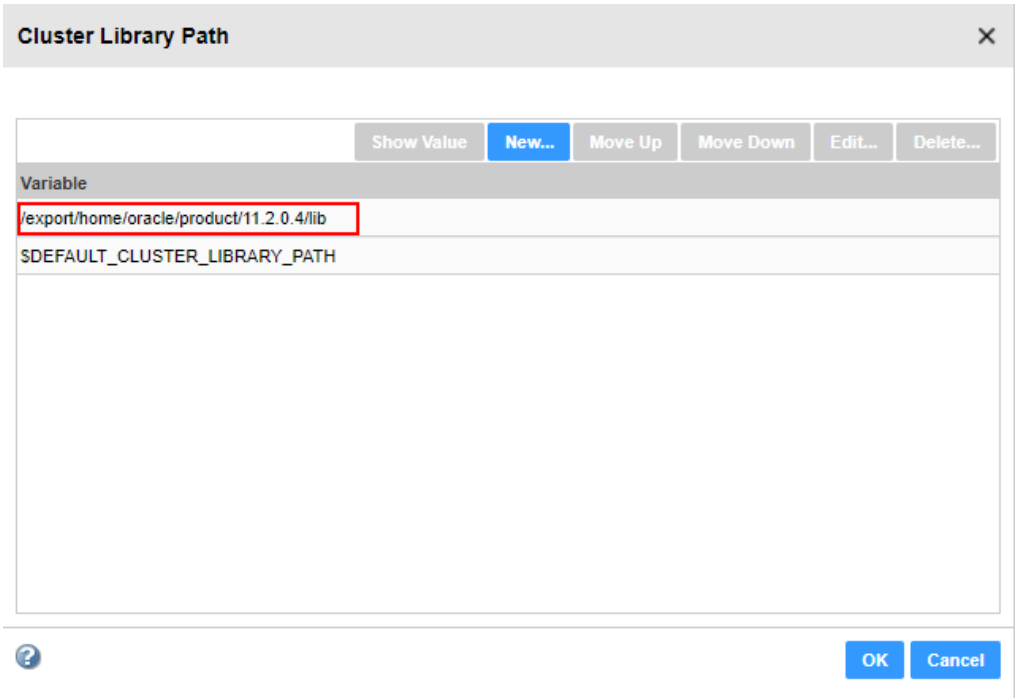
   The **Cluster Environment Variables** window closes.

8. Click the **Edit** icon for the **Cluster Library Path** property.

   The **Cluster Library Path** window appears.

9. Click **New**.

10. Enter the Oracle library path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/lib`



11. Click **OK**.

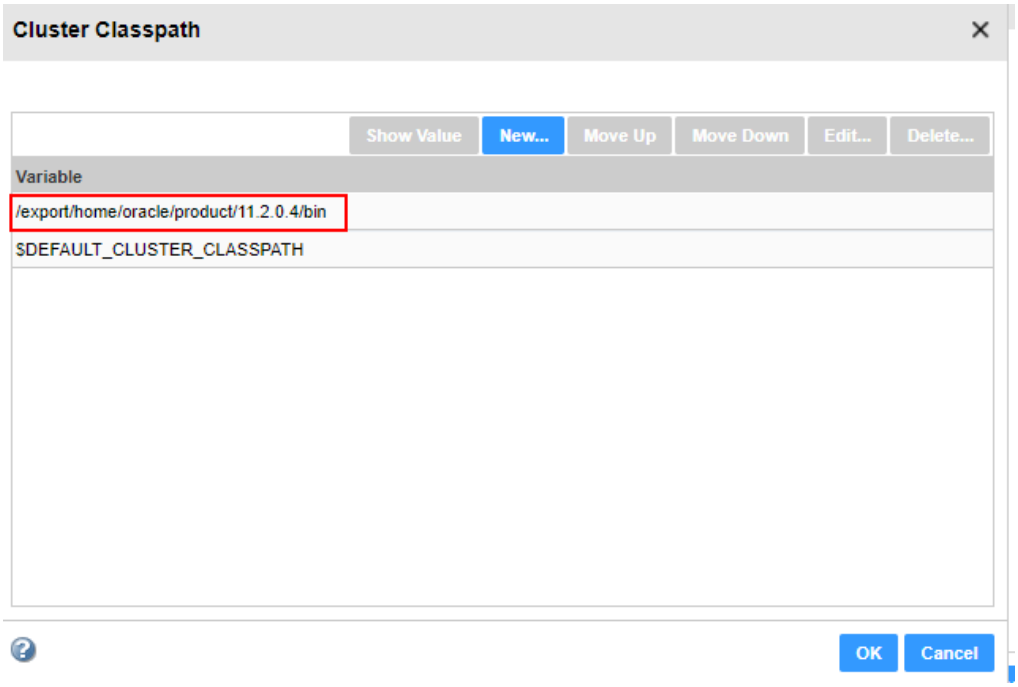
    The **Cluster Library Path** window closes.

12. Click the **Edit** icon for the **Cluster Classpath** property.

    The **Cluster Classpath** window appears.

13. Click **New**.

14. Enter the Oracle class path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/bin`



15. Click **OK**.

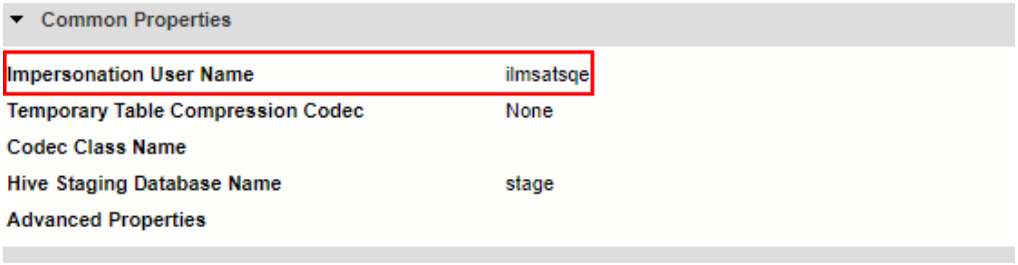
    The **Cluster Library Path** window closes.

16. Click **OK**.

    The **Edit Hadoop Cluster Properties** window closes.

17. Locate the **Common Properties** section and click the **Edit** icon.

18. In the **Impersonation User Name** field, enter the name of the user who can access the Hive database.



19. Click **OK**.

Step 5. Create a Hive Data Store in Data Privacy Management

To connect to the Cloudera Hive source, create a data store in Data Privacy Management and configure properties.

Before you create the data store, download the `com.infa.sats.hivecdh-connectionprovider-impl.jar` file and copy it to the following location: `<Informatica installation directory>/services/SecureAtSourceService/ROOT/WEB-INF/eclipse/plugins`
Contact Informatica Global Customer Service to access the `.jar` file.

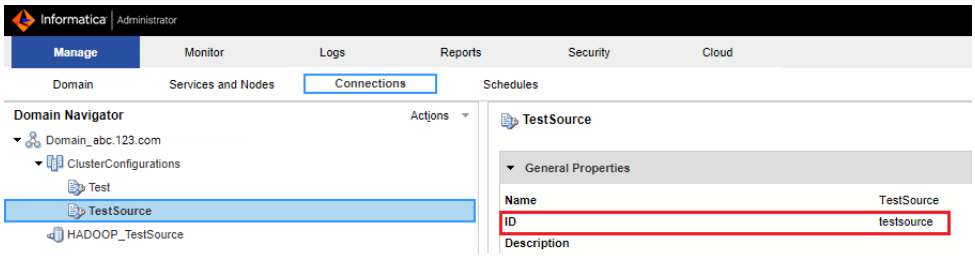
Recycle the Data Privacy Management Service after you copy the file.

1. Log in to Data Privacy Management and click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Click **New**.

   The **New_Data_Store** page appears.
3. Enter a name for the data store.
4. Select **Big Data** as the **Category**.
5. Select **Hive** as the **Data Store Type**.
6. Select **Cloudera** as the **Hadoop Distribution**.
7. Enter or select values for all required fields.
8. To determine the **Cluster Configuration ID**, perform the following steps:
   a. In the Administrator tool, select the **Manage** > **Connections** tab.
   b. In the **Domain Navigator** pane, select the cluster configuration object you created.

      The **General Properties** pane appears on the right.
   c. Make a note of the ID.



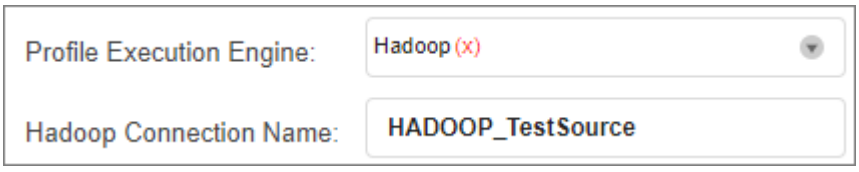
   Note: The **ID** value is automatically generated as the lower-case version of the cluster configuration name. For example, if the cluster configuration name is `TestSource`, then the ID is `testsource`.

   In Data Privacy Management, enter the ID in the **Cluster Configuration ID** field.
9. Select one of the following **Profile Execution Engine** types:
   - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. See the "Hadoop Profile Execution Engine" on page 124 section for configuration steps.
   - **Hive (Deprecated)**. Data Privacy Management uses the Hive engine to run profiles.

     Note: Informatica does not recommend this option because it slows performance.

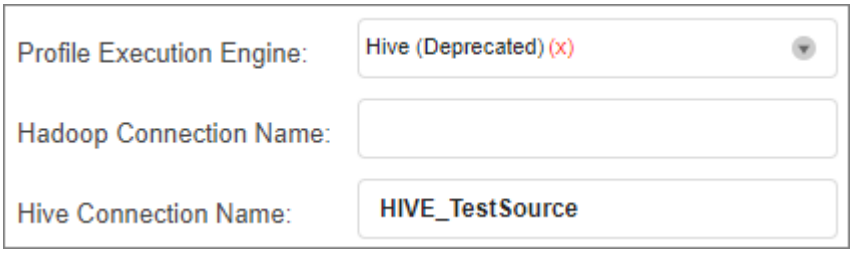
     See the "Hive Profile Execution Engine" on page 124 section for prerequisite steps.
   - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. This is the default option and does not require any additional configuration.
10. If you selected **Hadoop** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    The Hadoop connection name appears on the **Connections** tab in the Administrator tool.



11. If you selected **Hive** as the **Profile Execution Engine**, specify the **Hive Connection Name**.

    The Hive connection name appears on the **Connections** tab in the Administrator tool.



When you scan the Cloudera Hive data store, the Data Privacy Management Service creates a Big Data Scan job that connects to the data store. For more information, see the *Informatica Data Privacy Management User Guide*.

Hadoop Distribution Type: Hortonworks

Perform the following steps to connect to a Hive source with Hortonworks as the distribution type.

1. If the Hive data store is Kerberos-enabled, add the Keytab and Kerberos configuration files to the Informatica domain and Hadoop cluster.
2. Create a cluster configuration object.
3. Install the database client software.
4. Configure the environment variables.
5. Create the Hive data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Add the Keytab and the Kerberos Configuration File to the Informatica Domain and the Hadoop Cluster

Note: Perform the steps in this section only if the Hive source is enabled for Kerberos.

1. Get the keytab for a user that has access to Hive and the Hive source.
2. Copy the keytab to a common directory on the Data Privacy Management node and all nodes of the external cluster that Data Privacy Management uses. For example: `/opt/keytabs/hiveuser.keytab`

   The directory and filename must be the same on all Data Privacy Management and Data Privacy Management Hadoop cluster nodes.
   Also, make sure that the file is readable. Example: `chmod a+r hive.keytab`
3. Copy the `krb5.conf` file to the Data Privacy Management node:

   ```
   /etc/krb5.conf
       <INFA_HOME>/services/shared/security/krb5.conf
       <INFA_HOME>/java/lib/security/krb5.conf
   ```
4. Copy the `krb5.conf` file on all Hadoop nodes of the external cluster that Data Privacy Management uses:

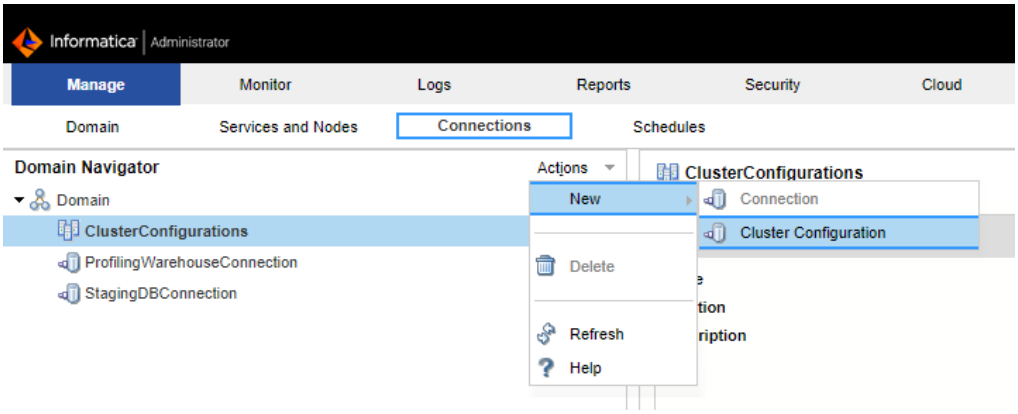
   ```
   /etc/krb5.conf
   ```
5. Verify that the Kerberos Key Distribution Center (KDC) server is known and accessible from all Data Privacy Management nodes and Data Privacy Management Hadoop cluster nodes.

   ```
   update /etc/hosts
   ```

6. If you want to select the **Create a new JDBC Profiling WareHouse Connection** option in the Administrator tool, enter the same user ID and password as the IBM Db2 Profiling Warehouse Connection.

Step 2. Create a Cluster Configuration Object

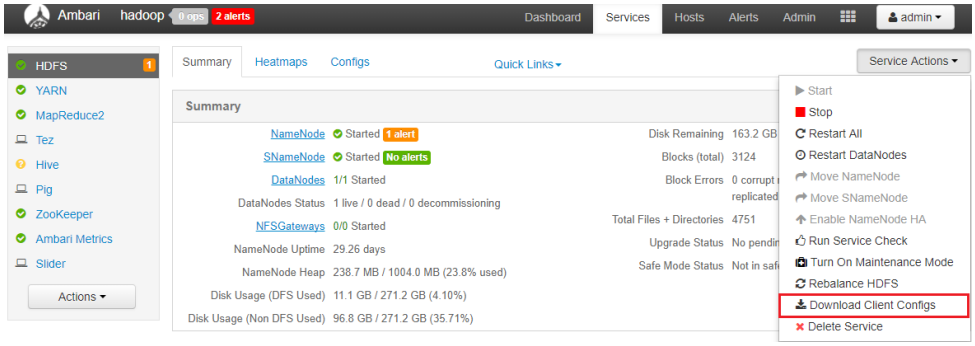
1. From the Administrator tool, select the **Connections** tab.
2. Expand **Domain** on the **Domain Navigator** pane.
3. Select **ClusterConfigurations** and click **Actions** > **New** > **Cluster Configuration**.



4. Enter a name for the cluster configuration.
5. Optionally, enter a description.
6. From the **Distribution Type** list, select **Hortonworks**.
7. Select one of the following methods to import cluster configuration details:
   - **Import from archive file**
   - **Import from cluster**

   If you selected **Import from archive file**, go to Step 8.

   If you selected **Import from cluster**, skip to Step 9.
8. To import from an archive file, log in to Ambari and perform the following steps:
   a. Click **HDFS** on the left pane.
   b. From the **Service Actions** menu, select **Download Client Configs**.



   c. Repeat steps a and b to download the client configurations from **YARN**, **MapReduce2**, **Tez** and **Hive** services.

d. Verify that the following .tar files download:
   - `HDFS_CLIENT-configs.tar.gz`
   - `YARN_CLIENT-configs.tar.gz`
   - `MAPREDUCE2_CLIENT-configs.tar.gz`
   - `TEZ_CLIENT-configs.tar.gz`
   - `HIVE_CLIENT-configs.tar.gz`

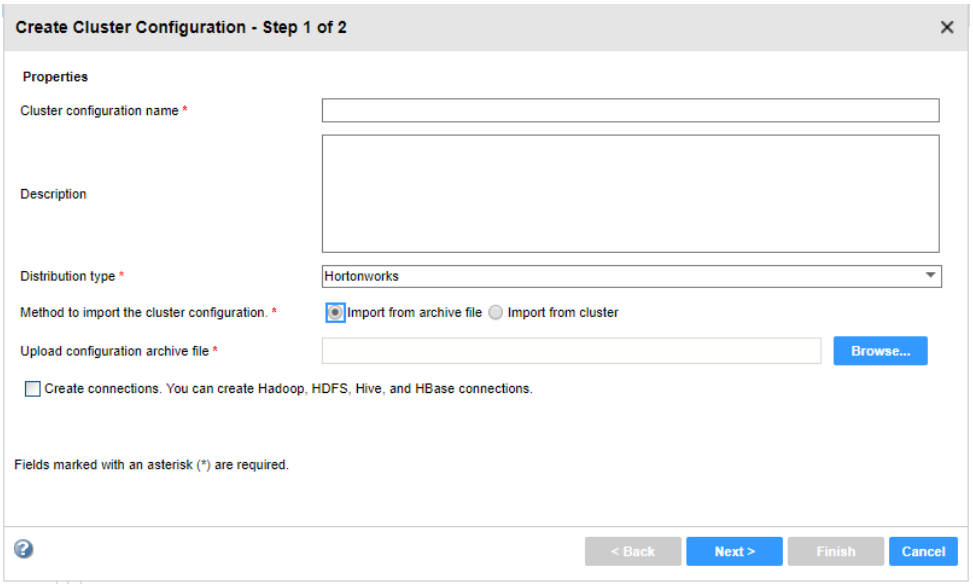
e. Extract each .zip file.

   Verify that the following files extract:
   - `hdfs-site.xml` and `core-site.xml` from the `HDFS_CLIENT-configs.tar.gz` file.
   - `yarn-site.xml` from the `YARN_CLIENT-configs.tar.gz` file.
   - `hive-site.xml` from the `HIVE_CLIENT-configs.tar.gz` file.
   - `mapred-site.xml` from the `MAPREDUCE2_CLIENT-configs.tar.gz` file.
   - `tez-site.xml` from the `tez_CLIENT-configs.tar.gz` file.

f. Select the extracted files and compress the files in a new .zip file.

g. Go back to the Administrator tool.

h. From the **Create Cluster Configuration** wizard, click **Browse** in the **Upload configuration archive file** field and upload the .zip file.



9. To import from the cluster, enter values for the following properties:

Name	Description
Ambari Host	The hostname of the machine on which Ambari is running.
Ambari Port	The port on which Ambari is running. Default is 8080.
Ambari user ID	The user name for logging into Ambari.

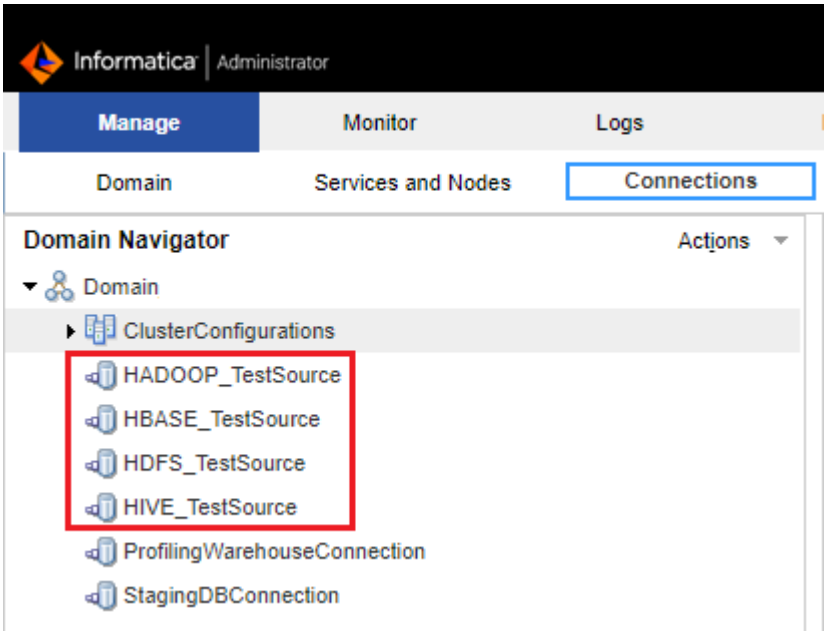
Name	Description
Ambari password	The password for the Ambari user.
Cluster name	Optional field to provide the cluster name in Ambari. Do not provide the value if Ambari is not configured for multiple clusters. If you still want to provide a value, get the exact cluster name from your administrator.

10. Check the **Create connections** option.
11. Click **Next**.
12. Review the information and click **Finish**.

    The Hadoop, HDFS, Hive, and HBase connections are created with the following name format:

    - `HADOOP_<Cluster Configuration Name>`
    - `HBASE_<Cluster Configuration Name>`
    - `HDFS_<Cluster Configuration Name>`
    - `HIVE_<Cluster Configuration Name>`

    For example, if the cluster configuration name is `TestSource`, the connection names would appear as in the following image:



Note: If you want to delete the cluster configuration object, you must first delete the four connections.

Step 3. Install the Database Client Software

If the Profiling Warehouse database in the associated Data Integration Service is on Oracle, install the database client software on the nodes within the Hive source cluster.

Note: If the Profiling Warehouse database in the associated Data Integration Service is on IBM Db2 or Microsoft SQL Server, use a JDBC connection and skip this step.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. Install the Oracle database client software in the same Oracle client path on all of the nodes within the Hive source cluster.
2. Add an entry in the `tnsnames.ora` file for the Profiling Warehouse database connection.

   For example:

```
ORAMAT=
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST =localhost)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = orcl)
    )
  )
```

Step 4. Configure the Environment Variables

Configure environment variables only if the Profiling Warehouse connection is Oracle and the profile execution engine for the data store is Hadoop or Hive.
Note: Do not configure environment variables in the following situations:

- The Profiling Warehouse connection is JDBC.
- The profile execution engine for the data store is native.

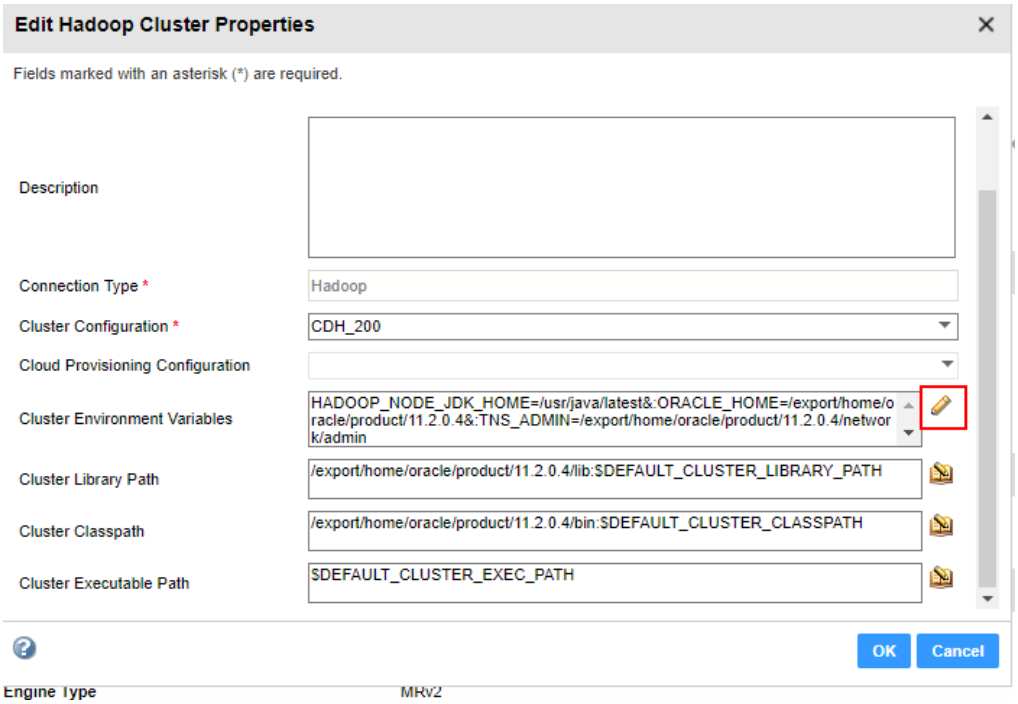
1. From the Administrator tool, click the **Manage** > **Connections** tab.
2. From the **Domain Navigator** pane on the left, select the Hadoop cluster connection.

   The Hadoop connection properties appear on the right.
3. Locate the **Hadoop Cluster Properties** section and click the **Edit** icon.



The **Edit Hadoop Cluster Properties** window appears.

4. Click the **Edit** icon for the **Cluster Environment Variables** property.

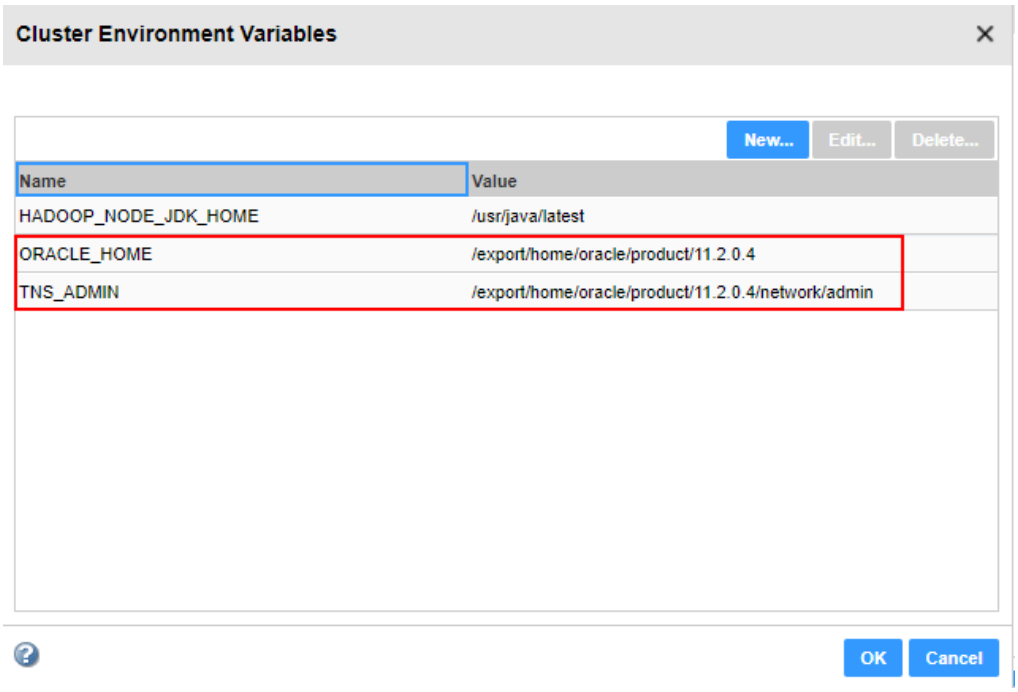


The **Cluster Environment Variables** window appears.

5. Click **New**.

6. Add the following environment variables:

Name	Value
ORACLE_HOME	`/export/home/oracle/product/11.2.0.4`
TNS_ADMIN	`/export/home/oracle/product/11.2.0.4/network/admin`



7. Click **OK**.

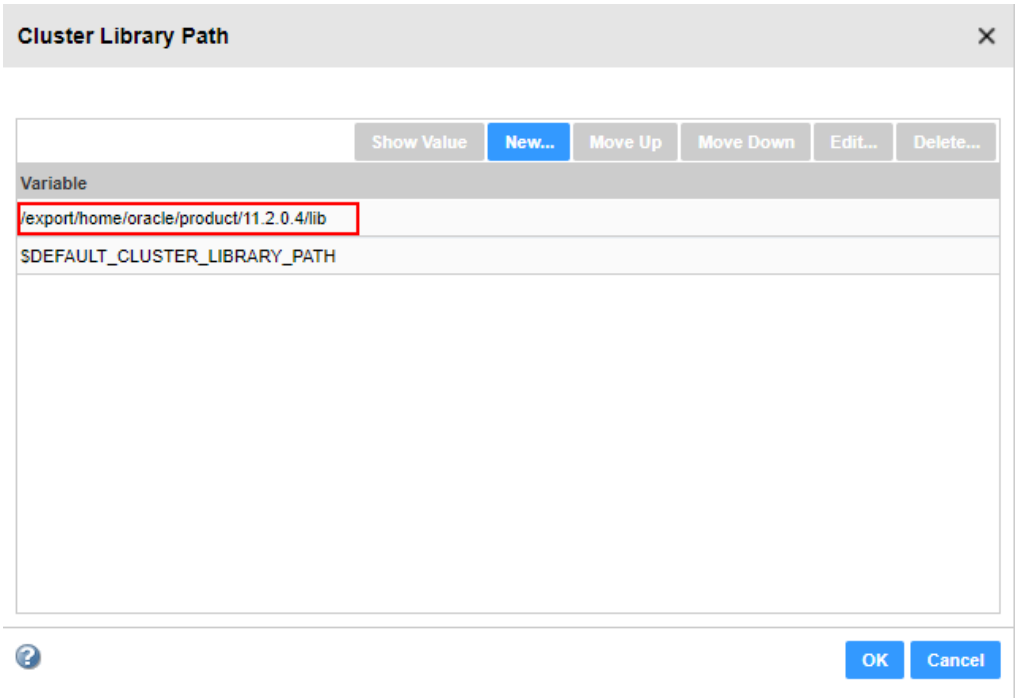
   The **Cluster Environment Variables** window closes.

8. Click the **Edit** icon for the **Cluster Library Path** property.

   The **Cluster Library Path** window appears.

9. Click **New**.

10. Enter the Oracle library path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/lib`



11. Click **OK**.

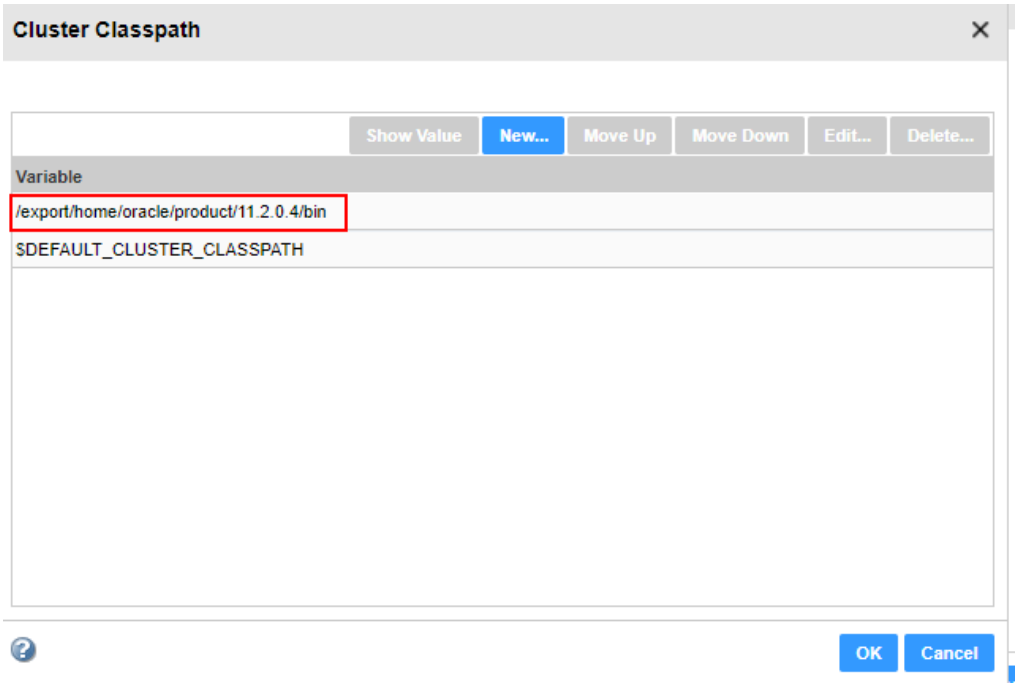
    The **Cluster Library Path** window closes.

12. Click the **Edit** icon for the **Cluster Classpath** property.

    The **Cluster Classpath** window appears.

13. Click **New**.

14. Enter the Oracle class path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/bin`



15. Click **OK**.

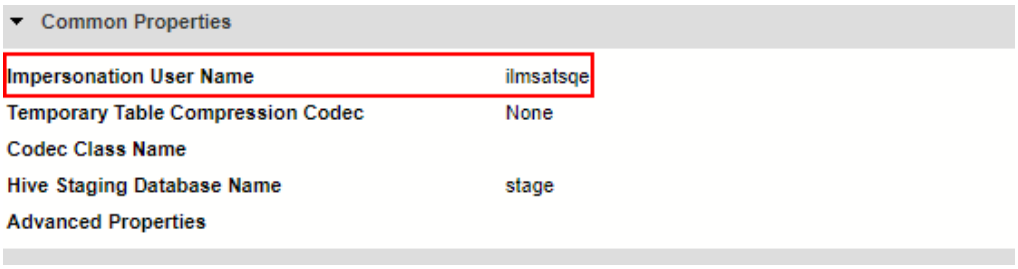
    The **Cluster Library Path** window closes.

16. Click **OK**.

    The **Edit Hadoop Cluster Properties** window closes.

17. Locate the **Common Properties** section and click the **Edit** icon.

18. In the **Impersonation User Name** field, enter the name of the user who can access the Hive database.



19. Click **OK**.

Step 5. Create a Hive Data Store in Data Privacy Management

Create a data store in Data Privacy Management and configure properties to connect the Hive source.

Before you create the data store, download the `com.infa.sats.hivehw-connectionprovider-impl.jar` file and copy it to the following location: `<Informatica installation directory>/services/SecureAtSourceService/ROOT/WEB-INF/eclipse/plugins`
Contact Informatica Global Customer Service to access the `.jar` file.

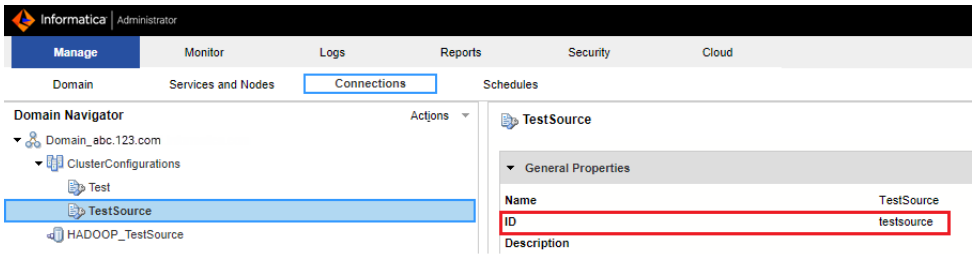
Recycle the Data Privacy Management Service after you copy the file.

1. Go to the Data Privacy Management user interface and click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Click **New**.

   The **New Data Store** page appears.
3. Enter a name for the data store.
4. Select **Big Data** as the **Category**.
5. Select **Hive** as the **Data Store Type**.
6. Select **Hortonworks** as the **Hadoop Distribution**.
7. Enter or select values for all required fields.
8. To determine the **Cluster Configuration ID**, perform the following steps:
   a. From the Administrator tool, select the **Manage** > **Connections** tab.
   b. From the **Domain Navigator** pane, select the cluster configuration object you created.

      The **General Properties** section appears on the right.
   c. Make a note of the ID.



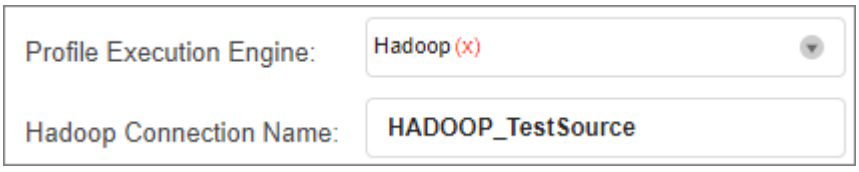
   Note: The **ID** value is automatically generated as the lowercase version of the cluster configuration name. For example, if the cluster configuration name is `TestSource`, then the ID is `testsource`.

   In Data Privacy Management, enter the ID in the **Cluster Configuration Id** field.
9. Specify the **Profile Execution Engine**. Select one of the following options:
   - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. See the "Hadoop Profile Execution Engine" on page 124 section for configuration steps.
   - **Hive**. Data Privacy Management uses the Hive engine to run profiles.

     Note: Informatica does not recommend this option because it slows down performance.

     See the "Hive Profile Execution Engine" on page 124 section for prerequisite steps.
   - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. This is the default option and does not require any additional configuration.
10. If you selected **Hadoop** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



11. If you selected **Hive** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



When you scan the data store, the Data Privacy Management Service creates a Big Data Scan job. The Scan job uses the connection properties in the data store to connect to the Hive data store. For more information, see the *Informatica Data Privacy Management User Guide*.

Hadoop Distribution Type: MapR

Perform the following steps to connect to a Hive source with MapR as the distribution type.

1. Complete the prerequisite steps.
2. Create a cluster configuration object.
3. Install the database client software.
4. Configure the environment variables.
5. Create the Hive data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Complete the Prerequisite Steps

Get the Data Integration Service User Details

Log in to the Data Integration Service machine and get the user details.

For example, a Data Integration Service user `ilmqa` might have the following properties:

```
[dtmqa@testmachine~]$ id dtmqa
uid=2483(dtmqa)
gid=128(qa)
```

Create a Data Integration Service User on MapR Cluster Nodes

After you verify the Data Integration Service user details, log in to each MapR cluster node as the root user and perform the following steps:

1. Create a user that has the same user ID and name as the Data Integration Service user. Use the following format: `sudo useradd <userName> -d <user_homedirectory> -u <userID> -g <gid>`

   Example: `sudo useradd dtmqa -d /home/dtmqa -u 2483 -g 128`
2. Verify the user details on all nodes of the cluster. Example:

```
id dtmqa
uid=2483(dtmqa) gid=128(qa) groups=128(qa)
```

Set the Hive Warehouse Directory Permissions

1. Log in to one of the cluster nodes as the MapR user.

2. Verify that the user you previously created has permissions for the Hive warehouse directory. The Hive warehouse directory is set in the following file: `/opt/mapr/hive/hive-<version>/conf/hive-site.xml`

   For example, if the warehouse directory is `/user/hive/warehouse`, run the following command to grant the user permissions for the directory: `hadoop fs -chmod -R 777 /user/hive/warehouse`

Set Up the HDFS User Directory

1. Log in to one of the cluster nodes as the MapR user.
2. Create a user directory in HDFS for the Data Integration Service user in the following format: `hadoop fs -mkdir /user/<userName>`

   Example: `hadoop fs -mkdir /user/dtmqa`
3. Set the Data Integration Service user as the HDFS directory owner in the following format: `hadoop fs -chown <userName> /user/<userName>`

   Example: `hadoop fs -chown -R dtmqa /user/dtmqa`

Configure the hive-site.xml File on the MapR Cluster

1. Log in to the MapR Control System and find the names of nodes where the HiveMeta and HiveServer2 services are running. By default, both services run on the same node.
2. Log in to the cluster node where the HiveMeta service is running as the root user.
3. Navigate to the following directory: `/opt/mapr/hive/hive-<version>/conf` and open the `hive-site.xml` file.
4. Add the following property if it is not already present:

```
<property>
<name>hive.metastore.execute.setugi</name>
<value>true</value>
</property>
```

5. Save the file.
6. Log in to the cluster node where the HiveServer2 service is running as the root user.
7. Navigate to the following directory: `/opt/mapr/hive/hive-<version>/conf` and open the `hive-site.xml` file.
8. Add the following property if it is not already present:

```
<property>
<name>hive.server2.enable.doAs</name>
<value>true</value>
</property>
```

9. Navigate to the MapR Control System and restart the HiveMeta and HiveServer2 services.

Configure the yarn-site.xml File on the MapR Cluster to Set Blaze Memory Requirements

1. Log in to the MapR Control System and find the names of nodes where Active and Standby Resource Managers are present.
2. Log in to each node and perform the following steps:
   a. Navigate to the following directory: `/opt/mapr/hadoop/hadoop-2.7.0/etc/hadoop` and open the `yarn-site.xml` file.
   b. Add or modify the following properties:

```
<property>
<name>yarn.nodemanager.resource.memory-mb</name>
<value>24000</value>
</property>
```

```
<property>
<name>yarn.nodemanager.resource.cpu-vcores</name>
<value>8</value>
</property>

<property>
<name>yarn.nodemanager.vmem-pmem-ratio</name>
<value>12</value>
</property>

<property>
<name>yarn.scheduler.minimum-allocation-mb</name>
<value>2048</value>
</property>
```

c. Save the file.

3. Navigate back to the MapR Control System and restart the ResourceManager and NodeManager services.

Configure the core-site.xml File on the MapR Cluster to Set the Impersonation Setting for the MapR User

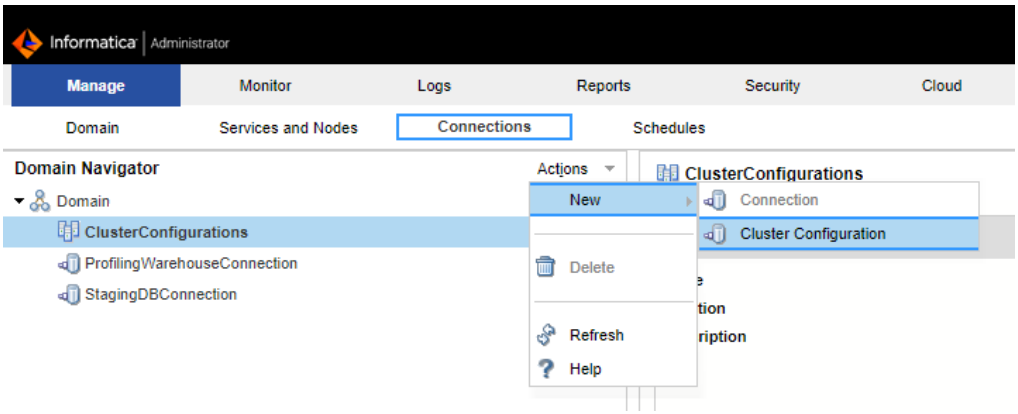
1. Log in to the MapR Control System and find the names of nodes where the CLDB service is running.
2. Log in to each node and perform the following steps:
   a. Navigate to the following directory: `/opt/mapr/hadoop/hadoop-2.7.0/etc/hadoop` and open the `core-site.xml` file.
   b. Add the following properties:

```
<property>
<name>hadoop.proxyuser.mapr.groups</name>
<value>*</value>
<description>Allow the superuser mapr to impersonate any member of any group</
description>
</property>
<property>
<name>hadoop.proxyuser.mapr.hosts</name>
<value>*</value>
<description>The superuser can connect from any host to impersonate a user</
description>
</property>
```

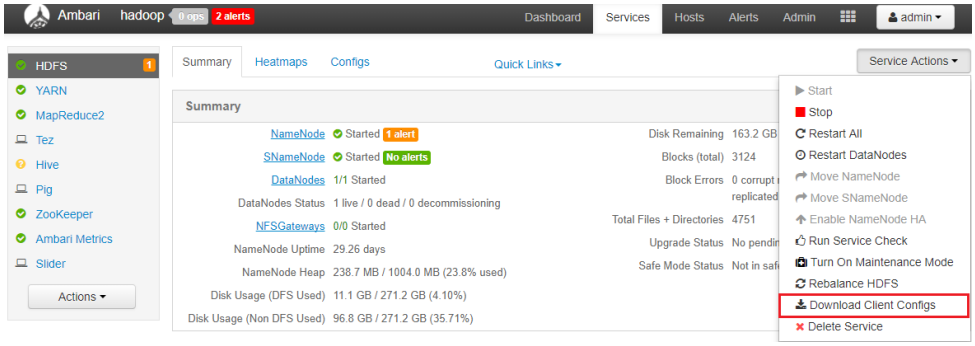
   c. Save the file.
3. Navigate back to the MapR Control system and restart the CLDB, HiveMeta, and HiveServer2 services.

Step 2. Create a Cluster Configuration Object

1. From the Administrator tool, select the **Connections** tab.
2. Expand **Domain** on the **Domain Navigator** pane.
3. Select **ClusterConfigurations** and click **Actions** > **New** > **Cluster Configuration**.



4. Enter a name for the cluster configuration.
5. Optionally, enter a description.
6. From the **Distribution Type** list, select **Amazon EMR**.
7. In the **Method to Import the Cluster Configuration** field, select **Import from archive file**.
8. Log in to Ambari and perform the following steps:
   a. Click **HDFS** on the left pane.
   b. From the **Service Actions** menu, select **Download Client Configs**.



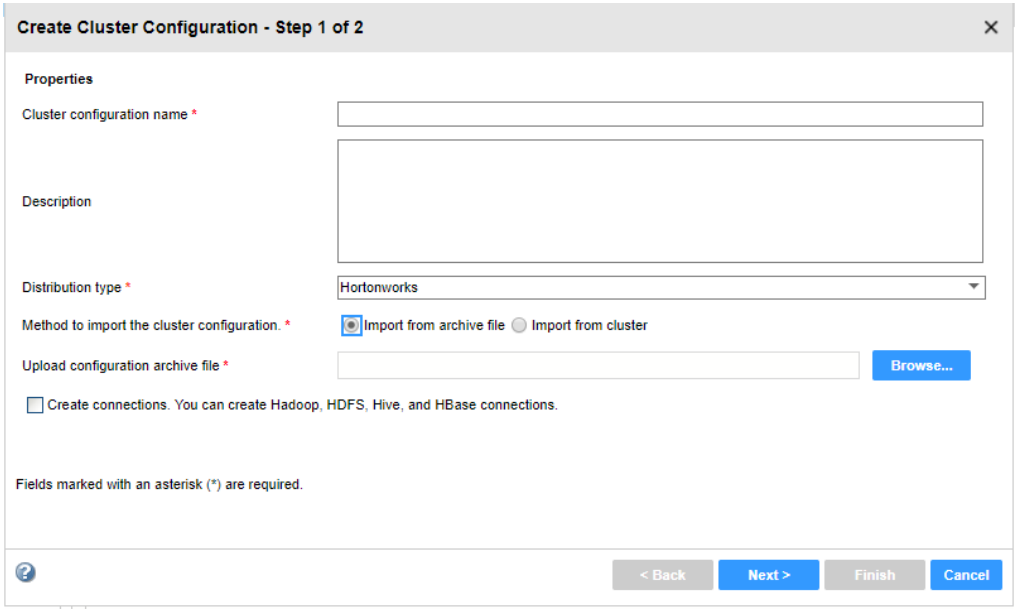
   c. Repeat steps a and b to download the client configurations from the **YARN**, **MapReduce2**, **Tez** and **Hive** services.
   d. Verify that the following .tar files download:
      - `HDFS_CLIENT-configs.tar.gz`
      - `YARN_CLIENT-configs.tar.gz`
      - `MAPREDUCE2_CLIENT-configs.tar.gz`
      - `TEZ_CLIENT-configs.tar.gz`
      - `HIVE_CLIENT-configs.tar.gz`
   e. Extract each .zip file.

      Verify that the following files extract:
      - `hdfs-site.xml` and `core-site.xml` from the `HDFS_CLIENT-configs.tar.gz` file.
      - `yarn-site.xml` from the `YARN_CLIENT-configs.tar.gz` file.

- `hive-site.xml` from the `HIVE_CLIENT-configs.tar.gz` file.
- `mapred-site.xml` from the `MAPREDUCE2_CLIENT-configs.tar.gz` file.
- `tez-site.xml` from the `tez_CLIENT-configs.tar.gz` file.

f. Select the extracted files and compress the files in a new .zip file.

9. Go back to the Administrator tool.
10. From the **Create Cluster Configuration** wizard, click **Browse** in the **Upload configuration archive file** field and upload the .zip file.

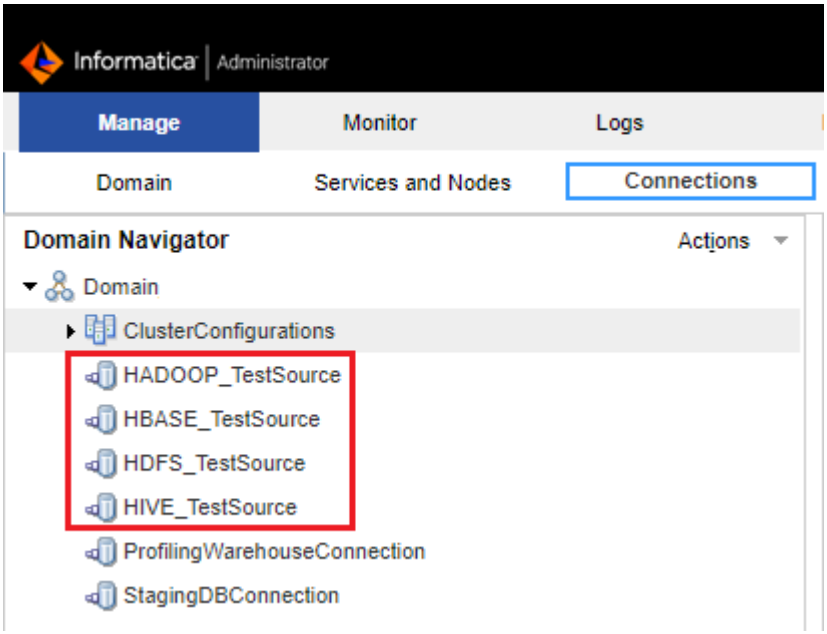


11. Select **Create connections**.
12. Click **Next**.
13. Review the information and click **Finish**.

    The Hadoop, HDFS, Hive, and HBase connections are created with the following name format:

    - `HADOOP_<Cluster Configuration Name>`
    - `HBASE_<Cluster Configuration Name>`
    - `HDFS_<Cluster Configuration Name>`
    - `HIVE_<Cluster Configuration Name>`

For example, if the cluster configuration name is `TestSource`, the connection names appear as shown in the following image:



Note: To delete the cluster configuration object, you must first delete the four connections.

Step 3. Install the Database Client Software

If the Profiling Warehouse database in the associated Data Integration Service is on Oracle, install the database client software on the nodes within the Hive source cluster.

Note: If the Profiling Warehouse database in the associated Data Integration Service is on IBM Db2 or Microsoft SQL Server, use a JDBC connection and skip this step.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. Install the Oracle database client software in the same Oracle client path on all of the nodes within the Hive source cluster.
2. Add an entry in the `tnsnames.ora` file for the Profiling Warehouse database connection.

   For example:

```
ORAMAT=
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST =localhost)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = orcl)
    )
  )
```

Step 4. Configure the Environment Variables

Configure environment variables only if the Profiling Warehouse connection is Oracle and the profile execution engine for the data store is Hadoop or Hive.

Note: Do not configure environment variables in the following situations:

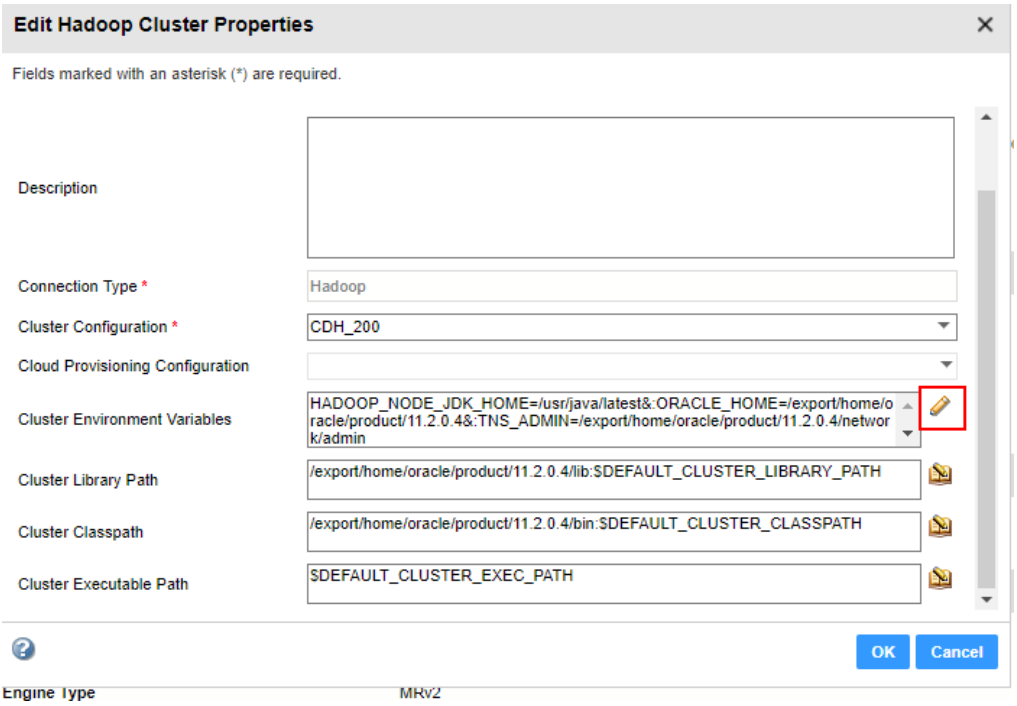
- The Profiling Warehouse connection is JDBC.
- The profile execution engine for the data store is native.

1. From the Administrator tool, click the **Manage** > **Connections** tab.
2. From the **Domain Navigator** pane on the left, select the Hadoop cluster connection.

   The Hadoop connection properties appear on the right.
3. Locate the **Hadoop Cluster Properties** section and click the **Edit** icon.



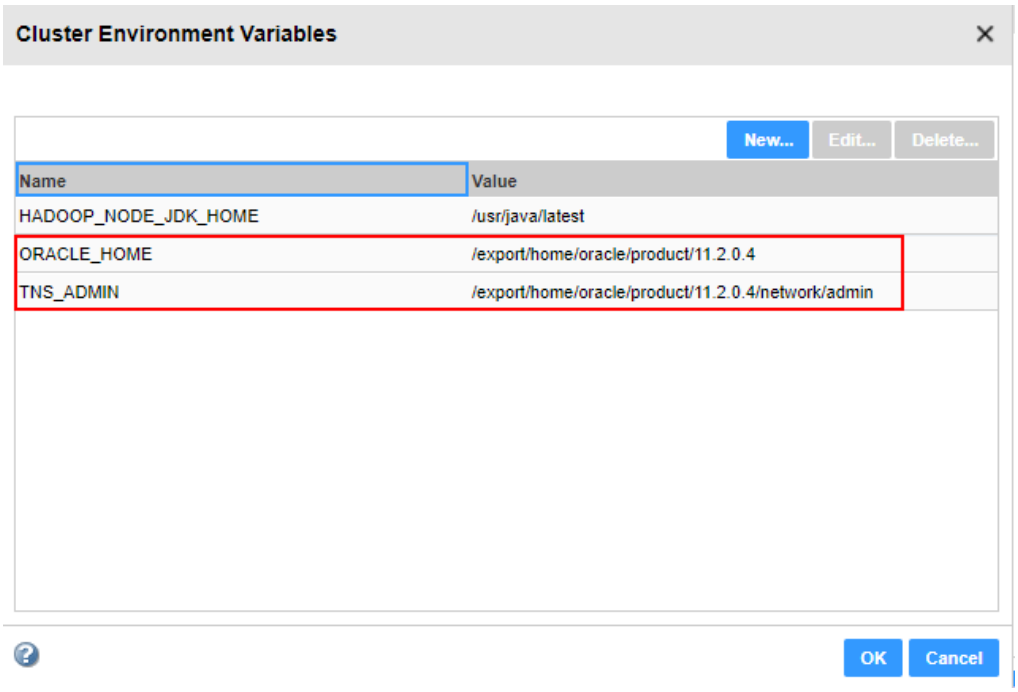
   The **Edit Hadoop Cluster Properties** window appears.
4. Click the **Edit** icon for the **Cluster Environment Variables** property.



   The **Cluster Environment Variables** window appears.
5. Click **New**.

6. Add the following environment variables:

Name	Value
ORACLE_HOME	/export/home/oracle/product/11.2.0.4
TNS_ADMIN	/export/home/oracle/product/11.2.0.4/network/admin



7. Click **OK**.

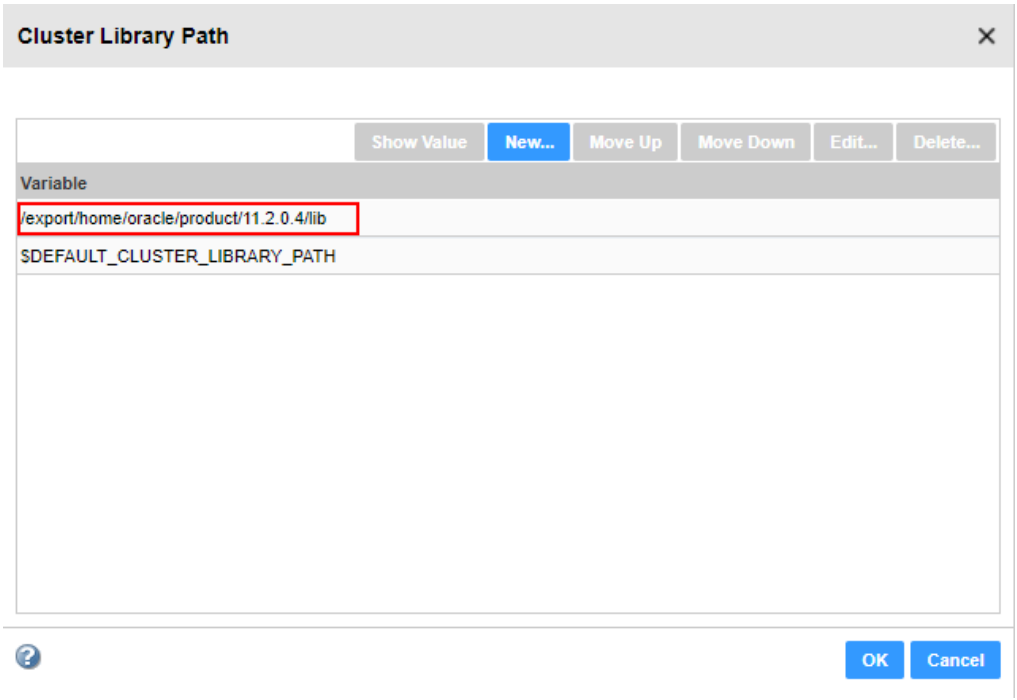
   The **Cluster Environment Variables** window closes.

8. Click the **Edit** icon for the **Cluster Library Path** property.

   The **Cluster Library Path** window appears.

9. Click **New**.

10. Enter the Oracle library path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/lib`



11. Click **OK**.

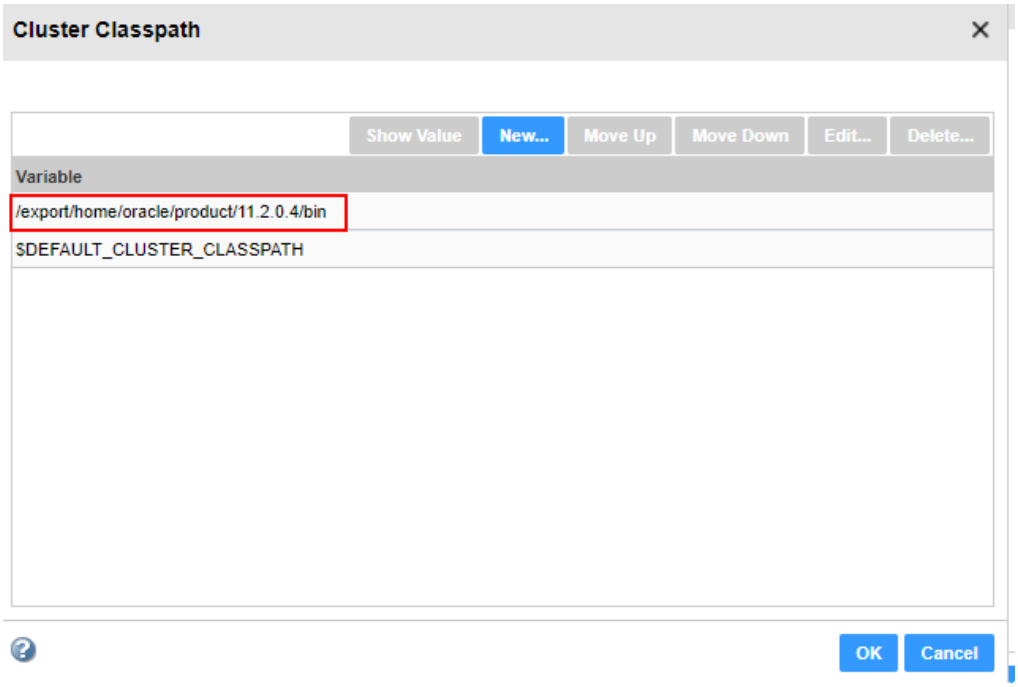
    The **Cluster Library Path** window closes.

12. Click the **Edit** icon for the **Cluster Classpath** property.

    The **Cluster Classpath** window appears.

13. Click **New**.

14. Enter the Oracle class path for the Hive cluster: `/export/home/oracle/product/11.2.0.4/bin`



15. Click **OK**.

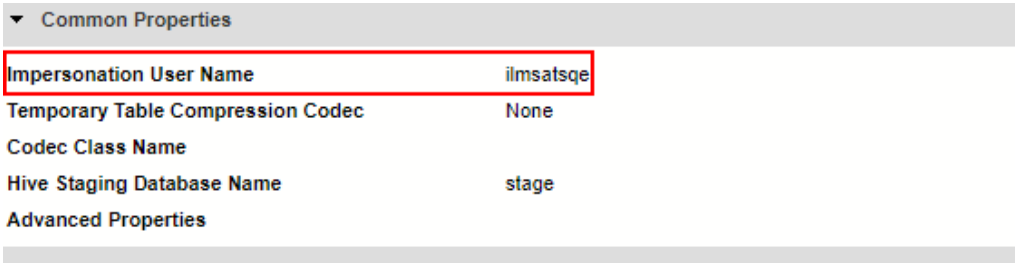
    The **Cluster Library Path** window closes.

16. Click **OK**.

    The **Edit Hadoop Cluster Properties** window closes.

17. Locate the **Common Properties** section and click the **Edit** icon.

18. In the **Impersonation User Name** field, enter the name of the user who can access the Hive database.



19. Click **OK**.

Step 5. Create a Hive Data Store in Data Privacy Management

Create a data store in Data Privacy Management and configure properties to connect to the Hive source.

When you scan the data store, theData Privacy Management Service creates a Big Data Scan job. The Scan job uses the connection properties in the data store or associated remote agent to connect to the Hive data store. For more information, see the *Informatica Data Privacy Management User Guide*.

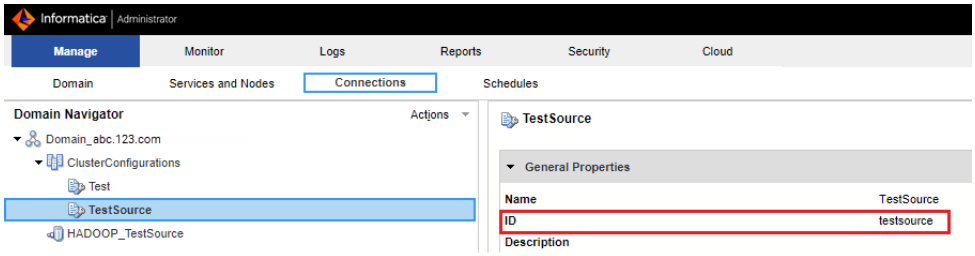
1. Go to the Data Privacy Management user interface and click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. Click **New**.

   The **New Data Store** page appears.
3. Enter a name for the data store.
4. Select **Big Data** as the **Category**.
5. Select **Hive** as the **Data Store Type**.
6. Select **MapR** as the **Hadoop Distribution**.
7. Enter or select values for all required fields.
8. To determine the **Cluster Configuration ID**, perform the following steps:
   a. From the Administrator tool, select the **Manage** > **Connections** tab.
   b. From the **Domain Navigator** pane, select the cluster configuration object you created.

      The **General Properties** section appears on the right.
   c. Make a note of the ID.



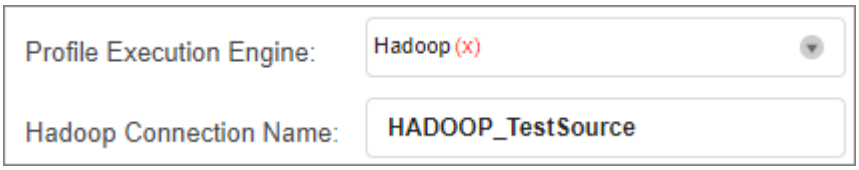
   Note: The **ID** value is automatically generated as the lowercase version of the cluster configuration name. For example, if the cluster configuration name is `TestSource`, the ID is `testsource`.

   In Data Privacy Management, enter the ID in the **Cluster Configuration Id** field.
9. Specify the **Profile Execution Engine**. Select one of the following options:
   - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. See the "Hadoop Profile Execution Engine" on page 124 section for configuration steps.
   - **Hive**. Data Privacy Management uses the Hive engine to run profiles.

     Note: Informatica does not recommend this option because it slows down performance.

     See the "Hive Profile Execution Engine" on page 124 section for prerequisite steps.
   - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. This is the default option and does not require any additional configuration.
10. If you selected **Hadoop** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



11. If you selected **Hive** as the **Profile Execution Engine**, specify the **Hadoop Connection Name**.

    You can find the Hadoop connection name in the Administrator tool, **Connections** tab.



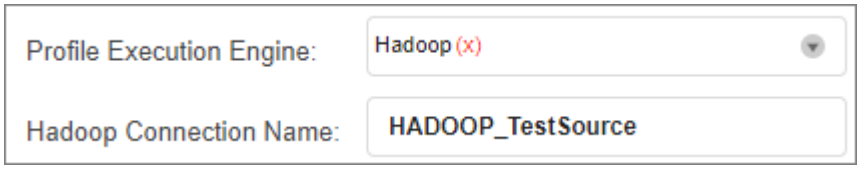
Hadoop Profile Execution Engine

To run profiles on a new Hive data store that specifies Hadoop as the Profile Execution Engine type in Data Privacy Management, configure the Blaze engine.

1. Log in to the HDFS that is configured for the Hadoop connection of the Hive cluster.
2. To create the Blaze working directory, run the following commands:

```
hdfs dfs -mkdir /blaze
hdfs dfs -mkdir /blaze/workdir
hdfs dfs -chmod -R 777 /blaze/workdir
```

3. Go to the Administrator tool.
4. Note the Hadoop connection name associated with the cluster configuration object.
   For example, if the cluster configuration name is `TestSource`, the Hadoop connection name is `HADOOP_TestSource`
5. InData Privacy Management, on the **New_Data_Store** page for the Hive data store, enter the Hadoop connection name in the **Hadoop Connection Name** field.



Hive Profile Execution Engine

To run profiles on a new Hive data store that specifies Hive as the Profile Execution Engine type in Data Privacy Management, verify the prerequisites.

- If the profiling warehouse connection is Oracle, verify that the Oracle database client is installed on all Hive cluster nodes.
- If the profiling warehouse connection is JDBC, do not install a database client.

Part IV: Configuring Connectivity to Cloud Sources

This part contains the following chapters:

- Configuring Connectivity to Microsoft Azure Blob Storage, 126
- Configuring Connectivity to Microsoft Azure Data Lake, 131
- Configuring Connectivity to Microsoft Azure SQL Data Warehouse, 133
- Configuring Connectivity to Microsoft Azure SQL Database, 135
- Configuring Connectivity to Amazon Redshift, 137
- Configuring Connectivity to Amazon S3, 140
- Configuring Connectivity to Google BigQuery, 142
- Configuring Connectivity to Salesforce, 145
- Configuring Connectivity to Snowflake, 149

CHAPTER 8

Configuring Connectivity to Microsoft Azure Blob Storage

This chapter includes the following topics:

- Configuring Microsoft Azure Blob Storage Connectivity Overview, 126
- Step 1. Import the Microsoft Azure Blob Storage Certificate, 126
- Step 2. Create a Microsoft Azure Blob Storage Data Store in Data Privacy Management, 130

Configuring Microsoft Azure Blob Storage Connectivity Overview

Configure connectivity to Microsoft Azure Blob Storage data stores before performing other operations in Data Privacy Management, such as running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

If the Microsoft Azure Blob Storage source is SSL enabled, you must configure connectivity to the source before you scan the data store. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Import the Microsoft Azure Blob Storage SSL certificates.
2. Create a Microsoft Azure Blob Storage data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Import the Microsoft Azure Blob Storage Certificate

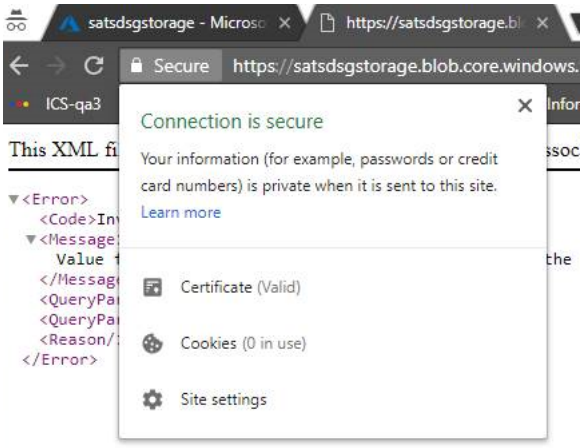
Perform the following steps to import the Microsoft Azure Blob Storage certificate.

1. Open the Azure Blob Storage end point URL in a browser.

2. Click **Secure** in the address field.



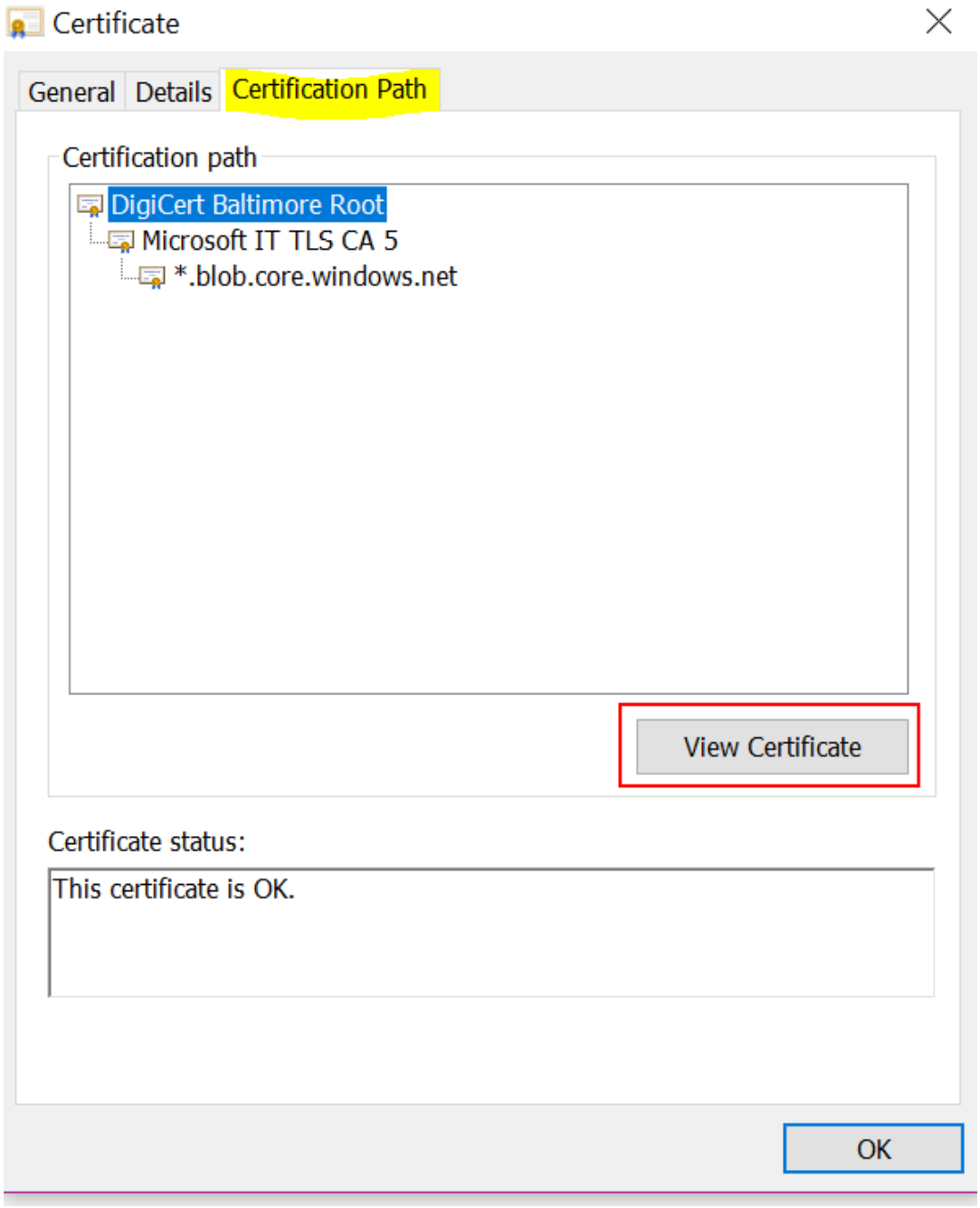
3. Click **Certificate**.



The **Certificate** dialog box appears.

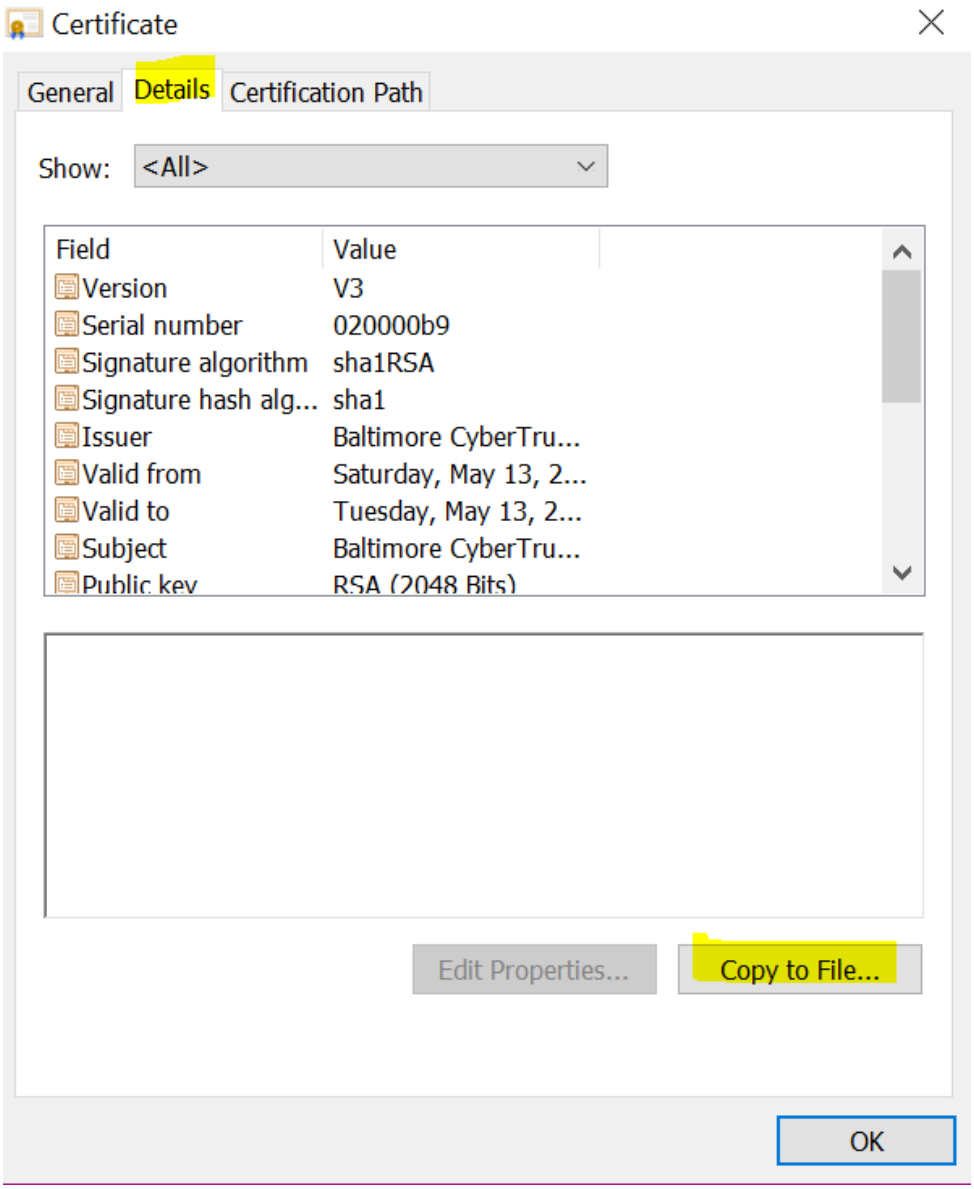
4. Click the **Certification Path** tab.
5. Perform the following steps for each certificate and at the leaf level within the hierarchy:
   a. Select the certificate.
   b. Click **View Certificate**.

c. Verify that the certificate is OK in the **Certificate status** field.



d. Click the **Details** tab.

e. Click **Copy to File**.



6. Restart the Informatica domain.

Commands to Import the Certificates

Refer to the following list of commands to import the certificates:

```
keytool -import -file /home/ilmsatsqe/configFiles/azureBlob_cert1.cer  -alias
azureblob_cert1_head -keystore
$INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL -
noprompt
keytool -import -file /home/ilmsatsqe/configFiles/azureBlob_cert2.cer  -alias
azureblob_cert2_imediate -keystore
$INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL -
noprompt
keytool -import -file /home/ilmsatsqe/configFiles/azureBlob_cert.cer  -alias
azureblob_cert_leaf -keystore
$INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL -
noprompt
```

Step 2. Create a Microsoft Azure Blob Storage Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft Azure Blob Storage.

When you create the data store, specify the category **Cloud** and the data store type **Microsoft Azure Blob Storage**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Microsoft Azure Blob Storage. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 9

Configuring Connectivity to Microsoft Azure Data Lake

This chapter includes the following topics:

- Configuring Microsoft Azure Data Lake Connectivity Overview, 131
- Step 1. Merging the Microsoft Azure Data Lake Certificates, 131
- Step 2. Create a Microsoft Azure Data Lake Data Store in Data Privacy Management, 132

Configuring Microsoft Azure Data Lake Connectivity Overview

Configure connectivity to Microsoft Azure Data Lake data stores before performing other operations in Data Privacy Management, such as running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

To configure connectivity, complete the following steps:

1. Merge the Microsoft Azure Data Lake certificates.
2. Create a Microsoft Azure Data Lake data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Merging the Microsoft Azure Data Lake Certificates

Perform the following step to merge the Microsoft Azure Data Lake certificates.

- Merge the certificates in `<INFA_HOME>/java/lib/security/cacerts` to the `infa_truststore.jks` file located in the `<INFA_HOME>/services/shared/security/` directory.

Step 2. Create a Microsoft Azure Data Lake Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft Azure Data Lake.

When you create the data store, specify the category **Cloud** and the data store type **Azure Data Lake**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Microsoft Azure Data Lake. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 10

Configuring Connectivity to Microsoft Azure SQL Data Warehouse

This chapter includes the following topics:

- Configuring Microsoft Azure SQL Data Warehouse Overview, 133
- Step 1. Access Information to Connect to Microsoft Azure SQL Data Warehouse, 134
- Step 2. Create a Microsoft Azure SQL Data Warehouse Data Store in Data Privacy Management, 134

Configuring Microsoft Azure SQL Data Warehouse Overview

Configure connectivity to Microsoft Azure SQL Data Warehouse data stores before performing other operations in Data Privacy Management.

To configure connectivity, perform the following steps:

1. Record the Microsoft Azure and new Azure Synapse Analytics or Data Warehouse account information. To create the data store in Data Privacy Management, you require the following account properties:

   Database server name

   Port

   Database Name

   Azure Blob Container Name

   Azure Blob Account Key
2. Create a Microsoft Azure SQL Data Warehouse data store on the **Data Stores** workspace in Data Privacy Management.

The drivers required to perform domain discovery are shipped with Data Privacy Management. To perform subject discovery scans, download the mssql-jdbc-7.4.1.jre8.jar file from the Microsoft download site. Copy the file to the following location: `<Data Privacy Management installation directory>/externaljdbcjars`

Step 1. Access Information to Connect to Microsoft Azure SQL Data Warehouse

Record the required Microsoft Azure SQL Data Warehouse account information before you create the data store.

Before you begin, you must have created a Microsoft Azure account, Azure Synapse Analytics or Azure Data Warehouse, and a Microsoft Azure Blob Storage account.

1. Log in to the Microsoft Azure account and search for Storage Accounts.
2. Open the required storage account.

   The **Azure Blob Container Name** contains the name of the storage container.
3. On the **Settings** tab on the left-hand side, click **Access Keys**.

   The **key1** field contains the Blob Account Key.
4. On the Microsoft Azure account home page, search for Azure Synapse Analytics.
5. Open the required Azure SQL Data Warehouse account.
6. On the **Essentials** tab, click **Connection Strings**.
7. Click JDBC and copy the values from the connection string.

   The JDBC connection string includes the database host, port, and database name information.

```
jdbc:sqlserver://<host name>:<port>;database=<Database name>;user=<Database user name>;password={your_password_here};encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;
```

Step 2. Create a Microsoft Azure SQL Data Warehouse Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft Azure SQL Data Warehouse.

When you create the data store, specify the category **Cloud** and the data store type **Microsoft Azure SQL Data Warehouse**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Microsoft Azure SQL Data Warehouse. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 11

Configuring Connectivity to Microsoft Azure SQL Database

This chapter includes the following topics:

- Configuring Microsoft Azure SQL Database Overview, 135
- Step 1. Access Information to Connect to the Microsoft Azure SQL Database, 136
- Step 2. Update the odbc.ini File, 136
- Step 3. Create a Microsoft Azure SQL Database Data Store in Data Privacy Management, 136

Configuring Microsoft Azure SQL Database Overview

Configure connectivity to Microsoft Azure SQL Database data stores before performing other operations in Data Privacy Management.

To configure connectivity, perform the following steps:

1. Record the Microsoft Azure SQL Database property information.
   You require the following database properties to create the data store in Data Privacy Management:

   Database server name

   Port

   Database Name
2. Add the entries to the `odbc.ini` file.
3. Create a Microsoft Azure SQL Data Warehouse data store on the **Data Stores** workspace in Data Privacy Management.

The drivers required to perform domain discovery are shipped with Data Privacy Management. To perform subject discovery scans, download the mssql-jdbc-7.4.1.jre8.jar file from the Microsoft download site. Copy the file to the following location: `<Data Privacy Management installation directory>/externaljdbcjars`

Step 1. Access Information to Connect to the Microsoft Azure SQL Database

Record the required Microsoft Azure SQL Database account information before you create the data store.

Before you begin, you must create the Microsoft Azure SQL Database.

1. Log in to the Microsoft Azure account and search for SQL databases.
2. Open the required Azure SQL database.
3. Click **Connection String** and select JDBC to copy the values.

   The JDBC connection string includes the database host, port, and database name information.

   ```
   jdbc:sqlserver://<host name>:<port>;database=<Database name>;user=<Database user name>;password={your_password_here};encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;
   ```

Step 2. Update the odbc.ini File

Add an entry for the Microsoft Azure SQL Database to the `odbc.ini` file.

1. Browse and open the following file: `<Data Privacy Management installation directory>/ODBC7.1/odbc.ini`
2. Add the following entry for the Microsoft Azure SQL Database:

   ```
   [AzureSQL DB]
   Driver=<Data Privacy Management installation directory>/ODBC7.1/lib/DWsqls28.so
   Description=DataDirect 7.1 SQL Server Wire Protocol
   HostName=<host>
   PortNumber=<port>
   Database= <Database name>
   EncryptionMethod=1
   ValidateServerCertificate=0
   PacketSize=32
   EnableQuotedIdentifiers=1
   ```

   Replace the placeholders with appropriate values.
3. Save and close the file.

Step 3. Create a Microsoft Azure SQL Database Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft Azure SQL Data Warehouse.

When you create the data store, specify the category **Cloud** and the data store type **Microsoft Azure SQL Database**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Microsoft Azure SQL Database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 12

Configuring Connectivity to Amazon Redshift

This chapter includes the following topics:

- Configuring Amazon Redshift Connectivity Overview, 137
- Step 1. Create an IAM Role, S3 Bucket, and Amazon Redshift Cluster, 138
- Step 2. Copy the Amazon Redshift .jar Files to the Data Privacy Management Machine, 138
- Step 3. Verify the CustomDeployer.xml File, 138
- Step 4. Restart the Informatica Domain, 139
- Step 5. Create an Amazon Redshift Data Store in Data Privacy Management, 139

Configuring Amazon Redshift Connectivity Overview

Configure connectivity to Amazon Redshift data stores before performing other operations in Data Privacy Management.

To configure connectivity, perform the following steps:

1. Create an IAM role, S3 bucket, and Amazon Redshift cluster.
2. Copy the Amazon Redshift .jar files to the Data Privacy Management machine.
3. Verify the `CustomDeployer.xml` File.
4. Restart the Informatica domain.
5. Create an Amazon Redshift data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Create an IAM Role, S3 Bucket, and Amazon Redshift Cluster

You require an IAM role, S3 bucket, and Amazon Redshift cluster to work with Amazon Redshift data stores in Data Privacy Management

1. Create an IAM role from the AWS Management page. Then attach the role to the cluster that you create.

   The role must have Administrator access and AmazonS3FullAccess. The role must allow Amazon Redshift to load data from Amazon S3 buckets.
2. Generate an access key ID and secret access key and save them.
3. Create an Amazon Redshift cluster.

   Verify details such as the cluster status and database health details.
4. Attach the IAM role to the cluster.
5. Create an Amazon S3 bucket from the Amazon S3 home page.

Step 2. Copy the Amazon Redshift .jar Files to the Data Privacy Management Machine

The Amazon Redshift .jar files must be present on the Data Privacy Management machine.

Download the `RedshiftJDBC42-1.2.55.1083.zip` file from the official Amazon S3 download site.

Extract the file and copy the redshift-jdbc42-1.2.1.1001.jar file to the following locations on the Data Privacy Management machine:

- `<Informatica installation directory>/externaljdbcjars`
- `<Informatica installation directory>/services/CatalogService/ScannerBinaries/Redshift.zip`

Step 3. Verify the CustomDeployer.xml File

The `CustomDeployer.xml` file must have the required properties to connect to the database.

Before you begin, verify that the `<Informatica installation directory>/services/CatalogService/ScannerBinaries` directory contains the `redshiftJars.zip`. If not, download the file and copy to the directory.

1. Browse to the following location and open the file: `<Informatica installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/CustomDeployer.xml`
2. Verify that the file contains the *ExecutionContextProperty*.

   If not, add the property to the file.

```
ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="redshiftJars.zip">
<PropertyName>RedShiftScanner_DriverLocation</PropertyName>
<PropertyValue>scanner_miti/RedShift/Drivers</PropertyValue>
```

```
</ExecutionContextProperty>
</ExecutionContextProperties>
```

3. Save the changes and close the file.

Step 4. Restart the Informatica Domain

Restart the Informatica domain to update the changes.

1. Run the following command to shut down the Informatica domain:

```
cd /<Data Privacy Management installation folder>/tomcat/bin
./infaservice.sh shutdown
```

2. Run the following command to verify that all Java processes are terminated: `ps -ef | grep java`

   The command must return 0 services.

3. Run the following command to start the Informatica domain:

```
cd /<Data Privacy Management installation folder>/tomcat/bin
./infaservice.sh startup
```

Step 5. Create an Amazon Redshift Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to the Amazon Redshift database.

When you create the data store, specify the category **Cloud** and the data store type **Amazon Redshift**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Amazon Redshift. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 13

Configuring Connectivity to Amazon S3

This chapter includes the following topics:

- Configuring Amazon S3 Connectivity Overview, 140
- Step 1. Create an Amazon S3 Bucket, 140
- Step 2. Create and Record the Required Amazon S3 Credentials, 141
- Step 3. Create an Amazon S3 Data Store in Data Privacy Management, 141

Configuring Amazon S3 Connectivity Overview

Configure connectivity to Amazon S3 data stores before performing other operations in Data Privacy Management.

To configure connectivity, perform the following steps:

1. Create an Amazon S3 bucket.
2. Create and get access to the Amazon Web Services Access Key ID and Secret Access Key.
3. Create an Amazon S3 data store on the **Data Stores** workspace in Data Privacy Management.

Note: If you create the Data Privacy Management repository on a Microsoft SQL Server or Microsoft Azure SQL database, you must configure the database as case sensitive. You cannot create an Amazon S3 data store if the Data Privacy Management repository is on a Microsoft SQL Server or Microsoft Azure SQL database configured as case insensitive.

Step 1. Create an Amazon S3 Bucket

You require an Amazon S3 bucket to work with Amazon S3 data stores in Data Privacy Management

To create an S3 bucket, you must have an Amazon account.

1. Log in to the Amazon Web Services Management Console.
2. From the Storage and Content Delivery, choose to open the Amazon S3 console.

   The Amazon S3 console dashboard opens.

3. Choose to create a bucket.

   Enter the required region and click **Create**.

Step 2. Create and Record the Required Amazon S3 Credentials

You must have an Amazon Web Services Access Key ID and Secret Access Key to work with an S3 bucket.

1. Log in to the Amazon Web Services Management Console.
2. From the Navigation panel, click **Users**.
3. Choose the required IAM user name.
4. Click **User Actions** > **Manage Access Keys** > **Create Access Key**.

   For example, an access key ID and secret access key: `AKIA4GH7IJR3XTD8C5WS` and `VelqCQzeSknKXPzE69Cbcm+089X9GXVnyYOGSZKA`
5. Click **Download Credentials** to download the information.

   Store the information in a secure location.

Step 3. Create an Amazon S3 Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to the Amazon S3 database.

When you create the data store, specify the category **Cloud** and the data store type **Amazon S3**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Amazon S3. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 14

Configuring Connectivity to Google BigQuery

This chapter includes the following topics:

- Configuring Google BigQuery Connectivity Overview, 142
- Step 1. Add the Google BigQuery Certificates to the Keystore and Truststore, 143
- Step 2. Update the Data Integration Service License, 143
- Step 3. Configure Google BigQuery Permissions, 143
- Step 4. Create a Google BigQuery Data Store in Data Privacy Management, 144

Configuring Google BigQuery Connectivity Overview

Configure connectivity to a Google BigQuery data store before you perform other operations, such as running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

To configure connectivity, complete the following tasks:

1. Add the Google BigQuery certificates to the truststore and keystore.
2. Update the Data Integration Service license.
3. Configure Google BigQuery user permissions.
4. Create a Google BigQuery data store on the **Data Stores** workspace in Data Privacy Management.

You can configure the following authentication methods for a Collect Row Count job that reads a Google BigQuery source:

- Service account credentials-based authentication
- Access token-based authentication

To enable access token-based authentication for a Google BigQuery source, add one of the following options in the Administrator page of Data Privacy Management:

-DEnableTokenAuthForGBQ=true

Add the property in the **Additional JVM Options**.

EnableTokenAuthForGBQ

Add the property in the **Custom Properties** and set the value to true.

Step 1. Add the Google BigQuery Certificates to the Keystore and Truststore

Before you can connect to a Google BigQuery database, copy the Google BigQuery certificates to the Informatica keystore and truststore.

1. Run the following query to copy the Google BigQuery certificates:

   ```
   keytool -import -file <GBQ_certificate location> -alias gbq -keystore <Informatica installation directory>/services/shared/security/infa_truststore.jks -storepass <password> -noprompt
   ```

2. Run the following query to copy CA certificates to the keystore and truststore:

   ```
   keytool -importkeystore -srckeystore <Informatica installation directory>/java/lib/security/cacerts -destkeystore <Informatica installation directory>/services/shared/security/infa_truststore.jks -srcstorepass changeit -deststorepass <password> -noprompt
   ```

3. Restart the Informatica domain.

Step 2. Update the Data Integration Service License

To run data scans on a Google BigQuery source, the Data Integration Service license must have the required connection options.

1. Log in to Informatica Administrator.
2. Select the Data Integration Service license.
3. Click **Options** to open the **Options** page.
4. On the **Connections** tab, verify that you have an entry for PowerExchange for Google BigQuery.

   If the license does not include the option, update the license to include the option.

See the Informatica Administrator Guide for information about how to update a license.

Step 3. Configure Google BigQuery Permissions

Configure the required Identity and Access Management permissions for the user account that you use to connect to Google BigQuery.

Assign the following roles to the service account that you use to connect to the database:

BigQuery Data Editor

BigQuery Job User

The roles include the following required permissions:

bigquery.tables.list

bigquery.jobs.get

Step 4. Create a Google BigQuery Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Google BigQuery.

When you create the data store, specify the category **Cloud** and the data store type **Google BigQuery**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Google BigQuery. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 15

Configuring Connectivity to Salesforce

This chapter includes the following topics:

- Configuring Salesforce Connectivity Overview, 145
- Step 1. Verify the Salesforce User Account, 145
- Step 2. Register the Salesforce SSL Certificates with the Informatica Domain, 146
- Step 3. Create a Salesforce Data Store in Data Privacy Management, 148

Configuring Salesforce Connectivity Overview

Configure connectivity to Salesforce data stores before performing other operations in Data Privacy Management, such as running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

Before you scan or import data from Salesforce, configure connectivity to the source. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Verify the Salesforce user account and privileges.
2. Register the Salesforce SSL certificates with the Informatica domain.
3. Create a Salesforce data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Verify the Salesforce User Account

When you scan or import data from Salesforce, the corresponding jobs require a Salesforce user account. You can create a new user account or use an existing user account. You specify the user account when you create a Salesforce data store in Data Privacy Management.

Verify that the user account includes the privileges and permissions required for the Import job. The Scan job does not require specific privileges.

Step 2. Register the Salesforce SSL Certificates with the Informatica Domain

If the Informatica domain is enabled for secure communication and the Salesforce instance uses the SSL protocol, export the Salesforce SSL certificates. Then, add the certificates to the Informatica domain truststore file.

If you create multiple data stores that connect to Salesforce, repeat this process for each unique Salesforce Service URL that you specify in the data store properties in Data Privacy Management.

Exporting the Salesforce Login SSL Certificate

Use a web browser to download the Salesforce.com SSL certificate from `https://login.salesforce.com/`. The specific steps depend on the web browser and the web browser version.

The following steps are an example of how to download the SSL certificate from Microsoft Internet Explorer:

1. In Microsoft Internet Explorer, open `https://login.salesforce.com/`.

   Note: Do not log in to Salesforce.
2. Click the SSL icon.
3. Click **View Certificate**.

   The **Certificate** dialog box appears.
4. Click the Details tab.
5. Click **Copy to File...**

   The **Certificate Export Wizard** appears. Click **Next**.
6. Select **Base-64 encoded X.509 (.CER)**. Click **Next**.
7. Browse to a temporary location to store the certificate and enter a file name. Click **Save**.
8. Click **Next**.

   The wizard confirms the settings that you selected.
9. Click **Finish**.

   The wizard exports the SSL certificate to the location you specified.

Exporting the Salesforce Instance SSL Certificate

Use a web browser to download the Salesforce.com SSL certificate after you log in to Salesforce. The specific steps depend on the web browser and the web browser version.

The following steps are an example of how to download the SSL certificate from Microsoft Internet Explorer:

1. In Microsoft Internet Explorer, open `https://login.salesforce.com/`.
2. Log in to Salesforce.

   Note that the URL changes after you log in. For example, the URL changes from `https://login.salesforce.com/` to `https://ap2.salesforce.com/`.
3. Click the SSL icon.
4. Click **View Certificate**.

   The **Certificate** dialog box appears.
5. Click the Details tab.

6. Click **Copy to File...**

   The **Certificate Export Wizard** appears. Click **Next**.
7. Select **Base-64 encoded X.509 (.CER)** and click **Next**.
8. Browse to a temporary location to store the certificate and enter a file name. Click **Save**.
9. Click **Next**.

   The wizard confirms the settings that you selected.
10. Click **Finish**.

    The wizard exports the SSL certificate to the location you specified.

Copying the Certificates to Data Privacy Management

Copy the two exported SSL certificates to the machine that hosts the Data Privacy Management Service.

Importing the Certificates to the Domain Truststore File

Import the SSL certificates to the Informatica domain truststore file.

1. From the command line, run the following command to import the certificate file that you exported from `login.salesforce.com`:

   ```
   $INFA_HOME/java/jre/bin/keytool -import -trustcacerts -alias <alias name> -file /tmp/
   salesforce_1.cer -keystore $INFA_HOME/services/shared/security/infa_truststore.jks
   ```

   Where *<alias name>* is the alias of the certificate file that you exported from `login.salesforce.com`.

   For example, if the exported certificate file name from `login.salesforce.com` is `sfdc_login`, run the following command:

   ```
   $INFA_HOME/java/jre/bin/keytool -import -trustcacerts -alias sfdc_login -file /tmp/
   salesforce_1.cer -keystore $INFA_HOME/services/shared/security/infa_truststore.jks
   ```
2. Enter the password for the truststore file.

   The certificate is added to the truststore file.
3. Run the following command to import the certificate file that you exported from the logged in Salesforce instance:

   ```
   $INFA_HOME/java/jre/bin/keytool -import -trustcacerts -alias <alias name> -file /tmp/
   salesforce_2.cer -keystore $INFA_HOME/services/shared/security/infa_truststore.jks
   ```

   Where *<alias name>* is the alias of the certificate file that you exported from the logged in instance, such as `https://ap2.salesforce.com/`.

   For example, if the exported certificate file name from `https://ap2.salesforce.com/` is `sfdc_login_ap2`, run the following command:

   ```
   $INFA_HOME/java/jre/bin/keytool -import -trustcacerts -alias sfdc_login_ap2 -
   file /tmp/salesforce_2.cer -keystore $INFA_HOME/services/shared/security/
   infa_truststore.jks
   ```
4. Enter the password for the truststore file.

   The certificate is added to the truststore file.
5. Run the following command to verify that the alias name of the certificate that you imported exists in the truststore file:

   ```
   $INFA_HOME/java/jre/bin/keytool -list -v -keystore $INFA_HOME/services/shared/
   security/infa_truststore.jks
   ```

Copying the Truststore File to Nodes in the Hadoop Cluster

After you import the certificates to the truststore file, copy the truststore file to all nodes in the Hadoop cluster of the Data Privacy Management deployment.

For an internal cluster deployment, copy the file to the truststore file location configured in the advanced options of the Informatica Cluster Service properties.

For an external cluster deployment, copy the file to the truststore location specified in the Hadoop cluster configuration.

Restarting the Informatica Domain

After you export the certificates from Salesforce and import the certificates to the truststore file, restart the Informatica domain.

Step 3. Create a Salesforce Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Salesforce.

When you create the data store, specify the category **Cloud** and the data store type **Salesforce**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Salesforce. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 16

Configuring Connectivity to Snowflake

This chapter includes the following topics:

- Configuring Native Snowflake Connectivity Overview, 149
- Step 1. Adding the Snowflake Truststore Certificates and JDBC JARs, 149
- Step 2. Create a Snowflake Data Store in Data Privacy Management, 150

Configuring Native Snowflake Connectivity Overview

Configure connectivity to Snowflake data stores before performing other operations in Data Privacy Management, such as running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

To configure connectivity, complete the following steps:

1. Add the Snowflake truststore certificates and JDBC JARs.
2. Create a Snowflake data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Adding the Snowflake Truststore Certificates and JDBC JARs

Perform the following steps to add the Snowflake truststore certificates and the JDBC JAR files:

1. Download the snowflake-jdbc-3.9.2.jar file from the official Snowflake download site.
2. Copy the snowflake-jdbc-3.9.2.jar file to the following locations:
   - `<Data Privacy Management installation diectory>/externaljdbcjars`
   - `<Informatica installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
3. Download all Snowflake truststore certificates from the Snowflake website.
4. Copy the certificates to the following location: `<Informatica installation directory>/services/shared/security`

5. Run the following command to add the certificates to Data Privacy Management:

   ```
   keytool -import -trustcacerts -file <certificate_name> -alias <alias_name> -keystore infa_truststore.jks -storepass <infa truststore password>
   ```

   For example:

   ```
   keytool -import -trustcacerts -file starfield.cer -alias starfield -keystore infa_truststore.jks -storepass abc@x12y
   ```

6. Run the following command to copy the certificates to the HDFS system:

   ```
   scp infa_truststore.jks ldmuser@<hdfs_machine_name>:/<HDFS TrustStore file location>
   ```

   For example:

   ```
   scp infa_truststore.jks ldmuser@<hdfs_machine_name>:/ opt/DomainTruststore
   ```

Step 2. Create a Snowflake Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to the Snowflake database.

When you create the data store, specify the category **Cloud** and the data store type **Snowflake**.

When you scan the data store, the Data Privacy Management Service creates a Cloud Scan job. The Scan job uses the connection properties in the data store to connect to Snowflake. For more information, see the *Informatica Data Privacy Management User Guide*.

Part V: Configuring Connectivity to Data Integration Sources

CHAPTER 17

Configuring Connectivity to SQL Server Integration Services

This chapter includes the following topics:

- Configuring SQL Server Integration Services Connectivity Overview, 152
- Step 1. Install the Enterprise Data Catalog Agent, 153
- Step 2. Verify the Microsoft SQL Server Database Availability, 153
- Step 3. Configure Privileges for the User Account, 153
- Step 4. Prepare the Package Files, 154
- Step 5. Prepare the Variable Values File, 154
- Step 6. Create an SQL Server Integration Services Data Store in Data Privacy Management, 155

Configuring SQL Server Integration Services Connectivity Overview

Before you run a scan on an SQL Server Integration Services data store, configure connectivity to the source. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Install the Enterprise Data Catalog agent.
2. Verify database availability.
3. Configure permissions and privileges for the user account.
4. Copy the package files.
5. Prepare the variable values file.
6. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the source.

No driver installation is required.

After you configure the connectivity, you can scan the data store.

Step 1. Install the Enterprise Data Catalog Agent

Install the Enterprise Data Catalog agent if the SQL Server Integration Service is not on the same machine as the Data Privacy Management Service. Install the Enterprise Data Catalog agent on a Windows machine. You specify the Enterprise Data Catalog agent URL when you create a SQL Server Integration Services data store in Data Privacy Management.

The Enterprise Data Catalog agent installation requirements depend on the location that hosts the packages. The following table describes the Enterprise Data Catalog agent installation requirements:

Package Location	Enterprise Data Catalog Agent Installation Requirements
Packages hosted in a Microsoft SQL Server Management Studio repository	Install the Enterprise Data Catalog agent on the same machine that hosts the SQL Server Integration Services server.
Packages not hosted in a Microsoft SQL Server Management Studio repository	If the package file is not password protected, install the Enterprise Data Catalog agent on any machine. If the package file is password protected, install the Enterprise Data Catalog agent on the same machine that hosts Visual Studio.

Step 2. Verify the Microsoft SQL Server Database Availability

To scan packages hosted in a Microsoft SQL Server Management Studio repository, verify that the Microsoft SQL Server Database Engine is available and running.

Step 3. Configure Privileges for the User Account

Configure privileges for the Windows user account that starts the Enterprise Data Catalog agent.

The user account privileges depend on the location that hosts the packages. The following table describes the user account privileges:

Package Location	User Account Privileges
Packages hosted in a Microsoft SQL Server Management Studio repository	The Enterprise Data Catalog agent uses Windows authentication to connect to SQL Server Integration Services. The Windows user account that starts the Enterprise Data Catalog agent must have read permission on the objects that the Enterprise Data Catalog agent extracts.
Packages not hosted in a Microsoft SQL Server Management Studio repository	The Windows user account that starts the Enterprise Data Catalog agent must have read permission on the files and on the directory that contains the files.

Step 4. Prepare the Package Files

This step is required to scan packages that you created in Microsoft Visual Studio, but do not exist in Microsoft SQL Server Management Studio. You can skip this step to scan packages that exist in a Microsoft SQL Server Management Studio.

When you create an SQL Server Integration Services data store in Data Privacy Management and you select **FILE** as the SSIS scanner type, you specify the full file path and name for the package files. Before you configure the data store, prepare the package files.

Review the following rules and guidelines to prepare the package files:

- To scan packages in a .dtsx or .zip file, copy the package files to the machine that hosts the Data Privacy Management Service.
- To include multiple package files in one data store, add the package files to one .zip file. Then, specify the .zip file in the data store connection properties in Data Privacy Management.
- You can create multiple SQL Server Integration Services data stores. However, each data store should connect to a different package. The connections in the mappings should also be unique.

  For example, two data stores connect to packages with mappings that include the same connection. When you scan the data stores, the Scan job might not properly identify the sensitive data proliferation for the connection that exists in multiple data stores.
- Although the package files are not stored in Microsoft SQL Server Management Studio, the Scan job supports package files that are compatible with Microsoft SQL Server Management Studio 2008, 2012, 2014, and 2016.

Step 5. Prepare the Variable Values File

You specify the full path and name of the variable values file when you create the SQL Server Integration Services data store. The Scan job uses the variable values file to resolve the variables. The Scan job can resolve variables that specify the table or view name in an OLE DB source or destination. The Scan job can also resolve variables that specify an SQL query in an OLE DB source. You cannot define SQL Server Integration Services parameters in a variable values file. The variables file must exist on the same machine that hosts the Data Privacy Management Service.

To define variables in the variable values file, group the variables by scope, either package-level or task-level. Define each variable on a separate line. You must also enter the namespace as a prefix.

Use the following syntax to define a variable that has a package-level scope:

```
[Package]
Namespace::Variable=Value
```

Use the following syntax to define a variable that has a task-level scope:

```
[Package.Task]
Namespace::Variable=Value
```

Variable names are case sensitive.

The following example shows a variable values file that defines variables for two tasks within a package:

```
[SSIS_Package_Configuration.Data Flow Task]
User::TargetTableName=PERSON_OUT
User::TableName=Person
```

Step 6. Create an SQL Server Integration Services Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft SQL Server Integration Services packages.

When you create the data store, specify the category **Data Integration** and the data store type **SQL Server Integration Services**.

When you scan an SQL Server Integration Services data store, the Data Privacy Management Service creates an SQL Server Integration Services Scan job. The Scan job uses the data store connection properties to scan data in the packages. For more information, see the *Informatica Data Privacy Management User Guide*.

Part VI: Configuring Connectivity to Database Management Sources

This part contains the following chapters:

- Configuring Connectivity to IBM Db2 Databases, 157
- Configuring Connectivity to IBM Db2 for i5/OS Databases, 160
- Configuring Connectivity to IBM Db2 for z/OS Databases, 163
- Configuring Connectivity to Microsoft SQL Server Databases, 167
- Configuring Connectivity to MongoDB Sources, 171
- Configuring Connectivity to Netezza Databases, 173
- Configuring Connectivity to Workday Sources, 179
- Configuring Connectivity to Oracle Databases, 182
- Configuring Connectivity to REST Services, 189
- Configuring Connectivity to SAP HANA Databases, 191
- Configuring Connectivity to Sybase Databases, 194
- Configuring Connectivity to Teradata Databases, 198

CHAPTER 18

Configuring Connectivity to IBM Db2 Databases

This chapter includes the following topics:

- Configuring IBM Db2 Database Connectivity Overview, 157
- Step 1. Install the IBM Db2 Database Client Software, 158
- Step 2. Configure Native Connectivity to an IBM Db2 Database, 158
- Step 3. Configure IBM Db2 Permissions, 158
- Step 4. Create an IBM Db2 Data Store in Data Privacy Management, 159

Configuring IBM Db2 Database Connectivity Overview

Before you scan an IBM Db2 database, configure connectivity to the database. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, perform the following steps:

1. Install the database client software.
2. Configure native connectivity to the database.
3. Configure IBM Db2 permissions.
4. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

The drivers required to perform domain discovery are shipped with Data Privacy Management. To perform subject discovery scans, download the db2jcc4.jar, db2jcc_license_cu.jar, and db2jcc_license_cisuz.jar files from the IBM download site. Copy the file to the following location: `<Data Privacy Management installation directory>/externaljdbcjars`

After you configure the connectivity, you can create and scan the data store.

Step 1. Install the IBM Db2 Database Client Software

Install the IBM Db2 database client software. To ensure compatibility between Data Privacy Management and databases, use the appropriate database client libraries.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. If the Data Privacy Management Service and the associated Catalog Service are running, disable the services.
2. Install the database client software on the machine that hosts the associated Catalog Service of the Data Privacy Management Service.

Step 2. Configure Native Connectivity to an IBM Db2 Database

After you install the database client software, configure the environment variables. Then, restart the Informatica domain.

1. Set the DB2_HOME environment variable in the user profile file, such as `~/.bashrc` or `~/.bash_profile`.

```
export DB2DIR= /<client installation directory> /db11inst/sqllib/bin
export DB2HOME=/opt/ibm/db2/V11.1
export DB2CODEPAGE=1208
```

2. Add the following entry to the user profile file:

```
if [ -f /<client installation directory> /sqllib/db2profile ]; then
. /<client installation directory> /sqllib/db2profile
fi
```

3. In the `/<DB2HOME/bin` directory, run the following commands to add an entry for the database schema:

```
db2cli
db2
catalog tcpip node <NodeName> remote <Hostname> server 60003
catalog database <DatabaseName> at node <NodeName>
connect to <DatabaseName> user <userName>
```

4. In a new session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

Step 3. Configure IBM Db2 Permissions

Configure the permissions for the IBM Db2 database user account that you use to connect to Db2.

The database user account must have CREATEIN, DROPIN, and ALTERIN privileges on the NULLID schema. Optionally, you can grant the BINDADD privilege to the user account.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the IBM Db2 database user account:

Schema Objects	System Tables
- Tables - Views - Indexes - Packages - Procedures - Functions - Sequences - Triggers - Synonyms	- SYSCAT.SCHEMATA - SYSCAT.TRIGGERS - SYSCAT.INDEXES - SYSCAT.INDEXCOLUSE - SYSCAT.REFERENCES - SYSCAT.TABCONST - SYSCAT.KEYCOLUSE - SYSCAT.COLCHECKS - SYSCAT.DATATYPES - SYSCAT.COLUMNS - SYSCAT.TABLES - SYSCAT.FUNCTIONS - SYSCAT.PROCEDURES - SYSCAT.PACKAGES - SYSCAT.TRIGGERS - SYSCAT.SEQUENCES - SYSCAT.INDEXES - SYSCAT.CHECKS - SYSCAT.VIEWS - SYSCAT.PACKAGES - SYSCAT.FUNCDEP - SYSCAT.TRIGDEP - SYSCAT.VIEWDEP

Step 4. Create an IBM Db2 Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to IBM Db2.

When you create the data store, specify the category **Database Management** and the data store type **IBM Db2**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the connection properties in the data store to connect to IBM Db2. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 19

Configuring Connectivity to IBM Db2 for i5/OS Databases

This chapter includes the following topics:

- Configuring IBM Db2 for i5/OS Connectivity Overview, 160
- Step 1. Install PowerExchange, 161
- Step 2. Configure PowerExchange, 161
- Step 3. Set the PowerExchange Directory as an Environment Variable, 162
- Step 4. Add the <jdbc_Db2>.jar File , 162
- Step 5. Create an IBM Db2 for i5/OS Data Store in Data Privacy Management, 162

Configuring IBM Db2 for i5/OS Connectivity Overview

Before you scan or import data from IBM Db2 for i5/OS, configure connectivity for the data store. If you do not correctly configure the connectivity, the Scan job can fail.

When you load the resource, Data Privacy Management uses PowerExchange for Db2 for i5/OS to access metadata from Db2 for i5/OS subsystems. The Data Privacy Management Catalog Service extracts the following types of IBM Db2 for i5/OS functions:

- Built-in
- User-defined
- Scalar
- Column
- Table

To access metadata in an IBM Db2 for i5/OS subsystem, complete the following tasks:

1. Install PowerExchange.
2. Configure PowerExchange.
3. Set the PowerExchange directory as an environment variable in the user profile file, such as `~/.bashrc` or `~/.bash_profile`.
4. Add the `<jdbc_DB2>.jar` file to the `$LDM_HOME` directory.
5. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can create and scan the data store.

Step 1. Install PowerExchange

Data Privacy Management uses PowerExchange to scan IBM Db2 for i5/OS systems. You must install the same version of PowerExchange as the Catalog Service of the Data Privacy Management Service. If the versions differ, the resource load can fail.

Install PowerExchange for Db2 for i5/OS in the machine where the Catalog Service of the Data Privacy Management Service runs.

Step 2. Configure PowerExchange

To access metadata from a Db2 for i5/OS subsystem, install PowerExchange on the machine where the Catalog Service of the Data Privacy Management Service runs. You must also install PowerExchange on the Db2 for i5/OS system where IBM Db2 runs.

To configure PowerExchange, complete the following tasks:

- Configure PowerExchange to connect to Db2 for i5/OS on the machine where the Catalog Service of the Data Privacy Management Service runs.
- Configure the dbmover.cfg file on the machine where the Catalog Service of the Data Privacy Management Service runs with a NODE statement that points to the PowerExchange Listener on i5. The following information shows an example of a dbmover.cfg file:

```
/*********************************************************************/
/*  PowerExchange Configuration File
/*********************************************************************/
LISTENER=(node1,TCPIP,2480,262144,262144,262144,262144)
NODE=(local,TCPIP,127.0.0.1,2480)
NODE=(node1,TCPIP,127.0.0.1,2480,262144,262144,262144,262144)
NODE=(default,TCPIP,x,2480)
NODE=(MERCURYDCOM,TCPIP,mhz890-2,40005)
NODE=(MERCURYIMS,TCPIP,mhz890-2,40006)
NODE=(MERCURYIMST,TCPIP,mhz890-2,40007)
NODE=(MERCURYIDMS,TCPIP,mhz890-2,40008)
NODE=(MERCURYI5OS,TCPIP,i5v7r2,40005)
APPBUFSIZE=256000
COLON=:
CONSOLE_TRACE=N
DECPOINT=.
DEFAULTCHAR=*
DEFAULTDATE=19800101
MAXTASKS=30
MSGPREFIX=PWX
NEGSIGN=-
PIPE=|
POLLTIME=1000
PRE861_COMPAT=N
EXT_CP_SUPPT=Y
```

Step 3. Set the PowerExchange Directory as an Environment Variable

Set the PowerExchange installation directory as an environment variable in the user profile file, such as `~/.bashrc` or `~/.bash_profile`.

1. Use the following syntax: `export PWX_HOME=<PowerExchangeInstalledDirectory>`
2. Set PWX_HOME in the shared library path environment variable. For example:

```
export LD_LIBRARY_PATH=$ODBCHOME/lib:/usr/lib64:/usr/lib:$INFA_HOME/server/shared/
bin:$INFA_HOME/server/bin:/usr/local/nz/lib64:$ORACLE_HOME/
lib:$ODBCINI:$TERADATA_HOME/lib64:$TERADATA_HOME/lib:$TERADATA_HOME/odbc_64/
lib:$PWX_HOME:/opt/ibm/db2/V11.1/lib64:$LD_LIBRARY_PATH
```

3. Restart the Informatica domain.

Step 4. Add the <jdbc_Db2>.jar File

Add the `<jdbc_Db2>.jar` file to a .zip file, and place the .zip in the `ScannerBinaries` folder inside the $EDC_HOME directory.

1. Create a .zip file named `genericJDBC.zip` and add the `<jdbc_Db2>.jar` file to the .zip file.
2. Add the .zip file to the following directory: `/$EDC_HOME/services/CatalogService/ScannerBinaries folder/`

   Note: If the PowerExchange Listener does not use Db2 security, configure SELECT privileges for the PowerExchange Listener user ID instead of the Db2 for i5/OS user account.

Step 5. Create an IBM Db2 for i5/OS Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to IBM Db2 for i5/OS.

When you create the data store, specify the category **Database Management** and the data store type **IBM Db2 for i5/OS**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the connection properties in the data store to connect to IBM Db2 for i5/OS. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 20

Configuring Connectivity to IBM Db2 for z/OS Databases

This chapter includes the following topics:

- Configuring IBM Db2 for z/OS Connectivity Overview, 163
- Install PowerExchange, 164
- Step 2. Configure PowerExchange, 164
- Step 3. Configure IBM Db2 for z/OS Privileges, 164
- Step 4. Set the PowerExchange Directory as an Environment Variable, 166
- Step 5. Create an IBM Db2 for z/OS Data Store in Data Privacy Management, 166
- IBM Db2 for z/OS Configuration Properties, 166

Configuring IBM Db2 for z/OS Connectivity Overview

IBM Db2 for z/OS is a relational database management system that runs on z/OS. You can create and configure an IBM Db2 for z/OS data store in Data Privacy Management to scan an IBM Db2 for z/OS database. If you do not correctly configure the connectivity, the Scan job can fail.

When you load the resource, Data Privacy Management uses PowerExchange for Db2 for z/OS to access metadata from z/OS subsystems. The Data Privacy Management Catalog Service extracts the following types of IBM Db2 for z/OS functions:

- Built-in
- User-defined
- Scalar
- Column
- Table

To access metadata in an IBM Db2 for z/OS subsystem, complete the following tasks:

1. Install PowerExchange.
2. Configure PowerExchange.
3. Configure IBM Db2 for z/OS privileges.
4. Set the PowerExchange directory as an environment variable in a user profile file, such as `~/.bashrc` or `~/.bash_profile`.

5. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can create and scan the data store.

Install PowerExchange

Data Privacy Management uses PowerExchange for IBM Db2 for z/OS to extract metadata from an IBM Db2 for z/OS subsystem. You must install the same version of PowerExchange as the Catalog Service of the Data Privacy Management Service. If the versions differ, the resource load can fail.

Install PowerExchange for Db2 for z/OS in the following locations:

- The machine where the Catalog Service of the Data Privacy Management Service runs.
- The z/OS system that hosts the IBM Db2 for z/OS database subsystem.

Step 2. Configure PowerExchange

To access metadata from a Db2 for z/OS subsystem, install PowerExchange on the machine where the Catalog Service of the Data Privacy Management Service runs. You must also install PowerExchange on the z/OS system where IBM Db2 runs.

To configure PowerExchange, complete the following tasks:

- Configure PowerExchange to connect to z/OS on the machine where the Catalog Service of the Data Privacy Management Service runs.
- Configure the dbmover.cfg file on the machine where the Catalog Service of the Data Privacy Management Service runs with a NODE statement that points to the PowerExchange Listener on z/OS.
- Configure the DBMOVER configuration member on the z/OS system to control access to the DB2 for z/OS system. Add the following SECURITY statement with option 2 to use an MVS user ID and password or PowerExchange passphrase, along with your z/OS security product:

```
SECURITY=(2,N)
```

By default, PowerExchange on z/OS accesses Db2 with the user ID of the PowerExchange Listener. To specify a different user ID for Db2 access, configure PowerExchange to use operating system security.

Step 3. Configure IBM Db2 for z/OS Privileges

Configure privileges for the IBM Db2 for z/OS user account that connects to the Db2 for z/OS database.

When you load an IBM Db2 for z/OS resource, Data Privacy Management accesses metadata from the IBM Db2 for z/OS subsystem. Data Privacy Management selects the metadata from the IBM Db2 catalog tables.

Configure SELECT privileges on the following tables in the IBM Db2 for z/OS user account that connects to the IBM Db2 for z/OS data source:

System Tables	Metadata Extracted
- SYSIBM.SYSSYNONYMS - SYSIBM.SYSTABLES - SYSIBM.SYSROUTINES - SYSIBM.SYSSEQUENCES - SYSIBM.SYSDATATYPES	Schemas
- SYSIBM.SYSTABLES - SYSIBM.SYSTABLESPACE - SYSIBM.SYSVIEWS	- Tables - Views - Alias
- SYSIBM.SYSTABLES - SYSIBM.SYSCOLUMNS - SYSIBM.SYSTABLESPACE	Columns
- SYSIBM.SYSKEYS - SYSIBM.SYSINDEXES	Primary keys
- SYSIBM.SYSRELS - SYSIBM.SYSFOREIGNKEYS - SYSIBM.SYSINDEXES	Foreign keys
- SYSIBM.SYSCHECKS - SYSIBM.SYSCHECKDEP	Check constraints
- SYSIBM.SYSINDEXES - SYSIBM.SYSKEYS	Indexes
- SYSIBM.SYSSYNONYMS - SYSIBM.SYSTABLES	Synonyms
SYSIBM.SYSTRIGGERS	Triggers
SYSIBM.SYSROUTINES	Stored procedures
SYSIBM.SYSROUTINES	Functions
SYSIBM.SYSSEQUENCES	Sequences
- SYSIBM.SYSSEQUENCESDEP - SYSIBM.SYSSEQUENCES	Identity
SYSIBM.SYSDATATYPES	User-defined data types

Note: If the PowerExchange Listener does not use IBM Db2 security, configure SELECT privileges for the PowerExchange Listener user ID instead of the IBM Db2 for z/OS user account.

Step 4. Set the PowerExchange Directory as an Environment Variable

Set the PowerExchange installation directory as an environment variable in the user profile file, such as `~/.bashrc` or `~/.bash_profile`.

1. Use the following syntax: `export PWX_HOME=<PowerExchangeInstalledDirectory>`
2. Set PWX_HOME in the shared library path environment variable. For example:

```
export LD_LIBRARY_PATH=$ODBCHOME/lib:/usr/lib64:/usr/lib:$INFA_HOME/server/shared/
bin:$INFA_HOME/server/bin:/usr/local/nz/lib64:$ORACLE_HOME/
lib:$ODBCINI:$TERADATA_HOME/lib64:$TERADATA_HOME/lib:$TERADATA_HOME/odbc_64/
lib:$PWX_HOME:/opt/ibm/db2/V11.1/lib64:$LD_LIBRARY_PATH
```

3. Restart the Informatica domain.

Step 5. Create an IBM Db2 for z/OS Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to IBM Db2 for z/OS.

When you create the data store, specify the category **Database Management** and the data store type **IBM Db2 for z/OS**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the connection properties in the data store to connect to IBM Db2 for z/OS. For more information, see the *Informatica Data Privacy Management User Guide*.

IBM Db2 for z/OS Configuration Properties

Update the configuration properties when you create or edit an IBM Db2 for z/OS resource.

The following table describes the IBM Db2 for z/OS configuration properties:

Property	Description
User name	z/OS user ID.
Password	Password for the z/OS user ID.
Location	Node name in the dbmover.cfg file on the machine where the Metadata Manager Service runs that points to the PowerExchange Listener on the z/OS system.
Subsystem ID	Identifier for the IBM Db2 for z/OS subsystem.
Encoding	Code page for the IBM Db2 for z/OS subsystem. Only code pages that are compatible with the code page of the Metadata Manager repository are displayed.

CHAPTER 21

Configuring Connectivity to Microsoft SQL Server Databases

This chapter includes the following topics:

- Configuring Microsoft SQL Server Database Connectivity Overview, 167
- Step 1. Configure ODBC Connectivity to Profile a Microsoft SQL Server Database, 168
- Step 1. Configure JDBC Connectivity to Profile a Microsoft SQL Server Database, 168
- Step 2. Configure Microsoft SQL Server Permissions, 170
- Step 3. Create a Microsoft SQL Server Data Store in Data Privacy Management, 170

Configuring Microsoft SQL Server Database Connectivity Overview

Before you run a scan on a Microsoft SQL Server database, configure connectivity to the database. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Configure one of the following database connectivity types:
   - For native mode, configure ODBC connectivity to the database.
   - For Hadoop mode, configure JDBC connectivity to the database.
2. Configure permissions and privileges for the database user account.
3. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Configure ODBC Connectivity to Profile a Microsoft SQL Server Database

The following steps provide a guideline for configuring ODBC connectivity. For specific instructions, see the database documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Verify if the ODBCINST environment variable is set to the directory that contains the `odbcinst.ini` file.
3. If the ODBCINST environment variable is not set to the directory that contains the `odbcinst.ini` file, set it.

   For example:

   ```
   export ODBCINST=$INFA_HOME/ODBC7.1/odbcinst.ini
   ```

   or

   ```
   export ODBCINST=$ODBCHOME/odbcinst.ini
   ```
4. Set the shared library path environment variable based on the operating system. The shared library path must contain the ODBC libraries.
   - For Solaris and Linux, set the LD_LIBRARY_PATH variable.
   - For AIX, set the LIBPATH variable.
   - For HP-UX, set the SHLIB_PATH variable.

   For example, use the following syntax for Linux or Solaris:

   ```
   export LD_LIBRARY_PATH=/usr/lib64:$ORACLE_HOME/lib64:$ODBCHOME/odbc_64/
   lib:$LD_LIBRARY_PATH
   export PATH=$JAVA_HOME/bin:$ORACLE_HOME/bin:$ODBCHOME:$ODBCHOME/
   bin:$LD_LIBRARY_PATH:$PATH
   ```
5. If you set any environment variable in this session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

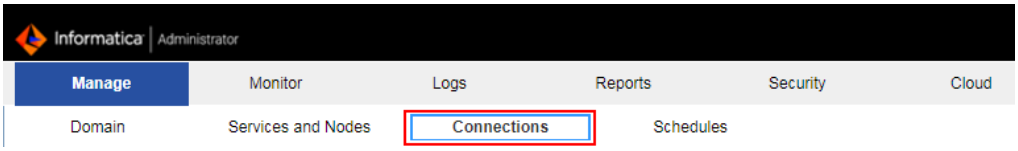
Step 1. Configure JDBC Connectivity to Profile a Microsoft SQL Server Database

If you profile the database in Hadoop mode, use a JDBC connection to access tables in a database. You must configure the JDBC connection with Sqoop arguments. You can create and manage a JDBC connection in the Administrator tool.

The following steps provide a guideline for configuring JDBC connectivity. For more information about Microsoft SQL Server connectivity, see the Microsoft SQL Server driver documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Download the mssql-jdbc-7.4.1.jre8.jar JDBC type 4 driver from the Microsoft download site.
3. Copy the .jar files to the following directory on the machine where the Data Integration Service runs:
   `<Data Privacy Management installation directory>/externaljdbcjars`

4. From the Administrator tool, select the **Connections** tab.



5. In the Domain Navigator pane, select the domain.
6. Click **Actions** > **New** > **Connection**.

   The **New Connection** window appears.
7. Scroll to the Databases category and select **JDBC**.
8. Click **OK**.

   The **New Connection - Step 1 of 2** window appears.
9. Configure the JDBC connection properties.

   The following table describes the JDBC connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Description of the connection that does not exceed 765 characters.
User Name	User name for the database account.
Password	Password for the database account.
JDBC Driver Class Name	Name of the JDBC driver class. Enter the DataDirect JDBC driver class name for Microsoft SQL Server. Example: `com.informatica.jdbc.sqlserver.SQLServerDriver`
Connection String	Connection string to connect to the database. Use the following connection string: `jdbc:informatica:sqlserver://<hostname>:<port>;databaseName=<databasename>`

10. Click **Next**.

    The **New Connection - Step 2 of 2** window appears.

    Note: Do not configure the following fields: Environmental SQL, Transaction SQL, Support Mixed-case Identifiers, SQL Identifier Character.
11. Select the Sqoop version from the **Use Sqoop Connector** property list.
12. Enter the Sqoop arguments in the **Sqoop Arguments** field.

    The Sqoop arguments that you specify take precedence over the JDBC connection properties. The Sqoop arguments define the connection string that the Sqoop program must use to connect to the database. Example:

    ```
    --connect jdbc:sqlserver://irw12dqd15:1433;databaseName=source -m 1
    ```
13. Click **Finish**.

14. Recycle the Data Integration Service.

    If you do not recycle the service, the Scan job cannot connect to the database.

Step 2. Configure Microsoft SQL Server Permissions

Configure the permissions for the Microsoft SQL Server database user account that you use to connect to Microsoft SQL Server.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Microsoft SQL Server database user account:

Schema Objects	System Tables
- Tables - Views - Indexes - Packages - Procedures - Functions - Sequences - Triggers - Synonyms	- sysobjects - sysusers - master.dbo.sysservers - syscolumns sysindexkeys - systypes - syscomments - sysreferences - sysdepends - sys.synonyms (Microsoft SQL Server 2005) - sysindexes - sys.sql_expression_dependencies

In addition, the user must have VIEW DEFINITION permission on the database.

Step 3. Create a Microsoft SQL Server Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft SQL Server databases.

When you create the data store, specify the category **Database Management** and the data store type **Microsoft SQL Server**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 22

Configuring Connectivity to MongoDB Sources

This chapter includes the following topics:

- Configuring MongoDB Connectivity Overview, 171
- Step 1. Download the CDATA Driver JAR Files for MongoDB and Update the File Entry, 171
- Step 2. Create a JDBC Data Store for MongoDB in Data Privacy Management, 172

Configuring MongoDB Connectivity Overview

You can perform domain discovery and Subject Registry scans on a MongoDB database. Before you scan a MongoDB database, configure connectivity to the database.

To configure connectivity, complete the following steps:

1. Download the required driver JAR files to the required locations on the Data Privacy Management machine and add an entry for the driver JAR files.
2. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Download the CDATA Driver JAR Files for MongoDB and Update the File Entry

Before you can create a JDBC data store for MongoDB in Data Privacy Management, you must install the required drivers.

1. Download and install the CData JDBC driver from the CData web site.

   Download the following files:

   - cdata.jdbc.mongodb.jar
   - cdata.jdbc.mongodb.lic

2. Copy the license and JAR files to the following locations:
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
     Add the file to the genericJDBC.zip file. Create the ZIP file if it does not exist.
3. Add an entry for the genericJDBC.zip file in the scannerDeployer.xml file in the following location: `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/`

   The following code is an example of an entry in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="genericJDBC.zip">
            <PropertyName>JDBCScanner_DriverLocation</PropertyName>
            <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue>
```

4. Verify that you can connect to the database.
5. In a new session, restart the Informatica domain.

Step 2. Create a JDBC Data Store for MongoDB in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to MongoDB databases.

When you create the data store, specify the category **Database Management** and the data store type **JDBC**.

Leave the connection string field blank when you create a JDBC data store for MongoDB. Scans fail if the data store properties include both a connection string and a JDBC URL. If you import a data store, ensure that the connection string field is blank.

You cannot perform a Data scan with the **First 10,000**, **First 100,000**, or **First 500,000 Sampling Technique** options. The CData MongoDB driver does not support choosing the first N rows. When you choose the options, the scan fails with a Malformed SQL Statement exception.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 23

Configuring Connectivity to Netezza Databases

This chapter includes the following topics:

- Configuring Netezza Database Connectivity Overview, 173
- Step 1. Install the Netezza Database Client Software, 173
- Step 2. Configure ODBC Connectivity to a Netezza Database, 174
- Step 2. Configure JDBC Connectivity to a Netezza Database, 176
- Step 3. Configure Netezza Permissions, 177
- Step 4. Create a Netezza Data Store in Data Privacy Management, 178

Configuring Netezza Database Connectivity Overview

Before you scan a Netezza database, configure connectivity to the database. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Install the database client software and ODBC drivers.
2. Configure one of the following types of connectivity to the database:
   - For native mode, configure ODBC connectivity to the database.
   - For Hadoop mode, configure JDBC connectivity to the database.
3. Configure permissions and privileges for the database user account.
4. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Install the Netezza Database Client Software

Install the Netezza database client software on the machine that hosts the associated Catalog Service of the Data Privacy Management Service. The database client software includes the ODBC drivers. To ensure

compatibility between Data Privacy Management and databases, use the appropriate database client libraries.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. If the Data Privacy Management Service and the associated Catalog Service are running, disable the services.
2. Install the 32-bit and 64-bit Netezza database client software on the machine that hosts the associated Catalog Service of the Data Privacy Management Service.
3. Create a .zip file named `netezza.zip` and add the `nzjdbc.jar` file.

   Add the .zip file to the following directory: `/$EDC_HOME/services/CatalogService/ScannerBinaries folder/`
4. Edit the following file: `$INFA_HOME/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml`
5. Add the following entry to the file:

```
<ExecutionContextProperty isLocationProperty="true" dependencyToUnpack="netezza.zip">
        <PropertyName>NetezzaScanner_DriverLocation</PropertyName>
        <PropertyValue>scanner_miti/netezza/Drivers</PropertyValue>
 </ExecutionContextProperty>
```

6. If you disabled the Data Privacy ManagementService and the associated Catalog Service, start the services.

Step 2. Configure ODBC Connectivity to a Netezza Database

After you install the database client software, configure the environment variables, ODBC drivers, and data sources. Then, recycle the Informatica services.

The following steps provide a guideline for configuring ODBC connectivity. For specific instructions, see the database documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Set the ODBCHOME, ODBCINI, and NZ_ODBC_INI_PATH environment variables:
   - Set the ODBCHOME variable to the ODBC installation directory. For example: `export ODBCHOME=$INFA_HOME/ODBC7.1`
   - Set the ODBCINI variable to the directory that contains the `odbc.ini` file. For example: `export ODBCINI=$INFA_HOME/ODBC7.1/odbc.ini`
   - Set the NZ_ODBC_INI_PATH variable to the directory that contains the `odbc.ini` file. For example: `export NZ_ODBC_INI_PATH=$INFA_HOME/ODBC7.1`
3. Set the shared library path environment variable. The shared library path must contain the following items:
   - ODBC libraries
   - Informatica services installation
   - Netezza 64-bit client installation

Set the shared library environment variable based on the operating system:

- For Solaris and Linux, set the LD_LIBRARY_PATH variable.
- For AIX, set the LIBPATH variable.
- For HP-UX, set the SHLIB_PATH variable.

For example, use the following syntax for Linux or Solaris:

```
export LD_LIBRARY_PATH=$INFA_HOME/ODBC7.1/lib:$INFA_HOME/server/shared/
bin:$INFA_HOME/server/bin:$LD_LIBRARY_PATH
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/local/nz/lib64
```

4. Edit the odbc.ini file in the $ODBCHOME directory.
   a. Add an entry for the Netezza data source under the section [ODBC Data Sources] and configure the data source.
   b. Configure the `Servername`, `Port`, and `Database` parameters. If you did not install the database client software in the default installation directory, configure the `Driver` parameter. You can use the default values for the rest of the parameters.

      For example:

```
[ODBC Data Sources]
NZSQL=NetezzaSQL ODBC

[NZSQL]
Driver = /usr/local/nz/lib64/libnzodbc.so
Description = NetezzaSQL ODBC
Servername = netezza-host
Port = 5480
Database = TESTDB
Username =
Password =
ReadOnly = false
FastSelect = false
ShowSystemTables = false
LegacySQLTables = false
LoginTimeout = 0
QueryTimeout = 0
DateFormat = 1
NumericAsChar = false
SQLBitOneZero = false
StripCRLF = false
securityLevel = preferredUnSecured
```

   For more information about Netezza connectivity, see the Netezza ODBC driver documentation.

5. In a new session, restart the Informatica domain.

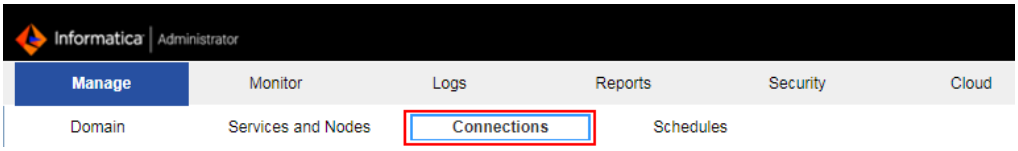
   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

Step 2. Configure JDBC Connectivity to a Netezza Database

If you run the Netezza database in Hadoop mode, use a JDBC connection to access tables in a database. You must configure the JDBC connection with Sqoop arguments. You can create and manage a JDBC connection in the Administrator tool.

The following steps provide a guideline for configuring JDBC connectivity. For more information about Netezza connectivity, see the Netezza driver documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Download the nzjdbc.jar JDBC type 4 driver that is specific to the database.
3. Copy the jar files to the `<Informatica Installation directory>/externaljdbcjars` folder on the machine where the Data Integration Service runs.
4. From the Administrator tool, select the **Connections** tab.



5. In the Domain Navigator pane, select the domain.
6. Click **Actions** > **New** > **Connection**.

   The **New Connection** window appears.
7. Scroll to the Databases category and select **JDBC**.
8. Click **OK**.

   The **New Connection - Step 1 of 2** window appears.
9. Configure the JDBC connection properties. The following table describes the JDBC connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Description of the connection that does not exceed 765 characters.
User Name	User name for the database account.
Password	Password for the database account.

Property	Description
JDBC Driver Class Name	Name of the JDBC driver class. Enter the following DataDirect JDBC driver class name for Netezza: `org.netezza.Driver`
Connection String	Connection string to connect to the database. Enter the following connection string: `jdbc:netezza://adaptersnz2:5480/ILM_DB`

10. Click **Next**.

    The **New Connection - Step 2 of 2** window appears.

    Note: Do not configure the following fields: Environmental SQL, Transaction SQL, Support Mixed-case Identifiers, and SQL Identifier Character.

11. Select the Sqoop version from the **Use Sqoop Connector** list.
12. Enter the Sqoop arguments in the **Sqoop Arguments** field.

    The Sqoop arguments that you specify take precedence over the JDBC connection properties. The Sqoop arguments define the connection string that the Sqoop program must use to connect to the database. Example:`--verbose`

13. Click **Finish**.

    A cluster configuration object for the Hadoop cluster is created.

14. Recycle the Data Integration Service.

    If you do not recycle the service, the Scan job cannot connect to the database.

Step 3. Configure Netezza Permissions

Configure the permissions for the Netezza database user account that you use to connect to Netezza.

Configure the LIST privilege for the Netezza user to extract metadata from a Netezza database. The LIST privilege grants the Netezza user access to all database objects in the Netezza database through Data Privacy Management. If required, you can assign database permissions to limit access to individual database objects.

Configure SELECT permissions for the Netezza user account on the following system views:

- _V_TABLE_DIST_MAP
- _V_TABLE
- _V_DATABASE
- _v_relation_keydata
- _v_relation_column
- _v_view
- _v_synonym
- _v_datatype
- _v_sequence

Step 4. Create a Netezza Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Netezza databases.

When you create the data store, specify the category **Database Management** and the data store type **Netezza**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 24

Configuring Connectivity to Workday Sources

This chapter includes the following topics:

- Overview, 179
- Step 1. Download the CDATA Driver JAR Files for Workday, 179
- Step 2. Download the Workday Certificates, 180
- Step 3. Create a JDBC Data Store for Workday in Data Privacy Management, 181

Overview

You can perform domain discovery and Subject Registry scans on Workday applications. Before you run a scan on a Workday application, configure connectivity to the Workday application.

To configure connectivity, complete the following tasks:

1. Download the driver JAR files to the required locations on the Data Privacy Management machine and add an entry for the driver JAR files in the scannerDeployer.xml file.
2. Download the Workday certificates.
3. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to Workday.

After you configure the connectivity, you can scan the data store.

Step 1. Download the CDATA Driver JAR Files for Workday

Before you can create a JDBC data store for Workday in Data Privacy Management, you must install the required drivers.

1. Download the following CData JDBC driver and license from the CData web site:
   - cdata.jdbc.workday.jar

- cdata.jdbc.workday.lic

2. Copy the files to the following locations:
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
     Add the file to the genericJDBC.zip file. Create the ZIP file if it does not exist.
3. Add an entry for the genericJDBC.zip file in the scannerDeployer.xml file in the following location: `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/`

   The following code is an example of an entry in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="genericJDBC.zip">
            <PropertyName>JDBCScanner_DriverLocation</PropertyName>
            <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue>
```

4. Verify that you can connect to the database.
5. In a new session, restart the Informatica domain.

Step 2. Download the Workday Certificates

Data Privacy Management requires an SSL handshake to run the Row Count job. The required certificates must be present for scans that include the Include Row Count option to run.

To ensure that Data Privacy Management can access the secret key required to communicate with the server, copy the parent and child Workday certificates to infa_truststore.jks.

Copy the following certificates:

- Parent certificate. Digicert Global Root G2
- Child certificate. GeoTrust TLS RSA CA G1

Perform the following steps to download and copy the certificates:

1. Log in to your Workday account.
2. Click the lock icon in the URL field.
3. On the Security information tab, click the **Certificate is valid** option to open the **Certificate** window.
4. Click the **Certification Path** tab and select the Digicert Global Root G2 certificate.
5. Click **View Certificate**.
6. Open the **Details** tab and click **Copy to File**.
7. Retain default options in the Certificate Export Wizard and download the certificates to any location on the machine.
8. Run the following command to copy the certificate:

```
<Informatica installation directory>/java/jre/bin/keytool -import -alias <Alias name
of certificate> -file <Location of certificate> -keystore <Informatica installation
directory>/services/shared/security/infa_truststore.jks -storepass <password> -
trustcacerts -noprompt
```

Repeat the steps to download and copy the child certificate.

Step 3. Create a JDBC Data Store for Workday in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Workday applications.

When you create the data store, specify the category **Database Management** and the data store type **JDBC**.

Leave the connection string field blank when you create a JDBC data store for Workday. Scans fail if the data store properties include both a connection string and a JDBC URL. If you import a data store, ensure that the connection string field is blank.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 25

Configuring Connectivity to Oracle Databases

This chapter includes the following topics:

- Configuring Oracle Database Connectivity Overview, 182
- Step 1. Install the Oracle Database Client Software, 183
- Step 2. Configure Native Connectivity to an Oracle Database, 183
- Step 2. Configure JDBC Connectivity to an Oracle Database, 184
- Step 3. Configure Oracle Permissions, 186
- Step 4. Create an Oracle Data Store in Data Privacy Management, 188

Configuring Oracle Database Connectivity Overview

Before you scan an Oracle database, configure connectivity to the database. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Install the database client software.
2. For native mode, configure native connectivity to the database. For Hadoop mode, configure JDBC connectivity to the database.
3. Configure permissions and privileges for the database user account.
4. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Install the Oracle Database Client Software

Install the full version of the Oracle database client software. Install the software on the machine that hosts the associated Catalog Service of the Data Privacy Management Service. To ensure compatibility between Data Privacy Management and the database, use the appropriate database client libraries.

Perform the installation with the same operating system user that performed the Data Privacy Management installation. When you install the full version, choose the Administrator installation type.

Important: Install the full version of the database client software. The instant client installation does not contain all of the libraries and verification tools that Data Privacy Management requires to connect to the database.

For specific instructions, see the database documentation.

Step 2. Configure Native Connectivity to an Oracle Database

After you install the database client software, configure the environment variables and configure the `tnsnames.ora` file. Then, recycle the Informatica services.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Set the ORACLE_HOME and PATH environment variables in the user profile file, such as `~/.bashrc` or `~/.bash_profile`.
   a. Set the ORACLE_HOME variable to the Oracle client installation directory. For example, if the client is installed in the `/app/oracle/product/12.1.0/` directory, set the following variable: `export ORACLE_HOME=/app/oracle/product/12.1.0/client_1`
   b. To run the Oracle command line programs, set the PATH variable to include the Oracle bin directory. For example: `export PATH=$ORACLE_HOME/bin:$PATH:$HOME/bin`
3. To locate the shared libraries during run time, set the shared library path environment variable in the user profile file. The shared library path must contain the Informatica services installation and the Oracle database client installation. Set the shared library environment variable based on the operating system:
   - For Solaris and Linux, set the LD_LIBRARY_PATH variable.
   - For AIX, set the LIBPATH variable.
   - For HP-UX, set the SHLIB_PATH variable.

   For example, use the following syntax for Linux or Solaris:

   ```
   export LD_LIBRARY_PATH=/usr/lib:$ORACLE_HOME:$ORACLE_HOME/lib:$ORACLE_HOME/
   bin:$LD_LIBRARY_PATH
   ```
4. Save the user profile file. Source the user profile, or close and open a new shell. For example, use one of the following commands to source the user profile:

   ```
   . ~/.bashrc
   source ~/.bashrc
   ```
5. Add the following entry in the `$ORACLE_HOME/network/admin/tnsnames.ora` file for each data store that connects to an Oracle database:

   ```
   <DB_ALIAS> =
     (DESCRIPTION =
   ```

```
    (ADDRESS = (PROTOCOL = TCP)(HOST = <DB_HOST>)(PORT = <DB_PORT>))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = <DB_SERVICE_NAME>)
    )
  )
```

Use the same database name for the <DB_ALIAS> entry in the tnsnames.ora file and the connection string in the Oracle data store connection properties. For example:

```
hrdb =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = hrhost.informatica.com)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = hr.informatica.com)
    )
  )
```

6. Verify that you can connect to the Oracle database.
   a. To verify the connection, run the following command from the command line: `tnsping <DB_ALIAS>`

      For example: `tnsping hrdb`

      If the connection is successful, you receive a message similar to the following text:

      ```
      TNS Ping Utility for Linux: Version 12.1.0.2.0 - Production on 11-FEB-2016
      20:11:45

      Copyright (c) 1997, 2014, Oracle.  All rights reserved.

      Used parameter files:
      $ORACLE_HOME/network/admin/sqlnet.ora

      Used TNSNAMES adapter to resolve the alias
      Attempting to contact (DESCRIPTION = (ADDRESS=(PROTOCOL=TCP)
      (HOST=hrhost.informatica.com)(port=1521))
      (CONNECT_DATA=(SERVICE_NAME=hr.informatica.com)))
      OK (0 msec)
      ```

   b. Run the following command from the command line: `sqlplus <DATA_STORE_USER>/<DATA_STORE_PASSWORD>@<DB_ALIAS>`

      For example: `sqlplus system/manager@hrdb`

      If the connection is successful, the user connects to the database.
7. In a new session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

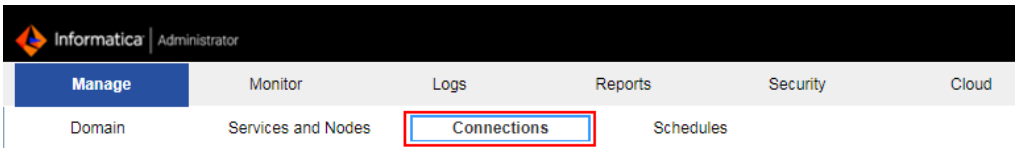
Step 2. Configure JDBC Connectivity to an Oracle Database

If you run the database in Hadoop mode, use a JDBC connection to access tables in a database. You must configure the JDBC connection with Sqoop arguments. You can create and manage a JDBC connection in the Administrator tool.

The following steps provide a guideline for configuring JDBC connectivity. For more information about Oracle connectivity, see the Oracle driver documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.

2. Download the ojdbc6.jar JDBC type 4 driver that is specific to the database.
3. Copy the jar files to the following directory on the machine where the Data Integration Service runs:
   `<Data Privacy Management installation directory>/externaljdbcjars`
4. From the Administrator tool, select the **Connections** tab.



5. In the Domain Navigator pane, select the domain.
6. Click **Actions** > **New** > **Connection**.

   The **New Connection** window appears.
7. Scroll to the Databases category and select **JDBC**.
8. Click **OK**.

   The **New Connection - Step 1 of 2** window appears.
9. Configure the JDBC connection properties.

   The following table describes the JDBC connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Description of the connection that does not exceed 765 characters.
User Name	User name for the database account.
Password	Password for the database account.
JDBC Driver Class Name	Name of the JDBC driver class. Enter the DataDirect JDBC driver class name for Oracle: `com.informatica.jdbc.oracle.OracleDriver`
Connection String	Connection string to connect to the database. Use the following connection string: `jdbc:informatica:oracle://<hostname>:<port>;SID=<sid>` Example: `jdbc:informatica:oracle://rac12c.informatica.com:1521;ServiceName=qa12crac.informatica.com`

10. Click **Next**.

    The **New Connection - Step 2 of 2** window appears.

    Note: Do not configure the following fields: Environmental SQL, Transaction SQL, Support Mixed-case Identifiers, and SQL Identifier Character.
11. Select the Sqoop version from the **Use Sqoop Connector** list.
12. Enter the Sqoop arguments in the **Sqoop Arguments** field.

The Sqoop arguments that you specify take precedence over the JDBC connection properties. The Sqoop arguments define the connection string that the Sqoop program must use to connect to the database.

Example for a RAC database:

```
-m 1 --connect jdbc:oracle:thin:@//rac12c.informatica.com:1521/
qa12crac.informatica.com
```

Example for a non-RAC database:

```
--verbose --connect jdbc:oracle:thin:@irl62dqd06.informatica.com:1521:scala11gr2 -m 1
```

13. Click **Finish**.

    A cluster configuration object for the Hadoop cluster is created.

14. Recycle the Data Integration Service.

    If you do not recycle the service, the Scan job cannot connect to the database.

Step 3. Configure Oracle Permissions

Configure the permissions for the Oracle user account that you use to connect to the Oracle database. Grant the privileges that Data Privacy Management requires to connect to the Oracle database and scan the data store.

You must configure the CONNECT, CREATE SESSION, and SELECT_CATALOG_ROLE permissions for the database user account that you use to access the Oracle database.

If you cannot configure the SELECT_CATALOG_ROLE permission for the user account, configure the following permissions:

- SELECT permissions on the v$version and v$instance views.
- CONNECT and SELECT permissions on the required schema objects and corresponding system tables.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Indexes
- Packages
- Procedures
- Functions
- Sequences
- Triggers
- Synonyms

Configure SELECT permissions on the following system tables:

- ALL_OBJECTS
- ALL_SOURCE
- ALL_USERS
- DBA_ARGUMENTS
- DBA_COL_COMMENTS

- DBA_CONS_COLUMNS
- DBA_CONSTRAINTS
- DBA_DEPENDENCIES
- DBA_IND_COLUMNS
- DBA_INDEXES
- DBA_MVIEWS
- DBA_OBJECTS
- DBA_PROCEDURES
- DBA_RECYCLEBIN
- DBA_SEQUENCES
- DBA_SOURCE
- DBA_SYNONYMS
- DBA_TAB_COLUMNS
- DBA_TAB_COMMENTS
- DBA_TAB_PARTITIONS
- DBA_TABLES
- DBA_TRIGGERS
- DBA_VIEWS
- SYS.DBA_ARGUMENTS
- SYS.DBA_CLU_COLUMNS
- SYS.DBA_CLUSTERS
- SYS.DBA_COL_COMMENTS
- SYS.DBA_CONS_COLUMNS
- SYS.DBA_CONSTRAINTS
- SYS.DBA_DEPENDENCIES
- SYS.DBA_DIRECTORIES
- SYS.DBA_EXTERNAL_LOCATIONS
- SYS.DBA_IND_COLUMNS
- SYS.DBA_INDEXES
- SYS.DBA_MVIEWS
- SYS.DBA_OBJECTS
- SYS.DBA_PROCEDURES
- SYS.DBA_RECYCLEBIN
- SYS.DBA_SEQUENCES
- SYS.DBA_SOURCE
- SYS.DBA_SYNONYMS
- SYS.DBA_TAB_COLUMNS
- SYS.DBA_TAB_COMMENTS
- SYS.DBA_TAB_PARTITIONS

- SYS.DBA_TABLES
- SYS.DBA_TRIGGERS
- SYS.DBA_USERS
- SYS.DBA_VIEWS
- SYS.V_$DATABASE
- SYS.V_$INSTANCE
- SYS.V_$VERSION

Either grant the SELECT ANY TABLE privilege or grant the direct SELECT privilege on all tables and views in the schema that you want to scan. If schemas have a table with a cluster defined on some columns, you must configure SELECT permission on the DBA_CLUSTERS system table.

Step 4. Create an Oracle Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Oracle databases.

When you create the data store, specify the category **Database Management** and the data store type **Oracle**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 26

Configuring Connectivity to REST Services

This chapter includes the following topics:

- Configuring REST Services Connectivity Overview, 189
- Step 1. Download the CDATA Driver JAR Files for REST Services and Update the File Entry, 189
- Step 2. Create a JDBC Data Store for REST Services in Data Privacy Management, 190

Configuring REST Services Connectivity Overview

You can connect to REST services to perform domain discovery and Subject Registry scans. Before you scan REST services, configure connectivity to the REST services.

To configure connectivity, complete the following steps:

1. Download the required driver JAR and license files to the required locations on the Data Privacy Management machine and add an entry for the driver JAR files.
2. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Download the CDATA Driver JAR Files for REST Services and Update the File Entry

Before you can create a JDBC data store for REST Services in Data Privacy Management, you must install the required drivers and update the required file entries.

1. Download and install the CData license and drivers for REST Services from the CData web site.

   Download the following files:

   - cdata.jdbc.rest.jar
   - cdata.jdbc.rest.lic

- rssbus.jar

2. Copy the license and JAR files to the following locations:
   - `<Data Privacy Management installation directory>/externaljdbcjars`
   - `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/genericJDBC.zip`
     Add the file to the genericJDBC.zip file. Create the ZIP file if it does not exist.
3. Add an entry for the genericJDBC.zip file in the scannerDeployer.xml file in the following location: `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/`

   The following code is an example of an entry in the file:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="genericJDBC.zip">
               <PropertyName>JDBCScanner_DriverLocation</PropertyName>
               <PropertyValue>scanner_miti/genericJDBC/Drivers</PropertyValue>
   ```
4. Verify that you can connect to the database.
5. In a new session, restart the Informatica domain.

Step 2. Create a JDBC Data Store for REST Services in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to REST services.

When you create the data store, specify the category **Database Management** and the data store type **JDBC**.

Leave the connection string field blank when you create a JDBC data store for REST Services. Scans fail if the data store properties include both a connection string and a JDBC URL. If you import a data store, ensure that the connection string field is blank.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 27

Configuring Connectivity to SAP HANA Databases

This chapter includes the following topics:

- Configuring SAP HANA Database Connectivity Overview, 191
- Step 1. Download the SAP HANA JDBC JAR Files and Client, 191
- Step 2. Update the ODBC.ini File, 192
- Step 3. Create an SAP HANA Data Store in Data Privacy Management, 193

Configuring SAP HANA Database Connectivity Overview

Before you scan an SAP HANA database, configure connectivity to the database.

To configure connectivity, complete the following steps:

1. Download the required JAR files and install the client.
2. Update the entry in the ODBC.ini file.
3. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Download the SAP HANA JDBC JAR Files and Client

Before you can create an SAP HANA data store in Data Privacy Management, you must install the required drivers and install the SAP HANA client.

1. Download the ngdbc-2.4.76.jar file for SAP HANA from the SAP web site.

2. Copy the JAR files to the following locations:
   - `<Data Privacy Management install directory>/externaljdbcjars`
   - `<Data Privacy Management install directory>/services/CatalogService/ScannerBinaries/saphanaJars.zip`
     Add the file to the genericJDBC.zip file.
3. Add an entry for the genericJDBC.zip file in the scannerDeployer.xml file in the following location: `<Data Privacy Management installation directory>/services/CatalogService/ScannerBinaries/CustomDeployer/`

   The following code is an example of an entry in the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="saphanaJars.zip">
          <PropertyName>SapHanaScanner_DriverLocation</PropertyName>
          <PropertyValue>scanner_miti/saphana/Drivers</PropertyValue>
        </ExecutionContextProperty>
```

4. Download and install the ODBC client for SAP HANA from the SAP web site.

   For specific instructions, see the database documentation.

Step 2. Update the ODBC.ini File

After you copy the JAR files and install the database client software, add an entry in the `odbc.ini` file.

1. Log in to the Data Privacy Management machine.
2. Open the `<Data Privacy Management installation directory>/ODBC7.1/odbc.ini` file and add an entry for the SAP HANA database.

   The following code is an example entry for SAP HANA:

```
[sapHana_conn]
Driver=/usr/sap/hdbclient/libodbcHDB.so
DriverUnicodeType=1
ServerNode=hanatestdb:30015
encrypt=0
#User=<user name>
#Password=<password>
ConnectionRetryCount=3
ConnectionRetryDelay=30
Trace=1
TraceFile=<Data Privacy Management installation folder>/server/bin/odbc_trace/
trace.log
TraceDll=<Data Privacy Management installation folder>/ODBC7.1/lib/DWtrc27.so
```

3. Verify that you can connect to the database.
4. In a new session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

Step 3. Create an SAP HANA Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to SAP HANA databases.

When you create the data store, specify the category **Database Management** and the data store type **SAP HANA**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 28

Configuring Connectivity to Sybase Databases

This chapter includes the following topics:

- Configuring Sybase Database Connectivity Overview, 194
- Step 1. Install the Sybase Database Client Software, 195
- Step 2. Configure ODBC Connectivity to a Sybase Database, 195
- Step 3. Configure Sybase Permissions, 197
- Step 4. Create a Sybase Data Store in Data Privacy Management, 197

Configuring Sybase Database Connectivity Overview

Before you scan on a Sybase database, configure connectivity to the database. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Install the Sybase database client software.
2. Configure ODBC connectivity to the database.
3. Configure permissions and privileges for the database user account.
4. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Install the Sybase Database Client Software

Install the Sybase database client software on the machine that hosts the associated Catalog Service of the Data Privacy Management Service. To ensure compatibility between Data Privacy Management and databases, use the appropriate database client libraries.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. If the Catalog Service is running, disable the service.
2. Create a .zip file named `sybase_jars.zip` and add the following driver .jar files to the .zip file:
   - `jconn3d.jar`
   - `jTDS3d.jar`

   Add the .zip file to the following directory: `/$LDM_HOME/services/CatalogService/ScannerBinaries/`
3. Edit the following file: `$INFA_HOME/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml`
4. Add the following entry to the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="sybase_jars.zip">
        <PropertyName>SybaseScanner_DriverLocation</PropertyName>
        <PropertyValue>scanner_miti/sybase_jars/Drivers</PropertyValue>
        </ExecutionContextProperty>
```

5. Restart the Catalog Service.

Step 2. Configure ODBC Connectivity to a Sybase Database

The following steps provide a guideline for configuring ODBC connectivity. For specific instructions, see the database documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Set the ODBCHOME and ODBCINI environment variables:
   - Set the ODBCHOME variable to the ODBC installation directory. For example: `export ODBCHOME=$INFA_HOME/ODBC7.1`
   - Set the ODBCINI variable to the directory that contains the `odbc.ini` file. For example: `export ODBCINI=$INFA_HOME/ODBC7.1/odbc.ini`
3. Set the shared library path environment variable based on the operating system. The shared library path must contain the ODBC libraries.
   - For Solaris and Linux, set the LD_LIBRARY_PATH variable.
   - For AIX, set the LIBPATH variable.
   - For HP-UX, set the SHLIB_PATH variable.

   For example, use the following syntax for Linux or Solaris:

```
export LD_LIBRARY_PATH=/usr/lib64:$ORACLE_HOME/lib64:$ODBCHOME/odbc_64/
lib:$ODBCINI:$LD_LIBRARY_PATH
```

```
export PATH=$JAVA_HOME/bin:$ORACLE_HOME/bin:$ODBCHOME:$ODBCHOME/
bin:$LD_LIBRARY_PATH:$PATH
```

4. Edit the `odbc.ini` file in the $ODBCHOME directory with the following changes:
   1. Add an entry for the Sybase data source under the section [ODBC Data Sources] and configure the data source.
   2. Configure the `Database` and `NetworkAddress` parameters. You can use the default values for the other parameters.

   Example:

```
[ODBC Data Sources]
Sybase Wire Protocol=DataDirect 7.1 Sybase Wire Protocol

[Sybase Wire Protocol]
Driver=/data/home/Informatica/SATS4.0/ODBC7.1/lib/DWsqls27.so
Description=DataDirect 7.1 Sybase Wire Protocol
AlternateServers=
ApplicationName=
ApplicationUsingThreads=1
ArraySize=50
AuthenticationMethod=0
BulkBinaryThreshold=32
BulkCharacterThreshold=-1
BulkLoadBatchSize=1024
BulkLoadFieldDelimiter=
BulkLoadRecordDelimiter=
Charset=
ConnectionReset=0
ConnectionRetryCount=0
ConnectionRetryDelay=3
CursorCacheSize=1
Database=<database_name>
DefaultLongDataBuffLen=1024
EnableBulkLoad=0
EnableDescribeParam=0
EnableQuotedIdentifiers=0
EncryptionMethod=0
FailoverGranularity=0
FailoverMode=0
FailoverPreconnect=0
GSSClient=native
HostNameInCertificate=
InitializationString=
Language=
LoadBalancing=0
LoadBalanceTimeout=0
LoginTimeout=15
LogonID=
MaxPoolSize=100
MinPoolSize=0
NetworkAddress=<Sybase_host,Sybase_server_port>
OptimizePrepare=1
PacketSize=0
Password=
Pooling=0
QueryTimeout=0
RaiseErrorPositionBehavior=0
ReportCodePageConversionErrors=0
SelectMethod=0
ServicePrincipalName=
TruncateTimeTypeFractions=0
TrustStore=
TrustStorePassword=
ValidateServerCertificate=1
WorkStationID=
```

For more information about Sybase connectivity, see the Sybase ODBC driver documentation.

5. In a new session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

Step 3. Configure Sybase Permissions

Configure the permissions for the Sybase database user account that you use to connect to the Sybase database.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Columns
- Procedures

Configure SELECT permissions on the following system tables:

- sysobjects
- syscomments
- sysdepends
- syscolumns
- sysindexes
- sysconstraints
- sysreferences
- sysusers
- systypes
- master.dbo.sysservers
- master.dbo.sysconfigures
- @@<servername>

Step 4. Create a Sybase Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Sybase databases.

When you create the data store, specify the category **Database Management** and the data store type **Sybase**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 29

Configuring Connectivity to Teradata Databases

This chapter includes the following topics:

- Configuring Teradata Database Connectivity Overview, 198
- Step 1. Install the Teradata Database Client Software, 199
- Step 2. Configure ODBC Connectivity to a Teradata Database, 199
- Step 2. Configure JDBC Connectivity to a Teradata Database, 201
- Step 3. Configure Teradata Permissions, 203
- Step 4. Create a Teradata Data Store in Data Privacy Management, 203

Configuring Teradata Database Connectivity Overview

Before you scan a Teradata database, configure connectivity to the database. If you do not correctly configure the connectivity, the Scan job can fail.

To configure connectivity, complete the following steps:

1. Install the database client software and ODBC drivers.
2. Configure one of the following types of connectivity to the database:
   - For native mode, configure ODBC connectivity to the database.
   - For Hadoop mode, configure JDBC connectivity to the database.
3. Configure permissions and privileges for the database user account.
4. Create a data store on the **Data Stores** workspace in Data Privacy Management and configure the connection properties to connect to the database.

After you configure the connectivity, you can scan the data store.

Step 1. Install the Teradata Database Client Software

Install the Teradata database client software. The database client software includes the ODBC drivers. To ensure compatibility between Data Privacy Management and databases, use the appropriate database client libraries.

The following steps provide a guideline for installing the database client software. For specific instructions, see the database documentation.

1. If the Data Privacy Management Service and the associated Catalog Service are running, disable the services.
2. Install the database client software on the machine that hosts the associated Catalog Service of the Data Privacy Management Service.
3. Create a .zip file named `teradataJars.zip` with the following two .jar files:

   ```
   /<client installation directory>/lib/terajdbc4.jar
   /<client installation directory>/lib/tdgssconfig.jar
   ```

4. Add the .zip file to the following directory: `/$EDC_HOME/services/CatalogService/ScannerBinaries folder/`
5. Edit the following file: `$INFA_HOME/services/CatalogService/ScannerBinaries/CustomDeployer/scannerDeployer.xml`
6. Uncomment the following entry in the file:

   ```
   <ExecutionContextProperty isLocationProperty="true"
   dependencyToUnpack="teradataJars.zip">
           <PropertyName>TeradataScanner_DriverLocation</PropertyName>
           <PropertyValue>scanner_agents/teradata/Drivers</PropertyValue>
    </ExecutionContextProperty>
   ```

7. If you disabled the Data Privacy ManagementService and the associated Catalog Service, start the services.

Step 2. Configure ODBC Connectivity to a Teradata Database

After you install the database client software, configure the environment variables, ODBC drivers, and data sources. Then, recycle the Informatica services.

The following steps provide a guideline for configuring ODBC connectivity. For specific instructions, see the database documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Set the following environment variables:
   - Set the ODBCHOME variable to the ODBC installation directory. For example: `export ODBCHOME=$INFA_HOME/ODBC<version>`
   - Set the ODBCINI variable to the directory that contains the `odbc.ini` file. For example: `export ODBCINI=$ODBCHOME/odbc.ini`

- Set the TERADATA_HOME variable to the Teradata driver installation directory. For example: `export TERADATA_HOME=/opt/teradata/client/<version>`
- Set the TD_ODBC_HOME variable to the Teradata ODBC installation directory. For example: `export TD_ODBC_HOME=/opt/teradata/client/<version>/odbc_64`
- Set the NLSPATH variable to the directory that contains the message catalog file for the ODBC driver. For example: `export NLSPATH=$TERADATA_HOME/odbc_64/msg`
- Set the PATH variable to the directory that contains the ODBC driver executables and scripts. For example: `export PATH=$INFA_HOME/server/bin:$INFA_HOME/isp/bin:$JAVA_HOME/bin:$ODBCHOME:$ODBCINI:$ODBCHOME/bin:/lib64:$PATH`

3. Set the shared library path environment variable based on the operating system:
   - For Solaris and Linux, set the LD_LIBRARY_PATH variable.
   - For AIX, set the LIBPATH variable.
   - For HP-UX, set the SHLIB_PATH variable.

   The shared library path must contain the following items:
   - ODBC libraries
   - Informatica services installation
   - Teradata database client installation

   For example, use the following syntax for Linux or Solaris:

   ```
   export LD_LIBRARY_PATH=$INFA_HOME/server/bin:$TD_ODBC_HOME/lib:$ODBCHOME/
   lib:$ODBCINI:/lib64:/usr/lib64:$TERADATA_HOME/lib64:$TERADATA_HOME/
   lib:$TERADATA_HOME/odbc_64/lib
   ```

4. Edit the `odbc.ini` file in the $ODBCHOME directory.
   a. Add an entry for the Teradata data source under the section [ODBC Data Sources] and configure the data source.
   b. If you did not install the database client software in the default installation directory, configure the `Driver` parameter.
   c. Set the `DateTimeFormat` to `IAA`.
   d. Set the `SessionMode` to `Teradata` or `ANSI`.

   For example:

   ```
   [ODBC Data Sources]
   TERADATA_ODBC=Teradata

   [TERADATA_ODBC]
   Driver=$TERADATA_HOME/odbc_64/lib/tdata.so
   Description=Teradata
   DBCName=10.11.11.11
   Database=rts_test
   DateTimeFormat=IAA
   SessionMode=[Teradata|ANSI]
   DefaultDatabase=rts_test
   Username=RTS_TEST
   Password=RTS_TEST
   ```

   For more information about Teradata connectivity, see the Teradata ODBC driver documentation.

5. In a new session, restart the Informatica domain.

   If you do not restart the Informatica domain, the Scan job cannot connect to the database.

Step 2. Configure JDBC Connectivity to a Teradata Database

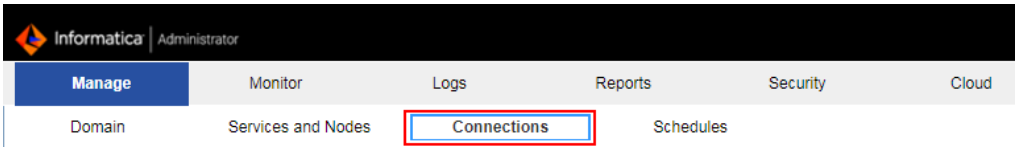
If you run the database in Hadoop mode, use a JDBC connection to access tables in a database. You must configure the JDBC connection with Sqoop arguments. You can create and manage a JDBC connection in the Administrator tool.

The following steps provide a guideline for configuring JDBC connectivity. For more information about Teradata connectivity, see the Teradata driver documentation.

1. Log in to the machine that hosts the associated Data Integration Service for the Data Privacy Management Service. Log in as a user who can start a service process.
2. Download the JDBC type 4 driver that is specific to the database.

   Download the following files from the Teradata download site:

   - terajdbc.jar
   - tdgssconfig.jar
3. Copy the .jar files to the following directory on the machine where the Data Integration Service runs:
   `<Informatica Installation directory>/externaljdbcjars`
4. From the Administrator tool, select the **Connections** tab.



5. In the **Domain Navigator** pane, select the domain.
6. Click **Actions** > **New** > **Connection**.

   The **New Connection** window appears.
7. Scroll to the Databases category and select **JDBC**.
8. Click **OK**.

   The **New Connection - Step 1 of 2** window appears.
9. Configure the JDBC connection properties.

   The following table describes the JDBC connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Description of the connection that does not exceed 765 characters.
User Name	User name for the database account.

Property	Description
Password	Password for the database account.
JDBC Driver Class Name	Name of the JDBC driver class.Enter the DataDirect JDBC driver class name for Teradata: `com.teradata.jdbc.TeraDriver`
Connection String	Connection string to connect to the database. Use the following connection string: `jdbc:teradata://IRSU11ADQ03/database=ide_user,tmode=ANSI,charset=UTF8`

10. Click **Next**.

    The **New Connection - Step 2 of 2** window appears.

    Note: Do not configure the following fields: Environmental SQL, Transaction SQL, Support Mixed-case Identifiers, and SQL Identifier Character.

11. Select the Sqoop version from the **Use Sqoop Connector** property drop-down list.
12. Enter the Sqoop arguments in the **Sqoop Arguments** field.

    The Sqoop arguments that you specify take precedence over the JDBC connection properties.
    The Sqoop arguments define the connection string that the Sqoop program must use to connect to the database.

    Use the following syntax for all Teradata databases regardless of cluster type:

    ```
    --driver com.teradata.jdbc.TeraDriver --verbose --connection-manager
    org.apache.sqoop.manager.GenericJdbcManager
    ```

    Use the following syntax based on the cluster type for Teradata databases:

    ```
    Dsqoop.connection.factories=com.<cluster
    type>.connector.teradata.TeradataManagerFactory
    ```

    Example:

    ```
    Dsqoop.connection.factories=com.cloudera.connector.teradata.TeradataManagerFactory
    ```

13. Click **Finish**.

    A cluster configuration object for the Hadoop cluster is created.

14. Recycle the Data Integration Service.

    If you do not recycle the service, the Scan job cannot connect to the database.

Step 3. Configure Teradata Permissions

Configure the permissions for the Teradata database user account that you use to connect to the Teradata database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Teradata database user account:

Schema Objects	System Tables
- Tables - Views - Indexes - Packages - Procedures - Functions - Sequences - Triggers - Synonyms	- DBC.ALL_RI_CHILDREN - DBC.COLUMNS - DBC.DATABASES - DBC.DBASE - DBC.DBCINFO - DBC.HOSTSINFO - DBC.INDICES - DBC.SHOWTBLCHECKS - DBC.TABLES - DBC.TABLETEXT - DBC.TRIGGERS - DBC.TVFIELDS - DBC.TVM - DBC.UDTINFO - TDQCD.STATSRECS

Step 4. Create a Teradata Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Teradata databases.

When you create the data store, specify the category **Database Management** and the data store type **Teradata**.

When you scan the data store, the Data Privacy Management Service creates a Database Scan job. The Scan job uses the data store connection properties to scan data in the database. For more information, see the *Informatica Data Privacy Management User Guide*.

Part VII: Configuring Connectivity to Email Server Sources

This part contains the following chapter:

- Configuring Connectivity to Office 365 Outlook, 205

CHAPTER 30

Configuring Connectivity to Office 365 Outlook

This chapter includes the following topics:

- Configuring Connectivity to Office 365 Outlook Overview, 205
- Step 1. Perform Prerequisite Tasks, 205
- Step 2. Create an Office 365 Outlook Data Store in Data Privacy Management, 206

Configuring Connectivity to Office 365 Outlook Overview

Configure connectivity to Office 365 Outlook data stores before you create a data store to run scans or perform associated tasks.

To configure connectivity, perform the following steps:

1. Perform prerequisite tasks.
2. Create an Office 365 Outlook data store on the **Data Stores** workspace in Data Privacy Management.

Office 365 Outlook data stores do not connect to Enterprise Data Catalog. When you scan the data store, the remote agent scanner maps sensitive files to data domains and adds subject data in the subject registry.

Step 1. Perform Prerequisite Tasks

Perform the following prerequisite tasks to configure connectivity to an Office 365 Outlook data store.

1. Create an account in the Azure Application Directory.

2. Verify that the Azure Application Directory has the following Application Level permissions granted to the Graph API:

   Directory.Read.All

   List all memberships of a user, including directory roles and administrative units. `GET /beta/users/{id}/memberOf`

   List all group members, including service principals. `GET /beta/groups/{id}/members`

   User.Read.All

   Read full profiles of all users. Allows the application to read the full set of profile properties, group membership, reports, and managers of other users in your organization without a signed-in user.

   Mail.Read

   Read mail in all mailboxes. Allows the application to read mail in all mailboxes without a signed-in user.

See the official Microsoft web site for more information or see the document: https://docs.microsoft.com/en-us/office/office-365-management-api/get-started-with-office-365-management-apis

Step 2. Create an Office 365 Outlook Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to the Office 365 Outlook mail server.

When you create the data store, specify the category **Email Server** and the data store type **Office 365 Outlook**.

When you scan the data store, the Informatica Discovery Agent creates an Email Server Scan job. The Scan job uses the connection properties in the data store to connect to the mail server. For more information, see the *Informatica Data Privacy Management User Guide*.

Part VIII: Configuring Connectivity to File Management Sources

This part contains the following chapters:

- Configuring Connectivity to a File System, 208
- Configuring Connectivity to Microsoft SharePoint, 210
- Configuring Connectivity to Microsoft OneDrive, 216
- Configuring Connectivity to Google Drive, 218

CHAPTER 31

Configuring Connectivity to a File System

This chapter includes the following topics:

- Configuring Connectivity to a File System Overview, 208
- Step 1. Set the INFA_HADOOP_DIST_DIR Variable, 208
- Step 2. Create a File System Data Store in Data Privacy Management, 209

Configuring Connectivity to a File System Overview

Before you scan a file system source, you must specify the file path of a Hadoop distribution directory on the Data Integration Service node. You specify the file path in the `INFA_HADOOP_DIST_DIR` Data Integration Service Process property.

To configure connectivity, complete the following steps:

1. Set the `INFA_HADOOP_DIST_DIR` variable.
2. Create a file system data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Set the INFA_HADOOP_DIST_DIR Variable

Perform the following steps to set the INFA_HADOOP_DIST_DIR variable:

1. Log in to the Administrator tool.
2. Click the **Services and Nodes** view.
3. Select the Data Integration Service from the **Domain Navigator** pane.
4. Select the **Processes** pane.
5. Scroll to the **Environment Variables** section.
6. Click the **Edit** icon.

   The **Edit Environment Variables** window opens.
7. Click **New**.

   The **New Environment Variables** window opens.

8. Enter values in the following fields:
    a. **Name**: INFA_HADOOP_DIST_DIR
    b. **Value**: `/<Informatica_installation_directory>/services/shared/hadoop/<Hadoop distribution directory>`

       Example: `/d01/infa/SATS_4.5/services/shared/hadoop/HDP_2.6`

       Example: `/d01/infa/SATS_4.5/services/shared/hadoop/CDH_5.13`
9. In the **New Environment Variables** window, click **OK**.
10. In the **Edit Environment Variables** window, click **OK** .
11. Click **Actions** > **Recycle Service** for the changes to take effect.

Step 2. Create a File System Data Store in Data Privacy Management

Create a data store and configure properties to connect to a file system. When you create the data store, specify the repository type **File Management** and the data store type **File System**.

When you run a scan on the data store, the Data Privacy Management Service creates a File Management Scan job. The Scan job uses the connection properties in the data store to connect to the file system. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 32

Configuring Connectivity to Microsoft SharePoint

This chapter includes the following topics:

- Configuring Microsoft SharePoint Connectivity Overview, 210
- Step 1. Import the Microsoft SharePoint Certificate, 210
- Step 2. Access the Required SharePoint Account Information, 214
- Step 3. Create a Microsoft SharePoint Data Store in Data Privacy Management, 214

Configuring Microsoft SharePoint Connectivity Overview

If the Microsoft SharePoint source is SSL enabled, then you must configure connectivity to the source before a scan. If you do not correctly configure the connectivity, the jobs can fail. You can configure the data store to use either a remote agent or Enterprise Data Catalog to run scans

You can configure a SharePoint data store to scan either a single account or multiple accounts in a tenant. To scan multiple accounts, the data store must use a remote agent.

To configure connectivity, complete the following steps:

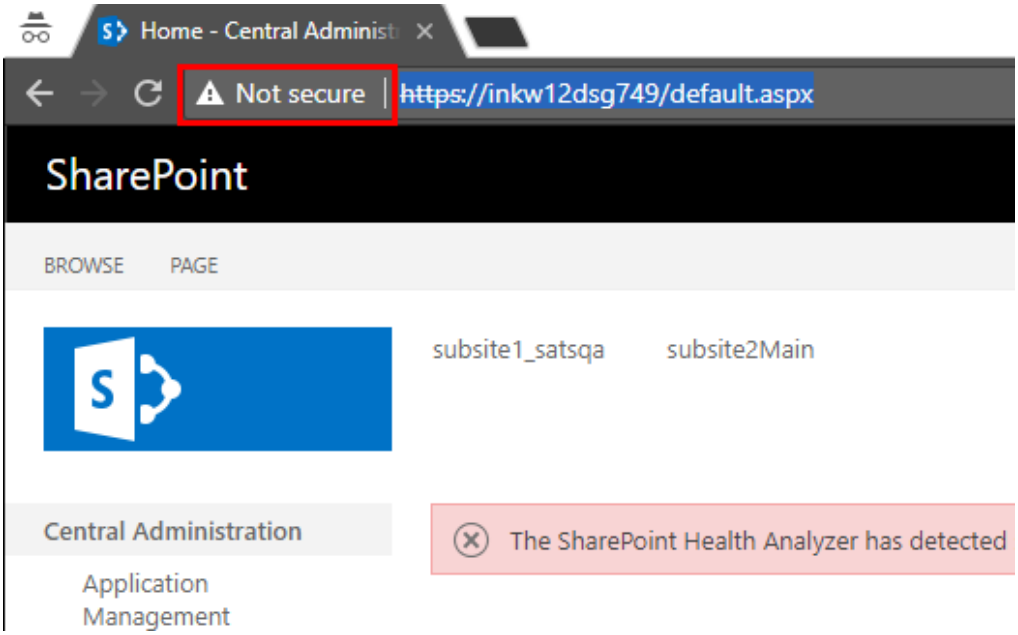
1. Import the Microsoft SharePoint SSL certificate.
2. Access the required SharePoint account information.
3. Create a Microsoft SharePoint data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Import the Microsoft SharePoint Certificate

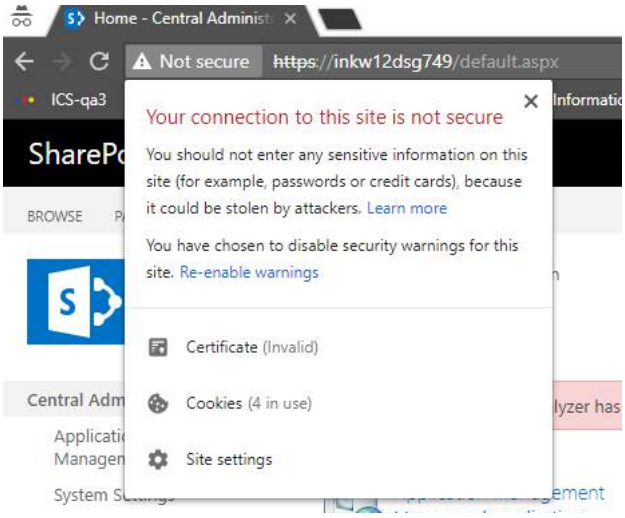
Perform the following steps to import the Microsoft SharePoint certificate.

1. Go to your SharePoint Central Administration site.

2. Click either **Not Secure** or **Secure** in the address field of the browser.



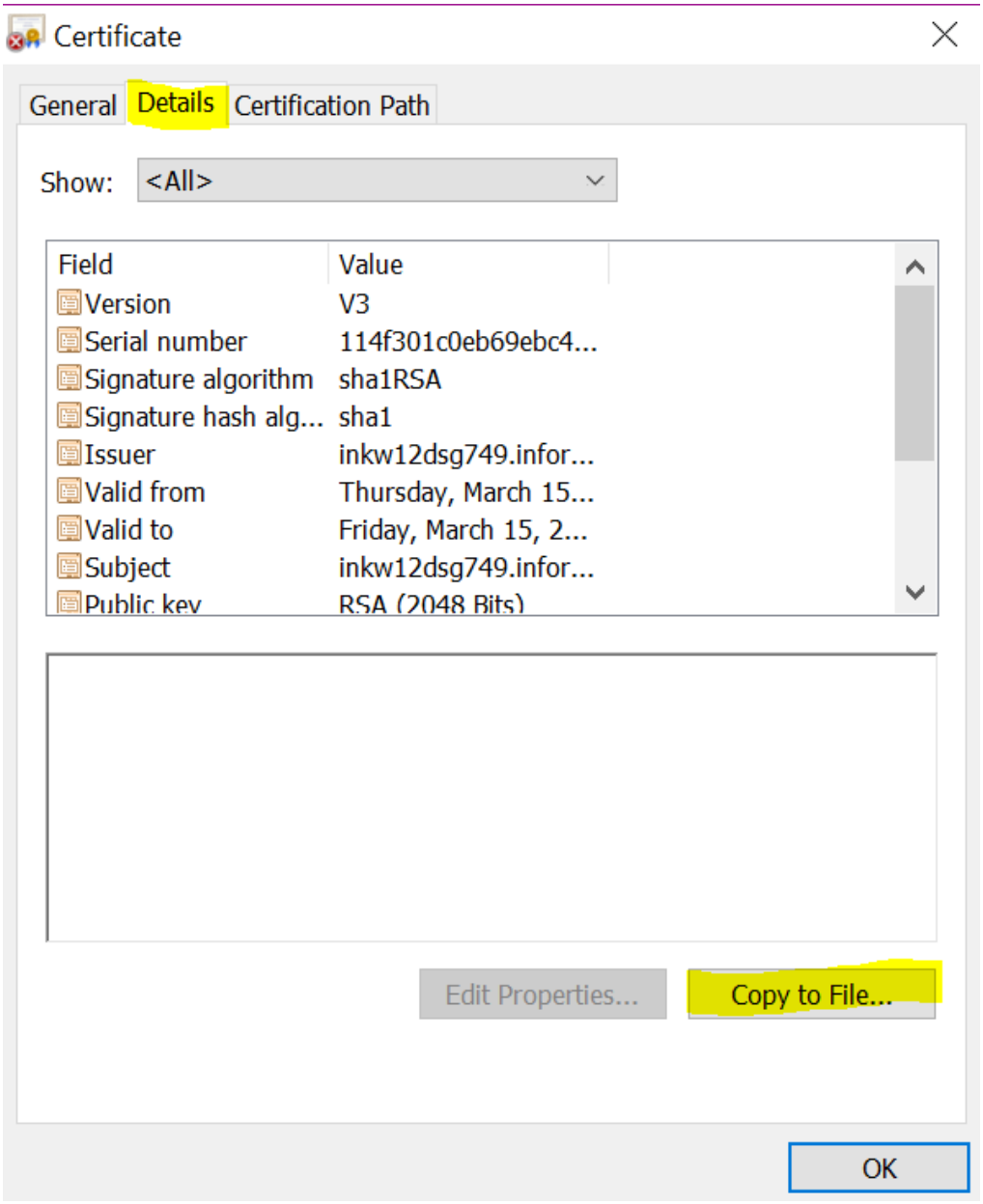
3. Click **Certificate**.



The **Certificate** dialog box appears.

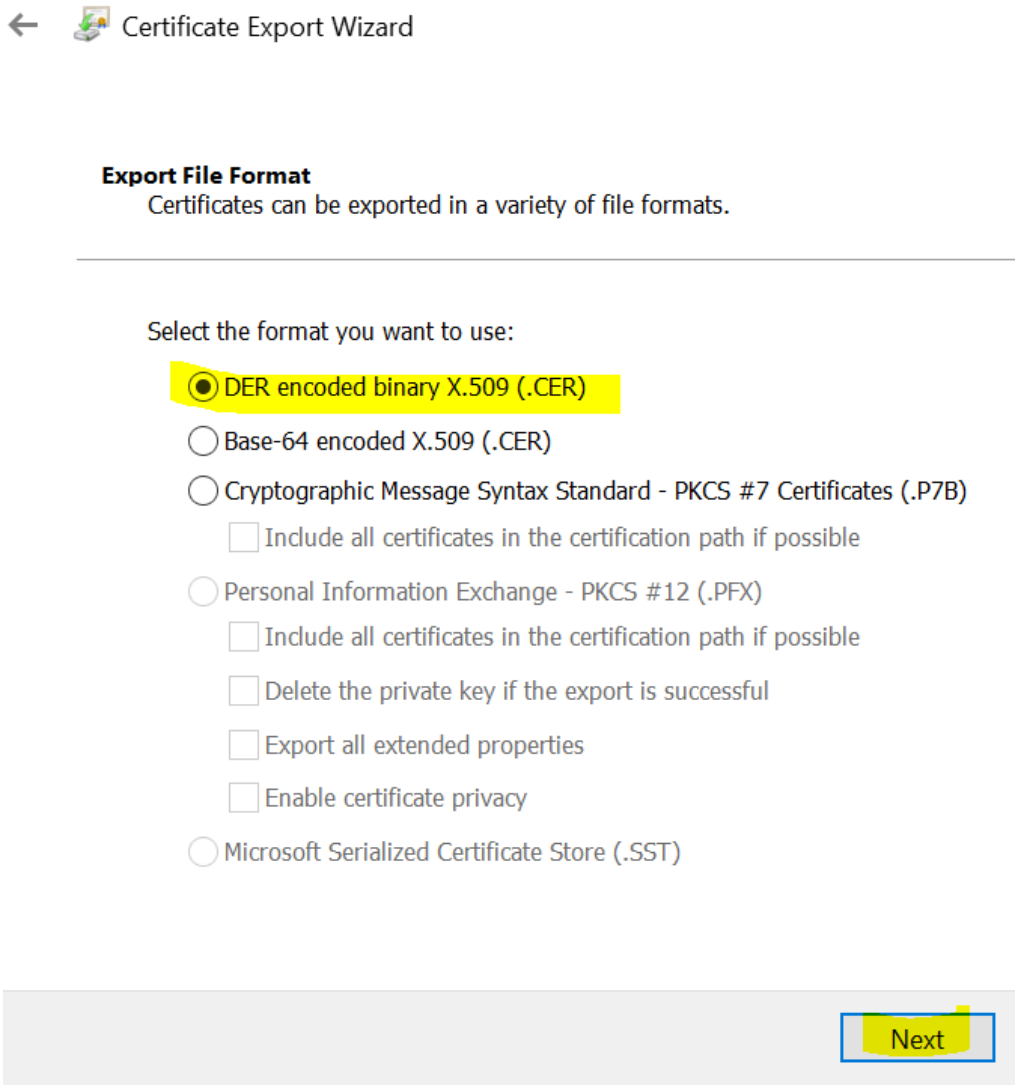
4. Click the **Details** tab.

5. Click **Copy to File**.

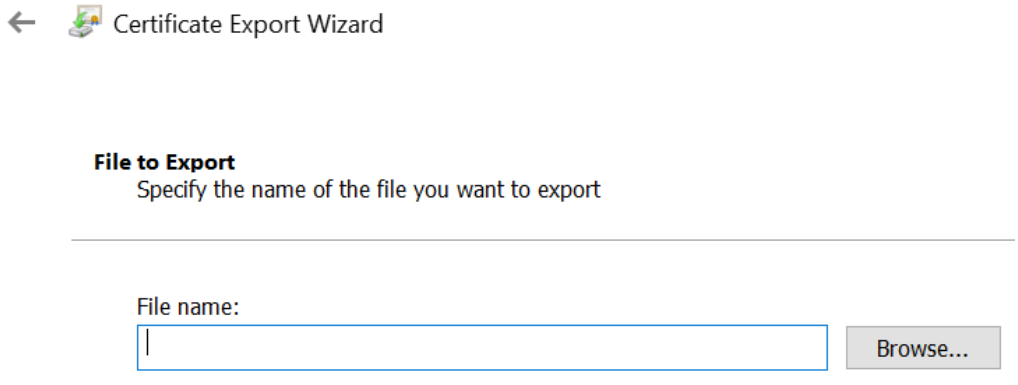


The **Certificate Export Wizard** appears.

6. Select the format for the certificate and click **Next**.



7. Enter a file name and select the folder where you want to save the certificate.



8. Click **Next**.
9. Restart the Informatica domain.

Commands to Import the Certificate

Refer to the following commands to import the certificate:

```
keytool -import -file <cer_path>/sharepoint.cer  -alias sharepoint_cert -keystore
$INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL -
noprompt
```

Step 2. Access the Required SharePoint Account Information

You need the SharePoint URL and tenant or user credentials to create a SharePoint data store.

You must create an Azure Active Directory application with the required permissions assigned. The application must have the Sites.Read.All and Files.Read.All API permissions.

1. To get the SharePoint URL, log in to the Microsoft Azure portal.
2. To access the required information to create a data store with tenant credentials, perform the following steps:
   a. Click **Azure Active Directory**.
   b. Click the required group to open the group page.

      The group must be a Microsoft Office 365 group with SharePoint enabled.
   c. From the **Group Links** section, click Sharepoint.
   d. From the page that opens, copy the first part of the URL.

      For example, if the URL is `https://abcorg1.sharepoint.com/sites/user1`, then the SharePoint URL is `abcorg1.sharepoint.com`.

      The URL might differ based on whether you configure the data store with tenant credentials or a single user details.
   e. Access the client details such as tenant ID, client ID, and client secret from the **Certificate & secrets** tab.

      You require client details to configure a data store to scan multiple user accounts.
3. To create a data store to scan a single user account, you require a user name and password. Ensure that you have the required details.

Step 3. Create a Microsoft SharePoint Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft SharePoint. When you create the data store, specify the category **File Management** and the data store type **Microsoft SharePoint**.

Enter either tenant details or user details depending on whether you create the data store to run service-level scans or to run scans on a single user account.

When you scan the data store, the Data Privacy Management Service creates a File Management Scan job. The Scan job uses the connection properties in the data store to connect to Microsoft SharePoint. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 33

Configuring Connectivity to Microsoft OneDrive

This chapter includes the following topics:

- Configuring Microsoft OneDrive Connectivity Overview, 216
- Access the Required OneDrive Account Information, 217
- Create a Microsoft OneDrive Data Store in Data Privacy Management, 217

Configuring Microsoft OneDrive Connectivity Overview

Before you can scan a OneDrive source you must configure connectivity to the source. You can configure the data store to use either a remote agent or Enterprise Data Catalog to run scans.

You can create a data store to scan either a single user account or to scan multiple user accounts in a tenant. To scan multiple user accounts, the data store must use a remote agent.

If you configure the data store to use a remote agent, ensure that you associate the agent with the data store. When you test the connection, by default Data Privacy Management uses the agent to test the connection.

If no agent is associated with the data store, Data Privacy Management uses Enterprise Data Catalog to test the connection of a single account data store. Data stores that you configure with tenant credentials generate an error if the agent is not associated to the data store.

To configure connectivity, complete the following steps:

1. Retrieve the required account or tenant information.
   The information that you require differs for a single-account data store and a tenant-level data store.
2. Create a Microsoft OneDrive data store on the **Data Stores** workspace in Data Privacy Management.

Access the Required OneDrive Account Information

To create a single-account data store, you must have a user name and the OneDrive URL. To create a tenant-level data store, you need the tenant details and credentials.

1. To access the required information to create a data store with tenant credentials, perform the following steps:
    a. Log in to the Microsoft Azure portal.
    b. Click **Azure Active Directory**.
    c. Click **App registrations** and create and register an application.
    d. Open the application page to access details such as tenant ID and client ID.
    e. Click API permissions to open the Request API permissions page.
    f. Click Application permissions and assign the following permissions: `Users.Read.All, Directory.Read.All, Files.Read.All`
    g. Click **Certificates & secrets** to open the **Certificates & secrets** page.
    h. Click **New client secret** to create a client secret.
2. To create a data store to scan a single user account, you require the specific OneDrive URL, user name, and password. Ensure that you have the required details.

Create a Microsoft OneDrive Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Microsoft OneDrive. When you create the data store, specify the category as **File Management** and the data store type as **Microsoft OneDrive**.

Enter either tenant details or user details depending on whether you create the data store to run tenant-level scans or to run scans on a single user account.

When you scan the data store, the Data Privacy Management Service creates a File Management Scan job. The Scan job uses the connection properties in the data store to connect to Microsoft OneDrive. For more information, see the *Informatica Data Privacy Management User Guide*.

CHAPTER 34

Configuring Connectivity to Google Drive

This chapter includes the following topics:

- Configuring Google Drive Connectivity Overview, 218
- Access the Required Google Drive Account Information, 219
- Create a Google Drive Data Store in Data Privacy Management, 219

Configuring Google Drive Connectivity Overview

Before you can scan a Google Drive source, you must configure connectivity to the source. You can create a data store to scan either a single user account or to scan multiple user accounts at the service level. All Google Drive data stores require a remote agent to run scans.

Ensure that you associate an agent with the data store. When you test the connection, by default, Data Privacy Management uses the agent to test the connection. The data stores generate an error if an agent is not associated with the data store.

To configure connectivity, complete the following steps:

1. Retrieve the required Google Drive user account or service account information.
2. Ensure that you generate and have access to the required credentials.
   The information that you require differs for a single-account data store and a service account-level data store.
3. Create a Google Drive data store on the **Data Stores** workspace in Data Privacy Management.

Access the Required Google Drive Account Information

To create a service account data store, you need the Google Drive service account details and credentials. To create a single user account data store, you must have the login credentials.

1. To access the required information to create a data store for a service account, perform the following steps:
   a. Log in to the Google Cloud console with a valid user account that has administrator privileges and create a service account.
   b. Generate the service account keys.
   c. Open the Service account details page on the Google Cloud console.
   d. Click the Keys tab and download the keys that you generated.

      The client ID, client email, private key and private key ID information is available in the Key file.
   e. Log in to the Google Workspace Administrator console.
   f. Click **Security** > **API Controls** > **Domain-wide Delegation** and enable domain-wide delegation for the service account.

      To access all user accounts in the service account, the service account must have domain-wide delegation enabled.
   g. Add the following drive and directory scopes:
      - DriveScopes.DRIVE
      - DriveScopes.DRIVE_APPDATA
      - DriveScopes.DRIVE_FILE
      - DriveScopes.DRIVE_READONLY
      - DriveScopes.DRIVE_METADATA_READONLY
      - DirectoryScopes.ADMIN_DIRECTORY_USER_READONLY
      - DirectoryScopes.ADMIN_DIRECTORY_GROUP_READONLY
2. To create a data store to scan a single user account, you require the following information for the account:
   - Client ID
   - Client secret
   - Access token and token expiry and refresh information.

Create a Google Drive Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to Google Drive. When you create the data store, specify the category as **File Management** and the data store type as **Google Drive**.

Enter either service account details or user account details depending on whether you create the data store to run service-level scans or to run scans on a single user account.

When you scan the data store, the Data Privacy Management Service creates a File Management Scan job. The Scan job uses the connection properties in the data store to connect to Google Drive. For more information, see the *Informatica Data Privacy Management User Guide*.

Part IX: Configuring Connectivity to NoSQL Sources

This part contains the following chapter:

- Configuring Connectivity to Apache Cassandra, 222

CHAPTER 35

Configuring Connectivity to Apache Cassandra

This chapter includes the following topics:

- Configuring Apache Cassandra Connectivity Overview, 222
- Step 1. Perform Prerequisite Tasks, 222
- Step 2. Create an Apache Cassandra Data Store in Data Privacy Management, 223

Configuring Apache Cassandra Connectivity Overview

Configure connectivity to Apache Cassandra data stores and perform tasks including running a scan, importing Enterprise Data Catalog metadata, importing the protection status of sensitive fields, or associating data domains with the data store.

To configure connectivity, perform the following steps:

1. Perform prerequisite tasks.
2. Create an Apache Cassandra data store on the **Data Stores** workspace in Data Privacy Management.

Step 1. Perform Prerequisite Tasks

Perform the following prerequisite tasks to configure connectivity to an Apache Cassandra data store.

1. Log in to Informatica Administrator.
2. Select the Data Integration Service.
3. On the **Advanced Profiling Properties** tab, verify that the *Maximum Column Heap Size* is set to at least 4 GB. If not set, set the value to 4 GB or 4096 MB.
4. On the **Custom Properties** tab, verify that the *ExecutionContextOptions.JVMMaxMemory* property is added and set to 1024M. If not added, add the property and value.

   ```
   ExecutionContextOptions.JVMMaxMemory=1024M
   ```

Step 2. Create an Apache Cassandra Data Store in Data Privacy Management

Create a data store on the **Data Stores** workspace in Data Privacy Management and configure properties to connect to the Apache Cassandra database.

When you create the data store, specify the category **NoSQL** and the data store type **Apache Cassandra**.

When you scan the data store, the Data Privacy Management Service creates a NoSQL Scan job. The Scan job uses the connection properties in the data store to connect to the database. For more information, see the *Informatica Data Privacy Management User Guide*.

Part X: Configuring Subject Registry

This part contains the following chapter:

- Configuring Subject Registry, 225

CHAPTER 36

Configuring Subject Registry

This chapter includes the following topics:

- Configuring Subject Registry Overview, 225
- Prerequisites, 226
- Configuring Subject Registry for Unstructured Data, 232
- Configuring Subject Registry for Structured and Unstructured Data, 236

Configuring Subject Registry Overview

For purposes of organizational or governmental compliance, you might need to search for records related to a specific individual such as a customer, employee, or vendor. If the search is to comply with a privacy regulation such as GDPR and CCPA, then you must act fast to pull together the data. Use Subject Registry to quickly locate personal information across your structured and unstructured data sources.

Before you can use Subject Registry to locate personal information, you must perform the following prerequisite tasks:

1. Create a data store for each data source you want to include in the Subject Registry search.
2. Note the data store ID of each data store.
3. Run a metadata or data scan on each data store.

If you want Subject Registry to locate personal information in unstructured files, perform the following tasks:

1. Install the remote agent and configure the certificates.
2. Update the Subject Registry configuration file `server.xml`.
3. Create the connection and add the unstructured data stores to the Subject Registry remote agent.

Perform the following tasks to configure Subject Registry:

1. Create the Subject Registry configuration file.
2. Configure an entity file for each subject type that you want to create. You can include multiple golden data stores in an entity file.
   Each entity represents a subject type. A subject registry can include multiple subject types. You choose the subject type that you want to search when you perform a subject registry search.
3. Create an SQL configuration file for each data store. Create a separate SQL file for each entity that uses the data store.
4. Run a Subject Registry scan on all data stores, including golden and transaction data stores.

After you configure Subject Registry, you can select a subject type and then search by one or more identifiers, such as Full Name or Customer ID. The search returns a list of individuals that match the search criteria. You can use other identifiable information to find the exact individual you seek. Then, drill down into the details to see the data stores, data domains, schemas, tables, and fields that contain personal data for the individual.

Prerequisites

Perform the following prerequisite steps for each data source you want to include in the Subject Registry search.

1. Create a data store on the **Data Stores** workspace in Data Privacy Management.
2. Note the data store ID.
3. Run a Domain Discovery scan to map data domains to sensitive data columns.

Step 1. Create a Data Store in Data Privacy Management

On the **Data Stores** workspace in Data Privacy Management, create a data store for golden data sources and each source you want to include in the Subject Registry search.

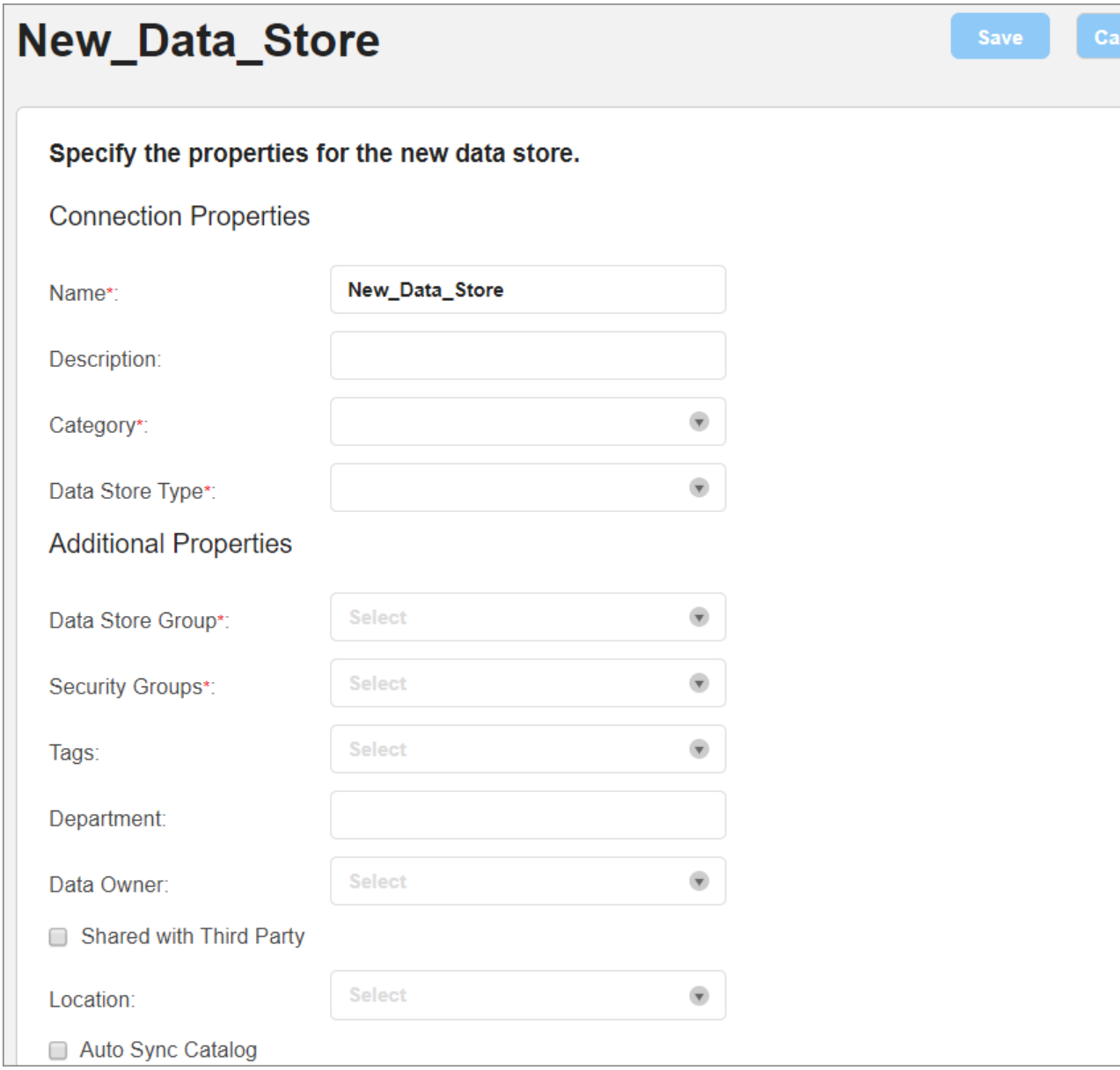
Before you create a data store, verify that you configured connectivity to the source. For example, if you create an Oracle data store, configure the `tnsnames.ora` file on the machine that hosts Enterprise Data Catalog. If you do not configure the connectivity, the connection test and the Database Scan job will fail.

For more information about connecting to the source, see the appropriate "Configuring Connectivity" chapter in this guide.

1. Click **Manage** > **Data Stores**.

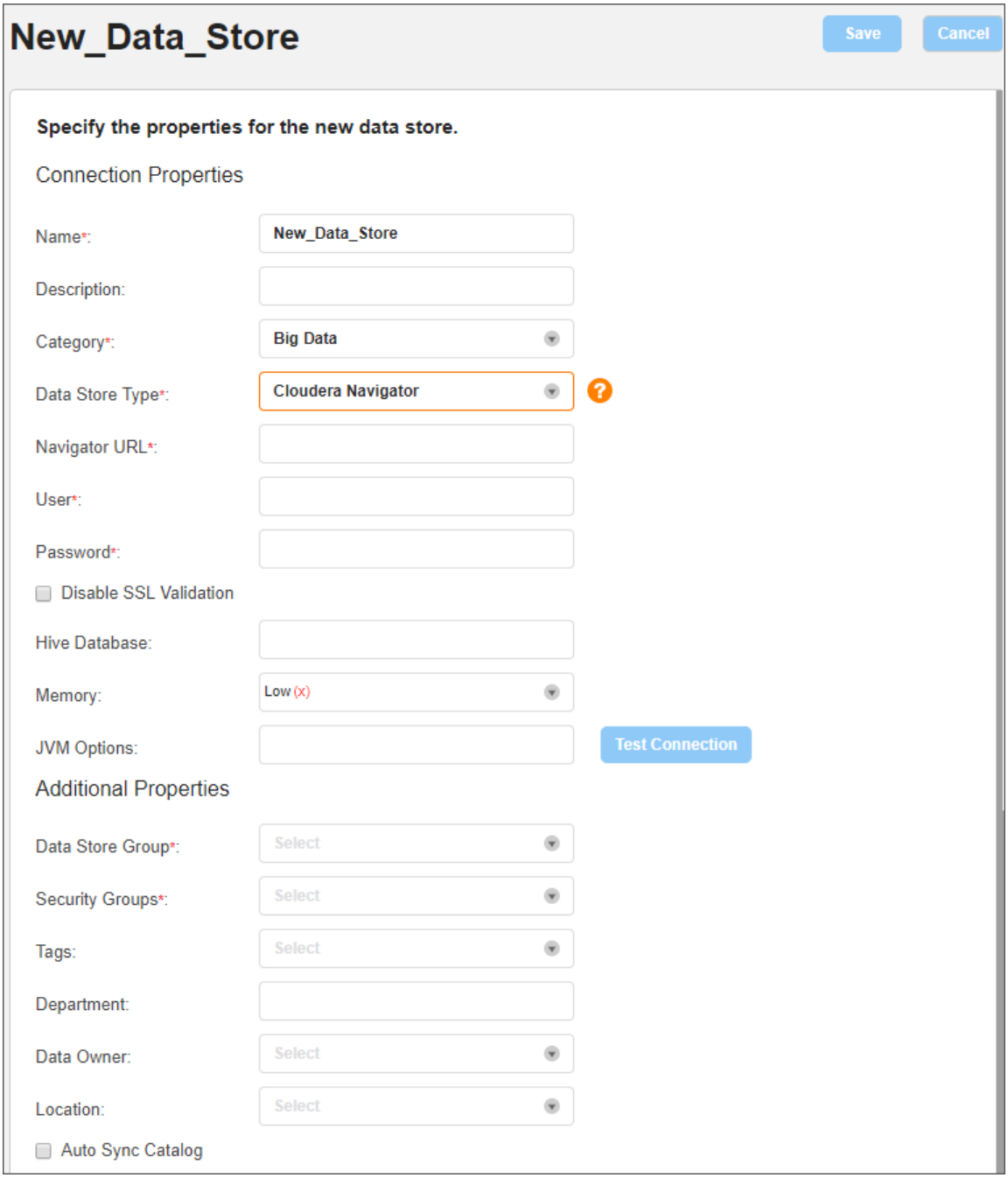
   The **Data Stores** workspace appears.
2. Click **New**.

The **New_Data_Store** page appears.



3. Enter a name for the data store.
4. Select the **Category** and **Data Store Type** for the source.

Connection properties specific to the data store type appear. The following image shows an example of the connection properties that appear if you select **Big Data** in the **Category** field and **Cloudera Navigator** in the **Data Store Type** field:



5. Enter the connection properties.
   For definitions of the connection properties, see the "Data Store Properties" chapter in the *Informatica Data Privacy Management User Guide*.
6. Click **Test Connection** to verify that Data Privacy Management can connect to the data store with the connection details and login.

Data Privacy Management validates that the connection properties are configured, the connection is available, and that the login credentials are valid.

Note: You can save a data store if the connection test fails. For example, you can configure a connection to a database if the database is down for maintenance. Data Privacy Management does not test the connection details when you save the data store.

7. Enter the additional properties.
8. Click **Save**.

   If there are no errors, Data Privacy Management saves the data store in the Data Privacy Management repository. An error can occur for the following reasons:

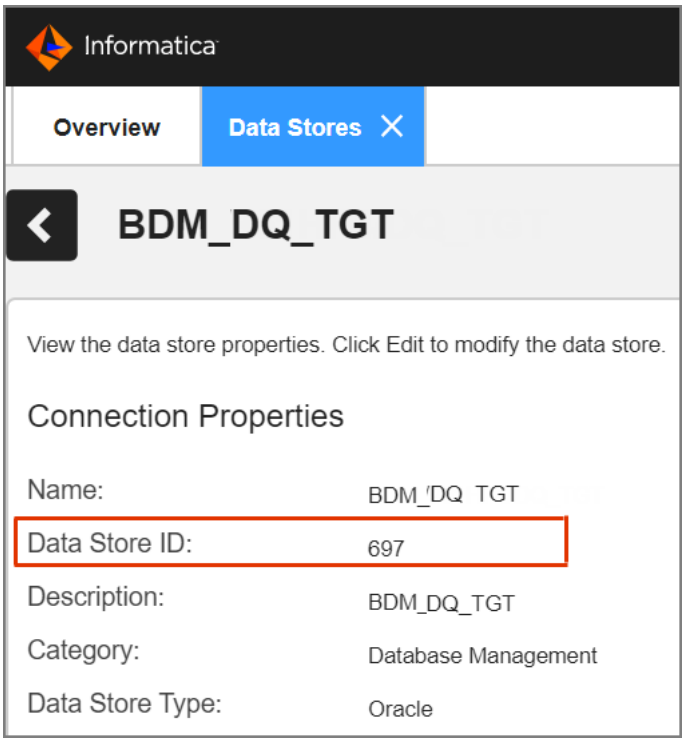
   - The connection properties are incomplete.
   - Another data store has an identical name.
   - Another data store has identical connection details.

Step 2. Note the Data Store ID

Data Privacy Management assigns a data store ID to each data store. Note the ID of the golden data store. You need this ID when you configure the entity file.

1. From the **Data Stores** workspace, click the name of the golden data store.

   The **Data Store Details** page appears.
2. Note the value in the **Data Store ID** field.



Step 3. Scan the Data Stores

Create and run a Domain Discovery scan on each data store that you want to include in the Subject Registry search. You must run scans on golden and transaction data stores.

When you create a scan, you specify the type of scan to run and the data stores to scan. You assign classification policies to the scan and schedule when the scan runs. When you run the scan, Data Privacy Management discovers and classifies sensitive data.

For more information about scans, see the *Informatica Data Privacy Management User Guide*.

1. Click **Manage** > **Scans**.

   The **Scans** workspace appears.

2. Click **New**.

   The **Define & Connect** panel appears.

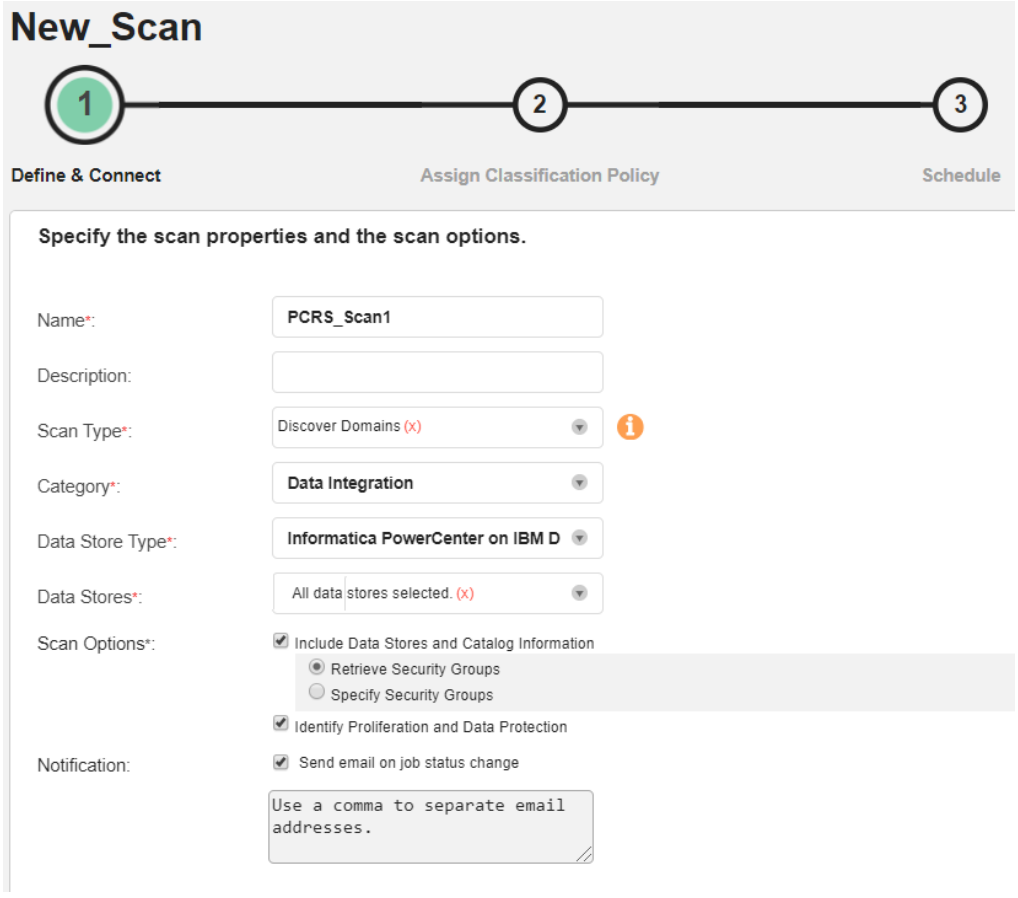
3. Enter a name and description.

4. From the **Scan Type** list, choose **Domain Discovery**.

5. Enter the scan properties.

   After you select the data store type, the scan options appear.

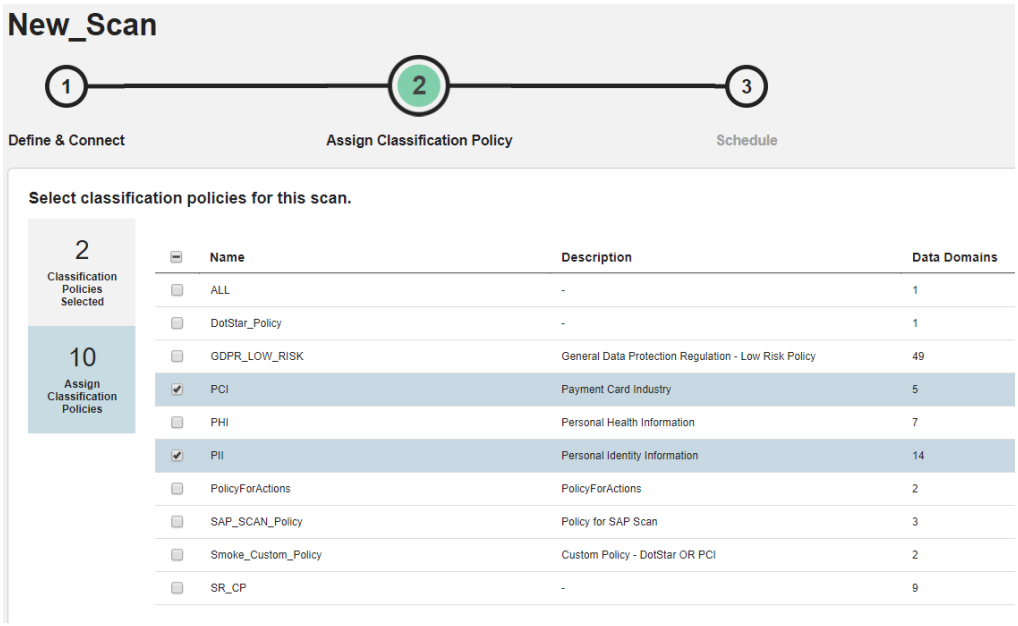
   The following image shows an example of the scan options that appear if you select Informatica PowerCenter on IBM DB2 as the data store type:



6. Enter the scan options.

7. Click **Next**.

The **Assign Classification Policy** panel appears.



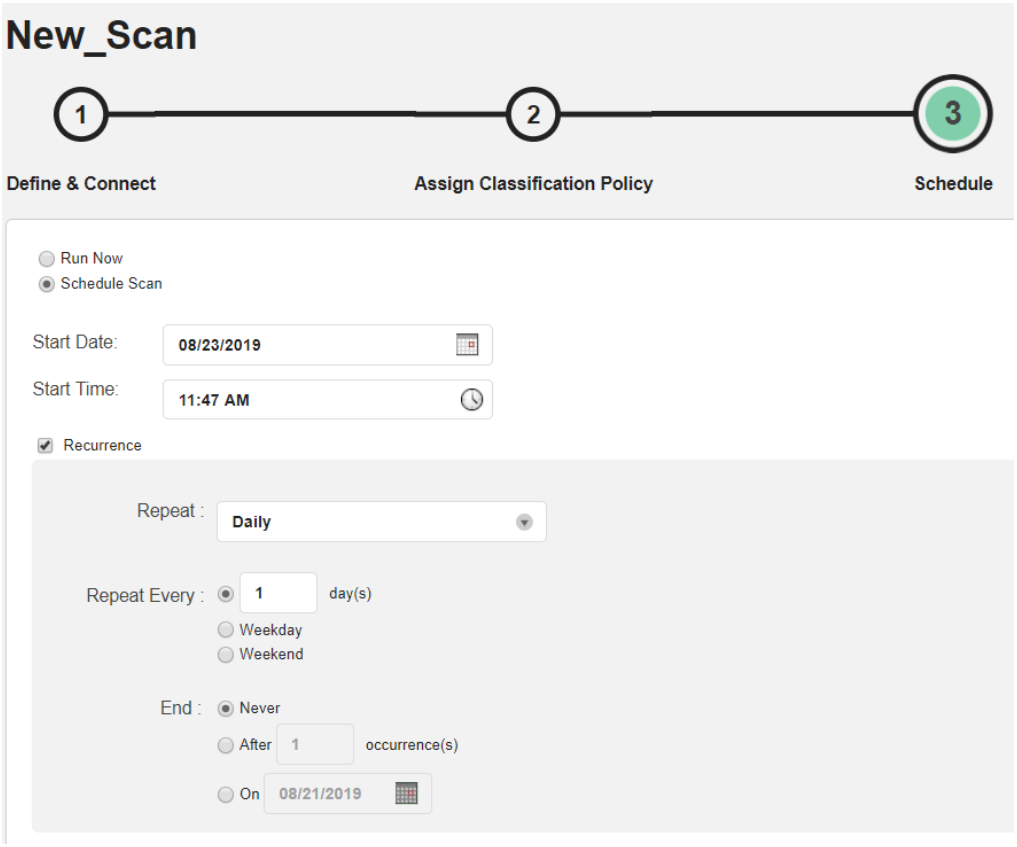
8. Select one or more classification policies to assign to the scan.

   You can use the following classification policy filters to manage the classification policy assignments:

   - **Classification Policies Selected**. Displays a list of the classification policies that are assigned to the scan. Use the filter to view or remove classification policies.
   - **Assign Classification Policies**. Displays a list of all classification policies in the Data Privacy Management repository. Use the filter to add or remove classification policies.

9. Click **Next**.

The **Schedule** panel appears.



10. Schedule when the scan runs.

    You can schedule the job to run immediately, in the future, or on a recurring schedule.

11. Click **Save**.

    The Data Privacy Management Service creates a Scan job. The job runs according to the schedule that you configured. Use the **Jobs** workspace to monitor the status of the job.

Configuring Subject Registry for Unstructured Data

Unstructured data such as emails, social media data, PDF, and Microsoft Word, PowerPoint, and Excel files might make up a large percentage of your source data. If you want to include unstructured data in a Subject Registry search, you must install and configure the Subject Registry remote agent.

Perform the following tasks to include unstructured data in the subject registry search:

1. Install the remote agent and configure the certificates.
2. Update the Subject Registry configuration file `server.xml`.
3. Create the connection and add the unstructured data stores to the Subject Registry remote agent.

Prerequisites

Perform the following tasks before you install the Informatica Discovery Agent:

1. Generate a keystore file for the remote agent.
2. Import the remote agent keystore certificate into the Informatica domain truststore.
3. Import the Data Privacy Management Service certificate into the Informatica Discovery Agent truststore.
4. At startup, the agent checks for all libraries that OCR requires and lists the libraries missing on the machine.
   To include image files in unstructured scans, the following files must be available on the agent machine:

   RHEL 7 or RHEL 8 agent machine

   The latest version of the `libpng15` shared object.

   SUSE Linux 12 SP 3 or SUSE Linux 15 agent machine

   The latest version of the `libpng15` and `libgomp`shared objects.

   Windows agent machine

   Microsoft Visual C++ 2015-2019 Redistributable Packages
   You can download the version from the Microsoft Support site.
5. Restart the agent if you install any missing files.

Install the Informatica Discovery Agent

Extract the agent files to create the agent.

You can install the agent on Windows and Linux machines. On Windows machines, use the `.bat` file to start the agent.

1. Navigate to the following folder: `<Informatica installation directory>/SecureAtSourceService/InformaticaDiscoveryAgent`
2. Extract the contents of the following file: `InformaticaDiscoveryAgent.zip`

   Extract the files to a folder. For example: `RemoteAgent`

Configure the Informatica Discovery Agent

Update the server port and truststore and keystore details in the following server configuration file: `server.xml`.

1. Navigate to the following folder: `<Remote agent>/tomcat/conf`
2. Open the `server.xml` file.
3. Locate the following lines in the text:

   ```
   <Connector port="8080" protocol="HTTP/1.1"
   connectionTimeout="20000"
   redirectPort="8443">
   ```
4. Comment out the lines.

   Add `<!--` before the section and `-->` after the section.

   ```
   <!-- <Connector port="8080" protocol="HTTP/1.1"
   connectionTimeout="20000"
   redirectPort="8443" /> -->
   ```
5. Find a port that is available for the remote agent.

You can use the netstat command to determine port availability. For example, use the following command to determine if port 9000 is available:

- Linux. `netstat -anp | grep 9000`
- Windows. `netstat -aon | find /i "listening" |find "9000"`

6. Add the following configuration information:

```
<Connector port="<port number>" protocol="org.apache.coyote.http11.Http11NioProtocol"
maxThreads="150" SSLEnabled="true" scheme="https" secure="true"
truststoreFile="<remote agent truststore file path>" truststorePass="<truststore
password>"
keystoreFile="<remote agent keystore file path>" keystorePass="<keystore password>"
keyAlias="<alias name of the certificate used for the remote agent>"
clientAuth="true" sslProtocol="TLS" />
```

7. Save the changes to `server.xml` file.

Starting and Stopping the Informatica Discovery Agent

You can start up and shut down the Informatica Discovery Agent. Before you run a scan, the agent must be up and running.

1. You can run the agent on Linux and Windows machines. To start the agent run the following command:

   - Linux.

     ```
     cd <remote agent>/bin
     ./siagent.sh startup
     ```

   - Windows.

     ```
     cd <remote agent>/bin
     siagent.bat startup
     ```

   The following message appears: `Tomcat started.`

2. To stop the agent run the following command:

   - Linux.

     ```
     cd <remote agent>/bin
     ./siagent.sh shutdown
     ```

   - Windows.

     ```
     cd <remote agent>/bin
     siagent.bat shutdown
     ```

   Note: You can view the Informatica Discovery Agent logs at the following location: `<remote agent>/tomcat/logs/catalina.out`

Create the Connection to the Subject Registry Remote Agent

On the **Remote Agents** workspace in Data Privacy Management, enter the connection details and add the unstructured data stores to the Subject Registry remote agent.

1. Log in to Data Privacy Management.
2. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace appears.

3. From the **Actions** menu, select **New** > **Subject Registry Agent**.

   The **New_Remote_Agent** page appears.

4. On the **Specify Properties** pane, enter a name for the remote agent.

   The name cannot exceed 255 characters.

5. Optionally, enter a description.

   The description cannot exceed 255 characters.
6. Enter the host address of the remote agent server.
7. Enter the administrator port number of the remote agent at the specified host address.
8. Optionally, test the connection to the remote agent server.
9. Click the **Next** icon.

   The **Associate Data Stores** pane appears.
10. Select the golden data store and other data stores you want to include in the Subject Registry search.

    To associate more data stores with the remote agent, click the plus sign (**+**).
11. When finished, click **Save**.

Step 4. Configuring the datastore.properties File and the siagent.sh File

Perform the following steps for HDFS and Azure WASB data stores that you want to use in Subject Registry.

1. Open the `datastore.properties` file from the following directory: `<SubjectRegistryAgent_Installation>/bin/`.
2. If you create an HDFS or Azure WASB data store and select the **Load From Configuration Archive File** option for **Cluster Configuration Details**, perform the following steps:
   a. If you install the remote agent on a different machine, copy the archive file to the machine on which you install the agent.
   b. Add the following property and value to the file:

      `<Data store name in Data Privacy Management>.archive.location=<complete path to the archive file>`
      For example: `HDFS_1.archive.location=/user/ilmsats/archives/hdfs.zip`
3. If you set the Kerberos Cluster property to Yes for an HDFS data store, add the following property and value to the file:

   `<Data store name in Data Privacy Management>.keytab.location=<complete path to the keytab file>`
   For example: `HDFS_1.keytab.location=/user/ilmsats/keytabs/hdfs.keytab`

   Verify that the machine has a keytab file and that the file is used for authentication.
4. Save and close the `datastore.properties` file.

   The agent application reads the values in the file during the connection phase.
5. If you set the Kerberos Cluster property to **Yes** for an HDFS data store, and the `krb5.conf` file is not present in the default `/etc/krb5.conf` location on the remote agent machine, perform the following steps:
   a. Shut down the remote agent.
   b. Open the following file: `<SubjectRegistryAgent_Installation>/bin/siagent.sh`
   c. Add the following JVM parameter:`INFA_JAVA_OPTS="-Djava.security.krb5.conf=<path to the Krb5.conf file>"`

      For example: `INFA_JAVA_OPTS="-Djava.security.krb5.conf=/home/ilmsats/Sun/krb5.conf"`

d. Add the following environment variable in the bash_profile: `export KRB5_CONFIG=/home/<path to the Krb5.conf file>`

e. Open a new session and start the remote agent.

6. If you create an SSL-enabled or a KMS-enabled HDFS source, perform the following tasks:

a. Import the certificates required to connect to the source into the truststore or keystore present in the Subject Registry Agent certificates folder.

b. Add the following properties in the `<SubjectRegistryAgent_Installation>/bin/siagent.sh` file:

```
INFA_JAVA_OPTS="-Xmx1024m ${INFA_JAVA_OPTS} ${_DEBUG} -XX:MaxMetaspaceSize=256m
-Djavax.net.ssl.keyStore=<SRAgent certificate folder>/SRAgentKeyStore.jks
-Djavax.net.ssl.keyStorePassword=<keystore password>
-Djavax.net.ssl.trustStore=<SRAgent certificate folder>/agentTrustStore.jks
-Djavax.net.ssl.trustStorePassword=<truststore password>"
```

Add the properties before the first `export INFA_JAVA_OPTS` property.

Configuring Subject Registry for Structured and Unstructured Data

Create a configuration file and an entity file to specify the search fields and the result fields on the **Subject Registry** workspace in Data Privacy Management. You must also create an SQL file and then run a Subject Registry scan on each data store.

Perform the following tasks:

1. Create the configuration file for Subject Registry.
2. Configure entity files for each subject type you want to create. Entity files list the golden data stores and configure search and result fields for a subject type.
3. Create an SQL configuration file for each structured data store.
4. Create an SOQL query file for Salesforce data stores.
5. Run a Subject Registry scan on all golden and transaction data stores.

Data Privacy Management validates the entities when a user loads the **Subject Registry** workspace or selects the required Subject Type from the list on the **Scans** workspace.

Step 1. Create the Subject Registry Configuration File

You must create a configuration file to specify the Name configurations for Subject Registry. The entries in the configuration file apply to all entities in Subject Registry.

The file contains a single section for Name3 configuration. All properties are required.

Configuration File Properties

The configuration file contains generic information about Name configuration for the Subject Registry. It applies to all entities that you create for the Subject Registry.

The following table lists the Name3Configuration properties:

Property	Description
SSAControls	Location of the Name3 population ysp file. Example: `SSAPR=/root/Informatica/10.5/SecureAtSourceService/name3/data`
PopulationPath	Enter the following value: `default`
Population	Type of population. Enter the following value: `usa`

Note: You can ignore the other fields in the Name3Configuration section of the configuration file.

The following text shows an example of an updated configuration file:

```
{
"Name3Configuration": {
"SSAControls": "SSAPR=/root/Informatica/10.4.0/SecureAtSourceService/name3/data",
"PopulationPath": "default",
"Population": "usa",
"KeySize": "8",
"KeyType": "HEX"
    }
}
```

Creating the Configuration File

1. Navigate to the following folder: `<Informatica installation directory>/SecureAtSourceService/config/samples`.
2. Locate the `sr_config.json` file.

   This is a sample file that you can use to create the Subject Registry configuration file.
3. Open the file in edit mode.
4. Update the properties.
5. Save the file to the `<Informatica installation directory>/SecureAtSourceService/config` location.

Step 2. Configure the Entity Files

You must configure entity files to specify the golden data stores that you want to use for each subject type.

You can create multiple entity files for a Subject Registry. Each entity file contains information for a subject type.
Configure the match rules that Subject Registry must use to identify matches and collect subject records. Configure fields that you want to display on the **Search** and **Results** panes of the **Subject Registry** workspace for the subject type.

You can use the properties to differentiate between fields that you want to include in DSAR reports or Subject Registry search pages, or both.

You can configure subject scans to be incremental on repeat or recurrent scans instead of complete. Configure the required field in the entity file and then choose the appropriate option on the scan page when you create the scan.

You can specify the columns that you want to use to determine residency.

Entity File Properties

Specify the fields you want to display on the Search and Results pane of the Subject Registry workspace in an entity file. Configure residency fields if you want to view residency information in the Subject Registry.

The entity file contains the following properties:

Header

Contains parameters related to the file name and golden-source data stores.

Field Names

Defines the column names and widths for the fixed-width input data. The column names that you define must match the names of the data domains that you give as aliases in the `<data store ID>_<Entity_Name>.sql` file. These are the columns that correspond to the data domains that you want to include in the Subject Registry search.

When you configure the column parameters, you provide the column name, the column length, and the display code. The display code determines how and where the data domain appears in the Subject Registry search results. Provide one of the following values for the display code in each column parameter:

- 0 - The column/corresponding data domain will not appear in either page of the Subject Registry user interface.
- 1 - The column/corresponding data domain will appear only on the initial search results page of the Subject Registry user interface.
- 2 - The column/corresponding data domain will appear only on the **Subject Details** page of the Subject Registry user interface.
- 3 - The column/corresponding data domain will appear on both the initial search results page and the **Subject Details** page of the Subject Registry user interface.

For example, if you want the subject Name to appear on both the initial search results page and the **Subject Details** page:

```
{"FieldName": "FullName",
"FieldLength"="30",
"DisplayCode"="03"}
```

When you configure the field names, you must also include an additional non-column parameter specific to the Subject Registry use case. This is the "COLUMN name="SOURCE" parameter, which you configure with the data store source, the length of the source, and the display code. For example:

```
{
"FieldName": "SOURCE",
"FieldLength": "40",
"DisplayCode": "00",
"IsPrimaryKey": "false"
}
```

Since the SOURCE parameter is not a data domain but is used internally by Data Privacy Management, set the display code to 0.

The length must be greater than the maximum length of the names of golden data stores included in the entity.

MatchConfigurations

Contains the SSA-NAME3 index parameters, such as the column name that you want to use to index the records. The values that you give for the <fieldname> under IndexConfiguration parameter appear as mandatory search fields in the Subject Registry user interface.

IndexConfiguration

Contains the properties that specify the fields that must be indexed in the Subject Registry tables.

SearchConfiguration

Contains the SSA-NAME3 search parameters, such as the field name based on which you want to search records.

MatchRules

Contains the SSA-NAME3 match parameters, such as the SSA-NAME3 purpose that you want to use to match the records.

The parameters that you include in each section and the values that you give will depend on unique factors. For example, the parameters might depend on the details of your Hadoop distribution, how you want the Subject Registry search to behave, and which data domains you want to include in the search.

When you finish configuring the file, save it in the following location: `$INFA_HOME/SecureAtSourceService/config`

Note: You must save the file as `entity_<Entity_Name>.json`, where `<Entity_Name>` is a name that you choose to represent the subject of the entity. For example, if you create an entity to search customer records in the Subject Registry, you might name the file `entity_Customer.json`.

The following text shows an example of a configured entity file:

```
{
"Version": "1",
"EntityName": <ENTITY_NAME>,
"GoldenDatastores": [
    {
      "name": "SR_ORCL_GOLDENDATA_HR"
    },
    {
      "name": "SR_ORCL_GOLDENDATA_CC"
    },
    {
      "name": "SR_ORCL_GOLDENDATA_INSURANCE"
    }
  ],
"SearchResultLimit":"50",
"EntityFields": [{
"FieldName": "ID",
"FieldLength": "10",
"DisplayCode": "00",
"IsPrimaryKey": "true"
        },
        {
"FieldName": "FullName",
"FieldLength": "30",
"IsSubjectName": "true",
"DisplayCode": "03",
        },
        {
"FieldName": "CompanyName",
"FieldLength": "30",
"DisplayCode": "03",
        },
        {
"FieldName": "Address",
"FieldLength": "30",
"DisplayCode": "00"

        },
        {
"FieldName": "City",
"FieldLength": "20",
"DisplayCode": "03",
"Residency": "true"
```

```
        },
        {
"FieldName": "PinCode",
"FieldLength": "10",
"DisplayCode": "02",
"Residency": "true"
        },
        {
"FieldName": "PhoneNumber",
"FieldLength": "20",
"IsMultiValued": "true",
"DisplayCode": "02"

        },
        {
"FieldName": "Salary",
"FieldLength": "10",
"DisplayCode": "02"
        },
        {
"FieldName": "DOJ",
"FieldLength": "10",
"DisplayCode": "00"
        },
        {
"FieldName": "Age",
"FieldLength": "10",
"DisplayCode": "01"
        },
        {
"FieldName": "SSN_General",
"FieldLength": "20",
"DisplayCode": "01"

        },
        {
"FieldName": "SOURCE",
"FieldLength": "20",
"DisplayCode": "00"
        }
    ],
"MatchConfigurations": [{
"MatchConfigurationName": "MatchConfiguration1",
"IndexConfiguration": {
"FieldName": "FullName",
"IndexType": "Fuzzy",
"IndexPurpose": "Person_Name",
"IndexLevel": "Standard"
        },
"SearchConfiguration": {
"FieldName": "FullName",
"SearchType": "Fuzzy",
"SearchPurpose": "Person_Name",
"SearchLevel": "Extreme"
        },
"MatchRules": [{
"MatchRuleName": "MatchConfig1",
"MatchPurpose": "Person_Name",
"MatchLevel": "Typical",
"MatchThreshold": "60",
"MatchRulePurposeCode": "3",
"MatchFields": [{
"FieldPurpose": "Person_Name",
"FieldName": "FullName",
"MatchType": "Fuzzy"
                },
                {
"FieldPurpose": "Address_Part2",
"FieldName": "City",
"MatchType": "Fuzzy"
                },
```

```
                {
"FieldPurpose": "Postal_Area",
"FieldName": "PinCode",
"MatchType": "Exact"
                }
            ]
        }]
    }]
```

Header Properties

Configure the following properties in the header section of the entity file:

EntityName

Name of entity. The entity name you enter here must match the entity file name. For example, if **EntityName** is `Employee`, then the entity file name must be `entity_Employee.json`.

GoldenDataStores

Names of the golden-source data stores. If you enter more than one data store, the first data store that you enter is considered the primary data store. During a subject registry scan, by default the scan uses the non-primary data stores for additional information or for information that is missing from the primary data store. You can set the `IsMultiValued` field attribute to change the default behavior for specific fields.

Example:

```
[
    {
      "name": "SR_ORCL_GOLDENDATA_HR"
    },
    {
      "name": "SR_ORCL_GOLDENDATA_CC"
    },
    {
      "name": "SR_ORCL_GOLDENDATA_INSURANCE"
    }
  ]
```

SearchResultLimit

Determines the maximum number of results to fetch and display.

IgnoreNullEmptyValues

Determines whether to ignore or reject a null or empty field value when you configure an index field for Exact type matching. If set to true, the scan ignores empty and null value fields during Link Subject scans.

Default is true.

Field Names

In the Field Names section of the entity file, configure the following set of properties for each column:

FieldName

Name of the column.

Example: `ID`

FieldLength

Length of the field. If you specify a length that is less than the length of the actual data, the data is truncated.

To ensure that a field appears in DSAR reports but is not included in scans and searches, set the field length to 0.

Note: If you set the field length to 0 to exclude the field from searches, ensure that you set the Display Code to 00 so that the empty field does not appear on the subject registry pages

Do not set the value to 0 for a field that you include in a match configuration, or for a field that you configure to be multivalued, to be the subject name, or to determine residency. Warning messages appear on the UI when you perform a search.

DisplayCode

Specifies if the field appears on the UI. You can specify that the field appear on the **Subject Registry Search** page, **Subject Registry Details** page, both pages, or does not appear on the UI.

The following table lists the DisplayCode values you can use:

DisplayCode Values	Display Condition
00	Do not display.
01	Display on the **Subject Registry Search** page.
02	Display on the **Subject Registry Details** page.
03	Display on both the **Subject Registry Search** and **Subject Registry Details** pages.

IsSubjectName

Optional field. Enter `true` or `false`. You can add this attribute for a single column.

The value of the column that you set to true appears at the upper left corner of the subject details page. For example, if you set the value of a `FirstName` column to true, the first name appears. If you set the value of a `FullName` column to true, the full name appears at the upper left corner of the subject details page. If you do not add the property to a column, the value defaults to false.

IsPrimaryKey

Specifies if the column is a primary key. Enter `true` or `false`.

You can mark one column as a primary key.

IsMultiValued

Optional field. Enter `true` or `false`.

If set to true, during a golden data store scan, the scan appends multiple field values for a single record instead of considering only the record in the primary golden data store. If a field contains values from multiple data stores, more records get linked during a subject registry scan to link subjects. A Link Subject scan associates data subjects in transaction data stores with subjects identified in golden data stores. During a subject registry search, the multiple field values appear separated by a pipe `"|"` symbol.

If you do not add the property to a column, the value defaults to false. If false, the scan considers values in non primary golden data stores only if the value is null in the primary golden data store. Multiple field values for the same subject attribute might be ignored during a subject registry search.

Residency

Links the column to Residency criteria. Enter `true` or `false`.

Optional field. If you do not add the property to a column, the value defaults to false. The values in columns that you set to true appear in the Residency field on the Subject Registry page. You can set multiple columns to true. The values appear separated by commas in the Residency field. The order of fields in the entity file determines the order of appearance in the Residency field.

For example, the following order of properties in the entity file results in `<City, State>` format in the Residency field:

```
{"FieldName": "FullName",
         "FieldLength": "30",
         "DisplayCode": "03"
         },

        {"FieldName": "City",
         "FieldLength": "20",
         "DisplayCode": "03",
         "Residency": "true"},

        {"FieldName": "State",
         "FieldLength": "10",
         "DisplayCode": "03",
         "Residency": "true"}
```

The following order of properties in the entity file results in `<State, City>` format in the Residency field:

```
{"FieldName": "FullName",
         "FieldLength": "30",
         "DisplayCode": "03"
         },

        {"FieldName": "State",
         "FieldLength": "10",
         "DisplayCode": "03",
         "Residency": "true"},

        {"FieldName": "City",
         "FieldLength": "20",
         "DisplayCode": "03",
         "Residency": "true"}
```

IncrementalScanField

Required to configure repeat scans of structured data stores as incremental scans. Enter `true` or `false`. Set this property to true for a field that contains a timestamp value. For example, LastModifiedTime. The scan uses the value in this field to determine when the data was last updated, and scans for changes post the date and time mentioned. The field must use the following format: `yyyy-MM-dd HH24:mm:ss` For example:

```
{
    "FieldName": "LastUpdatedTime",
    "FieldLength": "00",
    "DisplayCode": "00",
    "IsPrimaryKey": "false",
    "IncrementalScanField":"true"
    }
```

Read the following guidelines before you configure the flag:

- If you configure a subject scan for incremental scans and do not set the field to true for a valid field, the scan job fails.
- If you set the property to true for a field that does not contain a timestamp value in the `yyyy-MM-dd HH24:mm:ss` format, the scan job fails.

- If you set the value to true for more than one field, the scan uses the first field listed in the entity file. A warning message appears on the Subject Registry page and the scan creation page to indicate that the entity has multiple fields configured with the flag. If the first field is not the required format, the scan job fails.
- To ensure that the field is not added to the golden record table, set the FieldLength value to 00.
- To ensure that the field does not appear on search details pages, set the DisplayCode to 00.
- To ensure that the field is not included in DSAR reports, ensure that you do not have a data domain with the same name in Data Privacy Management. Use an alias in the SQL file if you have a data domain by the same name.

In addition to entering the set of properties for each column, you must enter a set of properties for the Source column. Specify the name and length. Specify 00 for the DisplayCode field.

Example:

```
"FieldName": "SOURCE",
      "FieldLength": "20",
      "DisplayCode": "00",
                        "IsPrimaryKey": "false"
```

Note: You cannot add the `Residency` property to the Source column.

MatchConfigurations

The properties in the MatchConfigurations section are organized in the following categories:

MatchConfigurationName

Name of the match rule.

Example: `MatchConfiguration1`

UsedIn

Indicates whether the match configuration is used in subject scans, searches, or both. Use one of the following values:

- Search
- Scan
- Scan, Search
- Search, Scan

If you do not enter a value, by default the match configuration is used in both scans and searches. If an entity file includes only one match configuration, and it is configured for only scan or only search, a warning message appears on the UI when you perform a search or configure a scan.

IndexConfiguration

Contains the properties that specify the fields that must be indexed in the Subject Registry tables.

The following table lists the IndexConfiguration properties:

Property	Description
FieldName	The name of the field to use for indexing. The field must be listed in the Field Names section. Example: `FullName`
IndexType	Type of matching. Use one of the following values: - If you want matching to be exact, enter: `Exact` - If you want matching to be fuzzy, enter: `Fuzzy`
IndexPurpose	Purpose from Name3 to be associated with indexing. Use one of following values: - Address_Part1 - Person_Name - Organization_Name - Generic_Field - Code - Telephone_Number - Date - CreditCard - VIN - ISBN10 - ISBN13 - Geocode - Company_Name
IndexLevel	Level of indexing by Name3. Use one of the following values: - Standard - Extended - Limited

SearchConfiguration

Specifies the fields that can be used to perform a Subject Registry search.

The following table lists the SearchConfiguration properties:

Property	Description
FieldName	The name of the field to use for searching. The field must be listed in the Field Names section. **Note:** The field names that you include must be included in the IndexConfguration field names. Example: `FullName`
SearchType	Type of matching. You must use one of the following values: - If you want matching to be exact, enter: `Exact` - If you want matching to be fuzzy, enter: `Fuzzy`

Property	Description
SearchPurpose	Purpose from Name3 to be associated with searching. Use one of the following values: - Address_Part1 - Person_Name - Organization_Name - Generic_Field - Code - Telephone_Number - Date - CreditCard - VIN - ISBN10 - ISBN13 - Geocode - Company_Name
SearchLevel	Level of search by Name3. Use one of the following values: - Typical - Exhaustive - Narrow - Extreme

MatchRules

Specifies how matching must be performed.

The following table lists the MatchRules properties:

Property	Description
MatchRuleName	Name of match configuration. Example: `MatchConfig1`
MatchPurpose	Purpose from Name3 to be associated with matching. Example: `Person_Name`
MatchLevel	Level of matching by Name3. Use one of the following values: - Typical - Conservative - Loose

Property	Description
MatchThreshold	Degree of similarity to which matching must be performed. Informatica recommends you set to 80.

MatchFields	Depends on the MatchPurpose that you use. If the MatchPurpose has mandatory match fields, a Subject Registry scan fails if you do not use the mandatory fields.
	Includes the following set of properties for each field:
	- `FieldPurpose`: Purpose from Name3 to be associated with a match field.
	Use one of the following values: Address_Part1, Address_Part2, Attribute1, Attribute2, Code,Company_Name,CreditCard, Date, Geocode, Id, ISBN10, ISBN13,Organization_Name,Person_Name,Postal_Area,Telephone_Number, VIN.
	- `FieldName`: The name of the field to use for matching. The field must be listed in the Field Names section.
	- `MatchType`: Type of matching. Enter either `Exact` or `Fuzzy`.
	Example:
	`"MatchFieldType": "Person_Name", (Purpose from Name3 to be associated with this match field *)`
	`"MatchFieldName": "FullName", (FieldName to be used for matching.)`
	`"MatchType": "Exact" (Same as IndexType in IndexConfiguration)`
	`},`
	`{`
	`"MatchFieldType": "Address_Part2",`
	`"MatchFieldName": "City",`
	`"MatchType": "Exact"`
	`},`
	`{`
	`"MatchFieldType": "Postal_Area",`
	`"MatchFieldName": "PinCode",`
	`"MatchType": "Exact"`

If you use the `Exact` match type configuration, note the following behavior changes with the Informatica 10.4.1.1 service pack:

- Exact field matching does not consider case.
- For unstructured sources, exact matching considers case.
- Exact comparison considers leading and trailing spaces. For example, an Exact comparison does not consider "John" and " John " to be a match.

If you used Exact matching for the *IndexType* and *SearchType* fields in version 10.4.1 and applied the Informatica 10.4.1.1 service pack, perform one of the following tasks:

- To retain the version 10.4.1 behavior, update the *Indextype* and *SearchType* fields in the entity file to `Fuzzy`.
- To implement the updated behavior, run the scan again without changes to the scan definition or entity file.

Creating the Entity File

1. Navigate to the following folder: `<INFA_HOME>/SecureAtSourceService/config/samples`
2. Find the `entity_[EntityName].json` file.
3. Rename and save the file to: `<INFA_HOME>/SecureAtSourceService/config/entity_<entity name>.json`

   The entity name that you use appears as the subject type on the Subject Registry page.

4. Open the file in edit mode.
5. Update the properties.
6. Save the file.

Step 3. Create the SQL File

You must create SQL files for each golden data store and each transaction data store that you use with an entity in Subject Registry. The SQL file contains the query that Subject Registry uses to query the data store with subject records.

The query specifies the columns associated with the data domains that you want to include in the search. If you use a data store with multiple entities either for subject discovery or for subject linking, create an SQL file for each entity.

Note: You cannot use the same data domain for multiple columns in an SQL query. Map each column in an SQL file to a different data domain.

For a Snowflake data store, provide the fully qualified name of the table: `<database name>.<schema name>.<table name>`

For example: `select ID, (FIRSTNAME || ' ' ||LASTNAME) as FullName,COMPANY CompanyName,ADDRESS Address,CITY City,PINCODE PinCode,PHONE PhoneNumber,SALARY Salary,DOJ DOJ,AGE Age,SSn SSN_General FROM DPM.TEST.CUSTOMER;`

Where `DPM` is the database name, `TEST` is the schema and `CUSTOMER` is the name of the table.

For a DB2 for z/OS data store, you cannot use the CONCAT operator to concatenate strings. Use the built-in CONCAT function to concatenate strings.

For example, the following query uses the built-in CONCAT function: `SELECT LASTNAME CONCAT(CONCAT(LASTNAME,','),FIRSTNAME) FROM EMP;`

Perform the following steps for each data store:

1. Create an SQL file with the following file name: `<data store ID>_<EntityName>.sql`.

   For example: `104_Customer.sql`

2. Enter a SELECT statement that returns data from the columns that you want to include in the subject registry search. For example:

```
SELECT ID, (FIRSTNAME || ' ' || LASTNAME) AS FullName, COMPANY CompanyName, ADDRESS
Address, CITY City, PINCODE PinCode, PHONE PhoneNumber, SALARY Salary, DOJ DOJ, AGE
Age, SSN SSN_General, FROM CUSTOMER;
```

```
select ('PROF_TABLE_101' ||'_'||ID) as ID,NAME as FullName,CITY as City,STREET as
Street,PHONE as PhoneNumber,SSN as SSN_General from PROF_TABLE_101 where rownum<=100
UNION ALL select ('PROF_TABLE_102' ||'_'||ID) as ID,NAME as FullName,CITY as
City,STREET as Street,PHONE as PhoneNumber,SSN as SSN_General from PROF_TABLE_102
where rownum<=100 UNION ALL select ('PROF_TABLE_103' ||'_'||ID) as ID,NAME as
FullName,CITY as City,STREET as Street,PHONE as PhoneNumber,SSN as SSN_General from
PROF_TABLE_103 where rownum<=100 UNION ALL select ('PROF_TABLE_104' ||'_'||ID) as
ID,NAME as FullName,CITY as City,STREET as Street,PHONE as PhoneNumber,SSN as
SSN_General from PROF_TABLE_104 where rownum<=100 UNION ALL select ('PROF_TABLE_105'
||'_'||ID) as ID,NAME as FullName,CITY as City,STREET as Street,PHONE as
PhoneNumber,SSN as SSN_General from PROF_TABLE_105 where rownum<=100 UNION ALL
select ('PROF_TABLE_106' ||'_'||ID) as ID,NAME as FullName,CITY as City,STREET as
Street,PHONE as PhoneNumber,SSN as SSN_General from PROF_TABLE_106 where rownum<=100
UNION ALL select ('PROF_TABLE_107' ||'_'||ID) as ID,NAME as FullName,CITY as
City,STREET as Street,PHONE as PhoneNumber,SSN as SSN_General from PROF_TABLE_107
where rownum<=100 UNION ALL select ('PROF_TABLE_108' ||'_'||ID) as ID,NAME as
FullName,CITY as City,STREET as Street,PHONE as PhoneNumber,SSN as SSN_General from
PROF_TABLE_108 where rownum<=100
```

Note: To configure incremental scans, include a field with timestamp value in the `yyyy-MM-dd HH24:mm:ss` format.

3. Save the file in the following location: `<INFA_HOME>/DPMService/config`

Create an SOQL Query File for a Salesforce Data Store

Use Salesforce Object Query Language (SOQL) to create the query for a Salesforce data store. Create a JSON file with the query. Subject Registry uses the query during a search for subject records. The query specifies the columns associated with the data domains that you want to include in the search.

All Salesforce objects contain an **ID** field with the ID field type. The field contains a unique identifier for each record in the object, similar to a primary column in relational databases. You must map the **ID** field of an object to the `field_domain` marked as a primary key.

You can use simple SOQL queries. You cannot use JOIN and WHERE clauses.

Note: You cannot use the same data domain for multiple columns in a query. Map each column in an SOQL file to a different data domain.

1. Navigate to the following folder: `<INFA_HOME>/DPMService/config/samples/salesforce_soql`
2. Save the `[RepoId]_[EntityName].json` file with the following file name: `<data store ID>_<EntityName>.json` Save the file in the following directory: `<INFA_HOME>/DPMService/config`

   You can also create a file that contains the same properties.
3. Enter a SELECT statement that returns data from the fields that you want to include in Subject Registry. For example:

```
select Id,SR_FULLNAME__c,SR_AGE__c,
SR_CITY__c,SR_DATE_OF_BIRTH__c ,SR_DESIGNATION__c ,SR_EMAILADDRESS__c,SR_GENDER__c,SR
_MARITALSTATUS__c,SR_STATE__c, SR_COUNTRY__c from SFDS_TRANSACTION__c
```

4. Use the `domain_field_mappings` property to map each field to a data domain. For example:

```
{
            "field_name": "SFDS_TRANSACTION__c.Id",
            "field_domain": "EMPLOYEE_ID"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_FULLNAME__c",
            "field_domain": "SR_FULLNAME"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_AGE__c",
            "field_domain": "SR_AGE"
        },
```

   Note: Salesforce object and field names are case sensitive. Enter the field names in the same case as in the database.

5. Save the file in the following location: `<INFA_HOME>/DPMService/config`

   The following code is an example of a Salesforce JSON file:

```
{
    "query_type": "SOQL",
    "search_query": "select Id,SR_FULLNAME__c,SR_AGE__c,
SR_CITY__c,SR_DATE_OF_BIRTH__c ,SR_DESIGNATION__c ,SR_EMAILADDRESS__c,SR_GENDER__c,SR
_STATE__c, SR_COUNTRY__c from SFDS_TRANSACTION__c",
    "domain_field_mappings": [
        {
            "field_name": "SFDS_TRANSACTION__c.Id",
            "field_domain": "EMPLOYEE_ID"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_FULLNAME__c",
```

```
            "field_domain": "SR_FULLNAME"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_AGE__c",
            "field_domain": "SR_AGE"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_CITY__c",
            "field_domain": "SR_CITY"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_DATE_OF_BIRTH__c",
            "field_domain": "SR_DATE_OF_BIRTH"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_DESIGNATION__c",
            "field_domain": "SR_DESIGNATION"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_EMAILADDRESS__c",
            "field_domain": "SR_EMAILADDRESS"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_GENDER__c",
            "field_domain": "SR_GENDER"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_STATE__c",
            "field_domain": "SR_STATE"
        },
        {
            "field_name": "SFDS_TRANSACTION__c.SR_COUNTRY__c",
            "field_domain": "SR_COUNTRY"
        }
    ]
}
```


Step 4. Run a Subject Registry Scan

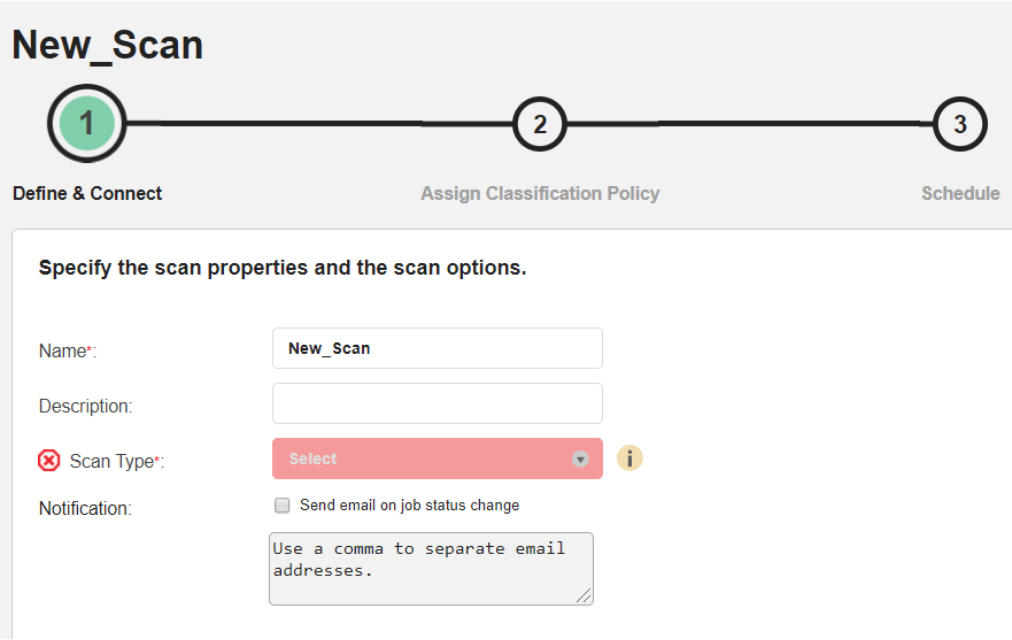
Run a Subject Registry scan to identify subjects in golden data stores and link subjects in transaction data stores with the golden data store subjects.

You can first run the scan to identify subjects in golden data stores, and then run the scan to link subjects in transaction data stores. You can choose to run both scans in a single Subject Registry scan.

1. Log in to Data Privacy Management.
2. Click **Manage** > **Scans**.

   The **Scans** workspace appears.
3. Click **New**.

The **Define & Connect** panel appears. The following image shows an example of the **New Scan** page:



4. Enter the scan name and description.
5. Select the required scan type.

   The following table lists the scan types with descriptions:

Option	Description
Domain Discovery	Maps columns to data domains. Include this type if you have not already run Domain Discovery scans on all data stores, or if you want to update the scan results.
Discover Subjects	Scans golden data stores to identify data subjects.
Link Subjects	Scans transaction data stores to link subjects with data subjects discovered in golden data stores. **Note:** You cannot run a scan to link subjects in transaction data stores before you discover subjects in golden data stores. If you have not already run a Discover Subjects scan to identify subjects, include a Discover Subjects scan with the Link Subjects scan.

6. Choose the subject type.

   Select an entity file that includes specific golden data stores. A Discover Subjects scan type scans the golden data stores listed in the entity file. A Link Subjects scan type links subjects in the transaction data stores with the subjects discovered in the golden data stores listed in the entity file.
7. Choose to configure repeat scans as incremental or full. Default is full.

   Note: Before you configure an incremental scan, read the rules and guidelines for incremental Subject Registry scans in the Informatica Data Privacy Management User Guide.
8. Select the category, data store type, and the data stores to include in the scan. Not applicable for scans that include only the Discover Subject scan type. If the scan includes the Link Subjects and the Discover Subjects scan types, choose data stores that you want to scan for subject linking from the list of data stores. The Subject Discovery scan uses data stores mentioned in the Subject Type.

After you select the data store type, the scan options appear. Enter the required scan options and click **Next**.

9. If applicable, assign classification policies.

   You can assign classification policies if the scan includes domain discovery.

10. Click **Next** to schedule or save and run the scan.

    The Data Privacy Management Service creates a Scan job. The job runs according to the schedule that you configured. On the **Jobs** workspace, you can monitor the status of the Subject Registry master job and child jobs.

Part XI: Configuring Encryption and Decryption

This part contains the following chapters:

- Encryption and Decryption Configuration, 254
- Encryption and Decryption Management , 269

CHAPTER 37

Encryption and Decryption Configuration

This chapter includes the following topics:

- Encryption and Decryption Configuration Overview, 254
- Configuration Steps to Enable Encryption and Decryption, 256
- Step 1. Install the Protection Remote Agent, 257
- Step 2. Configure Connectivity to the Hive Data Source, 259
- Step 3. Set the Data Privacy Management Service and Hadoop Cluster Properties, 260
- Step 4. Assign the Data Privacy Management Privileges to Encrypt and Decrypt Data, 263
- Step 5. Deploy the iCP Component, 264
- Step 6. Configure the Cloudera Hive User-Defined Functions, 265
- Data Privacy Management Tasks, 268

Encryption and Decryption Configuration Overview

To encrypt and decrypt data in Data Privacy Management, you must install the encryption and decryption components, configure the Informatica services, assign privileges, and create and configure the Hive data store.

The Data Privacy Management installation directory includes the following components:

- The installation files for the remote protection agent.
- The JAR and binary files for the Informatica Crypto Platform (iCP).

Encryption Components

The following components enable encryption:

SoftHSM

A utility that generates and stores the cryptographic key. The SoftHSM utility also performs encryption and decryption operations.

You access the SoftHSM utility through a PKCS #11 interface.

iCP daemon

A program that securely exchanges the cryptographic key between the remote protection agent and the Hive server. The iCP daemon includes the iCP binary files.

To install the SoftHSM utility and the iCP daemon, extract the files from the iCP bundle and deploy the files on the Hive server.

Encryption Process

The following steps occur during the encryption process:

1. From the Data Privacy Management user interface, the user configures a protection task. The protection task specifies the sensitive fields for encryption and the encryption rules. The protection task also specifies a cryptographic key.
2. The Data Privacy Management Service applies the encryption rules and encryption key to the specified sensitive fields.
3. The user runs the Protection job.
4. The Data Privacy Management Service encrypts the specified sensitive fields according to the encryption rules specified in the protection task.

Decryption Components

The following components facilitate decryption:

Remote Protection Agent

An agent that parses the query from the client and rewrites the query.

Hive User-Defined Functions (UDF)

Custom-defined functions that interact with the iCP libraries and decrypt the data in the database when the query is processed.

SoftHSM

A utility that generates and stores the cryptographic key. The SoftHSM utility also performs encryption and decryption operations.

You access the SoftHSM utility through a PKCS #11 interface.

iCP daemon

A program that securely exchanges the cryptographic key between the remote protection agent and the Hive server.

The iCP daemon includes the iCP binary files.

To install the SoftHSM utility and the iCP daemon, extract the files from the iCP bundle and deploy the files on the Hive server.

Decryption Process

The following steps occur during the decryption process:

1. From the Data Privacy Management user interface, the user creates a decryption security policy. The decryption security policy defines the conditions that must apply for the Data Privacy Management Service to decrypt and display the data in clear text.
2. The client sends a query to access data.

3. The remote protection agent performs the following tasks:
   - Intercepts the query before it reaches the Hive server.
   - Evaluates the context of the query.
   - If the query requests access to encrypted fields, the agent rewrites the query based on the decryption security policy defined in Data Privacy Management.
   - Sends the query to the Hive server.
   - Uses ZooKeeper to exchange the cryptographic key in secure mode with the iCP daemon on the Hive server.
4. The Hive UDFs interact with the iCP daemon to fetch the cryptographic key and decrypts data using the iCP core libraries.
5. The Hive source machine sends the decrypted data to the client.

Configuration Steps to Enable Encryption and Decryption

To enable users to encrypt and decrypt data, perform the following steps:

Step	Task	Location
1	Install the remote protection agent.	On any machine However, Informatica recommends that you install the agent on each machine that hosts a Hive data source.
2	Configure connectivity to the Hive data source.	Informatica Administrator tool
3	Set the Data Privacy Management Service and Hadoop cluster properties.	Informatica Administrator tool
4	Assign Data Privacy Management privileges to encrypt and dynamically decrypt data.	Informatica Administrator tool
5	Deploy the iCP components.	On each machine that hosts a Hive data source
6	Configure the Cloudera Hive UDF.	Cloudera Manager

After you complete the configuration tasks, create the Hive data store on the **Data Stores** workspace in Data Privacy Management. Then, encrypt and decrypt data.

For information on how to encrypt and decrypt data in Data Privacy Management, see the *Informatica Data Privacy Management User Guide*.

Step 1. Install the Protection Remote Agent

You can install the protection remote agent on any machine. However, Informatica recommends that you install the agent on each machine that hosts a Hive data source.

Perform the Prerequisite Tasks

Generate the keystore and export and import the certificate for the remote protection agent.

1. Generate a keystore.

   To generate the keystore, run the following command:

   ```
   keytool -genkey -alias <any-alias-name> -keyalg RSA -keysize 4096 -sigalg
   SHA256withRSA -dname CN=<Fully qualified hostname-of-protection-agent-machine> -
   keystore <keystoreName>.jks -storetype JKS -validity 7300 -ext SAN=DNS:<hostname-of-
   protection-agent-machine>,IP:<IP-address-of-protection-agent-machine> -ext
   EKU=serverAuth,clientAuth -ext BC=ca:true -storepass <storepass> -keypass <keypass>
   ```

   Example:

   ```
   keytool -genkey -alias abc123 -keyalg RSA -keysize 4096 -sigalg SHA256withRSA -dname
   CN=abc123.mycompany.com -keystore xyz.jks -storetype jks -validity

   abc123,IP: 10.10.100.100-ext EKU=serverAuth,clientAuth -ext BC=ca:true -storepass
   hadoop -keypass hadoop
   ```

2. Export the remote protection agent certificate.

   To export the certificate, run the following command:

   ```
   keytool -exportcert -alias <alias-of -keystore> -keystore <keystore file name>. jks -
   storetype JKS -file <certificate file name>.cer
   ```

   Example:

   ```
   keytool -exportcert -keystore abc123.jks -storepass hadoop -file abc123.cer -alias
   abc123
   ```

3. Import the certificate to the truststore on the machine that hosts Data Privacy Management.

   Perform the following steps:

   a. Copy the certificate file to the machine that hosts Data Privacy Management.

   b. Import the certificate to the Data Privacy Management truststore directory located in: `$INFA_HOME/services/shared/security/`

      To import the certificate, run the following command:

      ```
      keytool -importcert -keystore <keyStoreFile> -storepass <StorePassword> -file
      <cert-file> -alias <alias-name>
      ```

      Example:

      ```
      keytool -importcert -keystore infa_truststore.jks -storepass pass2038@infaSSL -
      file abc123.cer -alias abc123.informatica.com
      ```

4. Export the Data Privacy Management certificate from the Data Privacy Management keystore directory `$INFA_HOME/tomcat/conf/Default.keystore` and import it to the truststore directory of the remote protection agent.

   Perform the following steps:

   a. Copy the certificate file to the machine that hosts the remote protection agent.

   b. Import the certificate to the remote protection agent truststore.

      ```
      keytool -importcert -keystore <keyStoreFile> -storepass <StorePassword> -file
      <cert-file> -alias <alias-name>
      ```

Example:

```
keytool -importcert -keystore protectagent_truststore.jks -storepass hadoop -
file abc123.cer -alias abc123.mycompany.com
```

Note: You can use the same truststore located at `$INFA_HOME/services/shared/security/infa_truststore.jks` as the truststore for the remote protection agent. The certificate from the Data Privacy Management keystore is already imported to this truststore.

5. Locate the `ProtectionAgentInstaller.bin` file in the `$INFA_HOME/SecureAtSourceService/ProtectionAgent/ProtectionAgentInstaller.bin` directory.
6. Provide execute permission for the `ProtectionAgentInstaller.bin` file.

   Use the following command:

   ```
   chmod +x ProtectionAgentInstaller.bin
   ```
7. Run `ProtectionAgentInstaller.bin` file.
8. Press **Enter**.
9. Provide the absolute path of the remote protection agent installation directory or accept the default.
   For example: `/root/Informatica/ProtectionAgent598`
10. Provide the Data Privacy Management agentUri in the folllowing format: `hostname:port`
    For example: `abc123.mycompany.com:18295`
11. Provide the following keystore details:
    - The keystore path that you created.
    - The keystore type as `pkcs12` or `jks`.
    - The keystore alias.
    - The keystore password.
12. Provide the following truststore details:
    - The truststore path.
      You can copy the file from the Data Privacy Management machine at `$INFA_HOME/services/shared/security/infa_truststore.jks`
    - The truststore password.
13. Provide the license key.
14. Specify whether you want to enable Kerberos for the remote protection agent.
    - Enter `Y` if you want to enable Kerberos. See the section "Enable Kerberos for the Remote Protection Agent" on page 258.
    - Enter `N` if you do not want to enable Kerberos. The **Preinstallation Summary** panel appears. See the section "Install and Configure the Remote Protection Agent" on page 259.

Enable Kerberos for the Remote Protection Agent

If you want to enable Kerberos for the remote protection agent, perform the following steps:

1. Copy the `krb5.conf` and keytab files from the source Hive cluster to the machine that hosts the remote protection agent.
2. Provide the following kerberos details:
   - Provide the path to the `krb5.conf` file.
   - Provide the path to the keytab file.
   - Provide the Service Principal name.

The **Preinstallation Summary** panel appears. For next steps, see the section "Install and Configure the Remote Protection Agent" on page 259.

Install and Configure the Remote Protection Agent

1. Review the **Preinstallation Summary** panel of the installer and continue with the installation.

   The installer installs the remote protection agent. The **Post-Installation Summary** panel appears.
2. Note the port number that appears on the **Post-Installation Summary** panel.

   You need the port number to configure the remote protection agent in Data Privacy Management.
3. Stop the remote protection agent server.

   To stop the server, use the following command:

   ```
   ./server stop
   ```
4. Locate the following JAR files:
   If you are using Cloudera v5.15, you can find the JAR files in the following directory on the Hive source cluster: `/opt/cloudera/parcels/CDH-5.15.0-1.cdh5.15.0.p0.21/lib/hive/lib`

   - `hadoop-auth-2.6.0-cdh5.15.0.jar`
   - `hadoop-common-2.6.0-cdh5.15.1.jar`
   - `hive-jdbc-1.1.0-cdh5.15.1-standalone.jar`
5. Copy the JAR files to the following directory: `<protection agent installation directory>/lib/ext`
6. Create a file with the file name `jvm.params`.

   Add the following content in the file:

   ```
   -Djavax.net.ssl.keyStore=<KeyStore-File-path-on-protection-agent-machine> -
   Djavax.net.ssl.keyStorePassword=<keystore password> -Djavax.net.ssl.trustStore=
   <TrustStore-File-path-on-protection-agent-machine> -
   Djavax.net.ssl.trustStorePassword=<Truststore-password>
   ```

   Example:

   ```
   -Djavax.net.ssl.keyStore=/home/ora11g/Informatica/ProtectionAgent598/cfg/
   ProtectionAgentkeystore.jks -Djavax.net.ssl.keyStorePassword=changeit -
   Djavax.net.ssl.trustStore=/home/ora11g/Informatica/ProtectionAgent598/cfg/
   ProtectionAgenttruststore.jks -Djavax.net.ssl.trustStorePassword=pass2038@infaSSL
   ```
7. Save the `jvm.params` file to the remote protection agent installation directory.
8. Start the remote protection agent server.

   To start the server, use the following command:

   ```
   ./server start
   ```

Step 2. Configure Connectivity to the Hive Data Source

- See the *Configuring Connectivity to Hive Sources* chapter in this guide.

Step 3. Set the Data Privacy Management Service and Hadoop Cluster Properties

1. Log in to the Administrator tool.
2. Set the *customZooKeeperUri* Data Privacy Management property.

   For more information, see the section "Set the customZooKeeperUri Property" on page 260.
3. Set the *Cluster Environment Variables* Hadoop cluster property.

   For more information, see the section "Sett the Hadoop Cluster Environment Variables Connection Property" on page 261.

Set the customZooKeeperUri Property

Data Privacy Management communicates with the remote protection agent in an asynchronous fashion by writing to a designated node on the ZooKeeper server. You must set the *customZooKeeperUri* property to the URI of your custom ZooKeeper server. The remote protection agent listens to the changes on its designated node on the ZooKeeper server and retrieves information from Data Privacy Management.

Set the *customZooKeeperUri* property if one of the following conditions apply:

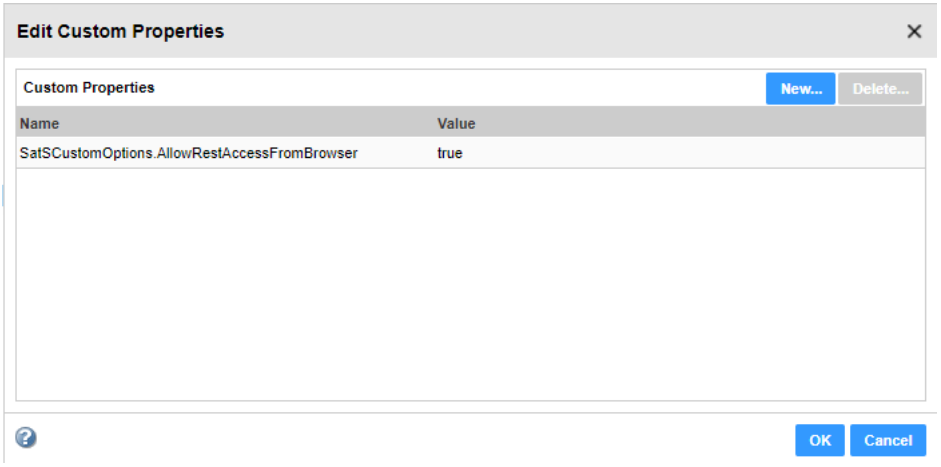
- You installed Data Privacy Management.
- You installed Data Privacy Management and the Catalog server uses an external Hadoop distribution. For this installation option, you must set the *customZooKeeperUri* property to the external pre-installed instance of ZooKeeper.

Note: If you installed Data Privacy Management and the Catalog server uses an embedded Hadoop distribution, do not set the *customZooKeeperUri* property.

You do not need to set the property because the Data Privacy Management Service uses the ZooKeeper server for the Catalog Service.

1. Click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. On the **Domain Navigator** pane, click the Data Privacy Management Service.
4. On the **Properties** pane, scroll to the **Custom Properties** section.
5. Click the **Edit** icon for the **Custom Properties** section.

The **Edit Custom Properties** window appears.

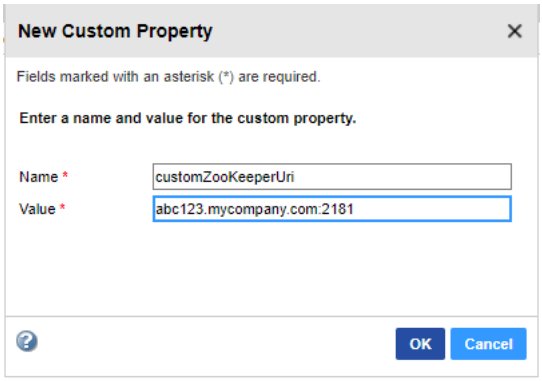


6. Click **New**.

   The **New Custom Property** window appears.

7. In the **Name** field, add the following name for the property: `customZooKeeperUri`
8. In the **Value** field, add the custom ZooKeeper URI in the following format: *<host name>:<port>*

   For example: `abc123.mycompany.com:2181`
9. Click **OK**.



   The **New Custom Property** window closes.

10. Click **OK** to close the **Edit Custom Properties** window.
11. Recycle the Data Privacy Management Service to apply the new property.

    To recycle, perform the following steps:

    a. Verify that the Data Privacy Management property is selected on the **Domain Navigator** pane.

    b. On the **Properties** pane, click **Actions** > **Recycle**.

Sett the Hadoop Cluster Environment Variables Connection Property

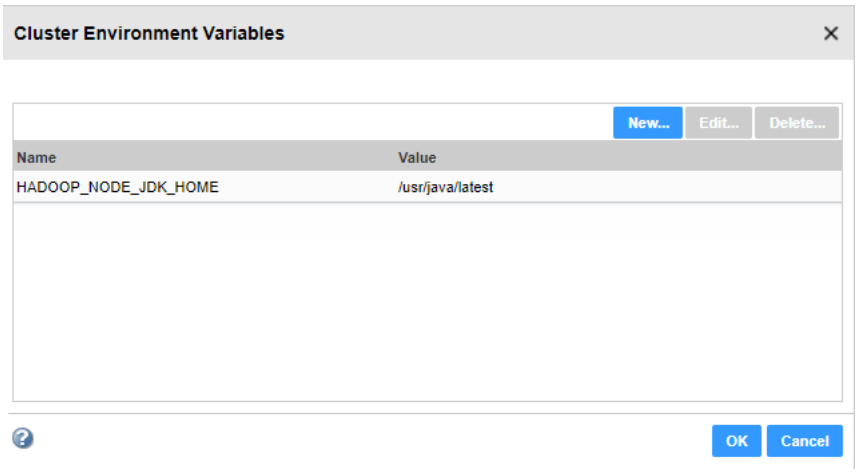
The **Cluster Environment Variables** property is a Hadoop cluster connection property. To encrypt data, you must set the property to the path of the `softhsm2.conf` file.

1. On the Hive data source machine, create a directory with the name `tokens`.

2. Copy the `softhsm2.conf` file from `$INFA_HOME/SecureAtSourceService/softhsm/cfg/` to a directory on the Hive source machine that has full permissions for the Hive impersonation user.
3. Edit the `softhsm2.conf` file.
   a. Open the `softhsm2.conf` file.
   b. Locate the **directories.tokendir** property.
   c. Enter the path of the `tokens` directory that you created in step 1 as the value for the **directories.tokendir** property.
   d. Save the `softhsm2.conf` file.
4. From the Administrator tool, click the **Manage** tab.
5. Click the **Connections** view.
6. On the **Domain Navigator** pane, click the Hadoop connection.
7. On the **Properties** pane, click the **Edit** icon for the **Hadoop Cluster Properties** section.

   The **Edit Hadoop Cluster Properties** window appears.
8. Locate the **Cluster Environment Variables** property and click the **Edit** icon for the property.

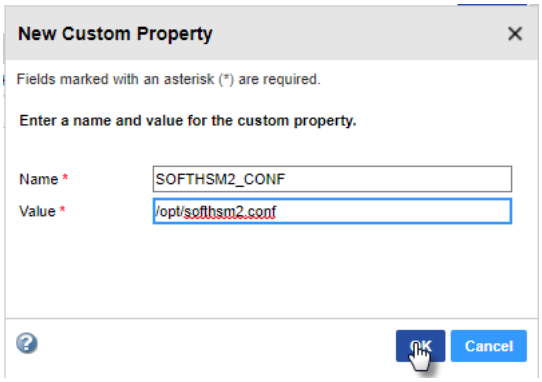
   The **Cluster Environment Variables** window appears.



9. Click **New**.

   The **New Custom Property** window appears.
10. In the **Name** field, add the name of the SoftHSM utility: `SOFTHSM2_CONF`
11. In the **Value** field, add the full path to the `softhsm2.conf` file that you copied in step 2. For example: `/opt/softhsm2.conf`

12. Click **OK**.



The **New Custom Property** window closes.

13. Click **OK** to close the **Cluster Environment Variables** window.
14. Click **OK** to close the **Edit Hadoop Cluster Properties** window.

Step 4. Assign the Data Privacy Management Privileges to Encrypt and Decrypt Data

▶ Assign the following privileges to users who perform the tasks to enable encryption and decryption:

Privilege	User Action
Add or Modify Encryption Rules	Create and edit encryption rules on the **Encryption Rules** workspace. - Create and edit encryption rules to specify the encryption technique. - Use an encryption rule to encrypt sensitive fields.
Add or Modify Proxies	Create and edit the connection to a remote agent on the **Remote Agents** workspace. - Create and edit connection properties for a subject registry remote agent and specify the unstructured data stores. - Create and edit connection properties for a protection remote agent and specify the Cloudera Hive data stores.
Add or Modify Task	- Create and edit custom, DSAR, email, protection, service management, and system log tasks for data domains, data stores, proliferations, classification policies, anomalies, and security policies. - Assign a task to another user.
Close Task	Close a custom, DSAR, email, protection, service management, or system log task.
Delete Encryption Rules	Delete an encryption rule.
Delete Proxies	Delete the connection to a subject registry or protection remote agent.
Delete Task	Delete a custom, DSAR, email, protection, service management, or system log task.

Privilege	User Action
Execute Task	Schedule a protection job for a task, or run a custom, email, service management, or system log task.
Manage Extensions	Add, edit, and delete custom, email, protection, service management, and system log extensions.
Manage Protection Task Configurations	- Mark a configuration as complete. - Change the protection extension for a data store. - Assign protection rules to sensitive columns. - Configure protection rules after assignment.
Risk Simulation and Manage Plans	Manage plans and indicators for simulating sensitive data risk.
View Encryption Rules	View the data type and encryption technique in an encryption rule.
View Extensions	View custom, email, protection, service management, and system log extensions.
View Proxies	View the connection details and the associated data stores for a subject registry or protection remote agent.
View Subject Index	Search for a data subject and view search results on the **Subject Registry** workspace.
View Task	- View tasks. - Add notes to tasks. - Upload attachments to tasks. - Reassign or reject a task. - View the Data Store Detail report. - View subject request information on the Privacy Dashboard.

Step 5. Deploy the iCP Component

The iCP component includes the iCP daemon. The iCP daemon is a standalone process that runs on each Hive node. When the client sends a query, the protection agent intercepts and rewrites the query. The iCP daemon then uses a cryptographic key to securely transfer the query between the protection remote agent and the machine with the Hive data source.

Extract and configure the iCP daemon files on the machine that hosts the Hive data source.

1. Navigate to the following directory on the machine that hosts Data Privacy Management: `$INFA_HOME/DPMService/hive/`
2. Copy the `DecryptionUtilities.zip` compressed file.
3. Place the copy of the `DecryptionUtilities.zip` file in a directory on the machine that hosts the Hive source.
4. Extract the contents of the `DecryptionUtilities.zip` compressed file.

   The compressed file contains the following directories:

   - `daemon`

- `lib`
- `tokens`

5. Open the `lib` directory and perform the following tasks:
   a. Open the `softhsm2.conf` file.
   b. Locate the *directories.tokendir* property.
   c. Enter the path to the `tokens` directory that you extracted in step 4.

      Note: The Hive source machine contains two directories with the name `tokens`. Ensure that you enter the path of the `tokens` folder that you extracted in step 4 and not the `tokens` folder you created to set the Hadoop **Cluster Environment Variables** property.
   d. Save and close the `softhsm2.conf` file.
   e. Open the `pkcs11.cfg` file.
   f. Locate the *library* property.
   g. Enter the path to the `libsofthsm2.so` file in the `lib` directory.
   h. Save and close the `pkcs11.cfg` file.
6. Open the `daemon` directory and perform the following tasks:
   a. Open the `icp-daemon.sh` file.
   b. Locate the *ZK_HOST_PORT* property in the `icp-daemon.sh` file.
   c. Change the default value for the property to the ZooKeeper URI value of the Hive data source.

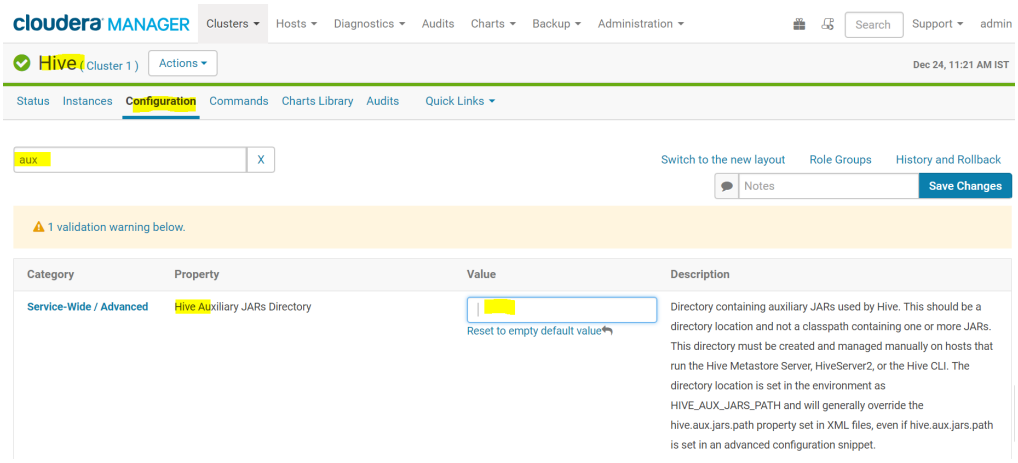
      Use the following format for the value: *<ZooKeeper URL>*:*<port>*
      For example: `abc123@mycompany.com:2181`
   d. Save and close the `icp-daemon.sh` file.
7. By default, the iCP daemon runs on port 9090. If you want to run the iCP daemon on a different port, such as port 9091, perform the following tasks:
   a. Create a text file with file name `icp-daemon.properties`.
   b. Enter the following text: `daemon.port=9091`
   c. Save the file in the `daemon` directory.

Step 6. Configure the Cloudera Hive User-Defined Functions

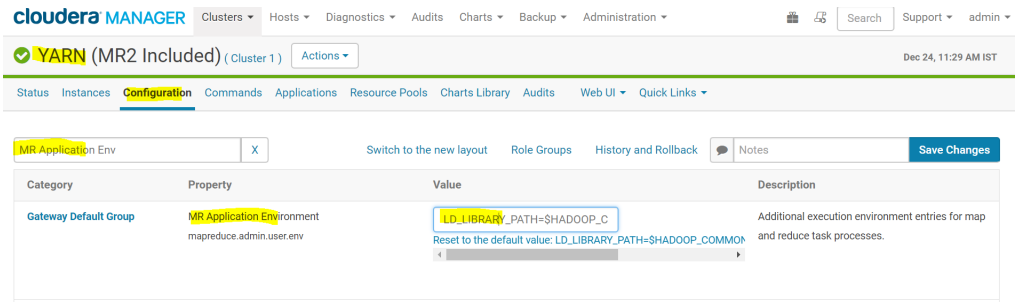
Use Cloudera Manager to configure the iCP daemon binaries in the library PATH and create the user-defined functions.

1. Log in to Cloudera Manager.
2. Copy the JAR files from the `lib/java` directory to the `hive/auxlib` directory.

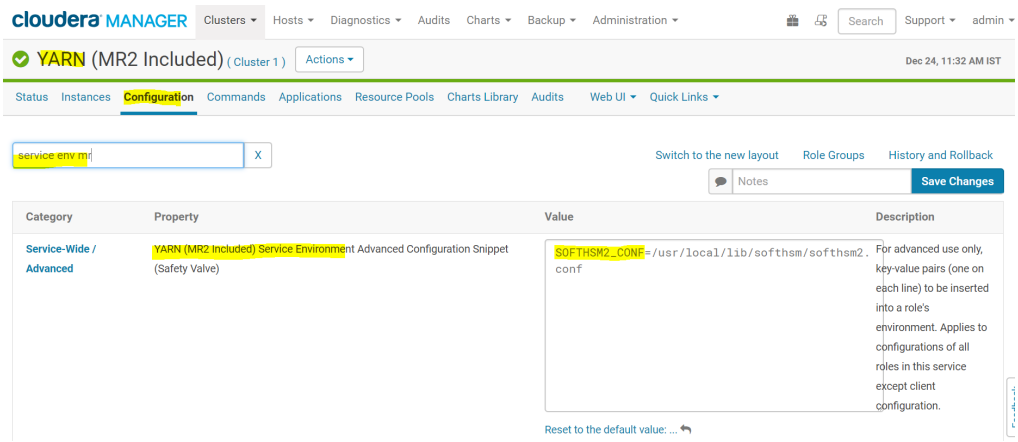
Alternatively, you can copy the JAR files to any directory. Then, provide the directory path in the **Configuration** tab.



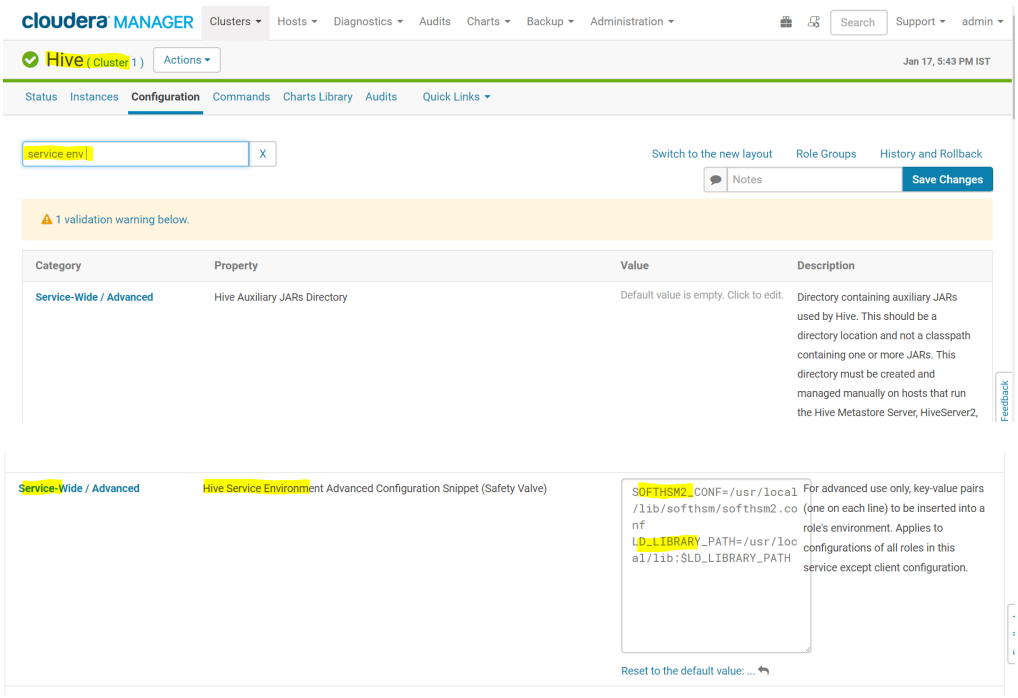
3. Insert `LD_LIBRARY_PATH` before the directory path of native libraries in YARN MR.



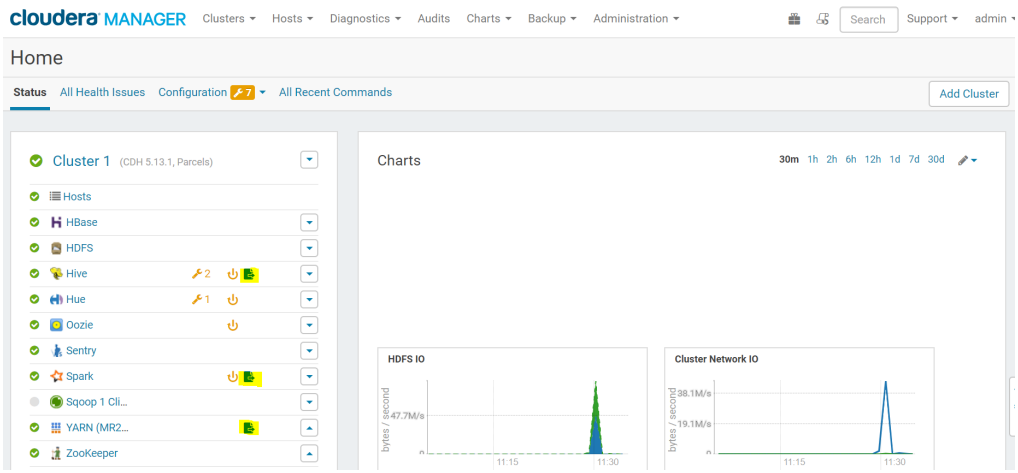
4. Export the `SOFTHSM2_CONF` file in YARN MR.



5. Insert `LD_LIBRARY_PATH` and `SOFTHSM2_CONF` in Hive.



6. Restart the cluster.
7. If it exists, deploy the stale configuration.



8. Use an SQL client such as Beeline or Hue to connect to the schema.
9. Create the following user-defined functions:
   - `drop function default.ff3encrypt;`
   - `drop function default.ff3decrypt;`
   - `create function default.ff3Encrypt as 'com.informatica.dsg.icp.udf.hive.FF3Encrypt';`
   - `create function default.ff3Decrypt as 'com.informatica.dsg.icp.udf.hive.FF3Decrypt';`

   Note: The `drop` and `create` functions work only when you change the signature for the user-defined functions.

10. To validate and verify that the configured UDFs are working, run the following encryption and decryption queries:

    a. Encryption query:

    ```
    select <schemaname>.ff3Encrypt ( '<input_data>', '<key>' ,
    '<Encrypt_Mode>', '<Transformation_Type>' , '<filter_chars>', <left_tweak> ,
    <right_tweak> , '<proxy_id>' );
    ```

    b. Decryption query:

    ```
    select ff3Decrypt ( '<encrypted_data>', '<key>' , '<encrypt_mode>',
    '<Transformation_Type>' , '<filter_chars>', <left_tweak> , <right_tweak> ,
    '<proxy_id>' );
    ```

    Example:

    ```
    select default.ff3Decrypt ( 'abcghj', 'osEE/
    Ualz2cuTsQdeArB423M11vYqng9BpkaMpB361c=' , '1', '' , 'bg', 1 , 1 , '' );
    ```

11. Start the iCP daemon.

    To start the daemon, navigate to the `/daemon` directory and use the following command:

    ```
    ./icp-daemon.sh start
    ```

Data Privacy Management Tasks

After you complete the configuration tasks, create the Hive data store on the **Data Stores** workspace in Data Privacy Management. For more information, see the *Data Store Properties* and the *Data Stores* chapters in the *Informatica Data Privacy Management User Guide*.

After you create the Hive data store, you can encrypt and decrypt data. For more information, see the *Protection* chapter of the *Informatica Data Privacy Management User Guide*.

CHAPTER 38

Encryption and Decryption Management

This chapter includes the following topic:

- Encryption and Decryption Management Overview, 269

Encryption and Decryption Management Overview

After you install and configure Data Privacy Management, you might need to reconfigure settings, such as change the user PIN to access the SoftHSM utility or modify the agentURI.

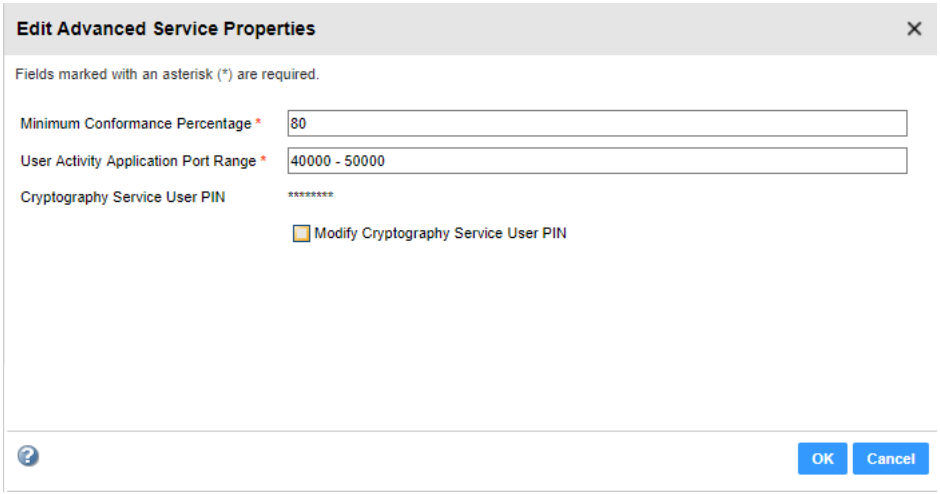
Changing the Cryptography Service User PIN

You must have a user PIN to access the SoftHSM utility. During installation, you specify whether you want to use a custom user PIN or the default user PIN. You can change the user PIN at any time.

To change the user PIN, set the Data Privacy Management Service **Modify Cryptography Service User PIN** property.

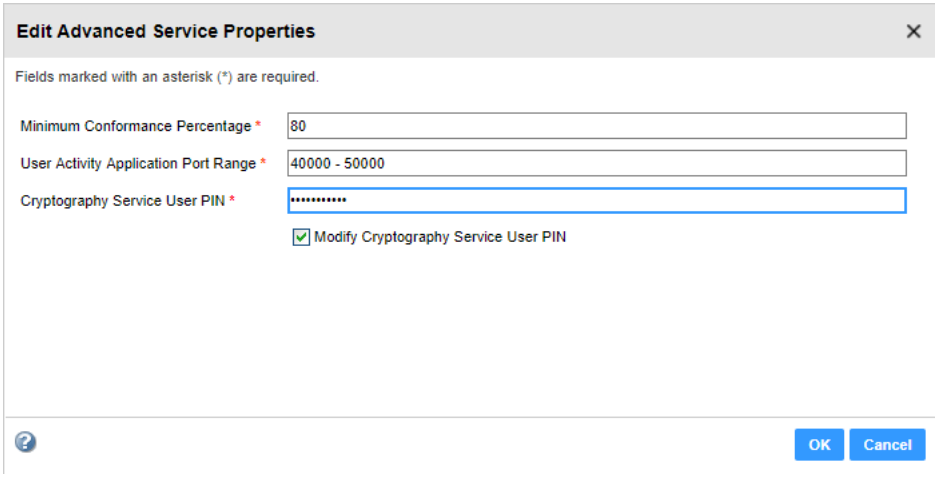
1. Log in to the Administrator tool.
2. Click the **Manage** tab.
3. Click the **Services and Nodes** view.
4. On the **Domain Navigator** pane, click the Data Privacy Management Service.
5. On the **Properties** pane, scroll to the **Advanced Service Properties** section.
6. Click the **Edit** icon for the **Advanced Service Properties** section.

The **Edit Advanced Service Properties** window appears.



7. Select the **Modify Cryptography Service User PIN** check box.
8. Enter the new user PIN in the **Cryptography Service User PIN** text box.

   The user PIN can be alphanumeric and can range from 4 through 255 characters in length.



9. Click **OK** to close the **Edit Advanced Service Properties** window.

   Make a note of the new user PIN and use it to log in to the SoftHSM utility.

Changing the URI for the Protection Remote Agent

If you want to configure an existing protection remote agent with a new or different Data Privacy Management server, perform the following steps:

1. Stop the protection remote agent.

   Use the following command:

   ```
   ./server stop
   ```

2. Change the agent URI.

   Use the following command:

   ```
   ./server config agentUri <Data Privacy Management URI>
   ```

Example:

```
./server config agentUri abc123.mycompany.com:18295
```

3. Start the protection remote agent.

   Use the following command:

   ```
   ./server start
   ```

Changing the Keystore File

1. Create the new keystore file.
2. Copy the new keystore file to the following directory: `<ProtectionAgent-Installation directory>/cfg`
3. Run the following command for the new keystore file to apply:

   ```
   ./server config keystore <keystore_file> <keystore_password> <alias> <store-type>
   ```

   Example:

   ```
   ./server config keystore abc123.jks hadoop abc123_new jks
   ```

Changing the Truststore File

1. Create the new truststore file.
2. Copy the new truststore file to the following directory: `<ProtectionAgent-Installation directory>/cfg`
3. Run the following command for the new truststore file to apply:

   ```
   ./server config truststore <truststore_file> <truststore_password>
   ```

   Example:

   ```
   ./server config truststore infa_truststore.jks pass2038@infaSSL
   ```

Managing the iCP Daemon

You can start, stop, and restart the iCP daemon.

Use the following commands:

- To start the ICP daemon: `./icp-daemon.sh start`
- To stop the ICP daemon: : `./icp-daemon.sh stop`
- To restart the ICP daemon: : `./icp-daemon.sh restart`

Initializing Tokens

The files in the `/tokens/slot` directory that you extract from the `DecryptionUtilities.zip` file are already initialized. However, if the original files are accidently deleted and you create new files, you must initialize the new files.

▶ To initialize the files in the `/tokens/slot` directory, use the following command:

```
softhsm2-util -init-token -slot 0 - label SoftHSMToken -so-pin <pin> -pin <pin>
```

Generating the Key for Encryption

Before you can encrypt data, you must generate a key that you use in the **Protection Extension** tab of the data domain definition to complete the encryption request.

1. Log in to the Data Privacy Management machine.
2. Navigate to the `$INFA_HOME/DataPrivacyManagement/softhsm` directory.
3. Log in to the SoftHSM tool with the default user PIN or the custom PIN you set up.
4. To generate the encryption key, enter the following command:

   ```
   ./hsm -g -keyName <keyName> -keyLength <keyLength>
   ```

   Where:

   - *<keyName>* is a name you specify for the key. You can include alphabets, numbers, and the hyphen and underscore characters in the name.
   - *<keyLength>* is the bit-length you specify for the key. You can specify a 16-, 24-, or 32-byte length.

APPENDIX A

Migrating the Data Privacy Management Repository from Oracle to PostgreSQL

This appendix includes the following topics:

- Overview, 273
- Prerequisites, 273
- Migrate the Repository, 280

Overview

You can migrate the Data Privacy Management repository to a different database.

Data Privacy Management includes a utility that you can use to migrate the Data Privacy Management repository from Oracle to PostgreSQL.

The utility migrates all the data on the Oracle database to the PostgreSQL database. The migration does not impact User Activity and Subject Registry data stored on ElasticSearch and MongoDB. There are no changes in functionality post migration.

Perform the prerequisite tasks before you run the utility.

Prerequisites

Before you run the utility, ensure that you perform the prerequisite tasks including update of the Data Privacy Management Service and creation of the required connections.

Perform the following tasks:

- Update the repository database details for the Data Privacy Management Service.
- Create the repository content and start and stop the service.
- Verify the Data Integration Service settings.
- Create an Oracle connection that points to the Oracle repository.

- Create an ODBC connection that points to the PostgreSQL database.
- Set the environment variables. Applicable if you use custom SSL certificates.

Update the Repository of the Data Privacy Management Service

The Data Privacy Management Service must point to the PostgreSQL database before you run the utility.

1. Log in to the Administrator tool.
2. Click the **Services and Nodes** tab.
3. From the Domain Navigator, select the Data Privacy Management Service.
4. Click Disable to stop the service.
5. Click **Edit** on the Data Privacy Management Repository tab.

   The **Edit Data Privacy Management Repository** window opens.
6. From the Database Type list, select PostgreSQL.
7. Enter the following database information:
   - Database Type. Select POSTGRESQL from the list.
   - URL. Enter the URL in the following format: `jdbc:informatica:postgresql://<host name>:<port number>;DatabaseName=`
   - Secure JDBC Parameters. If the repository database is secured with the SSL protocol, you must enter the secure database parameters. Enter the parameters as `name=value` pairs separated by semicolon characters (;). For example: `param1=value1;param2=value2`
   - User Name. The database user name for the repository.
   - Password. The repository database password for the database user.
   - Schema. The schema name for the database.
   - Tablespace. The tablespace name for the database.

   When you change the repository database, a prompt appears to create the repository content.
8. Click **Actions** > **Create Contents** to create the repository content.
9. Start the Data Privacy Management Service.

   Starting the service creates the required tables.

Stop the Data Privacy Management Service

1. Log in to Data Privacy Management.
2. Verify that all the default data domains and policies appear as expected.
3. Log in to the Administrator tool.
4. Click the **Services and Nodes** tab.
5. From the Domain Navigator, select the Data Privacy Management Service.
6. Click **Disable** to stop the service.

Verify the Data Integration Service Settings

The Data Integration Service runs the mappings included in the utility. To ensure that the service can run the mappings, verify the required property values.

1. Log in to the Administrator tool.
2. Click the **Services and Nodes** tab.
3. From the **Domain Navigator**, select the Data Integration Service.
4. Click **Edit** on the **Custom Properties** tab.

   The **Edit Custom Properties** window opens.
5. Verify the value of the `ExecutionContextOptions.JVMMaxMemory` property. Set the value to at least 2048M.

Create the Source Connection

Create a connection that points to the Oracle repository database. The utility runs mappings that migrate the repository data. The connection serves as the source connection in the mappings.

The mappings include the connection name. Ensure that you create the connection with the name as mentioned in the Connection Properties section. The ID must match the name.

1. Log in to the Administrator tool.
2. Click the **Connections** view.
3. In the Navigator, select the domain.
4. In the Navigator, click **Actions** > **New** > **Connection**.

   The **New Connection** dialog box appears.
5. In the **New Connection** dialog box, from the list of databases select Oracle as the connection type, and then click **OK**.

   The **New Connection** wizard appears.
6. Enter the connection properties.

   Click **Next** to go to the next page of the **New Connection** wizard.
7. When you finish entering connection properties, you can click **Test Connection** to test the connection.
8. Click **Finish**.

Oracle Connection Properties

Use an Oracle connection to connect to the Oracle database.

The following table describes Oracle connection properties:

Property	Description
Database Type	The database type.
Name	Name of the connection. Enter the following name: `DPMRepoSource` **Note:** The name is case sensitive.
ID	String that the Data Integration Service uses to identify the connection. The ID must be the same as the name.

Property	Description
Description	The description of the connection. The description cannot exceed 765 characters.
User Name	The database user name.
Password	The password for the database user name.
Pass-through security enabled	Enables pass-through security for the connection. When you enable pass-through security for a connection, the domain uses the client user name and password to log into the corresponding database, instead of the credentials defined in the connection object.

Metadata Access Properties: Connection String	Connection string used to access metadata from the database.
	Use the following connection string:
	`jdbc:informatica:oracle://<host_name>:<port>;SID=<database name>`
	Use the following connection string to connect to Oracle database through Oracle Connection Manager:
	`jdbc:informatica:oracle:TNSNamesFile=<fully qualified path to the tnsnames.ora file>;TNSServerName=<TNS server name>;`

Property	Description
AdvancedJDBCSecurityOptions	Database parameters for metadata access to a secure database. Informatica treats the value of the AdvancedJDBCSecurityOptions field as sensitive data and stores the parameter string encrypted. To connect to a secure database, include the following parameters: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to SSL. - ValidateServerCertificate. Optional. Indicates whether Informatica validates the certificate that is sent by the database server. If this parameter is set to True, Informatica validates the certificate that is sent by the database server. If you specify the HostNameInCertificate parameter, Informatica also validates the host name in the certificate. If this parameter is set to false, Informatica does not validate the certificate that is sent by the database server. Informatica ignores any truststore information that you specify. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Informatica validates the host name included in the connection string against the host name in the SSL certificate. - cryptoProtocolVersion. Optional. If you enable TLS for the Oracle instance, set the cryptoProtocolVersion parameter as follows: cryptoProtocolVersion=TLSv<version number>. For example, cryptoProtocolVersion=TLSv1.2 **Note:** The version number must be the same as the TLS version you configured for the server. - TrustStore. Required. Path and file name of the truststore file. **Note:** If you configure SSL or TLS and specify only the file name, you must copy the truststore file to the following directory to test the connection: `<Informatica server installation directory>/tomcat/bin` - TrustStorePassword. Required. Password for the truststore file for the secure database. - KeyStore. Required. Path and file name of the keystore file. - KeyStorePassword. Required. Password for the keystore file for the secure database. **Note:** Informatica appends the secure JDBC parameters to the connection string. If you include the secure JDBC parameters directly to the connection string, do not enter any parameters in the AdvancedJDBCSecurityOptions field.
Data Access Properties: Connection String	Use the following connection string: `<database name>.world`
Code Page	The code page used to read from a source database or to write to a target database or file.
Environment SQL	SQL commands to set the database environment when you connect to the database. The Data Integration Service runs the connection environment SQL each time it connects to the database.
Transaction SQL	SQL commands to set the database environment when you connect to the database. The Data Integration Service runs the transaction environment SQL at the beginning of each transaction.
Retry Period	This property is reserved for future use.

Property	Description
Enable Parallel Mode	Enables parallel processing when loading data into a table in bulk mode. By default, this option is cleared.
SQL Identifier Character	Type of character that the database uses to enclose delimited identifiers in SQL queries. The available characters depend on the database type. Select (None) if the database uses regular identifiers. When the Data Integration Service generates SQL queries, the service does not place delimited characters around any identifiers. Select a character if the database uses delimited identifiers. When the Data Integration Service generates SQL queries, the service encloses delimited identifiers within this character.
Support Mixed-case Identifiers	Enable if the database uses case-sensitive identifiers. When enabled, the Data Integration Service encloses all identifiers within the character selected for the **SQL Identifier Character** property. When the **SQL Identifier Character** property is set to none, the **Support Mixed-case Identifiers** property is disabled.

Create the Target Connection

Create an ODBC connection that points to the PostgreSQL database. The connection serves as the target connection in the mappings.

The mappings include the connection name. Ensure that you create the connection with the name as mentioned in the connection properties section. The ID must match the name.

Note: You can't use the native PostgreSQL ODBC driver to run the utility.

1. Open the following file:`<Informatica installation directory>/ODBC7.1/odbc.ini`
2. Update the entry for the PostgreSQL database to ensure that it points to the Data Direct PostgreSQL driver. The driver is shipped with the installer files.

   For example:

   ```
   [POSTGRES_infadsg1473]
   Driver=/home/INFA/Informatica/DPM10.5.1/ODBC7.1/lib/DWpsql27.so
   Description=DataDirect 7.1 PostgreSQL Wire Protocol
   Database=postgres
   HostName=<host name>
   PortNumber=<port number>
   EnableQuotedIdentifiers=1
   IANAAppCodePage=UTF-8
   CancelOnStatementClose=1
   SchemaName=schema
   ```
3. Log in to the Administrator tool.
4. Click the **Connections** view.
5. In the Navigator, select the domain.
6. In the Navigator, click **Actions** > **New** > **Connection**.

   The **New Connection** dialog box appears.
7. In the **New Connection** dialog box, from the list of databases select ODBC as the connection type, and then click **OK**.

   The **New Connection** wizard appears.
8. Enter the connection properties.

Click **Next** to go to the next page of the **New Connection** wizard.

9. When you finish entering connection properties, you can click **Test Connection** to test the connection.
10. Click **Finish**.

ODBC Connection Properties

Use an ODBC connection to connect to the PostgreSQL database.

The following table describes ODBC connection properties:

Property	Description
Database Type	The database type.
Name	Name of the connection. Enter the following name: `DPMRepoTarget` **Note:** The name is case sensitive.
ID	String that the Data Integration Service uses to identify the connection. The ID must match the name.
Description	The description of the connection. The description cannot exceed 765 characters.
User Name	The database user name.
Password	The password for the database user name.
Pass-through security enabled	Enables pass-through security for the connection. When you enable pass-through security for a connection, the domain uses the client user name and password to log into the corresponding database, instead of the credentials defined in the connection object.
Data Access Properties: Connection String	The ODBC connection URL used to access metadata from the database. `<data source name>`
Code Page	The code page used to read from a source database or to write to a target database or file.
Environment SQL	SQL commands to set the database environment when you connect to the database. The Data Integration Service runs the connection environment SQL each time it connects to the database.
Transaction SQL	SQL commands to set the database environment when you connect to the database. The Data Integration Service runs the transaction environment SQL at the beginning of each transaction.
Retry Period	This property is reserved for future use.
SQL Identifier Character	Type of character that the database uses to enclose delimited identifiers in SQL queries. The available characters depend on the database type. Select (None) if the database uses regular identifiers. When the Data Integration Service generates SQL queries, the service does not place delimited characters around any identifiers. Select a character if the database uses delimited identifiers. When the Data Integration Service generates SQL queries, the service encloses delimited identifiers within this character.

Property	Description
Support Mixed-case Identifiers	Enable if the database uses case-sensitive identifiers. When enabled, the Data Integration Service encloses all identifiers within the character selected for the **SQL Identifier Character** property. When the **SQL Identifier Character** property is set to none, the **Support Mixed-case Identifiers** property is disabled.
ODBC Provider	The type of database to which ODBC connects. For pushdown optimization, specify the database type to enable the Data Integration Service to generate native database SQL. The options are: - Other - Sybase - Microsoft_SQL_Server - Snowflake Default is Other.

Set the Environment Variables

If you use custom signed certificates, set the Truststore and Truststore password environment variables before you run the utility commands.

Set the following environment variables:

INFA_TRUSTORE

Set the variable to the custom Truststore file location.

```
export INFA_TRUSTSTORE=<Custom SSL Truststore file location>
```

INFA_TRUSTSORE_PASSWORD

Set the variable to the encrypted truststore password.

```
export INFA_TRUSTSTORE=<Custom SSL Truststore file location>
```

To get the encrypted password, run the following command:

```
<Informatica installation directory>/server/bin/pmpasswd <Custom SSL Truststore password>
```

Migrate the Repository

Use the migration utility to migrate a Data Privacy Management repository on Oracle to PostgreSQL. The installer includes the files required to run the utility.

The utility uses a mapping application with preconfigured mappings to migrate the data from source to target. The source is the Oracle repository database and the target is the PostgreSQL database.

To run the utility, you perform the following high-level tasks:

- Deploy the application.
- Run the mappings.
- Update the database sequences
- Perform a cleanup.

Note: The commands for each task require properties in a specific order. Enter the properties in the order as given in the sample command.

Command Reference

The command reference lists the properties used in the different commands that you run to migrate the repository data.

The following table lists the properties required to run the commands:

Property	Description
-DomainName -dn	Name of the Informatica domain.
-ServiceName -sn	Name of the service on which you run the command. To update the database sequences, enter the name of the Data Privacy Management Service. For the other commands, enter the name of the Data Integration Service.
-UserName -un	User name to connect to the domain.
-Password -pd	Password for the user name.

Deploy the Application

The utility first deploys the mappings to the Data Integration Service.

1. Open a command prompt.
2. Go to the following location: `<Informatica installation directory>/SecureAtSourceService/dmRepoMigration`
3. Run the following command:

   ```
   ./dpmRepoMigration.sh deployApplication -dn <domain_name> -sn <Data Integration Service name> -un <user name> -pd <password>
   ```

   The command deploys the mapping application to the Data Integration Service. The mapping application includes the mappings to migrate the repository data.
4. Verify that the application is deployed from the **Execution Statistics** tab on the **Monitor** page in the Administrator tool.

   The Application_DpmRepoMigration appears in the Data Integration Service details.

Run the Mappings

The mappings truncate existing data on the PostgreSQL database and copy the data from the Oracle database to the PostgreSQL database.

1. Open a command prompt.
2. Go to the following location: `<Informatica installation directory>/SecureAtSourceService/dmRepoMigration`

3. Run the following command:

```
./dpmRepoMigration.sh runMapping -dn <domain_name> -sn <Data Integration Service
name> -un <user name> -pd <password>
```

The command schedules the mappings in the application from the Data Integration Service. The application includes 255 mappings.

Note: Completion of the command indicates that the mappings are scheduled to run. The time taken to run the mappings depends on the volume of data.

4. Verify the status of the mappings from the **Execution Statistics** tab on the **Monitor** page in the Administrator tool.

Click **Deployed Mapping Jobs** in the Application_DpmMigration in the Data Integration Service details to view the status of the jobs.

5. If a mapping fails, verify the details and click **Reissue Selected Job** to run the mapping again.

The `act_re_deployment` job truncates the data of the `act_ge_bytearray` table. If the `act_re_deployment` job fails, perform the following steps:

a. Run the `act_re_deployment` mapping job again and ensure that it runs successfully.

b. Run the `act_ge_bytearray` mapping job again after the `act_re_deployment` job is successful.

Ensure that all 255 mappings run successfully before you proceed to the next step.

Update the Sequences

The migration utility migrates the repository data but does not update the sequence numbers in the database. Run the `infacmd sats updateSequences` command to update the sequences.

1. Open a command prompt.
2. Go to the following location: `<Informatica installation directory>/isp/bin`
3. Run the following command:

Optionally, you can add a debug level to the command.

```
infacmd.sh sats updateSequences -dn <domain_name> -un <username> -pd <password> -sn
<DPM_service_name>
```

The command updates the sequence numbers in the PostgreSQL repository database.

Perform a Cleanup

The utility includes a command that removes the mappings deployed to the Data Integration Service. The command also deletes the connections that you created to run the utility.

1. Open a command prompt.
2. Go to the following location: `<Informatica installation directory>/SecureAtSourceService/dmRepoMigration`
3. Run the following command:

```
./dpmRepoMigration.sh cleanup -dn <domain_name> -un <username> -pd <password> -sn
<DIS_service_name>
```

The command deletes the application deployed to the Data Integration Service and the connections.

Restart the Data Privacy Management Service

After you migrate the repository, you can continue to use Data Privacy Management as before the migration.

Log in to the Administrator tool and start the Data Privacy Management Service. The service starts and uses the PostgreSQL repository.

APPENDIX B

Source User Privileges for Jobs

This appendix includes the following topics:

- Source User Privileges for Jobs Overview, 284
- User Privileges for the Cloudera Navigator Scan Job, 284
- Database User Privileges for the Database Scan Job, 285
- User Privileges for the Big Data Scan Job, 290
- Database User Privileges for the Informatica PowerCenter Scan Job, 291
- User Privileges for Salesforce Import Jobs, 291
- User Privileges for the SAP Scan Job, 292

Source User Privileges for Jobs Overview

Data Privacy Management jobs require specific privileges for the user account that connects to the source. For example, the Informatica PowerCenter Scan job and the Database Scan job require specific privileges to connect to and run scans on databases. Before you scan or import a data store, verify that the user account configured in the data store or connection has the required privileges. If the user account does not have the required privileges, the job fails.

User Privileges for the Cloudera Navigator Scan Job

The Cloudera Navigator Scan job requires a Cloudera Manager user account that includes the Full Administrator or Navigator Administrator role. The roles include authorization to connect to and retrieve metadata from Cloudera Navigator. If the user account does not have one of these roles, the Scan job fails.

Database User Privileges for the Database Scan Job

The Database Scan job requires user account privileges to connect to and scan a database. When you create a data store, you specify the user account. When an Informatica PowerCenter Scan job imports data stores, the user account is populated from the connection attributes in the PowerCenter repository.

Before you run a database scan on a data store, verify that the user account in the data store has the required privileges. The required privileges depend on the database type.

IBM DB2 Database User Privileges

Verify the permissions for the database user account that you use to connect to an IBM DB2 database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the IBM DB2 database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	SYSCAT.SCHEMATA SYSCAT.TRIGGERS SYSCAT.INDEXES SYSCAT.INDEXCOLUSE SYSCAT.REFERENCES SYSCAT.TABCONST SYSCAT.KEYCOLUSE SYSCAT.COLCHECKS SYSCAT.DATATYPES SYSCAT.COLUMNS SYSCAT.TABLES SYSCAT.FUNCTIONS SYSCAT.PROCEDURES SYSCAT.PACKAGES SYSCAT.TRIGGERS SYSCAT.SEQUENCES SYSCAT.INDEXES SYSCAT.CHECKS SYSCAT.VIEWS SYSCAT.PACKAGES SYSCAT.FUNCDEP SYSCAT.TRIGDEP SYSCAT.VIEWDEP

In addition, the database user account must have CREATEIN, DROPIN, and ALTERIN privileges on the NULLID schema. Optionally, you can grant the BINDADD privilege to the user account.

IBM DB2 for z/OS Database User Privileges

Verify the permissions for the database user account that you use to connect to an IBM DB2 for z/OS database.

Configure SELECT privileges on the following tables in the DB2 for z/OS user account that you use to connect to the metadata source:

System Tables	Metadata Extracted
SYSIBM.SYSSYNONYMS SYSIBM.SYSTABLES SYSIBM.SYSROUTINES SYSIBM.SYSSEQUENCES SYSIBM.SYSDATATYPES	Schemas
SYSIBM.SYSTABLES SYSIBM.SYSTABLESPACE SYSIBM.SYSVIEWS	Tables Views Alias
SYSIBM.SYSTABLES SYSIBM.SYSCOLUMNS SYSIBM.SYSTABLESPACE	Columns
SYSIBM.SYSKEYS SYSIBM.SYSINDEXES	Primary keys
SYSIBM.SYSRELS SYSIBM.SYSFOREIGNKEYS SYSIBM.SYSINDEXES	Foreign keys
SYSIBM.SYSCHECKS SYSIBM.SYSCHECKDEP	Check constraints
SYSIBM.SYSINDEXES SYSIBM.SYSKEYS	Indexes
SYSIBM.SYSSYNONYMS SYSIBM.SYSTABLES	Synonyms
SYSIBM.SYSTRIGGERS	Triggers
SYSIBM.SYSROUTINES	Stored procedures
SYSIBM.SYSROUTINES	Functions
SYSIBM.SYSSEQUENCES	Sequences
SYSIBM.SYSSEQUENCESDEP SYSIBM.SYSSEQUENCES	Identity
SYSIBM.SYSDATATYPES	User-defined types

Note: If the PowerExchange Listener does not use DB2 security, configure SELECT privileges for the PowerExchange Listener user ID instead of the DB2 for z/OS user account.

Microsoft SQL Server Database User Privileges

Verify the permissions for the database user account that you use to connect to a Microsoft SQL Server database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Microsoft SQL Server database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	sysobjects sysusers master.dbo.sysservers syscolumns sysindexkeys systypes syscomments sysreferences sysdepends sys.synonyms (Microsoft SQL Server 2005) sysindexes sys.sql_expression_dependencies

In addition, the user must have VIEW DEFINITION permission on the database.

Netezza Database User Requirements

Verify the permissions for the database user account that you use to connect to a Netezza database. You specify the user account in the data store connection properties.

The user account must have the LIST privilege to extract metadata from a Netezza database.

Configure SELECT permissions for the Netezza user account on the following system views:

- _V_TABLE_DIST_MAP
- _V_TABLE
- _V_DATABASE
- _v_relation_keydata
- _v_relation_column
- _v_view
- _v_synonym
- _v_datatype
- _v_sequence

Oracle Database User Privileges

Verify the permissions for the database user account that you use to connect to an Oracle database. You specify the user account in the data store connection properties.

Configure CONNECT and SELECT_CATALOG_ROLE permissions for the user account.

If you cannot configure SELECT_CATALOG_ROLE permission for the user account, configure the following permissions:

- SELECT permissions on the v$version and v$instance views.
- CONNECT and SELECT permissions on the required schema objects and corresponding system tables.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Indexes
- Packages
- Procedures
- Functions
- Sequences
- Triggers
- Synonyms

Configure SELECT permissions on the following system tables:

- ALL_OBJECTS
- ALL_SOURCE
- ALL_USERS
- DBA_ARGUMENTS
- DBA_COL_COMMENTS
- DBA_CONS_COLUMNS
- DBA_CONSTRAINTS
- DBA_DEPENDENCIES
- DBA_IND_COLUMNS
- DBA_INDEXES
- DBA_MVIEWS
- DBA_OBJECTS
- DBA_PROCEDURES
- DBA_RECYCLEBIN
- DBA_SEQUENCES
- DBA_SOURCE
- DBA_SYNONYMS
- DBA_TAB_COLUMNS
- DBA_TAB_COMMENTS
- DBA_TAB_PARTITIONS

- DBA_TABLES
- DBA_TRIGGERS
- DBA_VIEWS
- SYS.V_$DATABASE
- SYS.DBA_USERS
- SYS.DBA_DIRECTORIES
- SYS.DBA_EXTERNAL_LOCATIONS
- SYS.DBA_CLU_COLUMNS

If schemas have a table with a cluster defined on some columns, you must configure SELECT permission on the DBA_CLUSTERS system table.

Sybase Database User Privileges

Verify the permissions for the database user account that you use to connect to a Sybase database.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Columns
- Procedures

Configure SELECT permissions on the following system tables:

- sysobjects
- syscomments
- sysdepends
- syscolumns
- sysindexes
- sysconstraints
- sysreferences
- sysusers
- systypes
- master.dbo.sysservers
- master.dbo.sysconfigures
- @@<servername>

Teradata Database User Privileges

Verify the permissions for the database user account that you use to connect to a Teradata database. You specify the user account in the data store connection properties.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Teradata database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	DBC.ALL_RI_CHILDREN DBC.COLUMNS DBC.DATABASES DBC.DBASE DBC.DBCINFO DBC.HOSTSINFO DBC.INDICES DBC.SHOWTBLCHECKS DBC.TABLES DBC.TABLETEXT DBC.TRIGGERS DBC.TVFIELDS DBC.TVM DBC.UDTINFO TDQCD.STATSRECS

User Privileges for the Big Data Scan Job

The Big Data Scan job requires the following privileges:

- The Big Data Scan job requires a user account that has read, write, and execute privileges on the directory configured in the **Hive Warehouse Directory On HDFS** data store property. If the user account does not have the required privileges, the Scan job fails.

  You specify the user in the Hive data store connection properties.
- By default, the Big Data Scan job uses the Blaze engine to run mappings in the Hadoop cluster. The Blaze user must have permissions to run mappings in the Hadoop cluster. You specify the Blaze user when you run the Hadoop Configuration Manager to create a Big Data connection in the Informatica domain.

Database User Privileges for the Informatica PowerCenter Scan Job

The Informatica PowerCenter Scan job requires database privileges to connect to and scan a PowerCenter repository. The Scan job also requires privileges for the PowerCenter user that logs in to the PowerCenter client tools. You specify the users when you create a data store that connects to Informatica PowerCenter.

Before you run an Informatica PowerCenter scan on a data store, verify that the user account and the PowerCenter user have the required privileges. If the users do not have the required privileges, the Scan job fails.

PowerCenter Repository Database User Account

The PowerCenter repository database user account must have the following permissions:

- Access Repository Manager privilege for the user who accesses the PowerCenter repository.
- Read permission on the PowerCenter data source for the user account that you use to access the PowerCenter data source.

Ensure that you run the pmrep ObjectExport command to export the mappings in PowerCenter.

PowerCenter User Account

The PowerCenter user account logs in to the PowerCenter client tools and accesses the PowerCenter repository.

The Informatica PowerCenter scan job uses the PowerCenter user account to create a folder in the PowerCenter repository. The folder includes a workflow that reads connection properties from database configuration files and PowerCenter parameter files. The workflow is required for the scan job to import the complete connection properties for all of the connections in the PowerCenter repository.

The PowerCenter user account must have PowerCenter Repository Service privileges for the following privilege groups:

- Folders
- Runtime objects
- Sources and targets
- Design objects

For more information about the privilege groups, see the *Informatica Security Guide*.

User Privileges for Salesforce Import Jobs

The Salesforce Import job requires org and user account privileges to connect to and import user metadata from Salesforce. You specify the user account in the Salesforce data store connection properties. Before you import user metadata from Salesforce, verify that the user account in the data store has the required privileges.

The Salesforce Import job requires the following privileges:

- The Salesforce org must have the Bulk API feature enabled.

- The user account must have the View All Data and the View Event Log Files privileges.

User Privileges for the SAP Scan Job

To scan SAP data stores, the user account must have the following authorizations:

Authorization Object	Activity	Purpose
S_DEVELOP	All activities.	Install and uninstall programs.
S_TRANSPRT	CREATE and CHANGE	Install and uninstall programs. Required for workbench requests TTYPE = DTRA.
S_TABU_DIS	READ	Extract data.
S_DATASET	READ and WRITE	Read/Write staging data to a file.
S_BATCH_JOB	DELE, LIST, PLAN, SHOW	Release background jobs.
S_RFC	All activities for RFC objects in the following function groups: - ZPMV - ZERP - ZPMH - ZPMR - ZPMD - ZPMI - ZPMF - SYST - RFC1 - /INFATRAN/* - SDIFRUNTIME - GSAC - SKEY - STR9 - SEU_COMPONENT - STRD - BTCH - SABC - KXUT - /1BCDWBEN/SEN4 - EDI1	Authorize RFC privileges.

APPENDIX C

System-Defined Data Domains

This appendix includes the following topics:

- System-Defined Data Domains, 293
- System-Defined Data Domains Column Name Rules, 300
- System-Defined Data Domains Data Rules, 302

System-Defined Data Domains

Use the system-defined data domains to perform data domain discovery and identify critical data within your data stores.

The following table describes the data domains available when you install Data Privacy Management:

Note: In the table, the asterisk (*) symbol is a wildcard character.

Name	Description	Dependent Rule Type	Data Domain Group
AccountNumber	Discovers column names that contain the "a*c*num" or "acc" string.	Column name rule	Account_Bank, PCI, PHI
Account_Status	Discovers column data that matches account status values in a reference table.	Data rule	Account_Bank
Age	Discovers column names that contain the "age" string or "dob" string and discovers the column data with values from 1 through 120.	Column name rule Data rule	PII
AlphaNumeric_SpecialCharacter	Discovers column data that contains unformatted alphanumeric data and special-character data.	Data rule	General
Austria_NationalID	Discovers column data that matches the national ID of Austria.	Data rule	NationalID

Name	Description	Dependent Rule Type	Data Domain Group
BIC_SwiftCode	Discovers column data that matches Bank Identifier Code (BIC) or Society for Worldwide Interbank Financial Telecommunication (SWIFT) code by pattern recognition and country code.	Data rule	Account_Bank
BinayValue	Discovers column data that contains binary values.	Data rule	General
BirthDay	Discovers column names that contain the "dob" string, "date*of*bir*" string, or "birth*da*" string and identifies the column data that matches valid birth dates.	Column name rule Data rule	PII
BirthPlace	Discovers column names that contain the "birth*place" string or "location*birth" string.	Column name rule	PII
Brazil_IDDoc	Discovers column data that matches the number format of the Brazilian ID card titled Registro Geral.	Data rule	NationalID
Brazil_NationalID	Discovers column data that matches the national ID of Brazil.	Data rule	NationalID
Bulagaria_NationalID	Discovers column data that matches the national ID of Bulgaria.	Data rule	NationalID
Canada_SIN	Discovers column data that matches the Canada Social Insurance Number format.	Data rule	NationalID
CertificateLicenseNumber	Discovers column names that contain the "cert*lic*number" string, "cert*lic*no*" string, "lic*nu*" string, or "lic*no*" string.	Column name rule	PHI
China_NationalID	Discovers column data that matches the national ID of China.	Data rule	NationalID
CompanyName	Discovers column names that contain the "company" string and identifies the column data that matches the organization-name values in a reference table.	Column name rule Data rule	Contact, PII
ComputerAddress	Discovers column data that matches computer address formats.	Data rule	General

Name	Description	Dependent Rule Type	Data Domain Group
Country	Discovers column names that contain the "iso*countr*code" string, "iso*country" string, or "countr*" string and identifies the column data that matches country names.	Column name rule Data rule	Address, PII
CountryCode_Phone	Discovers phone numbers based on international dialing codes.	Data rule	Contact
CreditCardNumber	Discovers column names that contain the "ccn" string, "cr*ca*nu" string, or "credit*no*" string and identifies the column data that matches the credit card number format of multiple credit card organizations.	Column name rule Data rule	Account_Bank, PCI, PII
CreditCard_AMEX	Discovers column data that matches the American Express credit card number format.	Data rule	Account_Bank
CreditCard_DinersCard	Discovers column data that matches the Diners Club International credit card number format.	Data rule	Account_Bank
CreditCard_DiscoverCard	Discovers column data that matches the Discover credit card number format.	Data rule	Account_Bank
CreditCard_JCB	Discovers column data that matches the JCB International credit card number format.	Data rule	Account_Bank
CreaditCard_MasterCard	Discovers column data that matches the MasterCard credit card number format.	Data rule	Account_Bank
CreditCard_Visa	Discovers column data that matches the Visa credit card number format.	Data rule	Account_Bank
Croatia_NationalID	Discovers column data that matches the national ID of Croatia.	Data rule	NationalID
Date_AllFormats	Discovers date values in the column data.	Data rule	General
Date_MM_DD_YYYY	Discovers date strings in the source data that appear in a single format in a date column. The default format is "MM/DD/YYYY."	Data rule	General

Name	Description	Dependent Rule Type	Data Domain Group
Denmark_NationalID	Discovers column data that matches the national ID of Denmark.	Data rule	NationalID
DeviceSerialNumber	Discovers column names that contain the "device*number" string, "device*no*" string, "serial*number" string, "serial*no*" string, or "device*identi*" string.	Column name rule	PHI
DriverLicense_Canada	Discovers column data that matches the Canada driver license numbers except for the provinces of British Columbia, Quebec, Manitoba, and Prince Edward Island.	Data rule	NationalID
DriverLicense_GBR	Discovers column data that matches the United Kingdom driver license numbers.	Data rule	NationalID
DriverLicense_USA	Discovers column data that matches the driver license numbers of most of the states in the United States.	Data rule	NationalID
DrivingLicenseNumber	Discovers column names that contain the "license" string or "driver*license" string.	Column name rule	PII
Email	Discovers column names that contain the "email" string and identifies the column data that matches a predefined email ID format.	Column name rule Data rule	Contact, PHI
ExpirationDate	Discovers column names that contain the "exp*da*" string or "cr*exp*" string and identifies the column data that matches expired credit card dates.	Column name rule Data rule	PCI
Finland_NationalID	Discovers column data that matches the national ID of Finland.	Data rule	NationalID
FirstName	Discovers column names that contain the "f*nam*" string and identifies the column data that matches values in a reference table of first names.	Column name rule Data rule	Contact, PCI, PII

Name	Description	Dependent Rule Type	Data Domain Group
France_INSEE	Discovers column data that matches the French Institut National de la Statistique et des Études Économiques (INSEE) number format.	Data rule	NationalID
Gender	Discovers column names that contain the "gender" string or strings such as "female" and "male" and identifies the column data that matches the gender values in a reference table.	Column name rule Data rule	Contact, PII
Geocode_Latitude	Discovers column names that contain the "latitude" string and identifies the column data that matches valid latitude coordinates.	Column name rule Data rule	Address, General
Geocode_LatitudeLongitude	Discovers column names that contain strings such as "latitude," "longitude," and "geocode" and identifies column data that matches valid latitude or longitude coordinates.	Column name rule Data rule	Address, General
Geocode_Longitude	Discovers column names that contain the "longitude" string and identifies the column data that matches valid longitude coordinates.	Column name rule Data rule	Address, General
Grade	Discovers column names that contain the "grade" string.	Column name rule	PII
GreatBritian_NINO	Discovers column data that matches the Great Britian National Insurance number format.	Data rule	NationalID
Hostname	Discovers column data that matches valid host names.	Data rule	General
IBAN	Discovers column data that matches the International Bank Account Number format of multiple European countries.	Data rule	Account_Bank
IPAddress	Discovers column names that contain the "ip" string or "inter*port*add" string and identifies the column data that matches a predefined IP address format.	Column name rule Data rule	PII

Name	Description	Dependent Rule Type	Data Domain Group
ISBN	Discovers column data that matches the International Standard Book Number format.	Data rule	General
India_NationalID	Discovers column data that matches the Indian Permanent Account Number format.	Data rule	NationalID
Italy_FiscalCode	Discovers column data that matches the Italian national ID format.	Data rule	NationalID
JobPosition	Discovers column names that contain the "title" string, "position" string, or "designation" string.	Column name rule	PII
Korea_NationalID	Discovers column data that matches the national ID of Korea.	Data rule	NationalID
LastName	Discovers column names that contain the "lname" string, "su*name" string, or "last*name" string and identifies the column data that matches values in a reference table of last names.	Column name rule Data rule	Contact, PCI, PII
Norway_NationalID	Discovers column data that matches the national ID of Norway.	Data rule	NationalID
Passport_DEU_MR	Discovers column data that matches the machine-readable, German passport number format.	Data rule	NationalID
Passport_GBR	Discovers column data that matches the United Kingdom passport number format.	Data rule	NationalID
Passport_India	Discovers column data that matches the India passport number format.	Data rule	NationalID
Passport_MachineReadable	Discovers column data that matches machine-readable passport numbers of all countries.	Data rule	NationalID
Passport_USA_MR	Discovers column data that matches machine-readable, United States passport number format.	Data rule	NationalID

Name	Description	Dependent Rule Type	Data Domain Group
PhoneNumber	Discovers column names that contain the "phone" string or "fax" string and identifies the column data that matches the United States phone number format.	Column name rule Data rule	Contact, PHI
Postcode	Discovers column data that matches the postal codes of multiple countries.	Data rule	Address
Romania_NationalID	Discovers column data that matches the national ID of Romania.	Data rule	NationalID
SSN	Discovers column names that contain the "SSN" string, "social*sec*no" string, or "social* sec*num*" string and identifies the column data that matches the Social Security number format.	Column name rule Data rule	NationalID, PHI
SSN_General	Discovers columns data that matches the Social Security number format.	Data rule	NationalID, PII
Salary	Discovers column names that contain the "compensation" string, "salary" string, or "wages" string.	Column name rule	PII
SouthAfrica_NationalID	Discovers column data that matches the national ID of South Africa.	Data rule	NationalID
State	Discovers column names that contain the "add*sta" string, "state" string, or "us*sta*" string and identifies the column data that matches the state names in the United States.	Column name rule Data rule	PII
Street	Discovers column names that contain one of the following strings: - street - road - lane - court - avenue - way - blvd - boule*ard	Column name rule	PII

Name	Description	Dependent Rule Type	Data Domain Group
Sweden_NationalID	Discovers column data that matches the national ID of Sweden.	Data rule	NationalID
Taiwan_NationalID	Discovers column data that matches the national ID of Taiwan.	Data rule	NationalID
UPC	Discovers column data that matches a Universal product Code.	Data rule	General
URL	Discovers column names that contain the "uni*res*loc" string, "URL" string, or "web" string and identifies the column data that matches predefined URL formats.	Column name rule Data rule	PHI
USZip_5digit	Discovers column data that matches United States ZIP codes.	Data rule	Address
UniqueIdentifyingNumber	Discovers column names that contain the "unique*iden*number" string or "iden*num" string.	Column name rule	PHI
VehicleRegPlateNumber	Discovers column names that contain the "registration" string, "number*plate" string, "license*plate" string, or "vehicle*registration" string.	Column name rule	PII
ZipCode	Discovers column names that contain the "zip" string or "pin" string and identifies the column data that matches United States ZIP codes.	Column name rule Data rule	PII

System-Defined Data Domains Column Name Rules

Use the data domain column name rules to identify columns by column name.

The column name rules are located in the Model repository, in the following repository location:

```
[Informatica_DQ_Content]\Domain_Discovery\MetaData_Rules
```

The following table describes the column name rules available when you install Data Privacy Management.

Note: In the table, the asterisk (*) symbol is a wildcard character.

Name	Description
dataDomain_MetaDataRule_AccountNumber	Discovers a column name that contains the "a*c*num" string or "acc" string.
dataDomain_MetaDataRule_Age	Discovers a column name that contains the "age" string or "dob" string.
dataDomain_MetaDataRule_BirthDay	Discovers a column name that contains the "dob" string, "date*of*bir*" string, or "birth*da*" string.
dataDomain_MetaDataRule_BirthPlace	Discovers a column name that contains the "birth*place*" string or "location*birth*" string.
dataDomain_MetaDataRule_CertificateLicenseNumber	Discovers a column name that contains the "*cert*lic*number*" string, "*cert*lic*no*" string, or "*cert*lic*nuo*" string.
dataDomain_MetaDataRule_CompanyName	Discovers a column name that contains the "company" string.
dataDomain_MetaDataRule_Country	Discovers a column name that contains the "*iso*countr*code*" string, "*iso*country*" string, or "countr*" string.
dataDomain_MetaDataRule_CreditCardNumber	Discovers a column name that contains the "ccn" string, "cr*ca*nu*" string, or "credit*no*" string.
dataDomain_MetaDataRule_DeviceSerialNumber	Discovers a column name that contains the "*device*number*" string, "*device*no*" string, "*serial*number*" string, "*serial*no*" string, or "*device*identi*" string.
dataDomain_MetaDataRule_DrivingLicenseNumber	Discovers a column name that contains the "license*" string or "driver*license" string.
dataDomain_MetaDataRule_Email	Discovers a column name that contains the "email" string.
dataDomain_MetaDataRule_ExpirationDate	Discovers a column name that contains the "exp*da*" string or "cr*exp*" string.
dataDomain_MetaDataRule_FirstName	Discovers a column name that contains the "f*nam*" string.
dataDomain_MetaDataRule_Gender	Discovers a column name that contains the "gender" string.
dataDomain_MetaDataRule_Grade	Discovers a column name that contains the "grade" string.
dataDomain_MetaDataRule_IPAddress	Discovers a column name that contains the "ip" string.
dataDomain_MetaDataRule_JobPosition	Discovers a column name that contains the "title" string, "position" string, or "designation" string.
dataDomain_MetaDataRule_LastName	Discovers a column name that contains the "lname*" string, "su*nam*" string, or "last*name*" string.
dataDomain_MetaDataRule_Latitude	Discovers a column name that contains the "latitude" string.

Name	Description
dataDomain_MetaDataRule_LatitudeLongitude	Discovers a column name that contains one of the following strings: - latitude - longitude - geocode - geocoding - geocoordinate
dataDomain_MetaDataRule_Longitude	Discovers a column name that contains the "longitude" string.
dataDomain_MetaDataRule_PhoneNumber	Discovers a column name that contains the "phone" string or "fax" string.
dataDomain_MetaDataRule_Salary	Discovers a column name that contains the "compensation" string, "salary" string, or "wages" string.
dataDomain_MetaDataRule_SSN	Discovers a column name that contains the "*SSN*" string, "*social*sec*no*" string, or "*social*sec*num*" string.
dataDomain_MetaDataRule_State	Discovers a column name that contains the "add*sta*" string, "state" string, or "us*sta*" string.
dataDomain_MetaDataRule_Street	Discovers a column name that contains one of the following strings: - street - road - lane - court - avenue - way - blvd - boule*ard
dataDomain_MetaDataRule_UniqueIdentifyingNumber	Discovers a column name that contains the "*unique*iden*number*" string or "*iden*num*" string.
dataDomain_MetaDataRule_URL	Discovers a column name that contains the "*uni*res*loc*" string, "url*" string, or "web*" string.
dataDomain_MetaDataRule_VehicleRegPlateNumber	Discovers a column name that contains the "*registration*" string, "*number*plate*" string, "license*plate*" string, or "*vehicle*registration*" string.
dataDomain_MetaDataRule_ZipCode	Discovers a column name that contains the "zip" string or "pin" string.

System-Defined Data Domains Data Rules

Use the data domain data rules to identify columns that contain data that match the rule criteria.

The data rules are located in the Model repository, in the following repository location:

```
[Informatica_DQ_Content]\Domain_Discovery\Data_Rules
```

The following table describes the data rules that are available when you install Data Privacy Management:

Name	Description
dataDomain_DataRule_Account_Status	Identifies the column data that matches account status values in a reference table.
dataDomain_DataRule_Alphanumeric_SpecialCharacter	Identifies column data that contains unformatted alphanumeric data and special-character data.
dataDomain_DataRule_AUT_NATID	Identifies the column data that matches the Austrian national ID format.
dataDomain_DataRule_BGR_NATID	Identifies the column data that matches the Bulgarian national ID format.
dataDomain_DataRule_BIC_SWIFTCode	Identifies the column data that matches Bank Identifier Code (BIC) or Society for Worldwide Interbank Financial Telecommunication (SWIFT) code by pattern recognition and country code.
dataDomain_DataRule_BinaryValues	Identifies the column data that contains binary values.
dataDomain_DataRule_BRA_IDDoc	Identifies the column data that matches the number format of the Brazilian ID card titled Registro Geral.
dataDomain_DataRule_BRA_Personal_ID	Identifies the column data that matches the Brazilian personal ID format.
dataDomain_DataRule_CHN_NATID	Identifies the column data that matches the Chinese national ID format.
dataDomain_DataRule_Computer_Address	Identifies the column data that matches the format of IP addresses and Mac addresses.
dataDomain_DataRule_CountryCode_Phone	Identifies phone numbers based on international dialing codes.
dataDomain_DataRule_CreditCard_AMEX	Identifies the column data that matches the American Express credit card number format.
dataDomain_DataRule_CreditCard_DinersCard	Identifies the column data that matches the Diners Club International credit card number format.
dataDomain_DataRule_CreditCard_DiscoverCard	Identifies the column data that matches the Discover credit card number format.
dataDomain_DataRule_CreditCard_JCB	Identifies the column data that matches the JCB International credit card number format.
dataDomain_DataRule_CreditCard_MasterCard	Identifies the column data that matches the MasterCard credit card number format.
dataDomain_DataRule_CreditCard_Visa	Identifies the column data that matches the Visa credit card number format.

Name	Description
dataDomain_DataRule_Date_Validation	Identifies the date strings in the source data that appear in a single format in a date column. To configure the date format that the rule uses for validation, open the dq_ValidateDate Expression transformation in the rule and update the In_Date_Format expression variable. The default format is "MM/DD/YYYY." The rule returns "Valid" or "Invalid."
dataDomain_DataRule_Date_Validation_All_Formats	Identifies the date values in the column data and standardizes the column data to a specific date format.
dataDomain_DataRule_DEU_Machine_Readable_Passport	Identifies the column data that matches the machine-readable, German passport number format.
dataDomain_DataRule_DNK_NATID	Identifies the column data that matches the Danish national ID format.
dataDomain_DataRule_DriversLicense	Identifies the column data that matches the United Kingdom, Unites States, and Canadian driver license numbers based on the length and pattern requirements.
dataDomain_DataRule_DriversLicense_Canada	Identifies the column data that matches the Canada driver license numbers except for the provinces of British Columbia, Quebec, Manitoba, and Prince Edward Island.
dataDomain_DataRule_DriversLicense_GBR	Identifies the column data that matches the United Kingdom driver license numbers.
dataDomain_DataRule_DriversLicense_USA	Identifies the column data that matches the driver license numbers of most of the states in the United States.
dataDomain_DataRule_FIN_NATID	Identifies the column data that matches the Finnish national ID format.
dataDomain_DataRule_FRA_INSEE	Identifies the column data that matches the French Institut National de la Statistique et des Études Économiques (INSEE) number format.
dataDomain_DataRule_GBR_NINO	Identifies the column data that matches the United Kingdom National Insurance number format.
dataDomain_DataRule_GBR_Passport_Number	Identifies the column data that matches the United Kingdom passport number format.
dataDomain_DataRule_HostName	Identifies the column data that matches valid host names.
dataDomain_DataRule_HRV_NATID	Identifies the column data that matches the Croatian national ID format.
dataDomain_DataRule_IBAN	Identifies the column data that matches the International Bank Account Number format of multiple European countries.
dataDomain_DataRule_IND_NATID	Identifies the column data that matches the Indian Permanent Account Number format.
dataDomain_DataRule_IND_Passport	Identifies the column data that matches the Indian passport number format.

Name	Description
dataDomain_DataRule_ISBN	Identifies the column data that matches the International Standard Book Number format.
dataDomain_DataRule_ItalyFiscalCode	Identifies the column data that matches the Italian national ID format.
dataDomain_DataRule_KOR_NATID	Identifies the column data that matches the Korean national ID format.
dataDomain_DataRule_Latitude	Identifies the column data that matches valid latitude coordinates.
dataDomain_DataRule_LatitudeLongitude	Identifies the column data that matches valid pairs of latitude and longitude coordinates, each pair separated by a semicolon.
dataDomain_DataRule_Longitude	Identifies the column data that matches valid longitude coordinates.
dataDomain_DataRule_Machine_Readable_Passport	Identifies the column data that matches machine-readable passport numbers of all countries.
dataDomain_DataRule_NOR_NATID	Identifies the column data that matches the Norwegian national ID format.
dataDomain_DataRule_PostCode	Identifies the column data that matches the postal codes of multiple countries.
dataDomain_DataRule_ROU_NATID	Identifies the column data that matches the Romanian national ID format.
dataDomain_DataRule_SouthAfrica_NATID	Identifies the column data that matches the South African national ID format.
dataDomain_DataRule_SWE_NATID	Identifies the column data that matches the Swedish national ID format.
dataDomain_DataRule_TWN_NATID	Identifies the column data that matches the Taiwanese national ID format.
dataDomain_DataRule_URL	Identifies the column data that matches predefined URL formats.
dataDomain_DataRule_US_Zip5	Identifies the column data that matches United States ZIP codes.
dataDomain_DataRule_USA_Machine_Readable_Passport	Identifies the column data that matches machine-readable, United States passport number format.
dataDomain_DataRule_USA_SSN_post_2011June	Identifies the column data that matches the Social Security number format in length, numeric values, and minimum and maximum values of the area, group, and serial number sections. Based on the SSN Randomization effective June 25, 2011, the rule does not verify the issuance of a Social Security number and the group and area number combination.

APPENDIX D

System-Defined Classification Policies

This appendix includes the following topics:

- System-Defined Classification Policies Overview, 306
- PCI Classification Policy, 307
- PHI Classification Policy, 307
- PII Classification Policy, 308

System-Defined Classification Policies Overview

The Data Privacy Management installation includes a set of commonly used classification policies that identify sensitive data and classify a data store based on industry or organization data security standards.

You can customize system-defined classification policies. For example, you can add or remove a data domain or change the data domain match condition.

The following table describes the system-defined classification policies:

Classification Policy	Description
PCI	The Payment Card Information (PCI) classification policy contains data domains that identify fields with credit or debit card information. If a data store matches at least one of the data domains in this classification policy, Data Privacy Management classifies the data store as PCI.
PHI	The Protected Health Information (PHI) classification policy contains data domains that identify fields with patient-specific information. If a data store matches at least one of the data domains in this classification policy, Data Privacy Management classifies the data store as PHI.
PII	The Personally Identifiable Information (PII) classification policy contains data domains that identify fields with personal details such as birth date and Social Security number. If a data store matches at least one of the data domains in this classification policy, Data Privacy Management classifies the data store as PII.

PCI Classification Policy

Use the system-defined PCI classification policy to discover and classify data stores that contain credit or debit card details.

The following table lists the data domains and the metadata and data match rules in the PCI classification policy:

Data Domain	Metadata Match Rule	Data Match Rule
AccoutNumber	dataDomain_MetaDataRule_AccountNumber	-
CreditCardNumber	dataDomain_MetaDataRule_CreditCardNumber	dataDomain_DataRule_CreditCardNumber1
ExpirationDate	dataDomain_MetaDataRule_ExpirationDate	dataDomain_DataRule_ExpirationDate
FirstName	dataDomain_MetaDataRule_FirstName	dataDomain_DataRule_FirstName1
LastName	dataDomain_MetaDataRule_LastName	dataDomain_DataRule_LastName1

PHI Classification Policy

Use the system-defined PHI classification policy to discover and classify data stores that contain patient-specific details.

The following table lists the data domains and the metadata and data match rules in the PHI classification policy:

Data Domain	Metadata Match Rule	Data Match Rule
AccountNumber	dataDomain_MetaDataRule_AccountNumber	-
CertificateLicenseNumber	dataDomain_MetaDataRule_CertificateLicenseNumber	-
DeviceSerialNumber	dataDomain_MetaDataRule_DeviceSerialNumber	-
Email	dataDomain_MetaDataRule_Email	dataDomain_DataRule_Email1
PhoneNumber	dataDomain_MetaDataRule_PhoneNumber	dataDomain_DataRule_PhoneNumber1
SSN	dataDomain_MetaDataRule_SSN	dataDomain_DataRule_SSN1
UniqueIdentifyingNumber	dataDomain_MetaDataRule_UniqueIdentifyingNumber	-
URL	dataDomain_MetaDataRule_URL	dataDomain_DataRule_URL

PII Classification Policy

Use the system-defined PII classification policy to discover and classify data stores that contain personal information.

The following table lists the data domains and the metadata and data match rules in the PII classification policy:

Data Domain	Metadata Match Rule	Data Match Rule
BirthDay	dataDomain_MetaDataRule_BirthDay	dataDomain_DataRule_BirthDay1
BirthPlace	dataDomain_MetaDataRule_BirthPlace	-
CompanyName	dataDomain_MetaDataRule_CompanyName	dataDomain_DataRule_CompanyName1
Country	dataDomain_MetaDataRule_Country	dataDomain_DataRule_Country1
CreditCardNumber	dataDomain_MetaDataRule_CreditCardNumber	dataDomain_DataRule_CreditCardNumber1
DrivingLicenseNumber	dataDomain_MetaDataRule_DrivingLicenseNumber	dataDomain_DataRule_DriversLicense
FirstName	dataDomain_MetaDataRule_FirstName	dataDomain_DataRule_FirstName1
Gender	dataDomain_MetaDataRule_Gender	dataDomain_DataRule_Gender
Grade	dataDomain_MetaDataRule_Grade	-
IPAddress	dataDomain_MetaDataRule_IPAddress	dataDomain_DataRule_IPAddress
JobPosition	dataDomain_MetaDataRule_JobPosition	-
LastName	dataDomain_MetaDataRule_LastName	dataDomain_DataRule_LastName1
Salary	dataDomain_MetaDataRule_Salary	-
SSN_General	-	dataDomain_DataRule_USA_SSN_post_2011June
State	dataDomain_MetaDataRule_State	dataDomain_DataRule_State1
Street	dataDomain_MetaDataRule_Street	-
VehicleRegPlateNumber	dataDomain_MetaDataRule_VehicleRegPlateNumber	-
ZipCode	dataDomain_MetaDataRule_ZipCode	dataDomain_DataRule_ZipCode

APPENDIX E

Troubleshooting

This appendix includes the following topic:

- Subject Registry, 309

Subject Registry

If you encounter a problem while creating or updating the Subject Registry configuration, review the following list of previously reported issues and their solutions before contacting Informatica Global Customer Support.

When I select the Discover Subject or Link Subject options from the Scan Type list, I get a warning message.

Data Privacy Management validates the Subject Registry configuration in the following ways:

- Data Privacy Management verifies settings that have a predefined set of values and numeric value settings that are bound by a specific range. For example, Data Privacy Management verifies that the MatchType setting is specified with a valid value such as "Fuzzy." If the setting is not valid, a warning message appears.
- Data Privacy Management verifies references to other settings in the entity file. For example, it verifies that the field name that is specified in the MatchFields section of the entity file, such as FieldName, matches a field that is defined in EntityFields section of the entity file. If Data Privacy Management cannot find a match, a warning message appears.
- Data Privacy Management verifies the configuration against other objects defined in Data Privacy Management. For example, it verifies that the FieldName specified in the EntityFields section of the entity file is a valid data domain name. If the data domain that matches the field name is not present, a warning message appears.

When a warning message appears, review the following Subject Registry configuration settings in the entity file:

- Verify that settings that have predefined values are set to a valid value.
- Verify that numeric value settings are specified within the valid range.
- Verify references to other settings within the entity file.
- Verify that data domain names and other object names are valid.

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